PRIME BOOKS

Humanities

Year 9

Cambridge Lower Secondary

Student Manual

Humanities

Year 9

Cambridge Lower Secondary

Student Book · Prime Books

ABOUT THIS BOOK

One subject, in ten parts. Part one builds your position on the planet: the lines, the coordinates, the climates and the two ways of halving a world. Parts two to ten then run one unbroken causal line from the Industrial Revolution to the world you live in. Thirty-four chapters, 176 numbered topics, and seventeen Portugal in the global story sections, with the answers, a glossary, the sources and an index at the back.

Prime Books is the student book series of Prime School.

First edition, 2026.

Written in British English. Distances are metric, money is in euros and times are given on the 24 hour clock.

This book was written for the pupils of Prime School by the Prime School Humanities Department. Every fact, date and figure in it was checked against a named source before it was printed, and those sources are listed at the back of this book, where you can check them too.

EDITORIAL BOARD

The Prime Books Subject Series has been developed under the guidance of the Pedagogical Academic Group for each subject and coordinated by the Pedagogical Team of Prime School.

This publication reflects our shared commitment to academic excellence, educational quality, professional integrity, and the continuous advancement of teaching and learning. Through the collective expertise and dedication of our educators, we aim to provide meaningful and engaging learning resources that support students' academic growth.

We extend our sincere appreciation to the Pedagogical Department and the Content Creation Team of Prime School for their invaluable contributions, dedication, and collaborative efforts in making this publication possible.

HOW TO CHECK THIS BOOK

Everything in these pages can be checked, and the book is arranged so that you can.

- √ Every **figure, map, graph and table** carries a number, a title and a caption saying what to look for. A figure numbered 7.2 is the second figure of chapter 7.
- √ Every **date, figure and statistic** was verified against a source before it was printed, and every source is named in the Sources section at the back.
- √ Every **square code** leads to a checked English-language resource from a museum, an archive, a national survey or a public database. No code appears twice, and the full web address is printed beside each one, so the page still works without a device.
- √ Every **closed question** has an answer at the back. Open questions have acceptance criteria instead: what a good answer must contain, rather than the words it must use.

If you find something in this book that is wrong, tell your teacher. That is how the next edition gets better, and it is also the habit this subject is trying to teach you.

Introduction: why study Humanities?

This book is about two questions that look separate and are not: where things are, and how they came to be that way. Geography answers the first. History answers the second. Neither answer is complete without the other.

Geography helps you understand where places are, why environments differ, how climate shapes the way people live, why populations are spread as unevenly as they are, how natural resources shape economies, and why location matters.

History helps you understand how societies change, why industry began where it did, why new political ideas appeared, why nations went to war, and how the international system you live in was built.

The two are joined, and the joins are the interesting part. Location, climate, resources, oceans, borders and distance shaped what happened. War, industrialisation, migration, technology and political decisions then reshaped territories, cities and populations. So the course has one central idea, and it runs through every chapter:

> **THE IDEA THIS BOOK IS BUILT ON**
>
> **Place shapes people. People reshape place.**
>
> Or, set out as this book teaches it:
>
> place, environment, resources, people, technology, economy, ideas, power, conflict, change, connection.

Where this book sits in your Humanities course

Year 9 completes a journey you began two years ago.

In **Year 7** you asked: *how do civilisations begin?* You went from the formation of the Earth to continents and oceans, to human settlement, agriculture, cities and the first civilisations.

In **Year 8** you asked: *how do states, empires and networks develop?* From Greek city-states and Alexander, through the Roman republic and empire, to medieval kingdoms, trade networks, stronger monarchies, the Renaissance and maritime expansion.

This year you ask the question those two were leading to:

> **THE YEAR 9 QUESTION**
>
> **How was the modern world created?**

The line this book follows

Part one puts you on the planet. Everything after it is one causal chain, and no link in it is skipped:

Industrial Revolution → technological innovation → industrial capitalism → new political ideologies → nationalism and imperial rivalry → **the First World War** → interwar instability → **the Second World War** → a divided world → **the Cold War** → the nuclear and space races → the end of the Cold War → the modern connected world.

That chain is the reason the book does not jump from the Industrial Revolution to

1 Parts three and four exist so that when you reach the First World War you

can explain **why** it happened, not only when.

Portugal in the global story

At the end of every historical chapter with a Portuguese dimension there is a section called **Portugal in the global story**. It always answers the same four questions:

- √ What was happening in Portugal?
- √ What was happening in the world at the same time?
- √ What part did geography play?
- √ Was Portugal participating, resisting, following later, developing

differently, or influencing events elsewhere?

Portugal is never a separate national history bolted on at the end. It is the same chronology, seen from here.

Contents

Every unit and every numbered topic in this book, with the page it really starts on.

How to use this book

Every page of this book is arranged the same way, and once you know the arrangement you can find anything in it in seconds.

The shape of a chapter

Every chapter opens with a full page that carries five things: an illustration, the chapter number and title, **the question the chapter answers**, what you will be able to do by the end, and the list of topics in the order you meet them. That page also carries the chapter's **square code**.

Then the topics run in sequence. Each one begins with its own label, giving the chapter and the topic number, so you always know where you are.

The panels, and what each is for

PANEL	WHAT IT IS FOR
Before you read	One question from ordinary life, before any teaching
The account	The teaching itself, in continuous prose
Worked enquiry	One question answered slowly, showing the reasoning
Your turn	Numbered practice. Closed questions have answers at the back
Go further	The stretch, for when you want more
From the archive	A verified, sourced fact worth knowing
On the map	The geography inside a history topic, and the reverse
Read the source	A source, with origin, purpose, value and limitation
Careful	The common mistake, named before you make it
The enquiry	The chapter's big question, to be argued rather than answered
What you can now do	Your own check, at the end of every chapter

Table 0.1 · The eleven panels.
Learn the first column. Every one of them appears in every part of the book, and they always do the same job.

The figures

Every map, diagram, graph, timeline and table carries a **number**, a **title** and a caption saying **what to look for**. A figure numbered 12.3 is the third figure of chapter 12. Nothing in a figure of this book was invented: every plotted number came from a source named at the back.

How to use the answers honestly

The answers to every closed question are at the back. Used well they are the most useful pages in the book; used badly they are the most useless.

- √ Attempt the question first, in writing, even if you are unsure.
- √ Then check, and if you were wrong, find out **why** before you move on.
- √ For open questions there is no single answer, so the back of the book gives

acceptance criteria instead: what a good answer has to contain.

The square codes

Each chapter carries one code, and it leads to a checked English-language resource from a museum, an archive, a national survey or a public database. No code appears twice in this book. The full address is printed beside each one, so the page works with no device in the room.

Ask your teacher before you scan. The codes are extension, never a substitute for the pages.

The two centuries at a glance

This is the whole book on one page: the world above, Portugal below. Come back to it whenever you lose your place in the chronology.

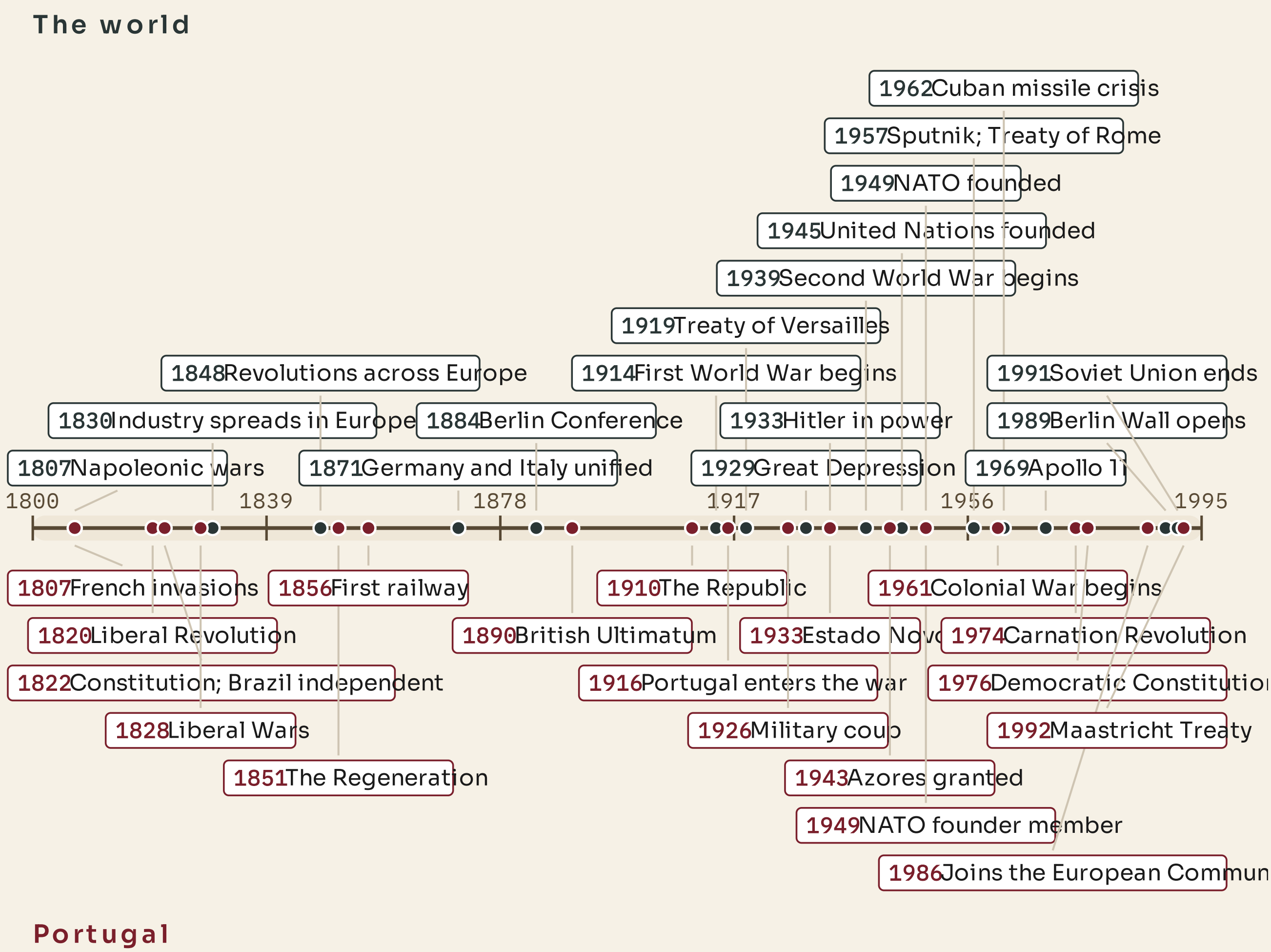


Figure 0.1 · Two centuries, two lines, one chronology.
Read the upper line, then the lower. Then read any year vertically, and ask what was happening in both places at once. You will meet this figure again on the last pages of the book, when you can explain every entry on it.

The skills this year is built on

Six things are being taught alongside the content, and they are what the questions in this book are really testing.

Chronology. Putting events in order, and knowing what was happening at the same time somewhere else.

Cause and consequence. Distinguishing a trigger from a cause, ranking causes, and saying what followed from what.

Source analysis. Asking of any source: who made it, for whom, why, what it shows, and what it had reason **not** to say.

Interpretation. Recognising that historians disagree, and that disagreement is usually about weight rather than about facts.

Map and data skills. Reading a map, a graph and a table, and saying what each does and does not show.

Geographical explanation. Explaining why something happened **where** it happened.

> **THE HABIT THIS BOOK IS TRYING TO BUILD**
>
> When you meet any event in this book, ask two questions in this order. **Where did this happen, and why there?** Then: **what changed because of it?**
>
> If you can do that reliably by the last page, you have learned Humanities rather than a list of dates.

PART ONE

Understanding our global position

Everything in the rest of this book happens somewhere. Before you can explain why industry began where it did, why one country was fought over and another was left alone, or why a small state on the edge of Europe mattered to two world wars, you need to be able to say exactly where anything is and what its position does to it. That is what these four chapters are for.

PART ONE · CHAPTER 1

The global grid

Five lines nobody can see, and what each one decides

THE QUESTION THIS CHAPTER ANSWERS

How do a few imaginary lines let us say exactly where anything on Earth is?

IN THIS CHAPTER, YOU WILL

- √ Say where the Equator, the Tropics and the polar circles are, and why.
- √ Explain why equatorial places are hot all year and polar places are not.
- √ Name the continents the Equator crosses, from a map and then from memory.
- √ Tell the Arctic and Antarctica apart, and say which is ocean and which land.
- √ Explain what a meridian is and why one of them was chosen as zero.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Equator. The line this chapter is built on, and why its position decides how much solar energy the ground below it receives.

`www.britannica.com/science/Equator`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Somebody sends you a message saying only: "Meet me at the beach." In pairs, write down every question you would have to ask before you could actually go. Now do the same for "Meet me at fifteen degrees north, sixty-one degrees west." Which instruction is more useful, and why? Keep your answers. You will come back to them at the end of chapter 2.

CHAPTER 1 · TOPIC 1.1

The Equator

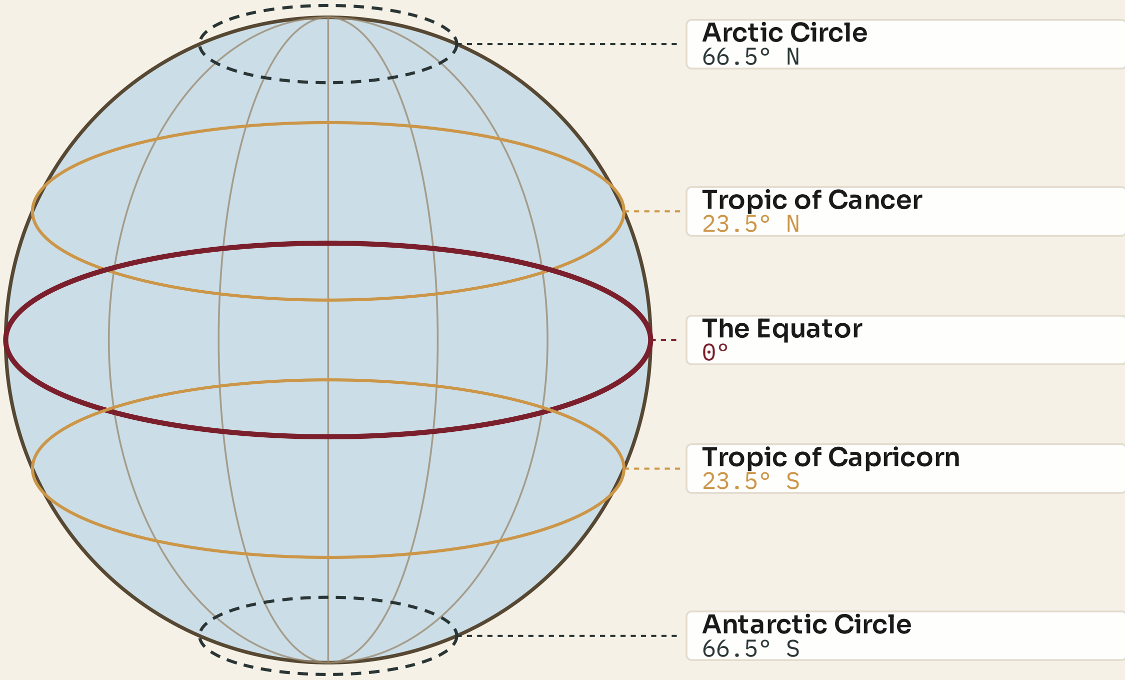
The account

The Equator is a line drawn exactly half way between the North Pole and the South Pole. Its position is not a matter of opinion or agreement: it is the one line the shape of the Earth decides for us. Everywhere on it is the same distance from both poles, and it divides the planet into a northern half and a southern half. Its latitude is zero degrees, and every other latitude on Earth is measured from it.

You cannot see it. There is no mark on the ground, no rope across the ocean, nothing painted on the sea. Yet an enormous amount follows from where it is.

The reason is sunlight. Because the Earth is a ball, the Sun's rays do not strike every part of the surface at the same angle. Near the Equator they arrive close to straight down, so a beam of a given width lands on a small patch of ground and heats it strongly. Near the poles the same beam arrives at a shallow angle and is smeared across a much larger patch, so it heats the ground far less. Nobody near the Equator has hotter sunshine. They have more concentrated sunshine.

Two consequences follow, and both matter for the rest of this book. First, equatorial places are warm all year rather than warm in one season, because their angle to the Sun barely changes. Second, warm air rises, and rising air cools and drops its moistur as rain. That is why a belt of very heavy rainfall, and the rainforests that live on it, sits along the Equator rather than anywhere else.



Five lines nobody can see, and every one of them is a statement about sunlight

Figure 1.1 · The five named lines, and the angles that put them there.
Follow the Equator round first. Notice that it is the only one of the five that is a complete circle at full size, and that the four others are placed by the tilt of the Earth's axis rather than chosen.

The Equator crosses three continents: South America, Africa and Asia, if the islands of Indonesia are counted as Asia. It also crosses a great deal of ocean. Thirteen countries have territory on it, and the list is worth looking at because it is a list of places whose climate, farming and history are all shaped by their position.

CAREFUL

The Equator is the hottest **belt** on average, but it is not where the highest air temperature has ever been measured. The record heat is found in the desert belts near the Tropics, where there is no cloud to shade the ground and no daily rain to cool it. Concentrated sunlight and extreme heat are not quite the same thing, and the difference is explained by rainfall.

YOUR TURN

1. In one sentence, say what the Equator is, without using the word "middle".
2. Explain why sunlight is more concentrated at the Equator than at the poles.
   Draw the two beams if it helps.
3. Name the three continents the Equator crosses.
4. A pupil writes: "It is hot at the Equator because it is closer to the Sun."
   The distance from Earth to the Sun is about 150 million kilometres, and the Earth's radius is about 6 400 kilometres. Use those two figures to explain why the pupil is wrong.

CHAPTER 1 · TOPIC 1.2

The Tropics

The account

The Earth spins on an axis, and that axis is tilted. It leans at about twenty-three and a half degrees away from upright, and, crucially, it keeps leaning the same way all year as the Earth travels round the Sun. Nothing about the tilt changes. What changes is which half of the planet happens to be leaning toward the Sun at that point in the orbit.

That single fact produces the seasons, and it also produces two more named lines.

The Tropic of Cancer sits at about twenty-three and a half degrees north. It marks the furthest north that the Sun can ever be directly overhead. The Tropic of Capricorn sits at the same angle south, and marks the furthest south. Between those two lines, there is at least one day each year when the Sun is exactly overhead at noon and an upright post casts no shadow at all. Outside them, that never happens, anywhere, ever.

Notice that the number is not a coincidence. The Tropics are at twenty-three and a half degrees because the axis is tilted by twenty-three and a half degrees. The lines are a direct measurement of the lean.

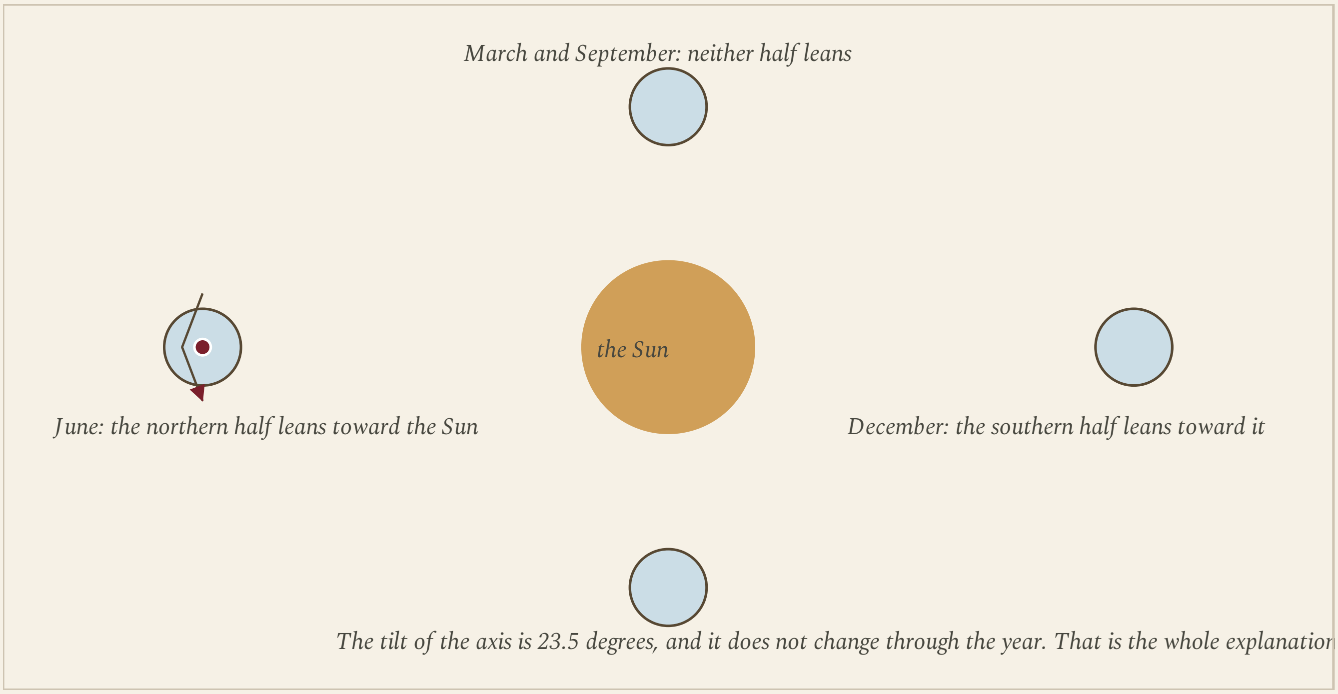


Figure 1.2 · The tilt, and why it makes seasons.
The orbit is drawn round rather than stretched, because the seasons are not caused by the Earth being nearer the Sun at some times of year. Follow the axis: it points the same way at all four positions.

The belt between the Tropics is called the tropical zone, and it is the part of the Earth that receives the most solar energy over a year. It is not uniform. It contains rainforest, savanna and some of the driest desert on the planet, because rainfall varies within it even where temperature does not. Chapter 3 takes that apart properly.

ON THE MAP

Find the Tropic of Cancer on a world map and follow it east from the Atlantic. Note the countries it passes through in north Africa and Asia. Now do the same for the Tropic of Capricorn in the southern hemisphere. Write one sentence saying what the land along both lines has in common.

YOUR TURN

1. What is the tilt of the Earth's axis, and what does it stay doing all year?
2. Explain, in your own words, what the Tropic of Cancer marks.
3. Why is the Tropic of Capricorn at the same angle as the Tropic of Cancer?
4. In which of these places can the Sun be directly overhead at noon: Lisbon, Nairobi, Sydney, Singapore? Use the latitudes to decide, and say how you did it.

CHAPTER 1 · TOPIC 1.3

The polar circles

The account

The same tilt that puts the Tropics where they are also puts two further lines near the poles. The Arctic Circle lies at about sixty-six and a half degrees north; the Antarctic Circle at about sixty-six and a half degrees south. The number is simply ninety minus twenty-three and a half.

What these lines mark is not temperature but daylight. On or inside the Arctic Circle there is at least one day a year when the Sun does not set at all, and at least one when it does not rise. The further in you go, the more of those days there are, until at the pole itself the year is one very long day followed by one very long night. The same is true, in the opposite months, inside the Antarctic Circle.

People who live there call the summer version the midnight Sun. In northern Norway, Sweden, Finland and Russia, and across northern Canada, Alaska and Greenland, there are weeks in June when the Sun circles the sky without touching the horizon, and weeks in December of polar night, when it never clears it.

It is worth pausing on how strange this is if you have never seen it. The question "what time does it get dark?" has no answer in Tromso in June, and no answer in December either.

	The Arctic	Antarctica
What it is	An ocean, largely surrounded by continents.	A continent, entirely surrounded by ocean.
What is underfoot	Floating sea ice over deep water. There is no land at the North Pole.	Rock, buried under an ice sheet kilometres thi
Who lives there	Peoples have lived around it for thousands of years, in eight countries.	Nobody permanently. Only research stations, s in rotation.
Who governs it	Eight states have Arctic territory and coastline.	No state owns it. A treaty of 1959 reserves it for peaceful use.
Why the difference matters	Sea ice thins and moves, so the map of the Arctic changes with the season.	An ice sheet on rock stores water that would otherwise be sea.

Figure 1.3 · The Arctic and Antarctica are not two versions of the same place.
Read the rows across rather than down. The difference in the first row explains almost every difference in the rows below it.

The two polar regions are often treated as a matching pair, and they are not. The Arctic is an ocean with sea ice floating on it, ringed by the northern edges of three continents, and people have lived around it for thousands of years. Antarctica is a continent of rock buried under an ice sheet kilometres thick, ringed by ocean, with no permanent population at all. That difference is geography deciding history: one is a place with nations, towns and disputes over shipping routes, the other is governed by a treaty because nobody lives there to govern.

YOUR TURN

1. At what latitude does each polar circle sit, and where does that number come from?
2. Explain the midnight Sun to somebody who has never left Portugal.
3. Give two differences between the Arctic and Antarctica.
4. Why does Antarctica have no permanent population when the Arctic has had one for thousands of years? Give a geographical reason, not just a cold one.

CHAPTER 1 · TOPIC 1.4

Meridians

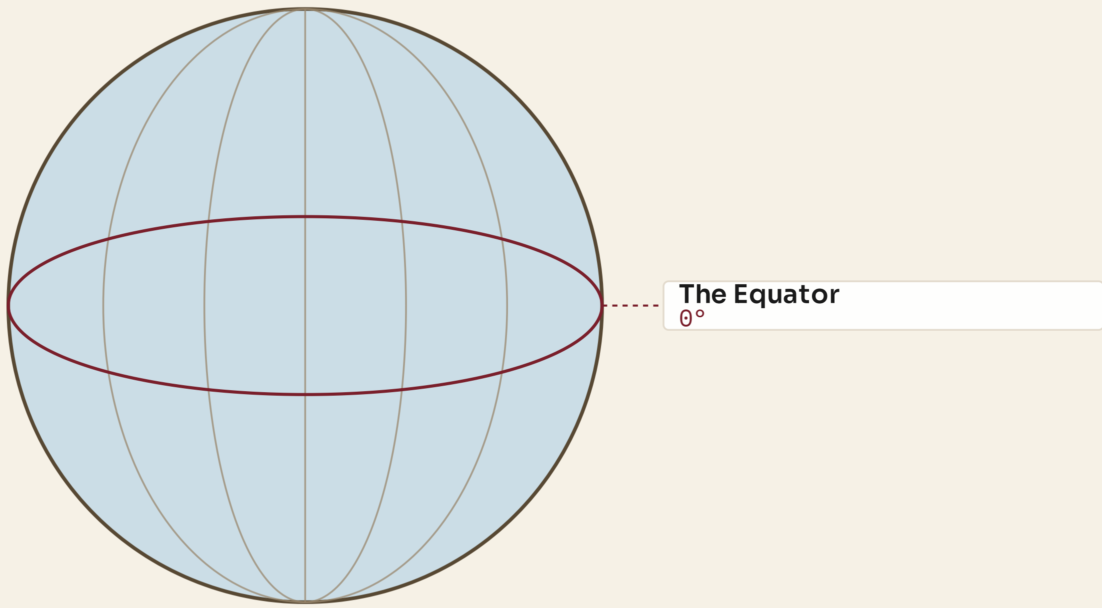
The account

Latitude alone cannot find a place. If you know only that somewhere is at forty degrees north, you know it sits on a circle running right round the planet, and that circle passes through Portugal, Spain, Italy, Greece, Turkey, China, Japan and the United States. You need a second measurement, running the other way.

That measurement is longitude, and the lines that carry it are called meridians. A meridian runs from the North Pole to the South Pole, crossing every parallel of latitude at a right angle. Every meridian is the same length as every other meridian, because each is half of a circle round the whole Earth. Parallels are not like that at all: the Equator is a full-sized circle, but a parallel near the pole is a small one.

Here is the difficulty. Latitude has a natural zero, because the Equator is fixed by the shape of the planet. Longitude has no natural zero whatsoever. One meridian is exactly as good as another. Somebody has to choose.

For centuries, different countries chose differently, and each printed charts measured from its own capital. A French chart measured from Paris, a Spanish chart from Cadiz, a Portuguese chart from Lisbon. A navigator carrying charts from two countries was carrying two incompatible grids, which is precisely the kind of problem that sinks ships.



Meridians all run pole to pole and are all the same length. Parallels are not: only the Equator is a full circle.

Figure 1.4 · Meridians all run pole to pole, and all are the same length.
Compare them with the parallels in the first figure of this chapter. Notice that meridians converge: they are furthest apart at the Equator and meet at the poles.

FROM THE ARCHIVE

In October 1884, delegates from twenty-five countries met in Washington at the International Meridian Conference and voted to adopt a single prime meridian for the world. Greenwich won largely because most of the world's shipping was already using charts measured from it. The vote was twenty-two in favour, one against, and two abstentions. France abstained and went on using Paris in its own charts for decades.

■ **YOUR TURN**

1. What is a meridian, and how is it different from a parallel?
2. Explain why latitude has a natural zero and longitude does not.
3. Why was it a practical problem for navigation that countries used different prime meridians?
4. Suggest one argument that a French delegate in 1884 might have made for Paris. Then say why it lost.

CHAPTER 1 · TOPIC 1.5

The Prime Meridian

The account

The line the 1884 conference chose runs through the Royal Observatory at Greenwich, in London, and it is called the Prime Meridian. Its longitude is zero degrees. Everything east of it is measured in degrees east, up to one hundred and eighty; everything west of it in degrees west, up to the same figure. The two meet on the far side of the planet at the one hundred and eightieth meridian, which is a single line that can be called either.

Choosing a prime meridian solved three problems at once.

It solved **navigation**, because every chart could now be drawn on the same grid, and a position reported by one ship meant the same thing to another.

It solved **mapping**, because a survey made in one country could be joined to a survey made in the next without a correction being applied by hand.

It solved **time**, and this is the part that reaches furthest into everyday life. Once the world agreed on a zero meridian, it could agree on a zero for clocks. Chapter 2 follows that through, but the link is worth seeing now: the reason your phone knows what time it is in Tokyo runs back to a decision about where to draw a line.

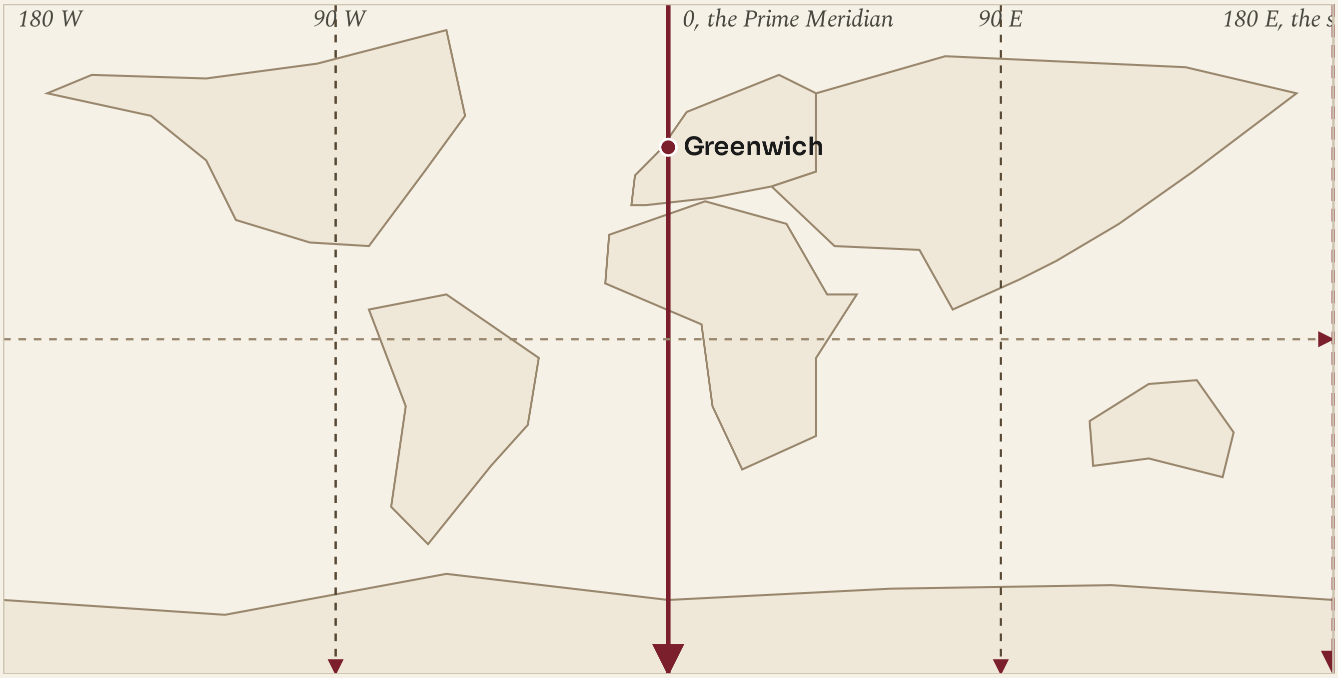


Figure 1.5 · The grid, and the one line somebody had to choose.
Follow the Prime Meridian from Greenwich down through western Europe, Africa and the Atlantic. Note how much of the map lies east of it, and how much west, and that this is an accident of the choice rather than a fact about the Earth.

Portugal sits just west of the Prime Meridian. Lisbon is at about nine degrees west, so mainland Portugal lies wholly in the western hemisphere as geographers define it, although it is in western Europe politically and culturally. Chapter 4 comes back to why those two statements are not the same statement.

WHAT YOU CAN NOW DO

- √ Say where the Equator is and explain why it decides so much about climate.
- √ Explain why the Tropics and the polar circles sit where they do.
- √ Describe the midnight Sun and the polar night, and say who experiences them.
- √ Give two real differences between the Arctic and Antarctica.
- √ Explain what a meridian is and why the world had to choose one as zero.

WORDS FOR THIS CHAPTER

Equator, latitude, longitude, meridian, parallel, axis, tilt, Tropic of Cancer, Tropic of Capricorn, Arctic Circle, Antarctic Circle, midnight Sun, polar night, Prime Meridian, hemisphere

CHAPTER 1 AT A GLANCE

The global grid

how do a few imaginary lines let us say exactly where anything on Earth is?

THE TOPICS YOU HAVE COVERED

1.1	The Equator
1.2	The Tropics
1.3	The polar circles
1.4	Meridians
1.5	The Prime Meridian

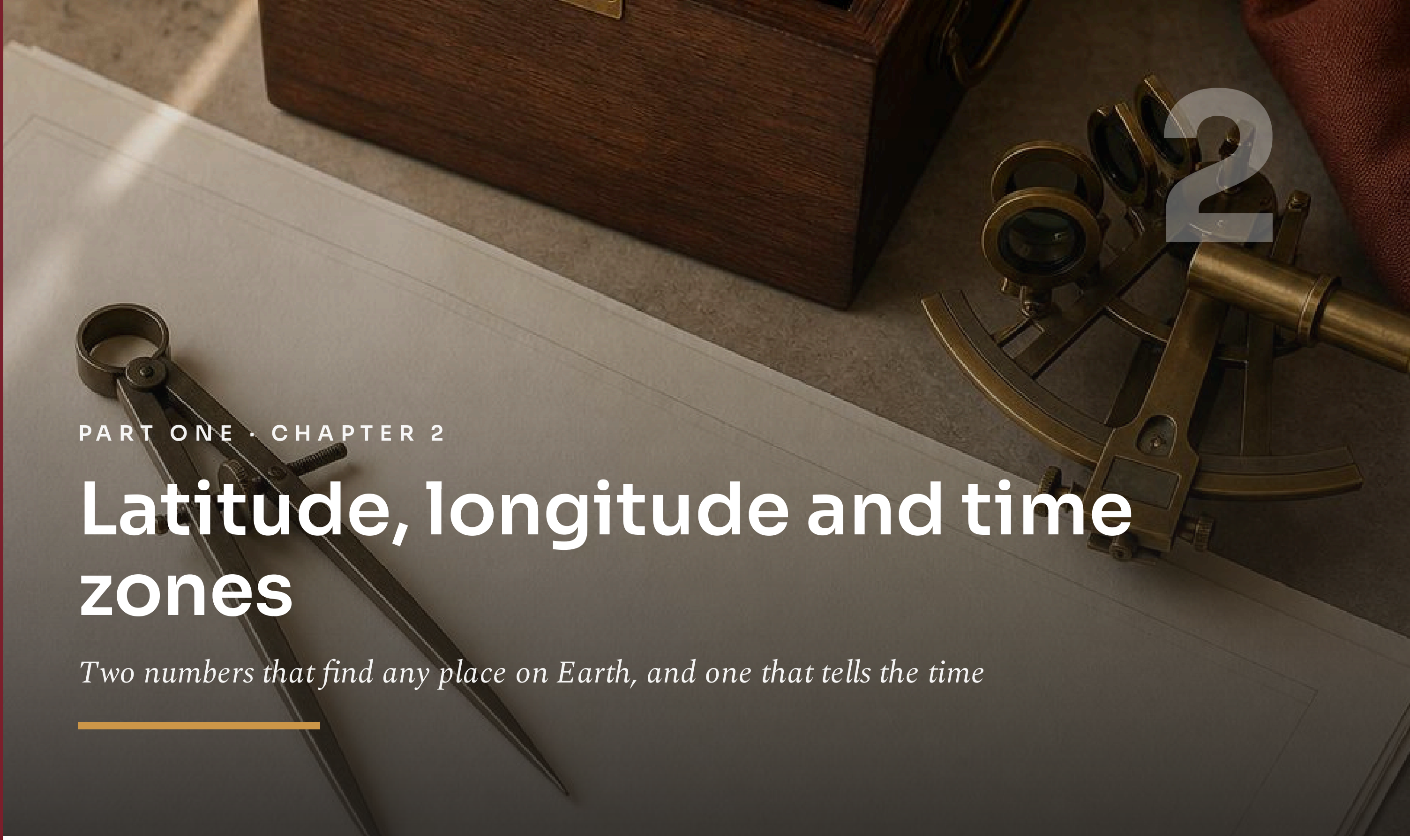
WHAT YOU CAN NOW DO

- √ Say where the Equator is and explain why it decides so much about climate.
- √ Explain why the Tropics and the polar circles sit where they do.
- √ Describe the midnight Sun and the polar night, and say who experiences them.
- √ Give two real differences between the Arctic and Antarctica.
- √ Explain what a meridian is and why the world had to choose one as zero.

THE WORDS THIS UNIT TAUGHT

Equator · latitude · longitude · meridian · parallel · axis · tilt · Tropic of Cancer · Tropic of Capricorn · Arctic Circle · Antarctic Circle · midnight Sun · polar night · Prime Meridian · hemisphere

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.



2

PART ONE · CHAPTER 2

Latitude, longitude and time zones

Two numbers that find any place on Earth, and one that tells the time

THE QUESTION THIS CHAPTER ANSWERS

If the Earth turns once in twenty-four hours, what does that do to where a place is and to what time it is there?

IN THIS CHAPTER, YOU WILL

- √ Read and write a position as latitude and then longitude.
- √ Find a named place from its coordinates, and give the coordinates of one.
- √ Explain why the Earth's rotation makes noon happen at different moments.
- √ Work out the time in another zone from the difference in hours.
- √ Explain why crossing the International Date Line changes the date.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Royal Observatory Greenwich, Royal Museums Greenwich. The observatory the Prime Meridian runs through, and the reason the world agreed to measure longitude and time from this one place.

www.rmg.co.uk/royal-observatory

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Your teacher will read out a pair of numbers. Using an atlas, find the place they name. Then, without telling anybody, choose a place yourself and write down its numbers for somebody else to find. If they end up somewhere else, work out together which of you made the mistake, and what kind of mistake it was.

CHAPTER 2 · TOPIC 2.1

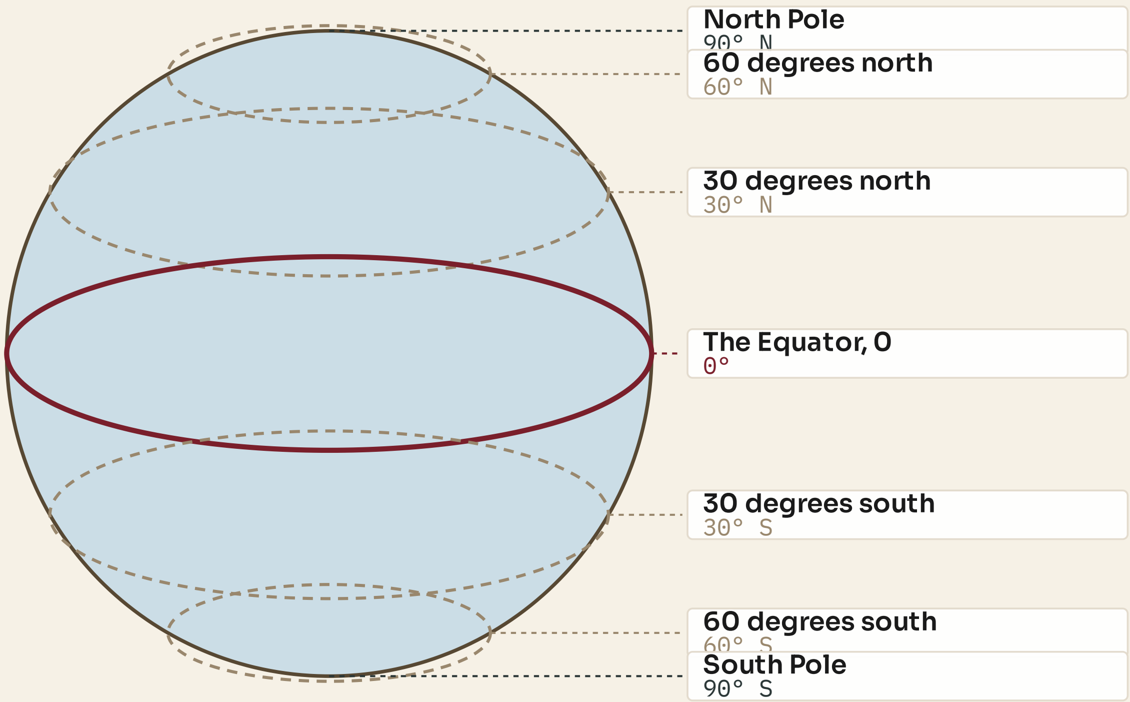
Latitude

The account

Latitude measures how far north or south of the Equator a place is. It is an angle, not a distance, and the angle is measured at the centre of the Earth between the Equator and the place itself.

The scale runs from zero at the Equator to ninety degrees at each pole, and it never goes further, because ninety degrees is already straight up. Every value carries a direction, north or south, and the direction is part of the reading. Forty degrees north and forty degrees south are two entirely different places with, as it happens, opposite seasons.

Lines of equal latitude are called parallels, and the name is accurate: they never meet. They do, however, get smaller as they approach the poles, because they are circles drawn round a ball rather than round a cylinder.



Latitude runs from 0 at the Equator to 90 at each pole, and never further

Figure 2.1 · The latitude scale, from nought to ninety.
Notice that the parallels are equally spaced in angle but not in the width of the circle they draw. The Equator is a full-sized circle; the parallel at sixty degrees is half the width.

Latitude is the single most useful number in physical geography, because it predicts so much. It tells you roughly how much solar energy a place receives across a year, which tells you roughly how warm it is, which goes a long way toward telling you what will grow there. The general rule is simple: the lower the latitude, the more direct the solar energy, and the warmer the place is likely to be.

It is a rule with exceptions, and the exceptions are interesting rather than inconvenient. Altitude changes it, because air cools as it rises: Quito sits almost on the Equator and is cool all year because it is high in the Andes. Ocean currents change it, because moving water carries heat: western Europe is far milder than its latitude alone would suggest. You will meet both again in chapter 3.

CAREFUL

Latitude is written before longitude, always, and each carries its letter. "38.7 N, 9.1 W" is Lisbon. "9.1 N, 38.7 E" is a spot in Ethiopia. The numbers are the same; the order and the letters are the whole meaning.

YOUR TURN

1. What does latitude measure, and between what two limits?
2. Why do parallels get smaller toward the poles when meridians do not?
3. Give the general rule linking latitude and temperature, then give one exception and explain it.
4. Two places are both at forty-two degrees. One is warm in July, the other cold. Explain how both can be true.

CHAPTER 2 · TOPIC 2.2

Longitude

The account

Longitude measures how far east or west of the Prime Meridian a place is. Like latitude it is an angle, and like latitude it carries a direction, but its scale runs to one hundred and eighty degrees rather than ninety, because it has to cover a full circle rather than a half one.

Going east from Greenwich, the readings rise to one hundred and eighty degrees east. Going west, they rise to one hundred and eighty degrees west. Both journeys arrive at the same meridian on the opposite side of the planet, which is why that line can be written either way.

The practical difficulty with longitude, for most of history, was not understanding it but measuring it. Latitude can be found from the height of the Sun at noon or of the Pole Star at night, with an instrument and a table, and sailors could do it reasonably well by the fifteenth century. Longitude cannot be read off the sky in the same way. It has to be worked out by comparing the local time where you are with the time at a known meridian, which means carrying an accurate clock through months of damp, heat, cold and pitching decks.

That problem was severe enough that governments offered prizes for solving it. It was solved in the eighteenth century by marine chronometers accurate enough to keep reference time on a long voyage. Notice what that means: **longitude and time are the same problem.** The next five topics are all consequences of that one sentence.

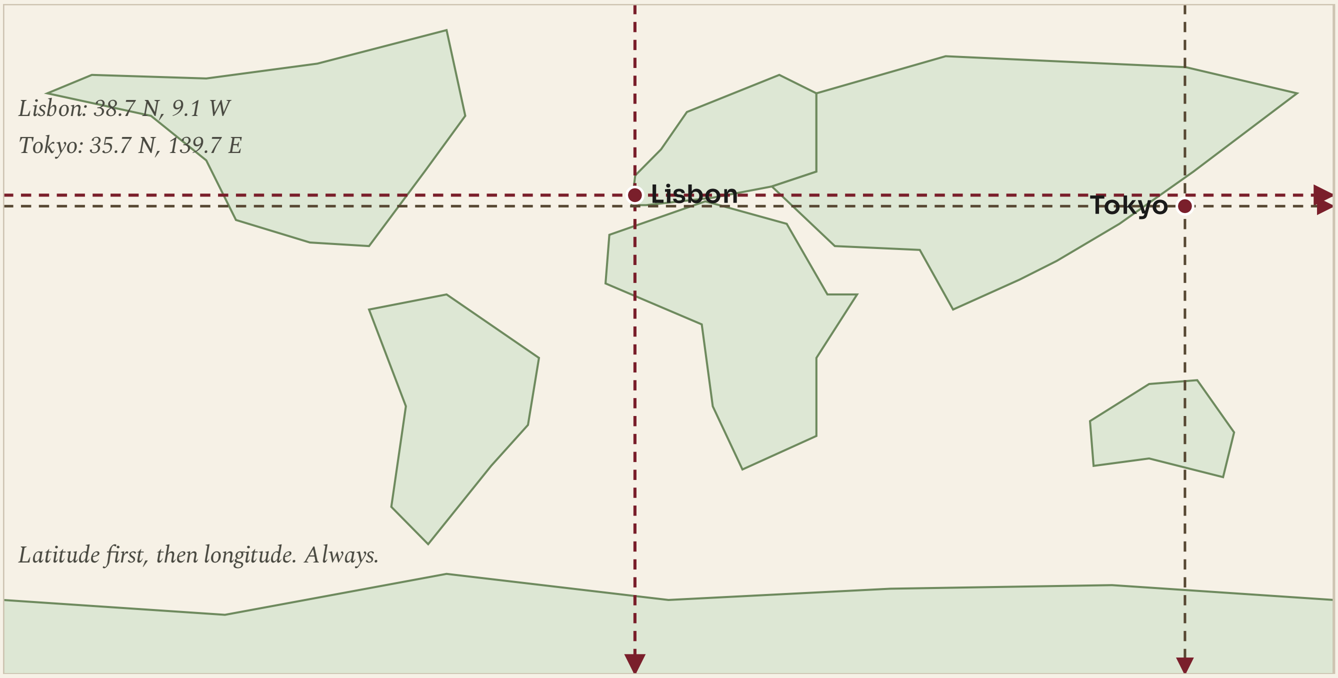


Figure 2.2 · Two numbers, one place.
Follow the parallel and the meridian for each city until they cross. The crossing is the only point on Earth with that pair of readings.

YOUR TURN

1. What does longitude measure, and what is its maximum value?
2. Why can the one hundred and eightieth meridian be described as either east or west?
3. Explain why finding longitude at sea was so much harder than finding latitude.
4. In one sentence, explain the link between longitude and time.

CHAPTER 2 · TOPIC 2.3

Absolute location

The account

Put the two measurements together and you have an absolute location: a pair of numbers that names one point on the surface of the Earth and no other. Latitude first, then longitude, each with its direction.

An absolute location does not depend on anything else being known. That is what makes it different from a relative location, which describes a place by its relationship to another place: "twenty kilometres north of Coimbra", "opposite the school", "the third house past the church". Relative locations are how people actually talk, and they are useless to somebody who does not already know the reference point.

Some absolute locations worth knowing, and worth checking in an atlas rather than taking on trust:

PLACE	LATITUDE	LONGITUDE
Lisbon	38.7 N	9.1 W
Porto	41.1 N	8.6 W
Funchal, Madeira	32.6 N	16.9 W
Ponta Delgada, Azores	37.7 N	25.7 W
London	51.5 N	0.1 W
New York	40.7 N	74.0 W
Tokyo	35.7 N	139.7 E
Sydney	33.9 S	151.2 E

Table 2.1 · Eight places, sixteen numbers.
Read down the latitude column first: four of these cities sit within four degrees of each other and are scattered across three continents. Latitude alone is never enough.

Look carefully at the Portuguese entries. Lisbon and Porto differ by less than three degrees of latitude, which is a small difference, and yet chapter 3 will show you that their climates differ noticeably. The Azores are more than sixteen degrees of longitude west of Lisbon, which is why they keep a different clock from the mainland, and, as chapter 18 will show, why they mattered enormously in a war fought across the Atlantic.

YOUR TURN

1. Define absolute location and relative location, and give an example of each for your own school.
2. Using the table, which two of the eight places are furthest apart in longitude?
3. Ponta Delgada is at 25.7 W and Lisbon at 9.1 W. How many degrees of longitude separate them?
4. Why might a rescue service insist on an absolute location rather than a description?

CHAPTER 2 · TOPIC 2.4

The Earth's rotation

The account

The Earth turns once on its axis in about twenty-four hours, and it turns from west to east. Everything in this topic follows from those two facts.

Because the Earth turns eastward, the Sun appears to move westward across the sky. It rises in the east, because that side of you is turning to face it, and sets in the west, because that side is turning away. The Sun is not going anywhere. You are.

At any instant, the Sun lights exactly half the planet. The line between the lit half and the dark half sweeps steadily westward across the surface, and as it passes a place, that place has dawn or dusk. Noon happens at a place when the Sun is highest in its sky, and that moment arrives at different instants in different places, because they are not all facing the Sun at once.

That is the whole origin of time differences. It is not a convention imposed on the world. It is a description of a spinning ball.

The Earth turns
One full turn on its axis takes about 24 hours, west to east.

→

Half is lit
At any moment the Sun light one half and the other half is in darkness.

→

Noon moves
The Sun is highest at different moments in different places, so noon travels west.

→

Clocks follow
Rather than every town keeping its own time, we agree zones about an hour wide.

Figure 2.3 · From one rotation to four different clocks.
Read the four boxes in order. The first three are physics; only the fourth is a decision people made.

Before railways, this caused nobody any trouble. Every town simply set its clocks so that noon was noon there, and since travelling between two towns took hours or days, a difference of a few minutes was invisible. Railways destroyed that. A train timetable has to say when the train leaves, and if every station keeps its own time the timetable means nothing. Britain adopted a single railway time in the 1840s, and the rest of the industrial world followed. The next topic is what they settled on.

YOUR TURN

1. In which direction does the Earth turn, and how does that explain sunrise in the east?
2. What fraction of the Earth is lit at any instant?
3. Explain why noon does not happen everywhere at once.
4. Why did railways make local time impossible to keep?

CHAPTER 2 · TOPIC 2.5

Time zones

The account

If the Earth turns three hundred and sixty degrees in twenty-four hours, then it turns fifteen degrees in one hour. That single division is the basis of the whole system:

360 degrees ÷ 24 hours = 15 degrees per hour

So the world is divided into zones roughly fifteen degrees of longitude wide, and all the clocks inside a zone are set to the same time. Move one zone east and clocks read one hour later; one zone west, one hour earlier.

Roughly is doing real work in that sentence. Zone boundaries are drawn by governments, not by geometry, and they bend around borders, islands and convenience. Some countries that are wide enough for several zones keep only one, so that the whole country shares a working day. Some zones are half an hour or three quarters of an hour off the pattern.

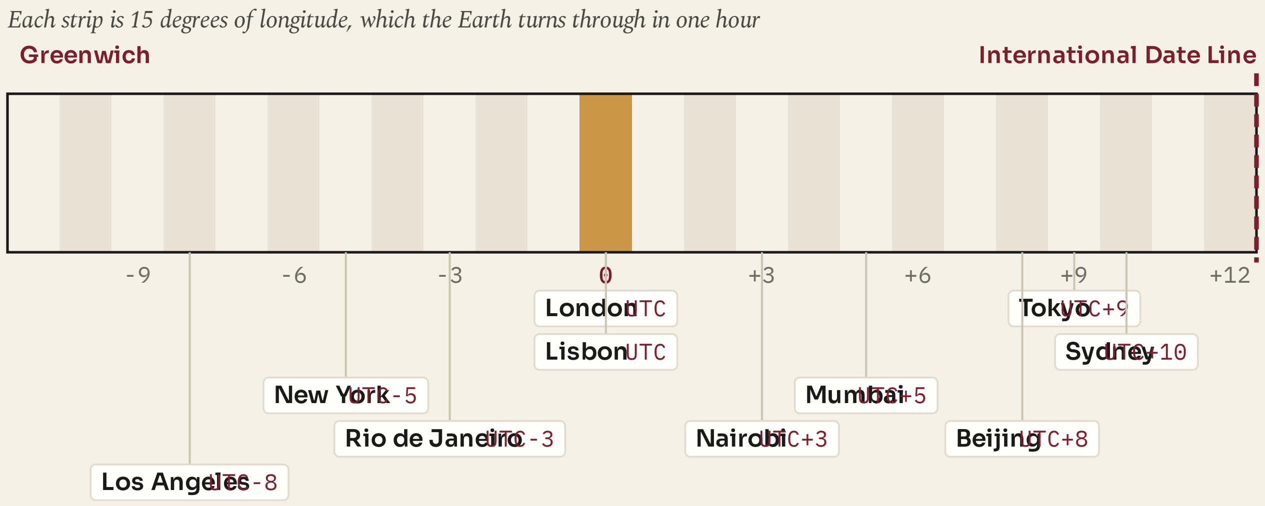


Figure 2.4 · The zones, and ten cities placed in the one their clocks really keep.
Read the offsets along the bottom. Then check a city you know against the strip it is standing in.

Here is the standard piece of arithmetic, done slowly.

◆ **WORKED ENQUIRY**

It is 12:00 in London. What time is it in Tokyo, New York and Sydney?

Work from the offsets, not from a feeling about which is "ahead".

- London in winter keeps UTC, so its offset is 0.
- Tokyo keeps UTC+9. The difference is 9 minus 0, which is +9. So 12:00 plus 9

hours is **21:00 in Tokyo**, the same day.

- New York keeps UTC−5. The difference is −5 minus 0, which is −5. So 12:00

minus 5 hours is **07:00 in New York**, the same day.

- Sydney keeps UTC+10 in its winter. The difference is +10, so **22:00 in

Sydney**.

And Lisbon, which keeps UTC in winter like London, reads **12:00**: the same as London, and one hour behind Madrid, which is only a short distance east but keeps UTC+1.

◆ **CAREFUL**

Many countries move their clocks forward in summer, so the offsets above change for part of the year. Always check whether summer time is in force before you answer, and say which you have assumed. A question that does not tell you should be answered for standard time, with a note.

YOUR TURN

1. Show the calculation that gives fifteen degrees per hour.
2. It is 09:00 in Lisbon in January. What time is it in Tokyo? Show your working.
3. It is 18:00 in New York. What time is it in London? Show your working.
4. Give two reasons why real zone boundaries are not straight lines.

CHAPTER 2 · TOPIC 2.6

Coordinated Universal Time

The account

If every clock is set relative to something, that something has to be defined very carefully. Today it is Coordinated Universal Time, written UTC, and it is the reference every other zone is described against.

UTC is not a time zone. That is the point most often missed. A time zone is where people live and what their clocks say; UTC is a standard, kept by a large number of atomic clocks in laboratories around the world, whose readings are averaged to produce a single scale. Every national time is then published as an offset from it: UTC+1, UTC−5, UTC+5:30.

It replaced Greenwich Mean Time as the world's reference, and the two are not quite the same thing, although they are close enough that people use the names interchangeably in ordinary speech.

	Greenwich Mean Time (GMT)	Coordinated Universal Time (UTC)
What it is	A time zone, kept by clocks in Britain in winter.	A time standard, kept by atomic clocks worldw
Where it comes from	The Sun crossing the meridian at Greenwich.	The average of hundreds of atomic clocks, corr to stay close to the Sun.
Used for	Everyday time in one country.	Aviation, shipping, computing and science everywhere.
In practice	Britain moves to summer time and leaves GMT.	UTC never moves. Offsets from it change inste

Figure 2.5 · One is a place's clock, the other is the world's ruler.
Read the last row first: it is the difference that actually matters in practice.

Greenwich still matters, and not only for history. The Prime Meridian is still at zero, longitude is still measured from it, and UTC is still kept close to the Sun's position over that meridian. When the Earth's rotation drifts very slightly out of step with atomic time, a leap second can be added to keep them together. The world runs on atomic clocks, but it still checks them against the sky.

ON THE MAP

Portugal keeps UTC in winter and UTC+1 in summer. Spain, immediately to the east, keeps UTC+1 and UTC+2. Look at where the two countries sit in longitude and explain why Portugal's choice fits the Sun better than Spain's does. Then suggest a reason a country might choose a zone that does not fit the Sun.

YOUR TURN

1. Explain the difference between a time standard and a time zone.
2. What keeps UTC accurate, and what keeps it close to the Sun?
3. Why is Portugal's winter offset zero?
4. A pilot flying from Lisbon to Rio files a flight plan in UTC rather than local time. Give two reasons why.

CHAPTER 2 · TOPIC 2.7

The International Date Line

The account

Follow the zones eastward from Greenwich and clocks run forward, hour by hour, until at one hundred and eighty degrees they are twelve hours ahead. Follow them westward and clocks run back, until at the same meridian they are twelve hours behind. Twelve hours ahead and twelve hours behind are twenty-four hours apart, which is a whole day.

So a line has to exist where the date changes, and by agreement it runs approximately along the one hundred and eightieth meridian. Cross it going west and you skip forward a day: Monday becomes Tuesday. Cross it going east and you repeat one: Tuesday becomes Monday again.

Approximately, again, is doing real work. The line bends, sometimes by hundreds of kilometres, to keep island groups and countries on one date rather than splitting them in half. A country that straddled the line would have neighbours whose weekends did not match, which is exactly the kind of thing governments move lines to avoid.

Cross the date line going west and you lose a day. Come back east and you get it again.

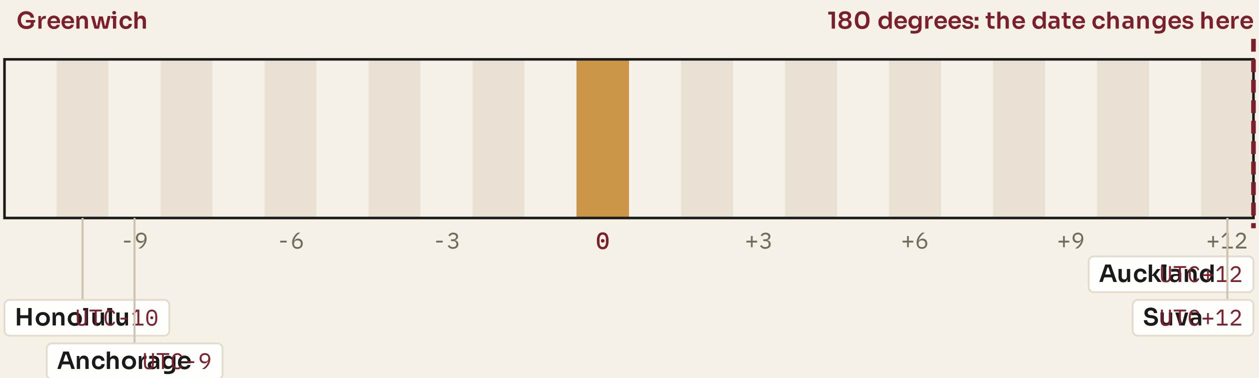


Figure 2.6 · Where the date changes, and four places that live with it.
Notice that the line is not the same thing as a time-zone boundary. Every zone boundary changes the hour; only this one changes the day.

FROM THE ARCHIVE

In 2011 Samoa moved from the eastern side of the line to the western side, to share working days with Australia and New Zealand rather than with the United States. To do it, the country skipped 30 December 2011 entirely. For Samoa that date does not exist: the calendar ran from Thursday 29 December straight to Saturday 31 December.

YOUR TURN

1. Explain, using the twelve-hour figures, why a date line has to exist.
2. What happens to the date when you cross the line travelling west?
3. Why does the line bend rather than run straight down the meridian?
4. Look again at the message task you did before this chapter. Rewrite "Meet me at the beach" as an instruction that would work anywhere in the world, and say what each part of it does.

WHAT YOU CAN NOW DO

- √ Read and write any position as latitude and then longitude, with directions.
- √ Explain why the Earth's rotation makes noon arrive at different moments.
- √ Calculate the time in another zone from its offset, and show the working.
- √ Explain the difference between UTC and a time zone.
- √ Explain why the International Date Line exists and why it is not straight.

WORDS FOR THIS CHAPTER

latitude, longitude, absolute location, relative location, rotation, axis, time zone, offset, Coordinated Universal Time, Greenwich Mean Time, summer time, International Date Line, meridian, parallel, chronometer

CHAPTER 2 AT A GLANCE

Latitude, longitude and time zones

if the Earth turns once in twenty-four hours, what does that do to where a place is and to what time it is there?

THE TOPICS YOU HAVE COVERED

2.1	Latitude
2.2	Longitude
2.3	Absolute location
2.4	The Earth's rotation
2.5	Time zones
2.6	Coordinated Universal Time
2.7	The International Date Line

WHAT YOU CAN NOW DO

- √ Read and write any position as latitude and then longitude, with directions.
- √ Explain why the Earth's rotation makes noon arrive at different moments.
- √ Calculate the time in another zone from its offset, and show the working.
- √ Explain the difference between UTC and a time zone.
- √ Explain why the International Date Line exists and why it is not straight.

THE WORDS THIS UNIT TAUGHT

latitude · longitude · absolute location · relative location · rotation · axis · time zone · offset · Coordinated Universal Time · Greenwich Mean Time · summer time · International Date Line · meridian · parallel · chronometer

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART ONE · CHAPTER 3

Climate zones and biomes

Why the living world changes as you move away from the Equator

THE QUESTION THIS CHAPTER ANSWERS

Why is the living world so different from one latitude to another?

IN THIS CHAPTER, YOU WILL

- √ Tell weather, climate and biome apart, and use each word correctly.
- √ Explain the chain from latitude through solar energy to vegetation.
- √ Read a climate graph and name the zone it belongs to.
- √ Describe the tropical, temperate, desert and polar zones and compare them.
- √ Give examples of climate shaping farming, building and travel.

THE TOPICS, IN THE ORDER YOU MEET THEM

3.1	Latitude and climate
3.2	Tropical zones
3.3	Temperate zones
3.4	Desert regions
3.5	Polar zones
3.6	Climate and human activity

WATCH, READ AND LISTEN

Encyclopaedia Britannica, biomes. A world biome map with each type described. Check it against the chain this chapter teaches, from latitude to vegetation.

www.britannica.com/science/biome

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Think of somewhere you have been where the plants were obviously different from the plants at home. Describe it to a partner without naming the place: say what grew there, how it felt to stand outside, and what people wore. Then see whether your partner can say roughly what latitude you were at. This chapter is about why that guess is possible at all.

CHAPTER 3 · TOPIC 3.1

Latitude and climate

The account

Three words get used as if they meant the same thing, and they do not.

Weather is what the atmosphere is doing now, or this week. It changes by the hour. It is forecast, and the forecast gets less reliable the further ahead it reaches.

Climate is the pattern that weather makes over a long period, usually taken as thirty years or more. It is not forecast; it is described. Saying "the climate of the Algarve is hot and dry in summer" is a statement about hundreds of summers, and it is not contradicted by one wet July.

Biome is a large region of the Earth defined by its climate, the vegetation that climate allows, and the animals that live in that vegetation. Rainforest is a biome. Tundra is a biome. Portugal is not a biome, because a country is a political unit and a biome is an ecological one.

Weather
What the atmosphere is doing now, or this week. It changes hour by hour and is forecast, not predicted.

Climate
The pattern that weather makes over thirty years or more. It is described, not forecast.

Biome
A large region defined by its climate, its vegetation and the animals that live in it.

Figure 3.1 · Three words, three different jobs.
The commonest mistake in this chapter is using the first word when you mean the second. Weather changes; climate is the pattern; a biome is the living result.

Now the chain that this whole chapter is built on. It starts with position and ends with people, and every link is a cause of the next.

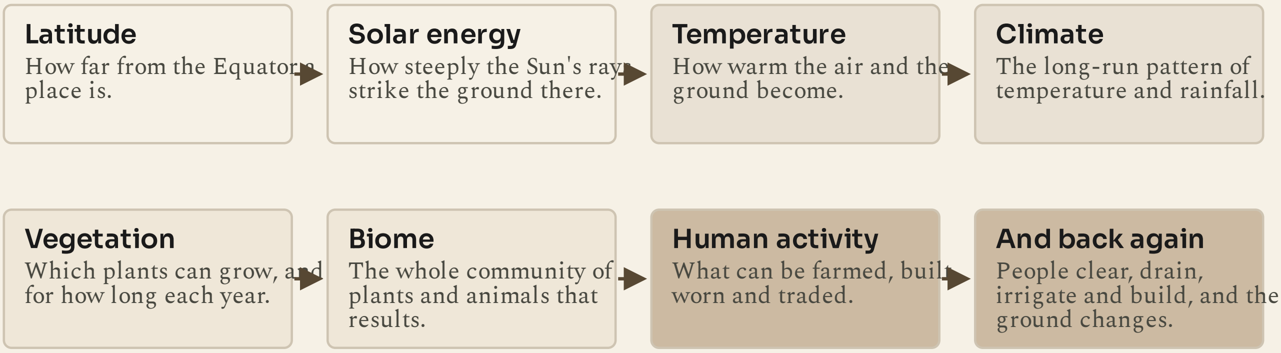


Figure 3.2 · From an angle to a way of life.
Read the boxes in order. The last arrow points back, because people change the ground as well as living on it.

The first link deserves care, because it is the one pupils most often get wrong. Latitude controls temperature not because low latitudes are nearer the Sun, but because of the **angle** at which sunlight arrives. A beam of sunlight has a fixed width. Arriving steeply, it lands on a small patch and delivers a lot of energy per square metre. Arriving at a shallow angle, the same beam is spread across a much wider patch. Shallow-angle light also travels further through the atmosphere before it lands, which absorbs and scatters more of it.

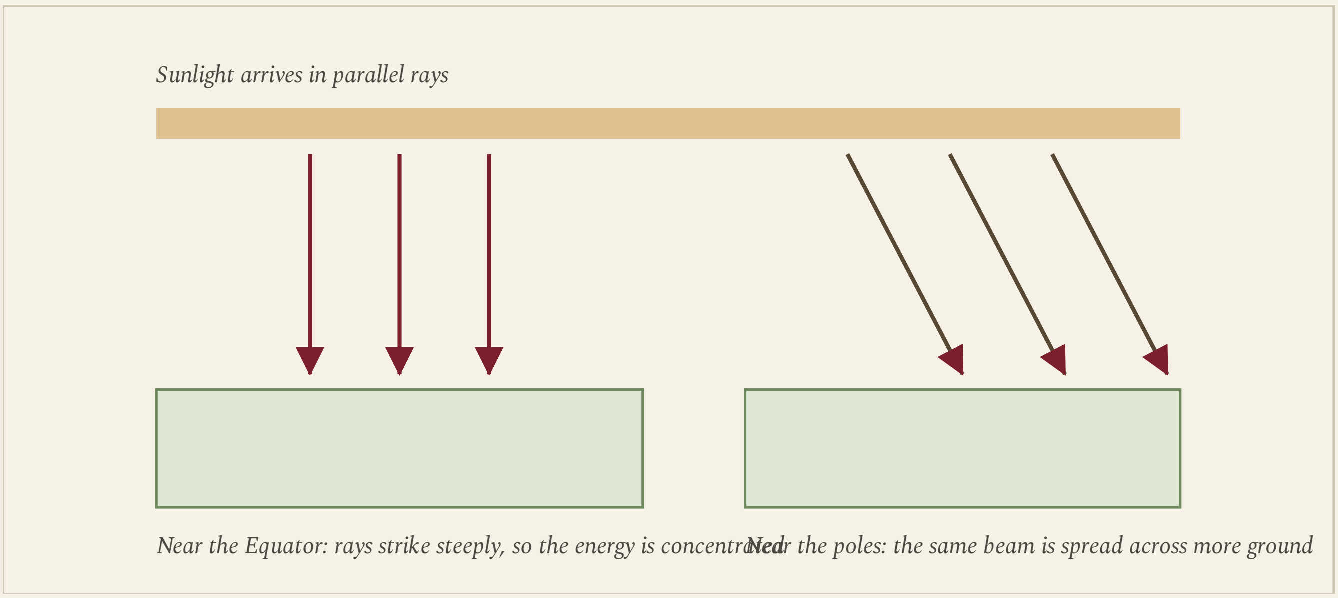


Figure 3.3 · The same beam, two different results.
Compare the width of the ground each set of rays covers. Nothing about the Sun has changed between the left and the right of this figure.

CAREFUL

Latitude is the strongest single control on climate, but it is not the only one. Altitude, distance from the sea, ocean currents, prevailing winds and mountain ranges all modify it. That is why two places at the same latitude can have very different climates, and why you should never explain a climate from latitude alone.

■ **YOUR TURN**

1. Give one-sentence definitions of weather, climate and biome.
2. Write out the chain from latitude to human activity, in order.
3. Explain, using angles, why low latitudes receive more energy per square metre.
4. Name three things other than latitude that affect a place's climate.

CHAPTER 3 · TOPIC 3.2

Tropical zones

The account

The tropical zone lies between the Tropic of Cancer and the Tropic of Capricorn. Across the whole of it, temperatures are high all year and vary little from month to month. What varies, and varies enormously, is rainfall, and rainfall is what divides the zone into different biomes.

Where rain falls heavily in every month, you get **tropical rainforest**. The Amazon basin, the Congo basin and the islands of south-east Asia are the three great blocks of it. The forest grows in layers, with a closed canopy overhead and very little light reaching the ground. It is the most biodiverse land environment on Earth: a single hectare can hold more tree species than the whole of temperate Europe.

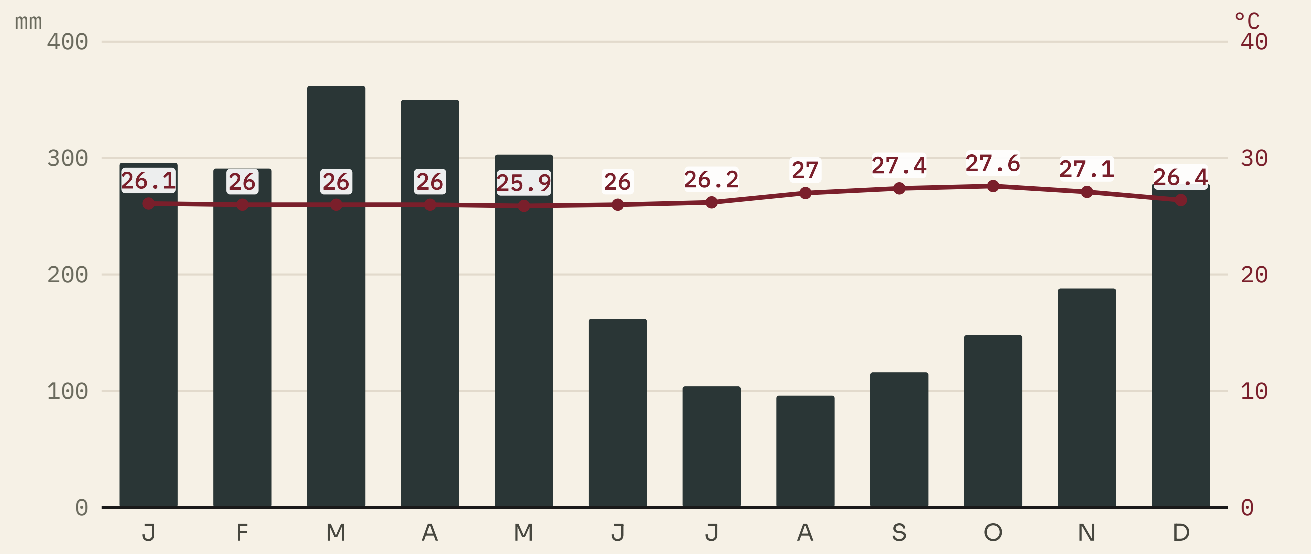


Figure 3.4 · Manaus, in the Amazon basin, 1991 to 2020.
Read the bars first: every month is wet, and even the driest is wetter than Lisbon's wettest. Then read the line: the warmest and coolest months differ by less than two degrees. This is what "warm all year" looks like on a graph.

Where the rain comes in a marked wet season followed by a marked dry season, you get **savanna**: grassland with scattered fire-resistant trees, across much of Africa, northern Australia and central Brazil. The grasses grow fast in the wet season and die back in the dry, and fire is a normal part of the system rather than a disaster in it.

There is a counter-intuitive point here that is worth stopping on. Rainforest soils are usually **poor**, not rich. Almost all the nutrients are locked up in the living plants, and the constant heavy rain washes minerals out of the soil. When forest is cleared for farming, the ash from burning gives one or two good harvests and then the soil is exhausted. That is a geographical fact with very large historical consequences, and you will meet it again in chapter 12.

ON THE MAP

On a world map, shade the rainforest belt along the Equator and the savanna belts on either side of it. Then mark the Amazon, Congo and south-east Asian blocks. Which of the three is the largest, and which countries share it?

YOUR TURN

1. What is constant across the tropical zone, and what varies?
2. What is the difference between rainforest and savanna, and what causes it?
3. Using the Manaus graph, give the wettest month, the driest month and the annual total.
4. Explain why rainforest soils are poor, and why that matters to farming.

CHAPTER 3 · TOPIC 3.3

Temperate zones

The account

The temperate zones lie between the Tropics and the polar circles, in both hemispheres. Their defining feature is not mildness but **seasons**: a real summer and a real winter, with noticeable differences in temperature and in day length between them.

Within the temperate zone there are several distinct climates, and three matter for this book.

Temperate forest climates have rain spread fairly evenly through the year and cold but not extreme winters. Trees are often deciduous, dropping their leaves in autumn because keeping them through a cold, dark winter costs more than it earns. Much of western and central Europe belongs here, and almost all of it was forest before people cleared it for farming.

Temperate grassland sits in continental interiors, far from the sea, where rainfall is lower. The Eurasian steppe, the North American prairie and the Argentine pampas are the great examples. These are the world's grain lands, and their deep, dark soils are among the most productive on the planet.

Mediterranean climates are the odd one out, and they are the ones a Portuguese pupil lives in. They have mild wet winters and hot dry summers, which is the reverse of the usual pattern of rain falling when plants most want it. Plants there are adapted to drought in the growing season: small tough leaves, deep roots, thick bark. Olive, cork oak, holm oak, vine and citrus are all Mediterranean plants, and all five are grown in Portugal.

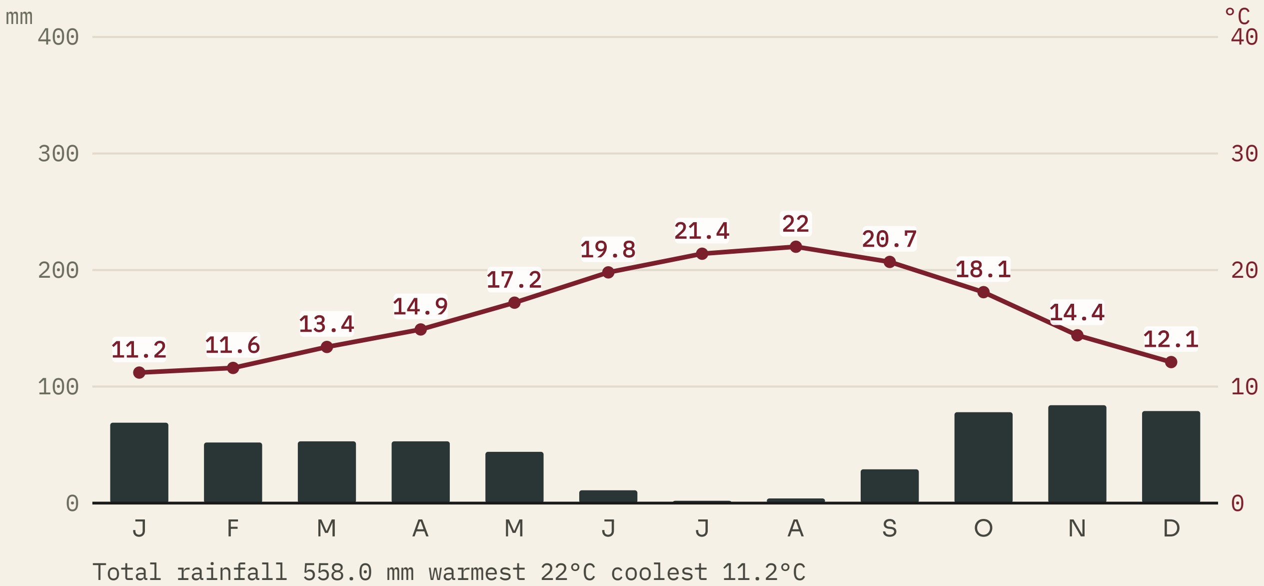


Figure 3.5 · Lisbon, 1991 to 2020.
Look at July and August: almost no rain at all, in the hottest months of the year. Then look at November and December. A Mediterranean climate delivers its water when the plants least need it, which is why the vegetation there is built to wait.

WHERE PORTUGAL FITS

Portugal is a good place to see that latitude is not the whole story. Porto is only about two and a half degrees north of Lisbon, yet it is noticeably wetter, because it faces the Atlantic more directly and catches more of the westerly weather systems. Move inland to Braganca and the summers are hotter and the winters colder, because the sea is too far away to moderate either. The same country, a few degrees, three noticeably different climates. Latitude, distance from the ocean and altitude are all at work at once.

YOUR TURN

1. What single feature defines the temperate zones?
2. Name the three temperate climates in this topic and give one example region for each.
3. Using the Lisbon graph, give the driest month and the annual rainfall total.
4. Explain why Mediterranean plants have small, tough leaves.

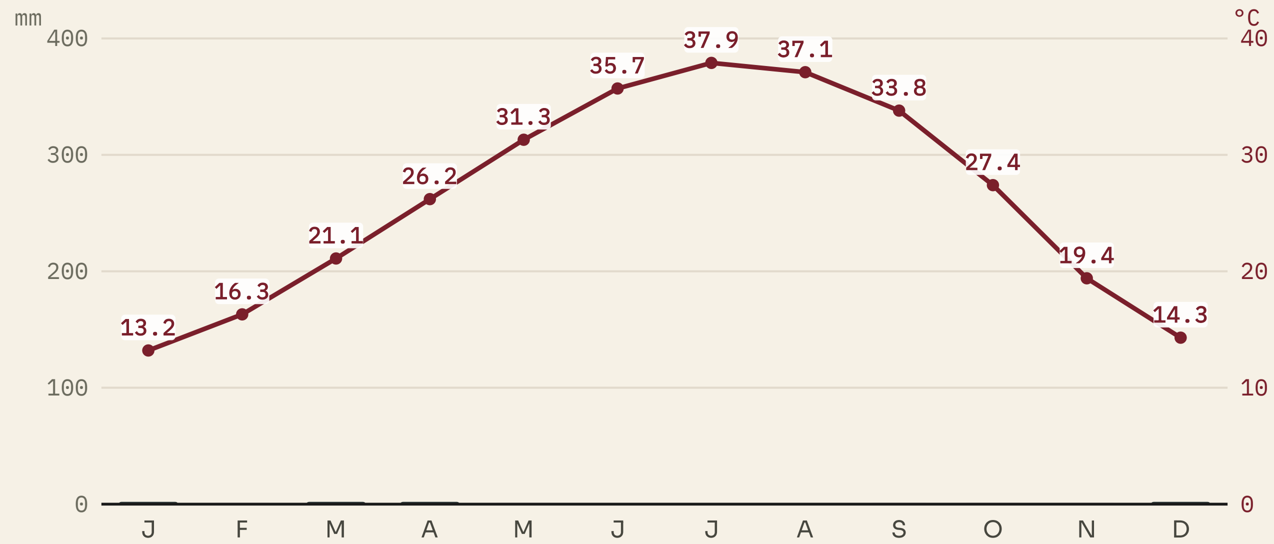
CHAPTER 3 · TOPIC 3.4

Desert regions

The account

A desert is defined by **rainfall**, not by temperature. The usual threshold is under about 250 millimetres a year. That is the whole definition. It says nothing at all about heat, which is why the largest desert on Earth is Antarctica.

Most hot deserts sit in two belts, at roughly twenty to thirty degrees north and south. That is not a coincidence either. Air that rose at the Equator, dropped its rain and travelled poleward high in the atmosphere sinks back down around those latitudes, and sinking air warms and dries. So the desert belts are the other end of the same circulation that makes the rainforest belt.



Total rainfall 13.0 mm warmest 37.9°C coolest 13.2°C

Figure 3.6 · In Salah, in the Algerian Sahara, 1991 to 2020.
The rainfall bars are almost invisible, and that is the point: the annual total is a fortnight of Lisbon's. Notice also the temperature range across the year, which is far wider than at Manaus.

Cold deserts work the same way for a different reason. The Gobi and the dry interior of central Asia are far from any ocean, so little moisture reaches them; some are also in the rain shadow of mountains, which strip the moisture out of air before it arrives. They are as dry as hot deserts and bitterly cold in winter.

	Tropical rainforest	Hot desert
Rainfall	More than 2000 mm a year, in every month.	Under 250 mm a year, and some years none at a
Temperature	Warm and almost unchanging: every month above 18 °C.	Very hot by day, and often cold at night.
Vegetation	Layered forest, evergreen, growing all year.	Sparse, spaced out, built to store water and lose little.
Soil	Thin and poor. The nutrients are in the living plants.	Thin, stony, and often salty where water evapor
People	Farming clears it, and the cleared soil is soon exhausted.	Settlement follows water: oases, wells and river from elsewhere.

Figure 3.7 · Two dry places, and what makes each of them dry.
Read the rainfall row first, because that is the row that makes both of them deserts. Everything else about them differs.

Life in a desert is built round water. Plants space themselves widely so that each has enough ground to draw from, store water in stems or leaves, and often lose their leaves entirely in the driest months. People settle where water is: at oases, over aquifers, or along rivers that rise somewhere wetter and simply pass through. Egypt is the classic case, a narrow ribbon of dense settlement along a river running through one of the driest places on Earth.

YOUR TURN

1. Give the definition of a desert. Which word in your answer is doing the work?
2. Explain why hot deserts cluster near twenty to thirty degrees of latitude.
3. Give two reasons a cold desert can be dry.
4. Using the In Salah graph, give the annual rainfall total, and compare it with Lisbon's.

CHAPTER 3 · TOPIC 3.5

Polar zones

The account

Beyond the polar circles, the defining conditions are cold and an extreme swing in daylight. Growing seasons are very short: in some places the ground is suitable for plant growth for only six to ten weeks a year.

Tundra is the biome of the ground that thaws at the surface in summer. Beneath it lies **permafrost**, ground that has stayed below freezing for at least two years and often for far longer. Because water cannot drain through frozen ground, tundra in summer is boggy, and because the growing season is so short, plants stay low: mosses, lichens, sedges, dwarf shrubs and, further south, stunted trees. There are no tall trees, because a tall tree cannot get water through frozen ground and cannot survive the wind.

South of the tundra, in the northern hemisphere, is the **taiga** or boreal forest, the largest forest on Earth, running across Canada, Scandinavia and Siberia. It is coniferous almost throughout: needles rather than leaves, because needles lose less water and can start work as soon as the light returns.

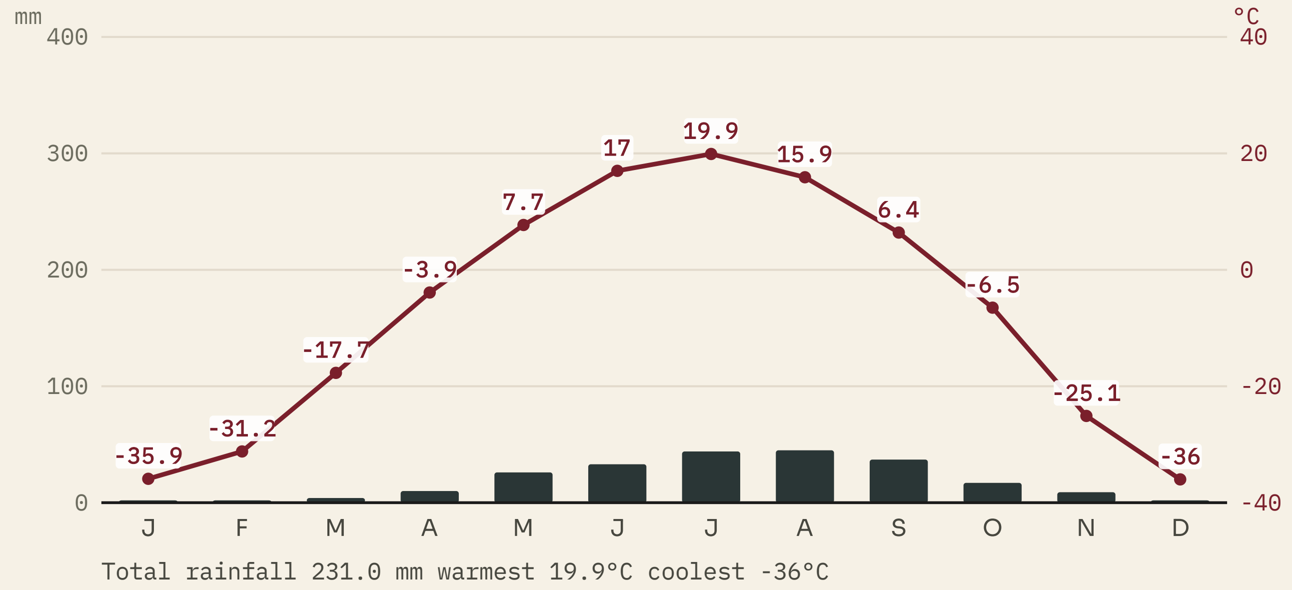


Figure 3.8 · Yakutsk, in eastern Siberia, 1991 to 2020.
This is the largest annual temperature range in this chapter: look at the difference between January and July. The rainfall total is small enough that some would call it a cold desert, which shows how the categories overlap at their edges.

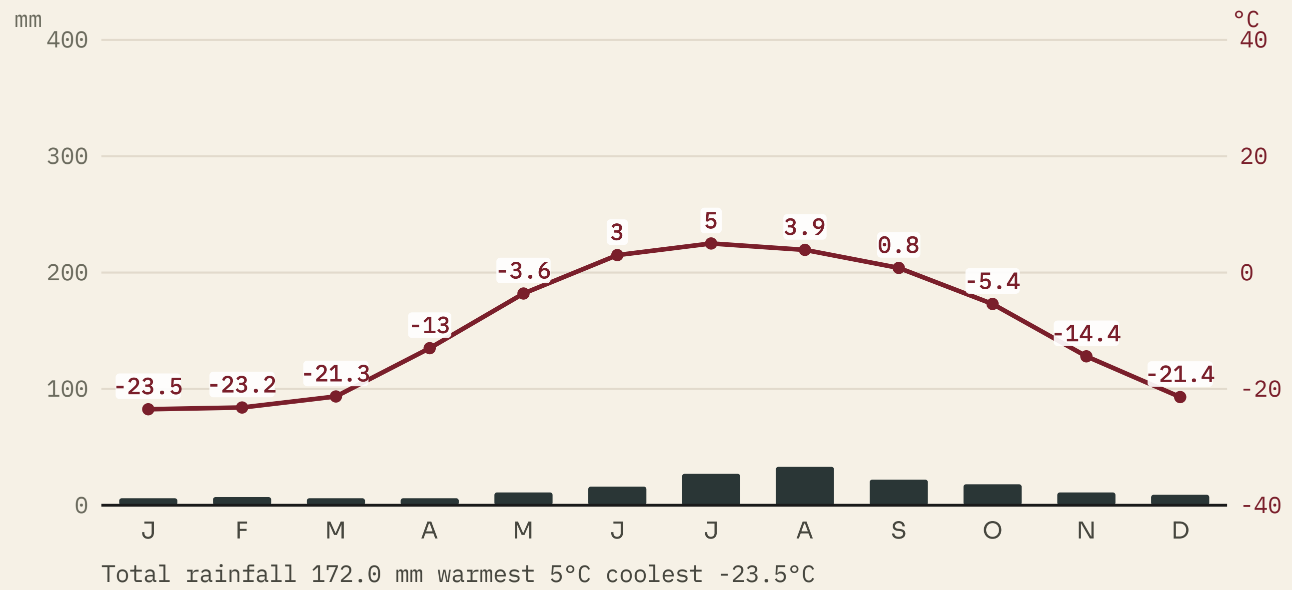


Figure 3.9 · Utqiagvik, on the Arctic coast of Alaska, 1991 to 2020.
Compare it with Yakutsk. It is far less extreme in winter, because it sits beside an ocean, and far cooler in summer, for the same reason. Distance from the sea is doing as much work here as latitude.

Adaptations are severe and specific. Animals grow thick insulating coats, change colour with the season, store fat, hibernate or migrate. Human settlement has historically depended on hunting, fishing and herding rather than on crops, because crops need a growing season the ground does not provide.

YOUR TURN

1. What is permafrost, and why does it make tundra boggy in summer?
2. Why are there no tall trees on the tundra? Give two reasons.
3. Compare the Yakutsk and Utqiagvik graphs. Which has the greater annual range, and why?
4. Explain why polar peoples have historically herded, hunted and fished rather than farmed.

CHAPTER 3 · TOPIC 3.6

Climate and human activity

The account

Climate does not decide what people do. It changes what things cost, and cost decides a great deal.

Farming follows climate most directly, because plants do. Rice needs warmth and standing water, so it is grown across monsoon Asia. Wheat needs a cooler, drier ripening season, so it belongs to temperate grasslands. Olives and vines need dry summers, so they belong to Mediterranean lands, which is why Portugal grows them and Norway does not.

Building follows climate too, and you can read the climate off a roof. Steep roofs shed snow, so they belong where snow falls. Flat roofs belong where it never does, and are used as working space. Thick walls and small windows keep heat out in hot dry lands and heat in in cold ones. Traditional Alentejo houses are painted white to reflect the summer sun and have small windows for the same reason.

Travel and trade follow climate in ways that turn up throughout the rest of this book. Rivers that freeze in winter close a trade route for months. Mountain passes fill with snow. Seas ice over. The monsoon winds of the Indian Ocean reverse twice a year, and for centuries the whole timetable of trade between east Africa, Arabia and India was set by them: sail one way in one season, wait, and sail back in the other.

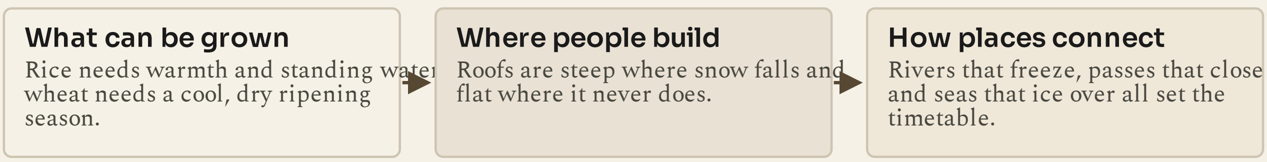


Figure 3.10 · Three ways a climate shows up in a life.
None of these is a rule about what people must do. Each is a statement about what is cheap and what is expensive where they live.

GO FURTHER

Climate sets costs, and people change those costs. Irrigation lets crops grow where rainfall would not allow them. Greenhouses move a climate indoors. Cheap refrigerated shipping means a country can eat food that could never be grown in it. So the honest statement is not "climate decides", nor "climate does not matter", but "climate sets the price, and technology and money can pay it". Keep that formula. You will need it in chapter 6, when you are asked why industry began in one country rather than another.

YOUR TURN

1. Give one crop each for a tropical, a temperate and a Mediterranean climate, and say why each belongs there.
2. What can you tell about a place's climate from the shape of its roofs?
3. Explain how the monsoon set the timetable for Indian Ocean trade.
4. "Climate decides how people live." Rewrite that sentence so that it is accurate, and explain what you changed.

WHAT YOU CAN NOW DO

- √ Use the words weather, climate and biome correctly and separately.
- √ Explain the chain from latitude through solar energy to human activity.
- √ Read a climate graph and say which zone it belongs to and why.
- √ Compare the tropical, temperate, desert and polar zones on real data.
- √ Explain how climate sets the cost of farming, building and travel.

WORDS FOR THIS CHAPTER

weather, climate, biome, solar energy, rainforest, savanna, temperate forest, grassland, Mediterranean climate, desert, rain shadow, tundra, permafrost, taiga, growing season, biodiversity

CHAPTER 3 AT A GLANCE

Climate zones and biomes

why is the living world so different from one latitude to another?

THE TOPICS YOU HAVE COVERED

3.1	Latitude and climate
3.2	Tropical zones
3.3	Temperate zones
3.4	Desert regions
3.5	Polar zones
3.6	Climate and human activity

WHAT YOU CAN NOW DO

- √ Use the words weather, climate and biome correctly and separately.
- √ Explain the chain from latitude through solar energy to human activity.
- √ Read a climate graph and say which zone it belongs to and why.
- √ Compare the tropical, temperate, desert and polar zones on real data.
- √ Explain how climate sets the cost of farming, building and travel.

THE WORDS THIS UNIT TAUGHT

weather · climate · biome · solar energy · rainforest · savanna · temperate forest · grassland · Mediterranean climate · desert · rain shadow · tundra · permafrost · taiga · growing season · biodiversity

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART ONE · CHAPTER 4

Hemispheres and global patterns

Two ways to cut the world in half, and what each cut shows

THE QUESTION THIS CHAPTER ANSWERS

How alike are the two halves of the planet, whichever way you cut it?

IN THIS CHAPTER, YOU WILL

- √ Compare the northern and southern hemispheres for land, ocean and people.
- √ Explain why the seasons are reversed between the two.
- √ Say where the eastern and western hemispheres are divided, and by what.
- √ Explain why the western hemisphere and the West are not the same idea.
- √ Give the main reasons people are distributed as unevenly as they are.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Our World in Data, world population growth. Where the world's people actually are, measured. Test the claim that the northern hemisphere holds the great majority of them.

ourworldindata.org/world-population-growth

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Without looking at a map, sketch the world as a circle and mark roughly where you think most people live. Then open an atlas and compare. Keep your sketch: the last topic of this chapter explains why almost everybody draws it wrong in the same direction.

CHAPTER 4 · TOPIC 4.1

The northern and southern hemispheres

The account

Cut the world along the Equator and the two halves are strikingly unlike each other, in ways that are not obvious until you look.

Land and ocean are not shared out evenly. The northern hemisphere holds most of the world's land; the southern is mostly water. The whole of Europe, almost all of Asia and North America, and most of Africa lie north of the Equator. The southern hemisphere has South America below the Equator, southern Africa, Australia and Antarctica, and a very great deal of ocean.

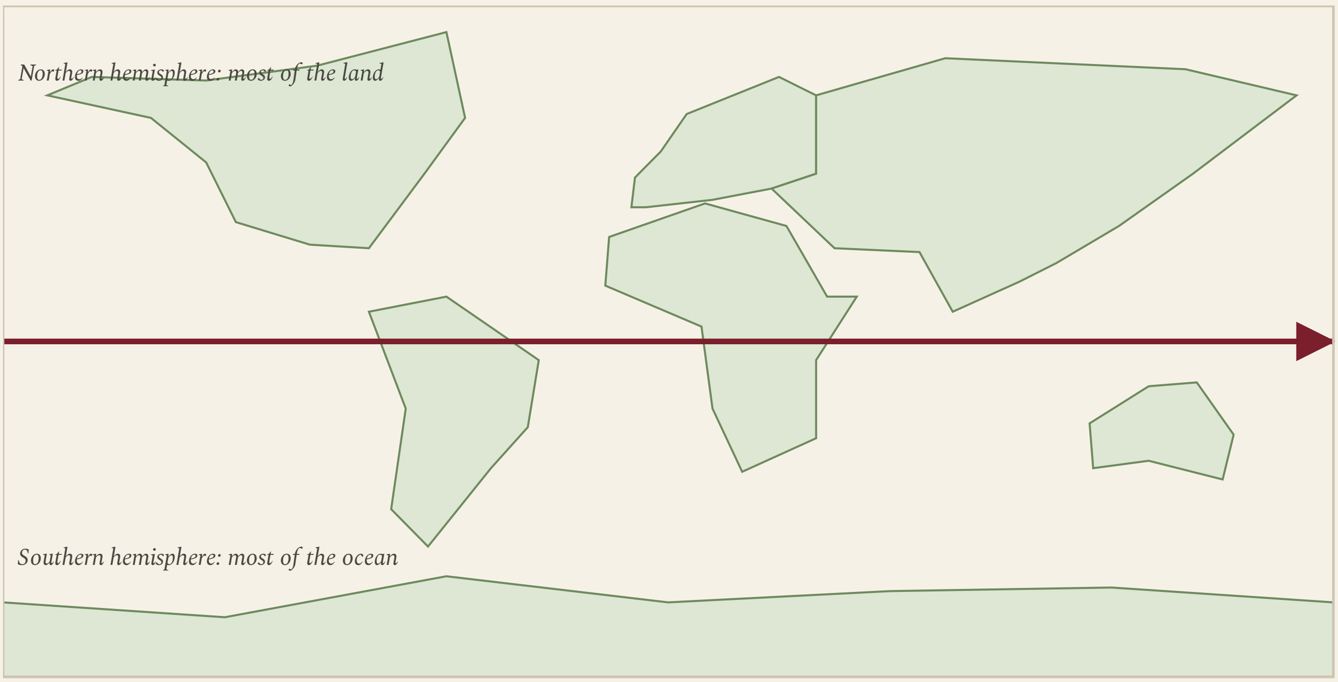


Figure 4.1 · One cut, two very different halves.
Compare the amount of land above the line with the amount below it. This single imbalance explains most of the differences in this topic.

People follow the land. Since people live on land rather than on water, and most of the land is northern, the great majority of the world's population is in the northern hemisphere. So are most of the largest cities, most of the industry and most of the world's political and economic weight, for reasons that are historical as well as geographical.

Climates behave differently. A hemisphere that is mostly ocean has milder extremes, because water heats and cools far more slowly than land does. Southern hemisphere temperate climates are generally less extreme than northern ones at the same latitude, and there is no southern equivalent of the bitter continental winter of Siberia or central Canada, because there is no southern landmass at that latitude to have one.

CAREFUL

"Most people live in the northern hemisphere" is a statement about where the land is, not a statement about wealth, development or importance. Chapter 9 and chapter 33 both deal with unevenness between countries, and neither is explained by which side of the Equator a country sits on.

YOUR TURN

1. Which hemisphere holds most of the world's land, and roughly what does the other one hold instead?
2. Explain why most people live in the northern hemisphere.
3. Why are southern temperate climates generally less extreme than northern ones at the same latitude?
4. Name one continent that lies wholly in the southern hemisphere and one that lies wholly in the northern.

CHAPTER 4 · TOPIC 4.2

The eastern and western hemispheres

The account

The second cut is made the other way, along a meridian, and this one is a human decision rather than a fact of the planet. The dividing lines are the Prime Meridian at zero degrees and the one hundred and eightieth meridian on the far side. Everything between them going east is the eastern hemisphere; everything going west is the western hemisphere.

The eastern hemisphere contains almost all of Europe, Africa and Asia, and Australia. The western hemisphere contains North and South America, and, because the Prime Meridian passes through western Europe and west Africa, a slice of those too.

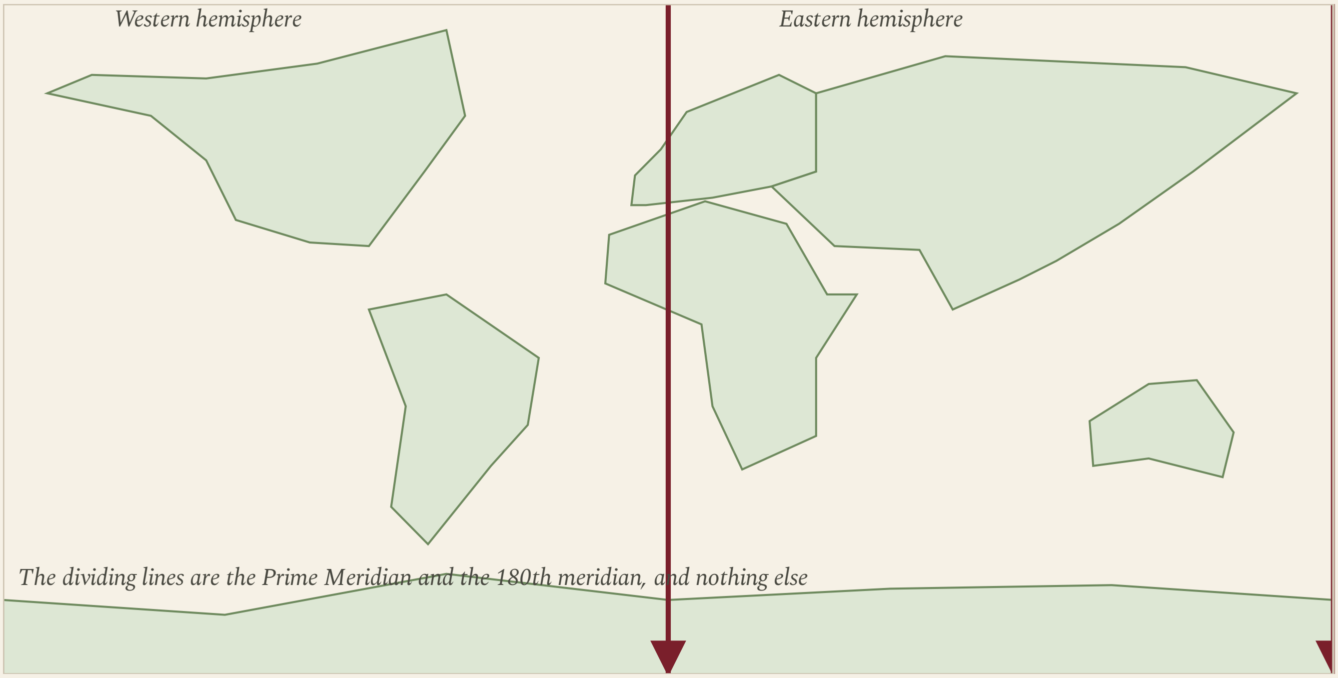


Figure 4.2 · The second cut, and where it falls.
Trace the Prime Meridian down through Britain, France, Spain, Algeria, Mali and Ghana. Everything to its left, as far as the far side of the Pacific, is the western hemisphere.

That last detail catches people out, and it is worth being exact about. Mainland Portugal lies entirely west of the Prime Meridian, so **Portugal is in the western hemisphere** in the geographical sense. So are Ireland, most of Britain, western France, Spain, Morocco, Senegal and Ghana. The line is at Greenwich, not at the Atlantic coast.

YOUR TURN

1. Which two meridians divide the eastern and western hemispheres?
2. Name three continents that lie mainly in the eastern hemisphere.
3. Is Portugal in the eastern or the western hemisphere? Explain how you know.
4. Why is this division a human decision when the Equator is not?

CHAPTER 4 · TOPIC 4.3

The geographical west and the political West

The account

Two phrases sound similar and mean completely different things, and confusing them causes a surprising amount of muddle.

The western hemisphere is a geographical term. It means everything between the Prime Meridian and the one hundred and eightieth meridian, going west. It is decided by an agreement of 1884 about where to measure longitude, and by nothing else.

The West is a political and cultural term. It is used to describe a group of countries thought of as sharing a broad political tradition, and it has changed meaning more than once. In the Cold War, which you will study in chapter 21, it meant the states aligned with the United States, whether or not they were democracies. Today it is used more loosely still.

	The western hemisphere	The West
What kind of term	Geographical. Everything west of the Prime Meridian as far as 180 degrees.	Political and cultural, and it has changed mean more than once.
Who decides it	An agreement of 1884 about where to measure longitude from.	Usage. No treaty defines it and no map fixes it.
Awkward case	Most of Africa and all of western Europe sit in it, including Portugal.	Australia and New Zealand are usually counte and they are about as far east as it is possible to be.
Why it matters	It tells you where a place is.	It tells you which side of an argument somebod thinks a country is on.

Figure 4.3 · Two phrases, two different jobs.
Read the last row first. One tells you where a place is; the other tells you which side of an argument somebody thinks it is on.

The awkward cases show the difference best. **Australia and New Zealand** are routinely counted as part of the West and are about as far east as it is possible to get. **Portugal** is in the western hemisphere geographically, and was also counted as part of the West during the Cold War, but at that time it was a dictatorship rather than a democracy, which is exactly the kind of complication chapter 21 exists to explain.

So the two terms can agree, disagree, or have nothing to do with each other. The rule for your own writing is simple: if you mean position, say hemisphere. If you mean politics, say what you actually mean and name the countries.

YOUR TURN

1. Define the western hemisphere and the West, and say what kind of term each is.
2. Give one country that is in one but not obviously in the other, and explain.
3. Why can no map settle an argument about whether a country is part of the West?
4. Rewrite this sentence so that it is precise: "Portugal has always been a western country."

CHAPTER 4 · TOPIC 4.4

Global population patterns

The account

People are not spread evenly over the Earth, and they never have been. Very large areas hold almost nobody, and small areas hold enormous numbers. Two regions alone, south Asia and east Asia, hold a large share of all the people alive.

The places that are nearly empty have things in common: they are too dry, too cold, too high, too steep, too wet underfoot, or too far from anywhere. The Sahara, the Arabian interior, central Australia, the Amazon basin, the Siberian north, Greenland, the high Himalaya and Antarctica between them cover a very large part of the land surface and hold a very small part of the population.

The places that are crowded also have things in common, and they are the reverse of that list.

Water
Rivers, wells and rainfall. Almost every dense population sits beside fresh water.

Growing season
Land that can feed people nearby, without moving food far.

Flat, workable land
Ploughs, roads and buildings all prefer it.

Work
Ports, mines, factories and offices pull people toward them.

Connection
A place on a route grows faster than a place off it.

History
People are also where earlier people were, which is why the map changes slowly.

Figure 4.4 · Six reasons people are where they are.
No single one of these explains a dense population; every dense population has several of them at once. The last box is the one people forget.

The last of those six is the one this book cares about most. **People are also where earlier people were.** A city does not move because the reason it was founded has stopped applying. Ports founded for sailing ships still hold cities in the age of container terminals. Towns founded at a river crossing still stand there long after bridges made the crossing unremarkable. Population maps change slowly, which is why a geography lesson keeps turning into a history lesson.

WHERE PORTUGAL FITS

Portugal's own population map shows the pattern clearly. The coastal strip from Braga through Porto to Lisbon and Setubal is densely settled; the interior, particularly the north-east and parts of the Alentejo, is thinly settled and in places has been losing people for decades. The reasons are the six above, all at once: the coast has the water, the milder climate, the flatter usable land, the ports, the work and the historical head start. The interior has fewer of each. You will meet the consequences again in chapter 7, when industry concentrates on the same coastal strip.

GO FURTHER

Look again at the sketch you drew before this chapter. Almost everybody draws the world's people too evenly, and almost everybody underestimates Asia. Ask yourself why. One likely reason is that the maps you see most often are political maps, where every country gets the same visual weight whatever its population, and news coverage that is not distributed by population either. Being able to notice that your mental map is skewed, and in which direction, is a genuine geographical skill.

YOUR TURN

1. Name four kinds of place that hold very few people, and say why in each case.
2. Give the six factors from the figure, and rank them for your own home region.
3. Explain, with an example, why population maps change slowly.
4. Compare your first sketch with an atlas. Which part of the world did you get most wrong, and can you say why?

WHAT YOU CAN NOW DO

√ Compare the northern and southern hemispheres for land, ocean, people and

climate, and explain the differences.

√ Say exactly where the eastern and western hemispheres are divided.

√ Explain why the western hemisphere and the West are different ideas.

√ Give the main reasons the world's population is distributed as it is.

√ Explain why a population map is partly a record of history.

WORDS FOR THIS CHAPTER

hemisphere, Equator, Prime Meridian, distribution, density, population, continental climate, maritime climate, the West, sparsely populated, densely populated

PART TWO

The Industrial Revolution

For almost all of human history, nearly everybody farmed, almost nothing moved faster than a horse, and a good year meant enough to eat. Then, over roughly a century, one country and then a handful of others changed how things were made, where people lived and how fast anything could travel. These four chapters ask what changed, why it began where it did, what it did to the people who lived through it, and where Portugal stood while it was happening.

5

PART TWO · CHAPTER 5

The world before industrialisation

The world that industry replaced, and why it could not simply grow

THE QUESTION THIS CHAPTER ANSWERS

What exactly was it that industry replaced?

IN THIS CHAPTER, YOU WILL

- √ Describe how most people lived and worked before industrialisation.
- √ Explain the domestic system and why its output could not rise quickly.
- √ Describe the Agricultural Revolution and name three of its changes.
- √ Explain how more food led to more people and to fewer of them farming.
- √ Judge who gained and who lost from enclosure.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Our World in Data, economic growth. How little output per person changed for centuries, and how sharply it turned after 1800. Find the year your own country turned.

`ourworldindata.org/economic-growth`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

List everything you have used today that was made in a factory. Then try to imagine the same day without any of it: where your clothes would come from, how food would reach you, what light you would have after dark. This chapter is about the world that list belongs to, and the one it replaced.

CHAPTER 5 · TOPIC 5.1

An agricultural world

The account

Around 1700, in Europe as in most of the world, the overwhelming majority of people worked the land. Estimates vary by country, but a figure of four people in five living in the countryside is a fair description of most of Europe. Towns existed, some of them large, but they were the exception rather than the setting of ordinary life.

Work in such a society has a particular shape, and it is worth understanding before you meet what replaced it.

It was **seasonal**. There were weeks of ferocious work at harvest and weeks in winter with little to do. The year had a rhythm set by weather and daylight rather than by a clock.

It was **local**. Most of what a household ate was grown within a few kilometres. Most of what it used was made nearby, often by the household itself. Heavy goods were expensive to move, so building materials, fuel and food came from close by unless there was water transport.

It was **precarious**. A bad harvest meant real hunger, and two bad harvests in a row meant famine. Because food could not easily be moved far, a shortage in one region could sit beside a surplus in another.

And it was **slow to change**. Output could rise if more land was brought into cultivation or more people worked it, but the amount each worker produced barely moved from one generation to the next.

	Around 1700	Around 1900
Where most people lived	In villages and small towns.	In cities, in the industrialised countries.
What most people did	Farmed, or made things at home between farming seasons.	Worked for wages, at a set time, in somebody el building.
Power	Muscle, water and wind.	Coal, steam and then electricity.
Speed of travel	As fast as a horse or a sailing ship.	As fast as a train or a steamship, and telegraphs faster still.
Who owned the tools	Often the worker did.	Usually the employer did.

Figure 5.1 · Two hundred years, and almost everything in this table changes.
Read the last row carefully. Who owned the tools turns out to matter as much as what the tools were.

CAREFUL

Do not read the pre-industrial world as simply miserable, or as simply peaceful. It was neither. People had more control over their own working day and far less security against hunger and disease. When you judge industrialisation later in this chapter and in chapter 7, you are comparing two sets of advantages and disadvantages, not progress against backwardness.

YOUR TURN

1. Roughly what proportion of people in Europe around 1700 lived in the countryside?
2. Give the four features of pre-industrial work described above, with an example of each.
3. Explain why a shortage in one region could sit beside a surplus in another.
4. Why did output per worker change so little from one generation to the next?

CHAPTER 5 · TOPIC 5.2

Domestic production

The account

Not everything was farming. A great deal of manufacturing already existed, and it happened in people's homes. Historians call it the domestic system, or the putting-out system, and in textiles it worked like this.

A merchant bought raw wool or cotton and carried it out to cottages across a district. The family there spun it into thread and wove it into cloth, on their own wheel and their own loom, in their own time. The merchant came back, collected the cloth, paid by the piece, and sold it on. Everybody in the family could take part, and the work fitted around the farming year: more of it in winter, less at harvest.

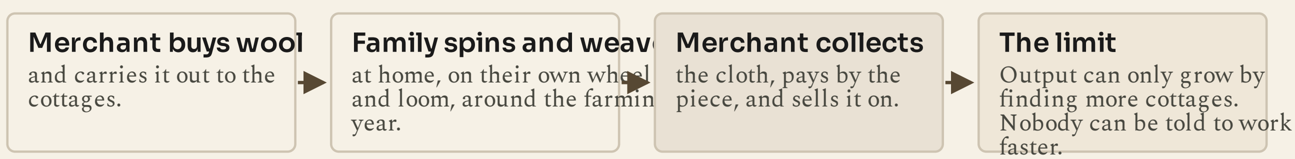


Figure 5.2 · The putting-out system, and the wall it runs into.
Follow the first three boxes round the cycle. The fourth box is the reason the system had to be replaced rather than expanded.

There were real advantages. The worker kept some control over hours. The tools belonged to the household. Income from cloth cushioned a bad farming year.

But there was a hard limit built into it, and it is the key to this whole part of the book. **Output could only grow by finding more cottages.** A merchant who wanted twice as much cloth had to ride to twice as many villages. He could not tell a family to work faster, could not stop them doing other things, could not guarantee that all the cloth would be the same quality, and could not stop some of the yarn quietly disappearing along the way.

Once demand for cloth began rising quickly, that limit became intolerable to the people making money from it. Everything in chapter 7 follows from the attempt to get round it.

YOUR TURN

1. Describe the putting-out system in your own words, in four steps.
2. Give two advantages of it for the family doing the work.
3. Explain the built-in limit on how fast output could grow.
4. Suggest two problems a merchant had that a factory owner would not have.

CHAPTER 5 · TOPIC 5.3

The Agricultural Revolution

The account

Before industry could take people off the land, agriculture had to be able to feed them without them. That is what the Agricultural Revolution did, over the course of the seventeenth and eighteenth centuries, chiefly in the Low Countries and then in Britain.

Three changes matter most.

Better rotations. In the older open-field system, a field was commonly left bare for a year to recover, which meant a third of the land grew nothing. Rotations that included clover and turnips replaced the bare year: clover restores nitrogen to the soil, and turnips could feed animals through the winter. More of the land was working, all of the time.

Selective breeding. Choosing which animals to breed from, deliberately and across generations, produced heavier cattle and sheep from the same pasture.

Enclosure. Open fields farmed in strips, and common land that villagers could use, were divided up, fenced and held as private farms. Enclosed land could be farmed in new ways without needing everybody in the village to agree.

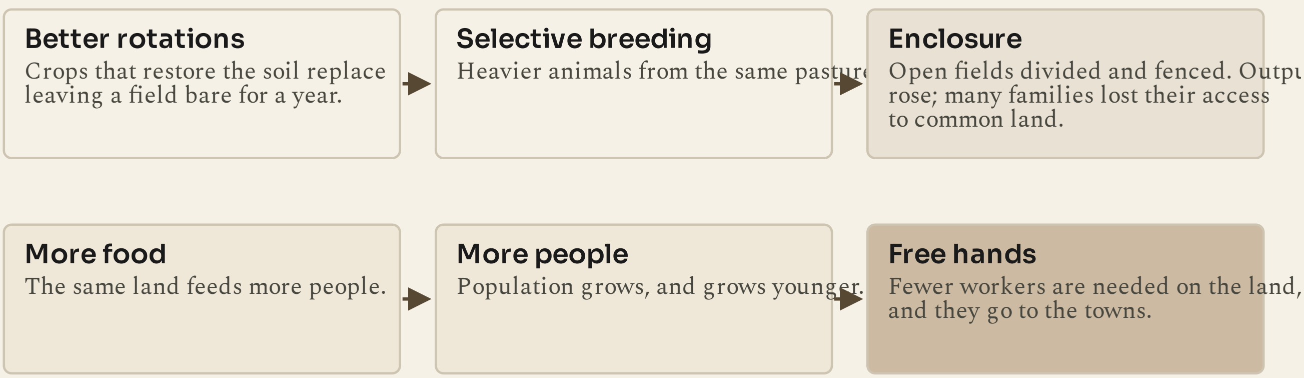


Figure 5.3 · From a change in farming to a supply of factory workers.
Read the six boxes in order, and notice that this chain has both a winner and a loser in it.

The consequences run in a chain, and it ends somewhere unexpected. More food meant more people, because fewer died young and more children survived. More people needed more work. But better farming needed **fewer** workers per hectare, not more. So a growing population found the land could not employ it, and moved toward the towns, where the new factories were.

Enclosure is where the honesty of the chain matters. It raised output, and it took away from many families the right to graze an animal or gather fuel on common land. Some families gained; many lost a resource they had relied on for generations and had no way of replacing. When people say the Industrial Revolution had winners and losers, this is where the losing starts, before a single factory has been built.

GO FURTHER

Try building the chain backwards. If enclosure had not happened, would there have been enough workers in the towns for the factories of chapter 7? If the population had not grown, would there have been enough buyers for what the factories made? Historians argue about how much weight to give agriculture in explaining industrialisation. Write two sentences: one arguing it was essential, one arguing it was only one condition among several. Keep both. You will need them in chapter 6.

YOUR TURN

1. Name and explain the three agricultural changes in this topic.
2. Why did leaving a field bare for a year waste so much?
3. Explain the chain from more food to more factory workers, in order.
4. "Enclosure was progress." Give one piece of evidence for that claim and one against it.

WHAT YOU CAN NOW DO

- √ Describe how most people lived and worked before industrialisation.
- √ Explain the domestic system and the limit built into it.
- √ Name and explain three changes of the Agricultural Revolution.
- √ Trace the chain from more food to a supply of workers in the towns.
- √ Judge enclosure with evidence on both sides.

WORDS FOR THIS CHAPTER

subsistence, domestic system, putting-out system, cottage industry, Agricultural Revolution, crop rotation, selective breeding, enclosure, common land, open field, output, rural, urban

CHAPTER 5 AT A GLANCE

The world before industrialisation

what exactly was it that industry replaced?

THE TOPICS YOU HAVE COVERED

5.1	An agricultural world
5.2	Domestic production
5.3	The Agricultural Revolution

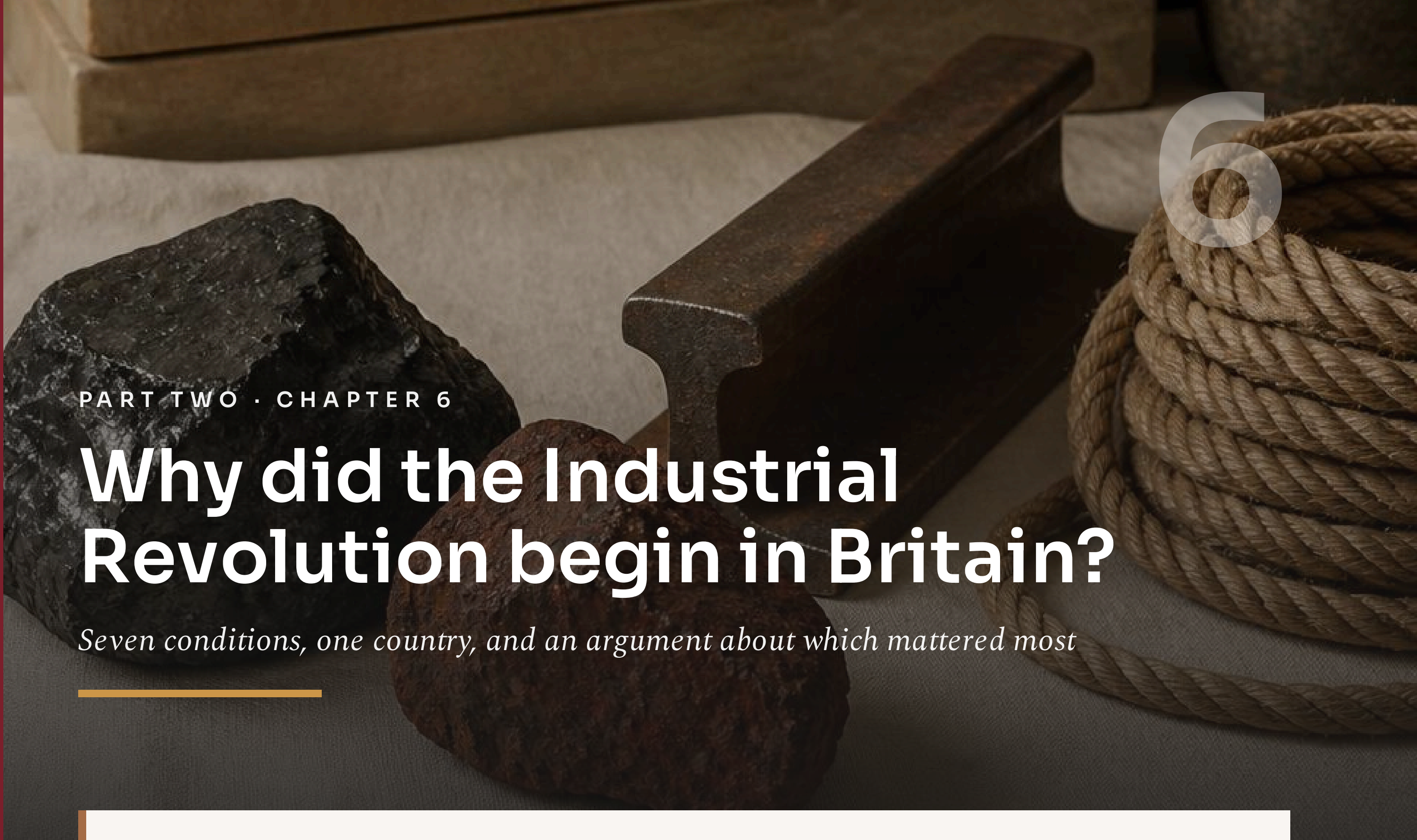
WHAT YOU CAN NOW DO

- √ Describe how most people lived and worked before industrialisation.
- √ Explain the domestic system and the limit built into it.
- √ Name and explain three changes of the Agricultural Revolution.
- √ Trace the chain from more food to a supply of workers in the towns.
- √ Judge enclosure with evidence on both sides.

THE WORDS THIS UNIT TAUGHT

subsistence · domestic system · putting-out system · cottage industry · Agricultural Revolution · crop rotation · selective breeding · enclosure · common land · open field · output · rural · urban

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.



6

PART TWO · CHAPTER 6

Why did the Industrial Revolution begin in Britain?

Seven conditions, one country, and an argument about which mattered most

THE QUESTION THIS CHAPTER ANSWERS

Why there, and why then, when every ingredient existed somewhere else?

IN THIS CHAPTER, YOU WILL

- √ List the conditions that made industrialisation possible in Britain.
- √ Explain why no single one of them is a sufficient explanation.
- √ Rank the conditions and defend your ranking with evidence.
- √ Explain why coal and iron in the same district mattered so much.
- √ Compare Britain's position around 1800 with Portugal's.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Industrial Revolution. Note every cause it gives for industry beginning in Britain, then rank them as you ranked them in this chapter.

www.britannica.com/event/Industrial-Revolution

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Every ingredient of industrialisation existed somewhere in the world in 1750. China had coal. France had capital. India had the largest textile industry on Earth. So the question is not "what was needed" but "where were they all in the same place at the same time". Keep that shape of question in your head for the whole chapter.

CHAPTER 6 · TOPIC 6.1

Coal

The account

Industry runs on concentrated energy, and until the eighteenth century the only sources were muscle, wind, falling water and wood. All four have hard limits. Muscle is weak. Wind is unreliable. Water power only exists beside fast-flowing water. Wood runs out, and Britain had already largely cut down its forests.

Coal solved the problem, because a tonne of coal releases far more heat than a tonne of wood and can be stored, stacked and carried.

Britain's advantage was not that it had coal, which several countries did, but **where** its coal was. Substantial coalfields lay near the surface, so early mining did not require deep shafts. They lay in the north and the midlands, near iron ore. And they lay close to navigable water, so coal could be moved by barge and by coastal ship rather than dragged overland, which for a heavy, cheap bulk cargo is the difference between a business and an impossibility.

There was a limit, and its solution runs through the next two chapters. The deeper a mine goes, the more water floods in. Pumping that water out by horse power set a maximum depth beyond which mining stopped paying. The steam engine was invented to solve exactly that problem, and you will meet it in chapter 8. Notice the shape of that: **coal created the problem whose solution then transformed everything else.**

YOUR TURN

1. Give the four sources of power available before coal, and the limit on each.
2. Britain was not the only country with coal. What was different about its coal?
3. Why did it matter so much that coalfields were near navigable water?
4. Explain the link between flooding in mines and the invention of the steam engine.

CHAPTER 6 · TOPIC 6.2

Iron

The account

If coal was the energy, iron was the material. Machines, rails, bridges, ships, boilers and tools all needed metal that was strong, could be shaped, and could be made in quantity.

The obstacle was fuel. Smelting iron ore traditionally used charcoal, made from wood, and a country short of forest was therefore short of iron. The change came when coke, made by baking coal, was used instead. That freed iron production from the supply of timber and tied it instead to the supply of coal.

The consequence is a geographical one. Because it takes several tonnes of coal and ore to make a tonne of iron, and because all of it is heavy, ironworks were built **where the coal and the ore were**, not where the customers were. Whole industrial districts grew on coalfields for that reason: south Wales, the west midlands, the north-east of England, and later the Ruhr in Germany and Pennsylvania in the United States.

That is one of the most useful ideas in this book, so it is worth stating plainly. **Heavy industry goes to its raw materials; light industry goes to its market.** You will use it again in chapter 9 and in chapter 34.

ON THE MAP

Find the main British coalfields on a map: south Wales, the midlands, Yorkshire, Lancashire, the north-east and central Scotland. Now find the cities that grew largest in the nineteenth century. How closely do the two maps match, and where do they not match? A place that grew without coal needs a different explanation, and finding one is the point of the exercise.

YOUR TURN

1. Why did iron production depend on forests before coke was used?
2. What changed when coke replaced charcoal in smelting?
3. Explain why ironworks were built near coalfields rather than near customers.
4. State the rule about heavy and light industry in your own words.

CHAPTER 6 · TOPIC 6.3

Rivers and ports

The account

Before railways, moving anything heavy overland was punishingly expensive. A cart on a poor road could carry very little and could be stopped by mud for weeks at a time. Water transport, by contrast, was cheap: one barge could carry what dozens of carts could not, and it did not need a road.

Britain's geography is unusually generous here. It is an island, so nowhere is far from the sea, and it has many navigable rivers and good natural harbours. From the middle of the eighteenth century a network of canals was cut to link coalfields, ironworks and towns, which brought the price of moving coal down sharply in the districts they reached.

Ports mattered twice over. They let raw cotton, which grows nowhere in Britain, arrive from across the Atlantic. And they let finished cloth leave for markets all over the world. An industry that imports its raw material and exports its product is an industry that lives or dies by its ports.

CAREFUL

It is tempting to say that Britain industrialised because it is an island. That is too strong. Being an island helped with transport and with security, but Japan is an island and did not industrialise until much later, and the Netherlands is not an island and had superb water transport. Geography made industrialisation cheaper in Britain. It did not make it inevitable.

■ YOUR TURN

1. Why was overland transport so expensive before railways?
2. Give two ways ports mattered to the British cotton industry.
3. What did canals do to the price of coal in the districts they reached?
4. Explain why "Britain industrialised because it is an island" is too strong a claim.

CHAPTER 6 · TOPIC 6.4

Capital

The account

A factory has to be paid for before it earns anything. Somebody has to buy the land, put up the building, buy the machines and pay the workers for months before a single piece of cloth is sold. That money is capital, and a country without it cannot industrialise however good its coal is.

Britain had accumulated capital from several sources at once: profitable agriculture, long-distance trade, colonial commerce including the Atlantic slave trade and the trade in the goods it produced, and a banking system that had grown up to serve all of them. There were country banks, insurance, a national debt that people trusted, and merchants who were used to putting money into ventures that might not pay for years.

That last point is easy to miss and is genuinely important. Capital is not only money; it is the **habit and the machinery of investment**. A society where wealth is held as land and passed down intact behaves quite differently from one where wealth is regularly risked in ventures.

◆ CAREFUL

The wealth that helped fund early industrialisation included profits from slavery and from colonial trade. Historians disagree about how large a share that was, and the argument is a real one rather than a settled figure. What is not in dispute is that it was part of the picture. A complete answer to "where did the capital come from" has to include it.

■ YOUR TURN

1. Explain why a factory needs money before it earns any.
2. Name three sources of capital in eighteenth-century Britain.
3. What is the difference between having wealth and having a habit of investment?
4. Why is the contribution of colonial and slave-trade profits described here as an argument rather than a figure?

CHAPTER 6 · TOPIC 6.5

Labour

The account

Factories need workers, in numbers, in one place, willing to work fixed hours for a wage. That is not a natural state of affairs. It has to be produced.

Chapter 5 showed how it was produced in Britain. The population was growing quickly. Better farming meant fewer hands were needed per hectare. Enclosure removed the common land that had let many families get by without a full wage. The result was a large number of people who had to sell their labour because they had no other way to live.

There is a second element, less obvious and just as important: **skill**. Britain had a large body of practical craftsmen, millwrights, clockmakers, instrument makers and blacksmiths, who could build a machine from a drawing and repair it when it broke. An invention is worthless without people who can make it work in a damp mill at six in the morning.

GO FURTHER

Notice that "labour" here is doing two jobs. It means the number of people available, and it means what those people can do. Countries that had plenty of the first and little of the second industrialised slowly. Hold that thought: it is one of the better answers to the question at the end of this chapter about why Portugal industrialised later than Britain.

YOUR TURN

1. Give three reasons a large workforce became available in Britain.
2. Why is a supply of skilled craftsmen as important as a supply of workers?
3. Explain why "people willing to work fixed hours for a wage" is not a natural state of affairs.
4. Which of the two meanings of labour do you think was harder for another country to copy? Defend your answer.

CHAPTER 6 · TOPIC 6.6

Markets

The account

Producing more only makes sense if somebody buys it. A factory that doubles its output and cannot sell the extra has simply ruined itself faster.

Britain had buyers on two scales. At **home**, a growing population with rising incomes bought cloth, pots, tools and coal. Domestic demand matters more than people expect: it is steady, close by, and cheap to supply.

Overseas, Britain had a very large trading network and an empire. Some of those markets were open competition; others were arrangements that favoured British goods and restricted local manufacturing, notably in India, whose own large textile industry was undercut and then displaced. That is part of the story of British industrial success and it should be stated rather than left out.

There is a loop here, and it is the engine of the whole period. Machines lower the cost of production. Lower costs mean lower prices. Lower prices mean more people can afford the goods. A larger market justifies more machines. Each turn of the loop makes the next turn easier, which is why industrialisation accelerates once it starts.

■ YOUR TURN

1. Give the two scales on which Britain had markets, and one advantage of each.
2. Describe the loop from machines to prices to markets and back.
3. Why does industrialisation speed up once it has begun?
4. Explain why a full account of British markets has to mention India.

CHAPTER 6 · TOPIC 6.7

Political and institutional conditions

The account

The last condition is the least visible and, several historians argue, among the most important: the rules.

Britain in the eighteenth century had reasonably secure **property rights**, so somebody who built a mill could expect to keep it. It had enforceable **contracts**, so agreements between strangers were worth making. It had a **patent system**, so an inventor had some prospect of profiting from an invention. It had a relatively stable government that had not been invaded, and a state that was interested in trade. It had no internal customs barriers, so goods moved inside the country without being taxed at every boundary, which was not true of France or the German lands at the time.

None of that makes anybody invent anything. What it does is change the odds. If you expect to keep what you build, you build.

Coal — Shallow, plentiful, and close to water for moving it. →
Iron — Ore and the fuel to smelt it, often in the same district. →
Rivers and ports — Nowhere in Britain is far from navigable water. →
Capital — Money from trade and farming, looking for a return.

Labour — A growing population, and fewer places for it on the land. →
Markets — Buyers at home and overseas including captive colonial markets. →
Institutions — Patents, contracts and a state that protected property and trade. →
Together — No single one of these is enough. The combination is the explanation.

Figure 6.1 · Seven conditions, and the argument they add up to.
Read the first seven boxes, then the last. The last box is the actual claim this chapter makes.

Resources and capital — Coal, iron and money in one place. →
Machines and factories — Production concentrated and mechanised. →
Markets and transport — Goods moved cheaply to buyers who wanted them.

Figure 6.2 · The same argument, compressed to three.
Use this version when you need to explain industrialisation in one sentence, and the seven-box version when you need to defend it.

THE ENQUIRY: WHY BRITAIN FIRST?

Here is the chapter's real task. You have seven conditions. Your job is to rank them.

1. Copy the seven into a list.
2. Rank them from most to least important, and write one sentence of justification for your top choice.
3. Now test your ranking. For your number one, find a country that had it and did not industrialise first. If you can, your factor is necessary but not sufficient, and you should say so.
4. Write a paragraph beginning: "The most important condition was X, because without it none of the others could have worked. However, X alone is not enough, because..."

There is no single correct ranking, and your teacher is marking the reasoning, not the order.

CHAPTER 6 · TOPIC 6.8

Portugal in the global story: Portugal at the beginning of the industrial age

What was happening in Portugal?

While Britain was building its first factories, Portugal remained an overwhelmingly rural and agricultural country, and the opening decades of the nineteenth century were among the most disrupted in its history.

In 1807 a French army invaded, as part of Napoleon's attempt to close the whole European coast to British trade. Rather than surrender, the Portuguese royal court sailed for Brazil under British naval escort, and the seat of the monarchy moved across the Atlantic. Two further French invasions followed, in 1809 and 1810, and the fighting and requisitioning did serious damage to farming, to towns and to trade.

The court's departure had a consequence nobody intended. With the king in Rio de Janeiro, Brazilian ports were opened to foreign trade, and Brazil's status rose until it was no longer plausibly a colony. When the king returned to Lisbon in 1821, the position could not be reversed, and in 1822 Brazil declared independence. Portugal lost the largest and richest territory in its empire in the same decade in which its own economy had been wrecked by invasion.

Figure 6.3 · Portugal, 1800 to 1830.
Read the cluster on the left as one event with three phases. Then note how close 1820 and 1822 are: a revolution at home and the loss of Brazil fall within two years of each other.

What part did geography play?

Portugal's position had been an enormous advantage for four centuries: an Atlantic seaboard, deep harbours and a route to Africa, Asia and Brazil. Against Britain's industrial conditions, that same position offered much less. Portugal had little accessible coal and modest iron ore. Its internal transport was difficult, with mountainous terrain and few navigable rivers reaching inland. Its domestic market was small and mostly poor. Its capital had been tied up in long-distance trade rather than in manufacturing, and much of that trade was now disrupted or lost.

What was the connection?

Portugal was not outside the industrial story. It was inside it as a **trading partner and a market** rather than as a producer. Britain sold manufactured goods into Portugal and bought Portuguese wine and other produce; the relationship was long established and formalised in commercial treaties. That is a real position in an industrial world, and it is a different one from Britain's.

KEY QUESTION

Why did industrialisation happen at different speeds in Britain and Portugal?

Take the seven conditions from topic 6.7 and go through them one at a time for Portugal in 1800: coal, iron, rivers and ports, capital, labour, markets, institutions. Award each a mark out of three. Then write a paragraph explaining which two mattered most in the difference, and why. Do not simply say Portugal lacked everything: it did not, and one of the seven is genuinely a Portuguese strength.

WHAT YOU CAN NOW DO

- √ List the conditions that made industrialisation possible in Britain.
- √ Explain why no single condition is a sufficient explanation on its own.
- √ Rank the conditions and defend the ranking with evidence.
- √ Explain why heavy industry locates near its raw materials.
- √ Compare Britain's position around 1800 with Portugal's, condition by condition.

WORDS FOR THIS CHAPTER

coal, coke, iron ore, smelting, canal, navigable, capital, investment, patent, property rights, domestic market, overseas market, labour supply, French invasions, Brazilian independence

PART TWO · CHAPTER 7

Factories, cities and social change

What the factory did to the working day, the city and the family

> **THE QUESTION THIS CHAPTER ANSWERS**
>
> **Did industrialisation make people's lives better?**

IN THIS CHAPTER, YOU WILL

- √ Explain what changed when work moved from the home to the factory.
- √ Describe how industrial cities grew, and what they failed to build in time.
- √ Use evidence to describe living and working conditions.
- √ Explain why children were employed and how that changed.
- √ Describe the two new social classes industry created.

THE TOPICS, IN THE ORDER YOU MEET THEM

7.1	The factory system
7.2	Industrial cities
7.3	Living conditions
7.4	Working conditions
7.5	Child labour
7.6	New social classes
7.7	Portugal in the global story: the beginnings of Portuguese industrialisation

WATCH, READ AND LISTEN

UK Parliament, early factory legislation. The Acts themselves, in the order they were passed. Count the years between the first limit on hours and the first inspectors.

www.parliament.uk/about/living-heritage/transformingsociety/livinglearning/19thcentury/overview/earlyfactorylegislation/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Work out how many hours a week you spend in lessons. Now imagine that number doubled, six days a week, from the age of nine, in a room full of moving machinery, with a fine deducted from your wages for arriving late. Write down the first three questions you would want answered before agreeing to it. This chapter tries to answer them.

CHAPTER 7 · TOPIC 7.1

The factory system

The account

A factory is not simply a large workshop. It is a different arrangement of work, and four things change at once when production moves into one.

The tools stop belonging to the worker. In the domestic system the family owned the wheel and the loom. A powered machine costs far more than a household can raise, so the employer owns it, and the worker brings only labour.

The pace stops being set by the worker. A spinning mule runs at the speed it runs. The person tending it works at the speed of the machine rather than the other way round.

The hours stop being negotiable. A steam engine is expensive whether it is running or not, so it is run for as long as possible, and everyone tending it must be there for those hours. The factory bell replaces daylight and the season.

The work is divided. Instead of one person making a whole article, each does one small part of it, repeatedly. That raises output per worker considerably, and it makes the work far more monotonous, and it means a worker learns a task rather than a trade.

	Work at home	Work in a factory
Who owned the tools	Usually the family.	The factory owner.
Who set the hours	The family, around daylight and the harvest.	The employer, by the clock and the bell.
What set the pace	The worker.	The machine.
Where the children were	In the same room as the adults.	In the factory too, and often on the machines.
What happened to output	It grew slowly, cottage by cottage.	It grew quickly, because more machines could added.

Figure 7.1 · The same work, in two places.
Read the middle row first: what set the pace is the change that all the others follow from.

Output rose enormously. So did the employer's control over how the day was spent. Both statements are true at once, and any honest account of the factory system has to hold them together.

YOUR TURN

1. Give the four things that change when production moves into a factory.
2. Explain why the employer, rather than the worker, owns the machine.
3. Why did the expense of a steam engine lead to long working hours?
4. Give one advantage and one disadvantage of dividing the work into small repeated tasks.

CHAPTER 7 · TOPIC 7.2

Industrial cities

The account

Factories concentrated work in one place, so people had to be within walking distance of it. There were no trams, no bicycles and no buses, so a worker lived close enough to walk to the mill gate before the bell. That single constraint produced the industrial city: dense, built quickly, and built right up against the works.

The cities that grew fastest grew where the conditions of chapter 6 were strongest. Manchester and the Lancashire towns grew on cotton, water and coal. Birmingham and the towns around it grew on metalworking. Glasgow, Newcastle and Belfast grew on shipbuilding and heavy engineering. Liverpool grew as the port that fed and emptied all of them.

The speed is the thing to grasp. Places that had been market towns became major cities within the lifetime of one person. Nothing that a city needs, and cities need a great deal, was built at that speed: not water supply, not sewers, not schools, not hospitals, not policing.

People arrive
Faster than anybody can build for them.

Housing
Built cheaply, back to back, with no space between.

Water and waste
Shared standpipes and cesspits, often side by side.

Disease
Cholera and typhus follow the water supply, not the air.

Figure 7.2 · What happens when people arrive faster than a city can be built.
Follow the four boxes in order. The fourth is a consequence of the third and not, as people at the time believed, of bad air.

ON THE MAP

Sketch the outline of Britain and mark Manchester, Liverpool, Birmingham, Leeds, Sheffield, Glasgow and Newcastle. Beside each, write the industry it grew on. Then mark the coalfields from chapter 6. How many of your seven sit on or beside one?

YOUR TURN

1. Why did industrial workers have to live close to the factory?
2. Name three British industrial cities and the industry each grew on.
3. Give three things a city needs that were not built fast enough.
4. Explain why the speed of growth, rather than the growth itself, caused most of the problems.

CHAPTER 7 · TOPIC 7.3

Living conditions

The account

Housing in the new districts was built to be cheap and built fast. Terraces were put up back to back, sharing side and rear walls, so that a house might have windows on one side only. Families, often large, lived in one or two rooms. Courtyards between blocks held a shared water tap and a shared privy for many households.

Water was the central problem. It came from standpipes running for a limited period each day, or from wells, and human waste was collected in cesspits or simply ran in the street. Where a cesspit sat near a well, the consequences were lethal.

Cholera epidemics swept European cities repeatedly in the nineteenth century. At the time most doctors believed disease came from bad air, which was a reasonable guess in a place that smelled as those districts did. It was wrong. In 1854, in Soho in London, the physician John Snow mapped cholera deaths street by street and showed that they clustered around one water pump in Broad Street. Removing the pump handle ended that outbreak. His map is one of the most famous documents in the history of both medicine and geography, and it is a piece of geographical reasoning: he found the cause by plotting where the cases were.

READ THE SOURCE

A public health report, England, 1842. Edwin Chadwick's *Report on the Sanitary Condition of the Labouring Population of Great Britain* argued that disease in the new towns was caused by filth, bad drainage and overcrowded housing, and that it cost the country more in lost work and support for widows and orphans than drainage would have cost to build.

- **Origin:** an official report by a government commissioner.
- **Purpose:** to persuade Parliament to legislate on drainage and water supply.
- **Value:** it is based on evidence gathered systematically across many towns,

and it tells you what a reforming official could establish and thought persuasive.

- **Limitation:** it argues a case. It also argues from cost as much as from

suffering, which tells you what its author expected to convince his readers.

Ask of every source in this book: who made it, for whom, and what were they trying to achieve?

■ YOUR TURN

1. Describe back-to-back housing and say why it was built that way.
2. Why was the arrangement of water supply and waste so dangerous?
3. What did John Snow do, and why is his map a piece of geography as well as medicine?
4. Using the source panel, give one value and one limitation of Chadwick's report.

CHAPTER 7 · TOPIC 7.4

Working conditions

The account

The working day in early factories commonly ran twelve hours or more, six days a week. Time was kept by the employer's clock, and lateness was punished by fines deducted from wages.

Machinery was unguarded. Belts and shafts ran overhead, close to hands and hair, and the machines did not stop for people. Injuries were frequent and compensation rare. In textile mills the air was kept warm and damp so that the thread would not snap, and it was thick with cotton dust, which over years damaged workers' lungs.

Wages were low, and a family generally needed more than one income to live. That is a large part of why women and children were in the mills at all.

Discipline was strict and unfamiliar. People used to working by task and season were now required to arrive at a fixed minute, stay until a fixed minute, and not leave the machine. Historians treat that shift in the experience of time as one of the deepest changes of the whole period, and one of the hardest for the first generation.

◆ CAREFUL

Long hours and dangerous work did not begin with factories. Farm work was exhausting and could be dangerous, and the domestic system could mean very long days too. What was new was the **concentration**: thousands of people under one roof, on one clock, at the pace of machinery. That concentration is also what eventually made it possible for workers to organise, which is chapter 11.

■ YOUR TURN

1. Describe the length and discipline of the working day in an early factory.
2. Give two specific dangers in a textile mill and explain each.
3. Why did families need more than one wage?
4. Explain what was genuinely new about factory work, given that farm work was also hard.

CHAPTER 7 · TOPIC 7.5

Child labour

The account

Children worked in factories and mines in large numbers, and it is worth being precise about why, because the reasons are not the ones pupils usually guess.

Children were employed because they were **cheap**, because families **needed** their wages to survive, because children had always worked in agriculture and in the domestic system so it did not seem like a new thing, and because in some tasks they were genuinely more useful: small enough to crawl under running machinery to clear waste, or to work in a low seam in a mine.

The work was correspondingly dangerous. Children lost fingers and hands in machinery, were injured by exhaustion, and in mines worked long hours underground.

Change came slowly, through investigation and then legislation. British parliamentary committees took evidence from workers and children themselves in the 1830s and 1840s, and the testimony was published. A succession of Factory Acts limited the hours children could work, set minimum ages, and required schooling. Crucially, later Acts appointed **inspectors**, because a law without anybody to enforce it changed very little.

GO FURTHER

Notice the sequence: evidence first, then law, then enforcement, then a real change on the ground, with years between each stage. That sequence turns up again and again in this book, and it is worth being able to name. When you reach chapter 11 and the trade unions, ask yourself which stage each of their victories belongs to.

ON THE MAP

Use the chapter's QR destination to find the Factory Acts in the order they were passed. Count the years between the first Act that limited children's hours and the first that appointed inspectors to check. Write one sentence on what that gap tells you about how change actually happens.

YOUR TURN

1. Give four reasons children were employed in factories and mines.
2. Why does the fact that children had always worked matter to the explanation?
3. What did the Factory Acts do, and why did inspectors matter so much?
4. Put these in order and explain each step: enforcement, evidence, law, change.

CHAPTER 7 · TOPIC 7.6

New social classes

The account

Industrial society produced two groups that had not existed in that form before, and the relationship between them shapes the politics of the next hundred years.

The **industrial middle class** owned the factories, the machines and the capital: mill owners, merchants, bankers, engineers, and the professionals who served them. Their income came from profit on what they owned. Many of them were newly wealthy rather than born to land, and they wanted political power to match their economic weight.

The **industrial working class** sold their labour for a wage. They owned no productive property. Their income depended on being employed, which depended on trade being good, which they did not control.

	Industrial middle class	Industrial working class
Who they were	Factory owners, merchants, bankers, engineers.	Men, women and children who worked for a wa
Where their income came from	Profit on capital they owned.	Wages for hours they worked.
Where they lived	Increasingly away from the factories, upwind.	Beside the works, because they walked to it.
What they wanted politically	The vote, and freedom to trade.	The vote, shorter hours, and the right to combi

Figure 7.3 · Two new groups, and what actually separates them.
The second row is the definition; everything else follows from it.

Two points prevent this becoming a caricature.

First, neither group was uniform. A skilled engineer and a child piecer were both working class and lived very differently. A great mill owner and a small shopkeeper were both middle class and had little in common.

Second, the older order did not vanish. Landowning aristocracies kept enormous influence throughout the nineteenth century, in Britain and even more so elsewhere in Europe. Industrial society was added to the old society before it replaced any of it.

YOUR TURN

1. Define the industrial middle class and the industrial working class by where their income came from.
2. Why did the new middle class want political power?
3. Give one way each group was internally divided.
4. Why is it wrong to say that industry swept the old landowning class away?

CHAPTER 7 · TOPIC 7.7

Portugal in the global story: the beginnings of Portuguese industrialisation

What was happening in Portugal?

Portuguese industry did develop in the nineteenth century, and it developed in particular places and particular trades rather than across the country.

The sectors that mattered were **textiles**, especially cotton and wool; **cork**, in which Portugal had and still has a genuine natural advantage, because the cork oak grows well in the Alentejo and the cork can be stripped without killing the tree; **ceramics and glass**; and **food processing**, including canned fish, which became an important export.

The geography of it was concentrated. Industry grew in and around **Lisbon**, in and around **Porto** and the northern towns of the Ave valley, and at **Setubal**, where the fish canneries were. Away from those districts, Portugal remained agricultural, and in much of the interior it remained agricultural in ways that had changed little for generations.

What was happening in the world?

Britain by mid-century was already an industrial society with more people in towns than in the countryside. Belgium, France and the German states were industrialising fast. Portugal was industrialising too, but from a smaller base, more slowly, and in fewer sectors.

What part did geography play?

The concentration is not an accident, and it is the same list of conditions from chapter 6, read for Portugal. Lisbon and Porto had ports, which brought in raw cotton and sent out finished goods. They had rivers, the Tejo and the Douro, and the labour of a coastal strip that was already the most populated part of the country, as chapter 4 showed. They had merchants with capital. The interior had none of those things, and mountainous terrain made moving heavy goods to it expensive.

What was the connection?

Portugal shows something the British story can hide: **industrialisation is regional before it is national.** A country does not industrialise all at once. It industrialises where the conditions are, and the gap between the industrialising districts and the rest can widen for decades. That was true of Portugal, and it was equally true of Italy, of Spain and, as chapter 9 will show, of the United States.

KEY QUESTION

Why might industrialisation concentrate in particular cities and regions?

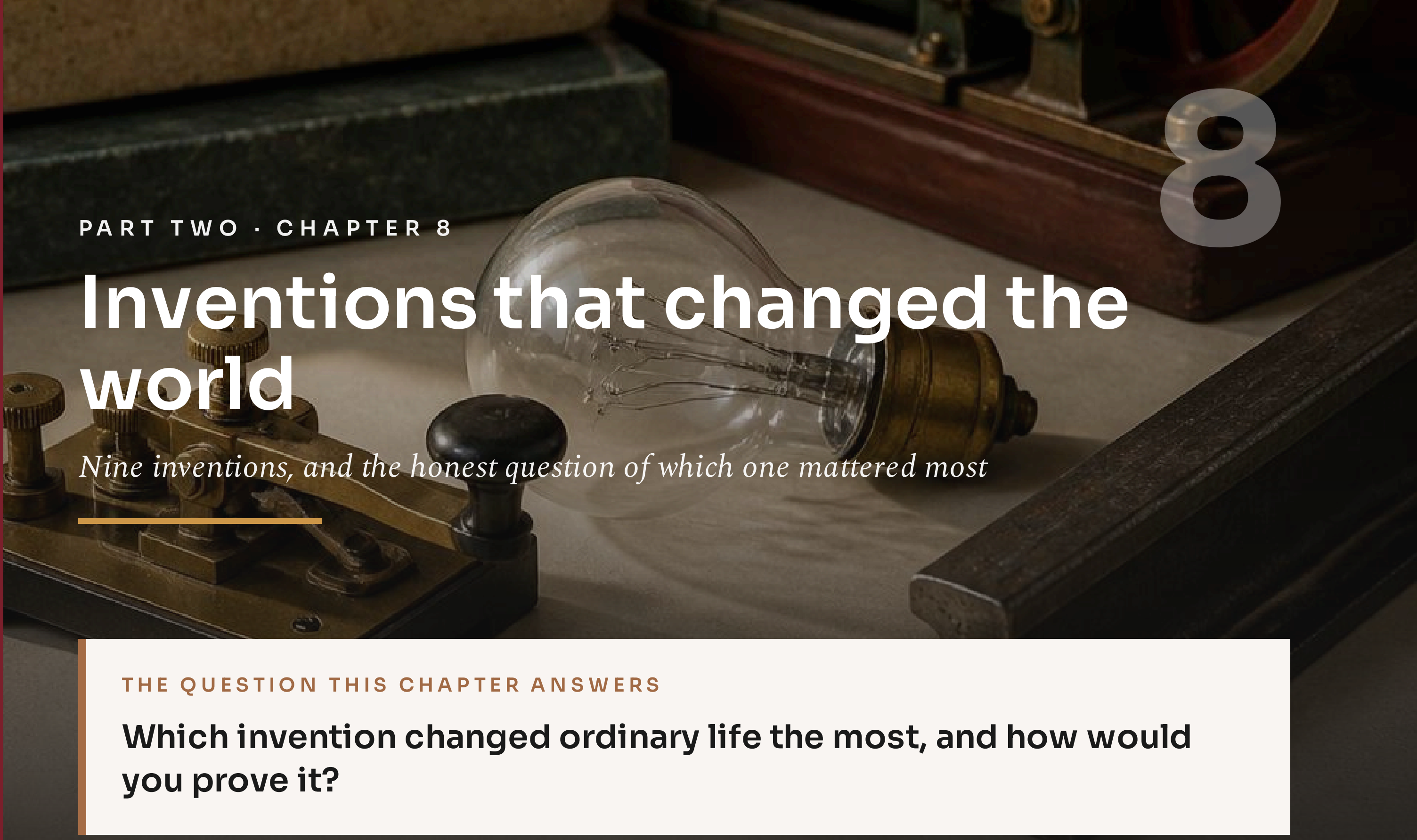
Use the coastal strip from Braga to Setubal as your case. List the conditions from chapter 6 that it had and the interior did not. Then answer this: if a government in 1880 had wanted to industrialise the interior, which single condition would it have had to supply first, and why that one?

WHAT YOU CAN NOW DO

- √ Explain what changed when work moved from the home into the factory.
- √ Describe how industrial cities grew and what they failed to build in time.
- √ Use evidence, including a source, to describe living and working conditions.
- √ Explain why children were employed and how and why that changed.
- √ Describe the two new social classes and what separates them.

WORDS FOR THIS CHAPTER

factory system, division of labour, urbanisation, back-to-back housing, sanitation, cholera, Factory Acts, inspector, industrial middle class, industrial working class, wage, profit, cork, cannery



PART TWO · CHAPTER 8

8

Inventions that changed the world

Nine inventions, and the honest question of which one mattered most

THE QUESTION THIS CHAPTER ANSWERS

Which invention changed ordinary life the most, and how would you prove it?

IN THIS CHAPTER, YOU WILL

- √ Explain, for each invention, the problem it solved and what changed.
- √ Describe how steam power freed factories from fast-flowing rivers.
- √ Explain why the telegraph separated information from transport.
- √ Trace the chain from invention to lower prices to new ways of living.
- √ Explain how a country can be transformed by a technology it did not invent.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the steam engine. Read what the separate condenser actually did, then say in one sentence why it saved fuel.

`www.britannica.com/technology/steam-engine`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Choose one object in this room and ask three questions about it: what problem does it solve, how does it work, and what would be different if it did not exist? Those three questions are the shape of this whole chapter, and you will answer them nine times.

CHAPTER 8 · TOPIC 8.1

Textile technology

The account

Cloth is where industrialisation began, and the reason is arithmetic. Making cloth has two stages: spinning fibre into thread, and weaving thread into cloth. For centuries the two were roughly in balance, with several spinners keeping one weaver supplied.

Then weaving got faster, and the balance broke. Weavers could work through the thread faster than spinners could make it, and the whole industry was held up waiting for yarn. That bottleneck is what the famous textile machines were built to clear.

Spinning jenny
One worker turns many spindles at once, so one pair of hands spins many threads.

→

Water frame
Powered by a water wheel, so it is too big for a cottage and needs a mill beside a river.

→

Power loom
Weaving is mechanised too, so the cloth can keep up with the thread.

Figure 8.1 · Three machines, and the problem each one solved.
Read them in order. Notice that the second machine is the one that forces production out of the cottage, because of what powers it.

The **spinning jenny** let one worker turn many spindles from a single wheel, so one pair of hands produced many threads at once. It was small enough to use at home.

The **water frame** produced stronger thread and was driven by a water wheel. That is the decisive change: a machine driven by falling water cannot be in a cottage. It has to be in a building beside a river, and the workers have to come to it. The first mills were built for this reason, before steam power existed.

The **power loom** mechanised weaving, so cloth production could keep pace with the new supply of thread.

Notice the pattern, because it repeats through the whole of industrial history: solving a bottleneck at one stage creates a bottleneck at the next.

YOUR TURN

1. What are the two stages of making cloth, and what broke the balance between them?
2. Explain what the spinning jenny did.
3. Why did the water frame force work out of the home?
4. Describe the bottleneck pattern in your own words, using these three machines.

CHAPTER 8 · TOPIC 8.2

Steam power

The account

Water power has one crippling limitation: it only exists where there is fast-flowing water. A mill has to be where the river is, however inconvenient that is for coal, workers or customers, and in a dry summer or a frozen winter it stops.

Steam removed that limitation. A steam engine burns fuel to boil water, and the pressure of the steam drives a piston. Because coal can be carried to the engine, the engine can be anywhere.

The first practical engines were built to pump water out of mines, which was the problem you met in chapter 6. They were enormously wasteful of fuel, which mattered little at a coal mine where fuel was underfoot and mattered a great deal anywhere else.

James Watt's improvements, from the 1760s onward, changed that. The most important was a **separate condenser**. In the earlier design the same cylinder was heated to fill with steam and then cooled to condense it, so an enormous amount of heat was spent reheating the cylinder on every stroke. Watt condensed the steam in a second vessel, so the working cylinder could stay hot. The engine used far less coal for the same work, which made steam power worth having away from the coalfield. Later developments turned the up-and-down motion into rotation, so an engine could drive machinery rather than only a pump.

Pumping mines
The first job: lifting water out so deeper coal could be cut.

→ **Driving factories**
Power wherever it was wanted, not only beside fast water.

→ **Moving trains**
A machine that carried its own power along the track with it.

→ **Driving ships**
A crossing that no longer waited for the wind.

Figure 8.2 · One machine, four jobs, in the order it took them on.
The first box is why it was invented. The three that follow are why it mattered.

CAREFUL

Watt did not invent the steam engine. Working engines existed before him and are associated with Thomas Savery and Thomas Newcomen. Watt made the engine efficient enough to be worth using away from a coal mine. In the history of technology, the person who makes something practical is often more important than the person who makes it first, and this is a good example to remember.

YOUR TURN

1. Give the crippling limitation of water power, in one sentence.
2. Explain what a separate condenser does and why it saved fuel.
3. Why did the first engines' inefficiency matter little at a coal mine?
4. "Watt invented the steam engine." Correct that sentence and explain the correction.

CHAPTER 8 · TOPIC 8.3

Railways

The account

Put a steam engine on wheels and set it on iron rails, and you have a machine that carries its own power with it. That is the railway, and it changed distance more than anything before it.

Rails matter as much as the engine. A smooth iron wheel on a smooth iron rail has very little friction, so one engine can pull an enormous load. That is why railways carry heavy freight so cheaply, and it is why the first tracks were built at collieries to move coal before passengers were considered at all.

The effects came in three waves.

Goods moved cheaply overland for the first time. Before railways, heavy freight went by water or it did not go. Afterwards, an inland town could be supplied as cheaply as a port, and a national market became possible.

People moved. Ordinary people could travel distances that had previously belonged to the wealthy. Seaside towns, commuting and national newspapers all follow from that.

Time itself was standardised. As chapter 2 explained, a timetable is impossible when every town keeps its own clock, so railways forced countries onto a single time.

GO FURTHER

Railways also created enormous demand for exactly the industries that built them: iron and then steel for rails, coal for engines, engineering for locomotives, and vast amounts of capital raised from investors. An industry that consumes its own inputs on that scale is an accelerator. Ask yourself, as you read chapter 9, why the United States is a country where this effect shows up even more dramatically than in Britain.

YOUR TURN

1. Why does an iron wheel on an iron rail let one engine pull so much?
2. Give the three waves of effect described above.
3. Explain why railways made national markets possible.
4. Why did railways force the standardisation of time?

CHAPTER 8 · TOPIC 8.4

Steamships

The account

A sailing ship goes where the wind allows and arrives when the wind decides. Whole trade routes were built round that fact: the Atlantic crossings, the Indian Ocean monsoon timetable you met in chapter 3, the seasonal rhythm of whole ports.

A steamship goes where it is pointed. It can keep a schedule, enter and leave harbour without waiting for a favourable wind, and go up a river against the current. The change is not mainly about speed. It is about **reliability**, and a merchant can plan around a reliable ship in a way that is impossible with a fast but unpredictable one.

Iron and then steel hulls made ships larger than timber ever allowed, and larger ships carry cargo more cheaply per tonne. Falling freight costs across the nineteenth century meant that heavy, low-value goods, grain, ore, coal and timber, could be shipped across oceans and still be worth selling. That is the beginning of a genuinely global market in ordinary commodities, and chapter 31 picks the story up again.

Steamships also needed coaling stations, because a ship burning coal must refuel. That gave certain islands and ports a strategic value they had never had before, which is a thread that runs directly into chapters 12, 18 and 21, and into the Azores.

ON THE MAP

Look at a map of the Atlantic and find the Azores. Now imagine you are routing a steamship from Europe to North America in 1900 and must refuel. Explain, in two sentences, why a group of islands in mid-ocean is worth more to you than it was to a sailing captain in 1700.

YOUR TURN

1. What is the main advantage of a steamship over a sailing ship? It is not speed.
2. Why did iron and steel hulls lower the cost of shipping per tonne?
3. What kinds of goods became worth shipping long distances, and why?
4. Explain why coaling stations gave some islands new strategic importance.

CHAPTER 8 · TOPIC 8.5

The telegraph

The account

Until the middle of the nineteenth century, a message travelled at the speed of the person carrying it. A letter went as fast as a horse, a ship or a train, and no faster. Information and transport were, for all practical purposes, the same thing.

The electric telegraph broke that link. A message became a pattern of electrical pulses along a wire, and it arrived in minutes over distances that took a ship weeks. Submarine cables followed, and by the later nineteenth century a network of wires connected the continents.

	Before the telegraph	After the telegraph
How fast news moved	As fast as the fastest horse, ship or train carrying it.	Nearly instantly, along a wire, over any distanc with a line.
What that meant for trade	A price could be days or weeks out of date when it arrived.	Markets in different countries could quote the price the same day.
What it meant for government	An order took as long to arrive as a person did.	A capital could instruct a distant official the sa afternoon.
The deep change	Information and transport were the same thing.	Information came loose from transport, for the time in history.

Figure 8.3 · The single deepest change in this chapter.
Read the last row first. Everything above it is a consequence of that one separation.

The consequences are easy to underestimate because they are so complete.

Markets changed. Traders in different countries could quote the same price on the same day. A price difference between two markets could be acted on immediately, which tends to close the difference.

Government changed. A capital could instruct a distant official the same afternoon rather than waiting three months for a reply. Empires became far more tightly controlled from the centre.

War changed. Orders, intelligence and news moved at a speed armies had never had to cope with, and newspapers could report a battle within a day.

YOUR TURN

1. What did "information and transport were the same thing" mean in practice?
2. Give three areas of life the telegraph changed, with one example of each.
3. Why did the telegraph tend to close price differences between markets?
4. Which do you think mattered more, the railway or the telegraph? Give one reason and one counter-argument.

CHAPTER 8 · TOPIC 8.6

The telephone

The account

The telegraph carried code. Somebody had to translate a message into pulses at one end and back into words at the other, which meant trained operators, offices and a charge per word that encouraged people to write as briefly as possible.

The telephone, developed in the 1870s and associated above all with **Alexander Graham Bell**, carried the voice itself. That removed the operator from the middle of the conversation and made instant communication something an ordinary person could do without training.

Two changes followed that are still with you. Communication became **conversational** rather than one-way: you could ask a question and get an answer while still on the line. And communication became **domestic**: it moved into homes and offices instead of being something you went to a telegraph office to do.

CAREFUL

Bell is the name in most textbooks, and the history is more crowded than that. Several people were working on transmitting speech electrically at the same time, and there were long legal disputes over priority. It is more accurate to say that the telephone emerged from the work of several inventors and that Bell secured the decisive patent and built the business.

YOUR TURN

1. Give two practical drawbacks of the telegraph that the telephone removed.
2. What does it mean to say communication became conversational?
3. Why did the telephone reach homes when the telegraph largely did not?
4. Why is "Bell invented the telephone" an incomplete statement?

CHAPTER 8 · TOPIC 8.7

Steel

The account

Iron comes in more than one form, and the differences decide what can be built. Cast iron is hard but brittle: it stands up to being squashed and snaps when bent. Wrought iron bends without snapping but is soft and was slow to make. **Steel** is iron with a carefully controlled small amount of carbon, and it is both hard and tough, which is exactly what engineering wants.

Steel had been made for centuries in small quantities, at great expense, for tools and weapons. The change in the nineteenth century was **process**: methods were developed, above all the Bessemer converter, that blew air through molten iron to burn off impurities and produced steel in tonnes rather than kilograms, cheaply.

Once steel was cheap, it went everywhere. Rails that lasted years instead of months. Bridges of unprecedented span. Ship hulls. Boilers that could hold higher pressures, which made better engines. Machine tools that could cut other metals accurately, which is what lets parts be interchangeable. Frames that let buildings rise far higher than masonry allowed.

WHERE PORTUGAL FITS

Cheap steel reached Portugal as an import and as engineering rather than as an industry. Two of the best-known structures in Porto are iron and steel bridges of this era: the Maria Pia railway bridge over the Douro, opened in 1877 and designed by the firm of Gustave Eiffel, and the Luis I bridge, opened in 1886. They are a good illustration of the point at the end of this chapter: a country can be visibly transformed by a technology it did not itself develop.

YOUR TURN

1. Give the difference between cast iron, wrought iron and steel.
2. What changed in the nineteenth century: the material or the process? Explain.
3. List four things cheap steel made possible.
4. Explain why better boilers led to better engines.

CHAPTER 8 · TOPIC 8.8

Electricity

The account

Steam power has to be delivered mechanically. A factory engine drove a long shaft running the length of the building, and every machine was linked to it by belts. The layout of the whole factory was therefore decided by the shaft, the belts were dangerous, and if the engine stopped, everything stopped.

Electricity separates where power is **generated** from where it is **used**. A generator can be somewhere else entirely, and current is delivered by wire to a motor on each machine. Machines can then be arranged in the order that suits the work, small workshops can have power without owning an engine, and one machine can stop without stopping the rest.

Electric lighting changed time as well as space. Gas lighting already existed and was dirty and dangerous; electric light was neither, and it made evening work, evening study and safer streets ordinary rather than exceptional.

And electricity made possible things that had no earlier version at all: electric trams and underground railways, electrolysis for refining metals, and eventually the whole of electronics.

GO FURTHER

Compare electricity with the telegraph. The telegraph separated information from transport. Electricity separated power from the place it was made. Both are separations, and both had enormous consequences precisely because two things that had always had to be in the same place no longer did. When you reach chapter 31, ask what the internet separated.

YOUR TURN

1. Describe how a steam-powered factory delivered power to its machines.
2. Give three advantages of electric power over a shaft and belts.
3. How did electric lighting change the working and studying day?
4. Name two things electricity made possible that had no earlier version.

CHAPTER 8 · TOPIC 8.9

The internal combustion engine

The account

A steam engine burns its fuel outside the cylinder, to boil water. An internal combustion engine burns fuel inside the cylinder itself, so the expanding gases push the piston directly. That removes the boiler, the water and a great deal of weight.

The consequence is that the engine can be **small enough to move itself and a useful load**. A steam locomotive is superb at hauling a heavy train along a fixed track; it is hopeless as a family vehicle. A petrol engine made the motor car, the lorry, the tractor and, once it was light enough, the aeroplane.

Three consequences reach directly into the rest of this book.

Transport became **individual and flexible**, not tied to a track or a timetable. Cities could spread outward in a way railways had not allowed.

Warfare changed. Lorries, tanks and aircraft all depend on it, and chapters 14 and 18 are unimaginable without it.

Oil became a strategic resource. A country that ran on coal could usually find coal at home; a country that runs on oil must have oil or must obtain it. That single sentence explains a great deal of twentieth-century foreign policy, and you will meet it again in chapters 18 and 23.

YOUR TURN

1. Give the essential difference between an external and an internal combustion engine.
2. Why does burning the fuel inside the cylinder make the engine so much lighter?
3. Give three things the internal combustion engine made possible.
4. Explain why oil became strategically important in a way coal had not been.

CHAPTER 8 · TOPIC 8.10

Invention, production and a new way of living

The account

Nine inventions have appeared in this chapter, and the point of the last topic is to join them into one argument rather than leave them as a list.

The chain runs like this. An **invention** raises what a worker can produce in an hour. Higher productivity allows **mass production**, because more can be made with the same effort. Mass production lowers the **cost** of each item, which lowers the **price**. Lower prices bring more people into the **market**, because things that were once luxuries become ordinary purchases. A larger market repays further investment in machinery, and the loop turns again.

What problem
Every invention in this chapter answers a problem somebody already had.

How it worked
The mechanism, in one sentence a pup can repeat.

What changed
Who could now do what they could not do before.

Figure 8.4 · Three questions, asked of every invention in this chapter.
Use this as a checklist when you revise: an invention you cannot answer all three questions about is one you have not learned yet.

What comes out at the end is not simply more goods. It is a **different way of living**. Ordinary households owned more clothes, ate food grown further away, had light in the evening, could travel, could send a message the same day, and worked to a clock rather than to the sun. None of those was true of the world described in chapter 5, and all of them were true within about a century.

■ **THE ENQUIRY: WHICH INVENTION MATTERED MOST?**

Choose one invention from this chapter and argue that it mattered more than the others.

1. State your choice and the three answers: what problem, how it worked, what changed.
2. Give two consequences that would not have happened without it.
3. Name the strongest rival to your choice and say honestly why somebody might pick it.
4. Explain why your choice still wins, or change your mind and say so. Changing your mind with a reason is a better answer than defending a weak claim.

CHAPTER 8 · TOPIC 8.11

Portugal in the global story: railways and modernisation

What was happening in Portugal?

Portugal did not originate the technologies of this chapter. It **adopted** them, and the clearest case is the railway.

On 28 October 1856 the first railway line in Portugal opened, running from Lisbon to Carregado, a distance of about thirty-six kilometres along the Tejo valley. From that short beginning the network was extended steadily: to the Spanish border in the 1860s, north to Porto, and eventually into the interior. Two great river crossings at Porto, the Maria Pia railway bridge of 1877 and the Luis I bridge of 1886, belong to the same effort.

Telegraph lines, improved roads, port works and urban transport followed the same pattern in the same decades.

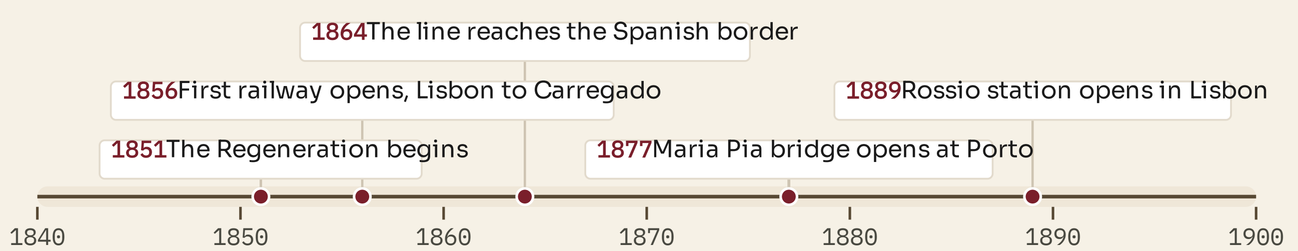


Figure 8.5 · Portugal's railway age.
Read the first entry alongside the others: the political decision of 1851 comes before the engineering, and chapter 10 explains why.

What was happening in the world?

By 1856 Britain already had a dense railway network and had been running trains for a generation. France, Belgium and the German states were well advanced. Portugal was arriving later, and building a smaller network, in more difficult terrain, with less capital.

What part did geography play?

Portuguese terrain is not kind to railways. Mountains in the north and centre mean cuttings, tunnels and viaducts, all of which are expensive, and the country had few navigable rivers reaching inland to compete with or to feed the lines. That is part of why the network cost so much and why the interior waited longest.

What was the connection?

Here is the lesson this section exists to teach, and it applies far beyond Portugal.

A country does not have to invent a technology in order to be transformed by it. The railway was British in origin, and its effects in Portugal were Portuguese: regions that had been days apart became hours apart, agricultural produce could reach Lisbon and Porto in a condition worth selling, and internal markets that had been separate became one. Adopting a technology is a real historical act, and it costs money, needs engineers, and requires a government willing to borrow. Chapter 10 is about who borrowed, and what it cost.

KEY QUESTION

How did railways change geographical distance within Portugal?

Choose two Portuguese places at least two hundred kilometres apart. Estimate the journey time between them in 1840, by road and on foot or by cart, and then by train in 1900. Then answer this: which changed, the distance or the time? Write one sentence explaining why geographers care about the difference between the two.

WHAT YOU CAN NOW DO

- √ Explain, for each invention, the problem it solved and what changed because of it.
- √ Explain why the water frame forced work out of the home.
- √ Explain how steam freed factories from fast-flowing rivers.
- √ Explain why the telegraph separated information from transport.
- √ Explain how a country can be transformed by a technology it did not invent.

WORDS FOR THIS CHAPTER

spinning jenny, water frame, power loom, bottleneck, steam engine, separate condenser, locomotive, steamship, coaling station, telegraph, submarine cable, telephone, steel, Bessemer converter, generator, internal combustion engine, mass production

PART THREE

Industrial capitalism and new ideas

Industry did not only change machines. It changed who owned things, how money was raised, how big a company could become, and how unequal a society could be. Those changes produced arguments, and the arguments produced ideas that people were still fighting over a century later. These three chapters follow industry out of the factory and into economics and politics.

9

PART THREE · CHAPTER 9

The United States becomes an industrial power

A continent, a market and the largest industrial economy on Earth

THE QUESTION THIS CHAPTER ANSWERS

How did a country with almost no industry in 1800 become the largest industrial economy on Earth?

IN THIS CHAPTER, YOU WILL

- √ Explain how resources, navigable water and scale combined in the United States.
- √ Explain why a single large internal market helps manufacturers.
- √ Describe what immigration gave American industry, and what it cost the migrants.
- √ Compare vertical integration with control of a choke point.
- √ Explain the assembly line and the loop from lower cost to a larger market.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the rise of big business in the United States. How a farming republic became the largest industrial economy on Earth in roughly forty years.

`www.britannica.com/place/United-States/The-rise-of-big-business`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Name a company whose products you have used this week. Do you know who owns it, where it is based, or how many people work for it? Most of us do not. That is a modern situation, and this chapter is about how it came about.

CHAPTER 9 · TOPIC 9.1

Geography and resources

The account

In 1800 the United States was a farming republic on the Atlantic seaboard with almost no industry. By 1900 it produced more manufactured goods than any other country on Earth. That transformation is the subject of this chapter, and the first part of the explanation is the ground itself.

The United States had, inside one set of borders, an unusual combination: enormous **coalfields**, particularly in Pennsylvania and the Appalachians; extensive **iron ore**, notably around Lake Superior; huge reserves of **timber**; **oil**, discovered in Pennsylvania in 1859 and later in far greater quantities elsewhere; and some of the most productive **agricultural land** anywhere, in the Mississippi basin and the prairies.

Two features multiply the value of that list.

The first is **navigable water**. The Mississippi and its tributaries drain a vast interior and reach the sea. The Great Lakes form an inland waterway connecting the ore of the north-west with the coal of the east. Moving iron ore to coal by water is exactly the arrangement chapter 6 showed to be decisive, and the United States had it on a continental scale.

The second is **scale itself**. Britain's advantages were real and were contained in a small island. The same advantages spread across a continent produce a different order of magnitude, and by the end of the century that difference had become a difference in world power.

CAREFUL

Resources do not industrialise a country by themselves. Several countries with comparable resources did not become industrial powers in this period, and the land the United States expanded across was not empty: it was taken, by treaty, purchase and force, from peoples who already lived there. A full account of American industrial growth includes that, and chapter 12 returns to the wider pattern it belongs to.

YOUR TURN

1. List five natural resources the United States had in quantity.
2. Explain why the Great Lakes and the Mississippi mattered so much.
3. Using chapter 6, explain why ore and coal being connected by water is so important.
4. Why is "they had resources" an incomplete explanation of American industrialisation?

CHAPTER 9 · TOPIC 9.2

A large internal market

The account

Chapter 6 showed that industry needs buyers. The United States generated its own buyers in a way no European country could, because its population grew extremely fast throughout the nineteenth century, and the whole of it lay inside a single customs area.

That last point matters more than it sounds. Goods moved from Massachusetts to Missouri without crossing a border, paying a tariff or being weighed by a foreign customs officer. In Europe at the same date, a manufacturer in one German state met a customs post at the edge of the next. A single large internal market lets a firm build one very large factory rather than several small ones, and the larger the run, the lower the cost of each item.

The population was also, on average, better off than most of Europe's. Land was available, wages were relatively high, and a family with money left over after food is a family that buys manufactured goods.

So the loop from chapter 6 turned faster here than anywhere: more buyers, larger factories, lower costs, lower prices, more buyers again.

YOUR TURN

1. Why does a single customs area help manufacturers?
2. Explain the link between the size of a factory and the cost of each item.
3. Why does a relatively well-paid population matter to industry?
4. Compare the American internal market with the German lands before unification.

CHAPTER 9 · TOPIC 9.3

Immigration

The account

Between the middle of the nineteenth century and the First World War, tens of millions of people crossed the Atlantic to the United States. They came from Ireland, Germany, Scandinavia, Italy, Poland, Russia, the Austro-Hungarian lands, Portugal and many other places, driven by hunger, poverty, land shortage, religious persecution and political repression, and drawn by wages, land and the reports of those who had already gone.

For American industry the effect was threefold. Immigration supplied **labour** in the quantities the new factories needed. It supplied **skills**, because migrants included trained craftsmen, engineers and miners. And it supplied **consumers**, because everybody who arrived needed clothes, food, housing, transport and fuel.

The experience of the migrants themselves was much harder than that summary suggests. Wages were low, hours long and housing crowded. Newly arrived groups were frequently met with hostility from those who had arrived a generation earlier, and were often given the worst-paid and most dangerous work.

WHERE PORTUGAL FITS

Portugal was one of the countries people left. Portuguese emigration in this period went overwhelmingly to **Brazil**, and also to the United States, where Portuguese communities grew in New England and in California, and to Hawaii. Emigration on this scale is a sign of an economy that could not employ its own population, and it had a double effect at home: it removed working-age people, and it sent back money that supported families and paid for land and houses. Chapter 25 shows the same pattern continuing into the twentieth century, with different destinations.

YOUR TURN

1. Give three things immigration supplied to American industry.
2. Give two reasons people left Europe in this period, and two reasons they chose the United States.
3. Why did newly arrived groups often get the worst work?
4. What does large-scale emigration tell you about the economy people are leaving?

CHAPTER 9 · TOPIC 9.4

Railways

The account

A continent is only a single market if goods can cross it. Railways are what made the United States one country economically as well as politically.

Track was laid at extraordinary speed through the second half of the century, and in 1869 the first transcontinental line was completed, joining the eastern network to the Pacific coast. The consequences ran in every direction at once.

Resources reached factories. Ore, coal and timber could be moved from where they were to where they were used.

Produce reached cities. Grain from the prairies and cattle from the plains reached Chicago and then the eastern cities, and refrigerated cars later meant meat could travel too.

Settlement followed the track. Towns grew where the railway put a station, and the railway companies, who had been granted enormous areas of land, had every reason to encourage people to settle along their lines.

The railway was itself an industry. It consumed steel, coal, timber and engineering on a colossal scale, and it raised capital from investors on a scale that changed American finance. Chapter 10 picks that up.

GO FURTHER

Railways also carried the settlement of the interior, and that settlement displaced the Native American nations who lived there, by treaty, by purchase and by force. The railway is a good example of a technology whose effects cannot be described as simply good or simply bad: the same lines that built a national market also carried the process that destroyed other people's countries. Hold both halves.

YOUR TURN

1. Give four effects of railway building in the United States.
2. Why did railway companies want people to settle along their lines?
3. Explain the sentence "the railway was itself an industry".
4. Why is the railway an example of a technology with effects that cannot be summed up in one word?

CHAPTER 9 · TOPIC 9.5

Industrial cities

The account

American industrial cities grew even faster than British ones, and each grew around a particular advantage.

Chicago stood where the railways of the interior met the Great Lakes, and became the point at which the produce of the prairies was gathered, processed and sent east. **Pittsburgh** sat on coal and at the meeting of rivers, and became the steel city. **Detroit** had the Great Lakes, engineering skill and, later, the motor industry. **New York** was the great port, the point of arrival for migrants and the centre of finance.

The problems of chapter 7 appeared again, at a larger scale: crowded housing, poor sanitation, dangerous work, and districts divided by which group had arrived when. So did the responses: public health measures, building regulations, and eventually organised labour.

One American difference is worth noting. Cheap steel and the safety lift made it possible to build upward, and from the 1880s the tall steel-framed building appears, first in Chicago. A city that cannot spread outward across a lake or a river spreads upward instead, and the skyline you recognise today is a consequence of that.

YOUR TURN

1. Match each of Chicago, Pittsburgh, Detroit and New York to its advantage.
2. Give two problems these cities shared with British industrial cities.
3. What two technologies made tall buildings possible?
4. Why did Chicago in particular grow upward?

CHAPTER 9 · TOPIC 9.6

Andrew Carnegie and steel

The account

Andrew Carnegie arrived in the United States from Scotland as a poor child, worked in a cotton mill and then for a railway, and by the end of the century controlled the largest steel business in the world. His career is worth studying not as a story about one man but as an example of how industrial fortunes were built.

His central method was **vertical integration**: owning every stage of production. Rather than buy iron ore from one firm, coke from another and transport from a third, his company owned the ore fields, the coke works, the ships and railways that moved them, and the mills. Every stage that would have been somebody else's profit became his own, and no supplier could hold him up.

He also invested heavily in the newest processes, replacing plant that still worked with plant that worked better, and drove costs down relentlessly.

The results were spectacular and the costs were real. Steelworkers worked extremely long shifts in dangerous conditions, and attempts to organise were resisted hard, most famously at the Homestead strike of 1892, where the dispute became violent and the union was broken.

Carnegie later gave away most of his fortune, endowing libraries, universities and concert halls, and argued in his own writing that the rich had a duty to distribute their wealth in their lifetime. Whether that answers the criticism of how the wealth was made is exactly the kind of question a historian is supposed to ask rather than settle.

YOUR TURN

1. Explain vertical integration and give an example from Carnegie's business.
2. Give two advantages vertical integration gave him over a rival.
3. What were the costs of his methods, and who paid them?
4. "His philanthropy answers the criticism of how he made the money." Argue for and against.

CHAPTER 9 · TOPIC 9.7

John D. Rockefeller and oil

The account

Oil in 1859 was a curiosity. By 1900 it lit homes, lubricated machinery and was about to fuel the internal combustion engine you met in chapter 8. John D. Rockefeller built the business that dominated it.

His method was different from Carnegie's. Rather than own every stage, he concentrated on **refining**, the stage through which everything had to pass, and then on the transport of oil. Controlling the point that everything must go through gives you power over everybody at the other stages.

Standard Oil grew by buying rivals, by negotiating rates with railways that its competitors could not match, and by relentless efficiency. By the 1880s it controlled the overwhelming majority of American refining, which is a **monopoly**: a market with effectively one seller.

Monopoly is where economics becomes politics. A monopolist can set prices, and customers have no alternative. Public anger led to laws intended to restrain it, and in 1911 the United States Supreme Court ordered Standard Oil to be broken into separate companies.

GO FURTHER

Compare the two methods. Carnegie controlled a **chain**; Rockefeller controlled a **choke point**. Both produced enormous firms. Ask yourself which is harder for a competitor to challenge, and which is harder for a government to regulate. Then look at a modern industry you know and decide which pattern it looks like.

YOUR TURN

1. What was Rockefeller's strategy, and how did it differ from Carnegie's?
2. Define monopoly, and explain why it worries governments.
3. Give two methods Standard Oil used to grow.
4. What happened to Standard Oil in 1911, and why?

CHAPTER 9 · TOPIC 9.8

Thomas Edison and invention as an industry

The account

Thomas Edison's importance is not only the things associated with his name, the practical electric light, the phonograph and much of the equipment of an electrical system. It is that he changed **how invention was done**.

Before him, inventing was largely the work of individuals in workshops. Edison built a **research laboratory** at Menlo Park, employing teams of skilled workers, chemists and engineers, funded deliberately to produce useful inventions. It was invention organised as a business: a target was chosen, teams worked on it systematically, and the results were patented.

He also understood that an invention is worthless without a **system** around it. An electric lamp needs generation, cables, switches, a way of measuring what each customer uses, and somebody to supply and bill for the current. Edison built the system as well as the lamp, including the first central generating stations.

Two consequences reach into your own life. Corporate research laboratories became normal, and are where most modern invention happens. And the pattern of inventing a product together with the system it needs is the pattern of every modern technology industry.

CAREFUL

Edison did not invent the light bulb on his own, and other inventors, including Joseph Swan in Britain, were working on it at the same time. Edison produced a practical, long-lasting lamp and, crucially, the electrical system to run it. As with Watt in chapter 8, "who made it work and made it usable" is often a more useful question than "who was first".

YOUR TURN

1. What did Edison change about the way invention was done?
2. Why is a research laboratory more productive than a lone inventor?
3. Explain why an electric lamp needs a system, and name three parts of it.
4. Why is "Edison invented the light bulb" an incomplete statement?

CHAPTER 9 · TOPIC 9.9

Henry Ford and the assembly line

The account

Henry Ford did not invent the motor car, and he did not invent the idea of interchangeable parts. What he did was combine existing ideas into a way of producing that changed manufacturing everywhere.

The **assembly line** moves the work past the worker, instead of moving the worker round the work. Each person performs one small operation, repeatedly, and the pace is set by the line. Combined with genuinely interchangeable parts, made accurately enough that any one will fit any car, it cut the time to build a car enormously.

The economics of it are the important part, and they follow the loop you already know. Lower cost per car means a lower price. A lower price means far more people can afford one. A far bigger market justifies still more investment in the line. Ford also paid comparatively high wages, which reduced the number of workers leaving, and which he argued made his own workers into customers.

The cost was the work itself. Assembly line work is repetitive, tightly paced and gives the worker almost no control over the day. That complaint is the industrial argument of chapter 7 again, in a twentieth-century form.

GO FURTHER

Mass production has a hidden condition: it only works if you can sell what you make. A factory built for very large runs is very efficient at full capacity and ruinous at half. Hold that thought until chapter 16, when demand collapses in the Great Depression and the most modern factories in the world stand idle.

YOUR TURN

1. Explain what an assembly line does and why it saves time.
2. Why are interchangeable parts essential to it?
3. Set out the loop from lower cost to a larger market.
4. Give one benefit and one cost of assembly line work for the worker.

CHAPTER 9 · TOPIC 9.10

Portugal in the global story: two different economic paths

What was happening in Portugal?

While the United States was becoming the largest industrial economy in the world, Portugal remained a small, largely agricultural economy with industry concentrated in a few coastal districts, as chapter 7 described. Its most visible connection to the American story was not trade. It was **people**.

Portuguese emigration rose steadily through the second half of the nineteenth century and into the twentieth. The largest destination by far was **Brazil**, where language and long historical ties made settlement easier. Substantial Portuguese communities also formed in the **United States**, particularly in New England and later in California, and in **Hawaii**. Many came from the Azores and Madeira, where land was short and opportunity limited.

What was happening in the world?

The same decades saw the greatest movement of people across the Atlantic in history. Portugal was one source among many, and its emigrants were part of the same labour supply that chapter 9 has just described as one of the causes of American industrial growth.

What part did geography play?

Compare the two countries condition by condition, and the difference is not mysterious.

	The United States	Portugal
Territory	A continent, crossed by navigable rivers and the Great Lakes.	A small country with mountainous interior and navigable rivers.
Coal and iron	Very large fields of both, connected to each other by water.	Little accessible coal; modest iron ore.
Internal market	Tens of millions of buyers inside one customs area, growing every year.	A small home market, and much of it poor.
People	Receiving millions of migrants who supplied labour, skill and demand.	Sending emigrants, chiefly to Brazil and also to United States.
Result by 1900	The largest industrial economy in the world.	Industry in a few coastal districts, inside a farming country.

Figure 9.1 · Two economies, side by side.
Read the rows across. Notice that the last row is the consequence of the rows above it, not a separate fact.

The United States had a continent, coal and ore connected by water, a market of tens of millions inside one customs area, and immigration adding to it every year. Portugal had a small territory, little accessible coal, difficult internal transport, a small and largely poor domestic market, and a population that was leaving.

What was the connection?

The connection is direct and it runs through people. **Portugal supplied some of the labour that built the American economy.** A Portuguese family in the Azores in 1890 and a steel mill in Pennsylvania are part of the same system, and the money sent home from one paid for houses and land in the other.

KEY QUESTION

Why did the United States become an industrial superpower while Portugal industrialised more slowly?

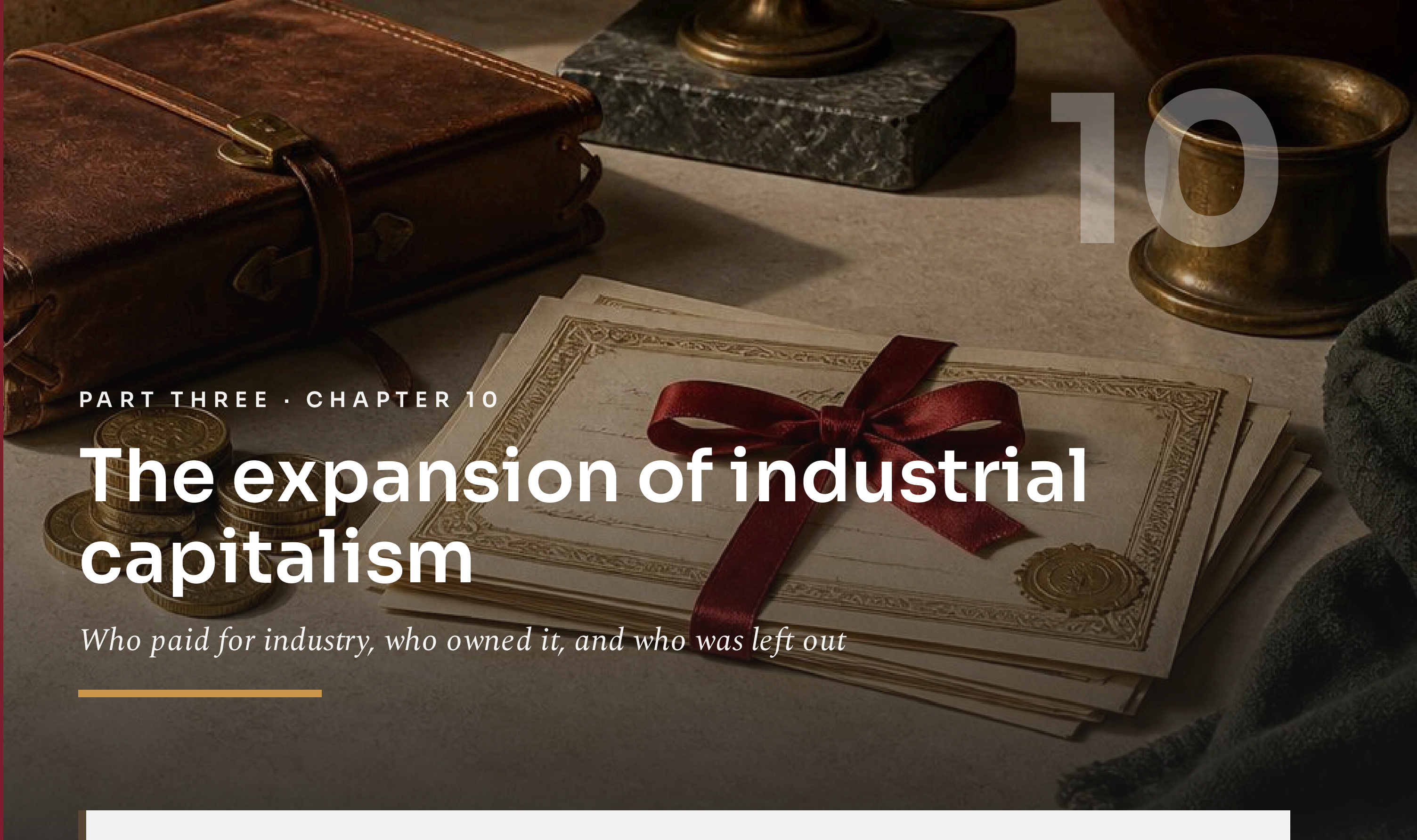
Build a table with the seven conditions from chapter 6 down the side and the two countries across the top. Fill it in. Then write a paragraph that does two things: names the two conditions where the gap was widest, and identifies one condition where Portugal was **not** at a disadvantage. A comparison that finds nothing in Portugal's favour has not been done carefully enough.

WHAT YOU CAN NOW DO

- √ Explain how resources, water and scale combined in the United States.
- √ Explain why a single large internal market matters to manufacturers.
- √ Describe what immigration supplied to American industry, and what it cost the migrants.
- √ Compare vertical integration with control of a choke point.
- √ Explain the assembly line and the loop from cost to market it drives.

WORDS FOR THIS CHAPTER

internal market, customs area, tariff, immigration, emigration, remittance, transcontinental, vertical integration, monopoly, refining, research laboratory, patent, assembly line, interchangeable parts, mass production



10

PART THREE · CHAPTER 10

The expansion of industrial capitalism

Who paid for industry, who owned it, and who was left out

THE QUESTION THIS CHAPTER ANSWERS

Industry changed the machines, but what did it change about money, work and ownership?

IN THIS CHAPTER, YOU WILL

- √ Define capital in the economic sense, and explain why it must be paid for first.
- √ Explain what follows from private ownership for risk, profit and decisions.
- √ Explain how banks, shares and bonds put many people's savings behind one project.
- √ Explain limited liability and why ownership separated from management.
- √ Judge, with evidence, whether industrial capitalism benefited everyone.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, capitalism. Read its definition before you argue about the word, then list which of its features existed in Portugal in 1880.

www.britannica.com/topic/capitalism

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

If you wanted to open a shop tomorrow, list everything you would need before you could sell anything, and put a rough price beside each. Now say where that money would come from. You have just described the central problem of this chapter.

CHAPTER 10 · TOPIC 10.1

What is capital?

The account

In everyday speech capital means money. In economics it means something more precise and more useful: **the things used to produce other things**.

A loom is capital. So is a factory building, a lorry, a railway line, a computer, a set of tools. None of them is wanted for its own sake; each is wanted because it helps produce something else. Money used to buy them is often called capital too, and that is where the everyday meaning comes from, but the idea underneath is about production.

Two features of capital explain a great deal of the nineteenth century.

Capital has to be paid for before it earns. A mill must be built, equipped and staffed before the first piece of cloth is sold. Somebody must therefore either have savings or borrow.

Capital multiplies what a worker can do. A weaver with a hand loom produces a certain amount of cloth in a day; the same weaver with a power loom produces many times more. That is why capital is worth paying for, and it is also why whoever owns the capital is in a strong bargaining position against whoever does not.

Industrialisation is, in this sense, the process of putting far more capital behind each worker. Everything else in this chapter follows from asking who supplies it and on what terms.

YOUR TURN

1. Give the economic definition of capital and three examples.
2. Explain why capital has to be paid for before it earns anything.
3. Explain how capital multiplies what a worker can produce.
4. Why does owning the capital give bargaining power?

CHAPTER 10 · TOPIC 10.2

Private property

The account

Capitalism takes its name from capital, and its defining feature is that productive property is **privately owned**: the factory, the machines, the land and the raw materials belong to individuals or to companies rather than to the state or to the community.

Three things follow, and they are worth separating.

The owner takes the **risk**. If nobody buys the cloth, the owner loses the money spent on making it.

The owner takes the **profit**. If the cloth sells for more than it cost, the difference belongs to the owner.

The owner makes the **decisions**. What to produce, how much, with what machinery and how many workers are the owner's choices.

For private property to function at all, the law has to protect it, which is why chapter 6 listed property rights among the conditions for industrialisation. A factory in a place where it might be seized will not be built.

The argument that runs through the rest of this part of the book begins here. The people who supply the labour do not own the capital, take the risk, receive the profit or make the decisions. Whether that is fair, whether it is efficient, and whether it could be otherwise are the questions chapter 11 is about.

YOUR TURN

1. What does private ownership of productive property mean?
2. Give the three things that follow from ownership, with an example of each.
3. Why does capitalism need law?
4. State, in one sentence each, the position of the owner and of the worker.

CHAPTER 10 · TOPIC 10.3

Entrepreneurship

The account

An entrepreneur is somebody who brings the parts together: raises the capital, chooses what to produce, hires the labour, buys the materials and takes responsibility for whether it works.

That is a real job, and it is worth saying what it consists of, because pupils often assume the entrepreneur simply owns things. The entrepreneur **combines** things, and the combination is the contribution: seeing that a new machine could be used for a product nobody has yet applied it to, or that a market exists which nobody is serving.

It is also a job that involves risk. Most new ventures in the nineteenth century failed, as most fail today. The successes are the ones history remembers, which makes entrepreneurship look far safer in a textbook than it was in a ledger.

A last point, easy to miss and important for judging the period fairly. Entrepreneurship needs **conditions**: available capital, a legal system, some education, a market, and a society that permits people to rise. Where those are missing, ability is wasted. The uncomfortable implication is that a country can lack entrepreneurs not because its people are less able but because its arrangements do not let ability find capital.

YOUR TURN

1. List the things an entrepreneur brings together.
2. Explain the difference between owning and combining.
3. Why does a textbook make entrepreneurship look safer than it was?
4. Give three conditions a society needs if ability is to find capital.

CHAPTER 10 · TOPIC 10.4

Investment

The account

Industry at the scale of chapters 8 and 9 needed sums no individual could provide. A railway costs more than a mill; a steelworks more than a railway station. The nineteenth century therefore built the machinery for gathering many people's savings and putting them behind one project.

Banks took deposits from people who had money spare and lent to people who wanted to build, taking the difference in interest as their income.

Stock exchanges let a company sell shares, so that thousands of people each owned a small part of an enterprise none of them could have funded alone.

Bonds let companies and governments borrow directly from investors, promising to repay with interest over years.

That machinery had two effects that shaped the century. It made very large projects possible, which is why railway networks could be built at all. And it tied the fortunes of ordinary savers to the success of distant enterprises, which means that when confidence collapses, it collapses for everybody at once. That is the mechanism you will need in chapter 16.

WHERE PORTUGAL FITS

Portugal used exactly this machinery, and topic 10.8 is about what it cost. The railways, roads and ports of the Regeneration were paid for by government borrowing, much of it raised abroad. Borrowing to build infrastructure is a perfectly ordinary policy: the works last for decades and can be paid for over decades. The difficulty is that the interest must be paid whether or not the works produce as much growth as was hoped.

YOUR TURN

1. Explain how a bank turns other people's savings into a factory.
2. What is a share, and what problem does it solve?
3. Why did railways in particular need this machinery?
4. Explain why linking savers to distant enterprises makes a collapse spread.

CHAPTER 10 · TOPIC 10.5

Corporations

The account

As enterprises grew, a new legal form became normal: the **company** or corporation, owned by shareholders and treated in law as a body of its own, separate from the people who own it.

The key feature is **limited liability**. If the company fails, a shareholder loses what they invested and no more. Without that, anybody buying a small share of a large enterprise would risk their house, and almost nobody would do it. Limited liability is what makes it possible to raise money from thousands of strangers.

Two consequences follow.

Ownership separates from management. The people who own a large company are shareholders scattered across a country; the people who run it are managers employed to do so. Whole professions exist because of that separation.

Companies can become very large and can outlive people. A company does not retire or die, and it can grow beyond any scale a family firm could reach.

Whether large companies are good is genuinely disputed. They can produce more cheaply than small firms and invest in research that small firms cannot afford. They can also dominate a market so completely that competition disappears, which is the monopoly problem of topic 9.7.

YOUR TURN

1. What is limited liability, and what problem does it solve?
2. Explain the separation of ownership from management.
3. Give one advantage and one danger of very large companies.
4. Why can a company grow larger than a family firm?

CHAPTER 10 · TOPIC 10.6

Competition

The account

In a market with several sellers, each has to attract buyers, and there are only so many ways to do it: sell more cheaply, sell something better, or sell something nobody else offers. All three push firms to improve.

That pressure explains a great deal of the nineteenth century. Firms adopted new machinery because rivals who adopted it could undercut them. Costs fell. Prices fell. The consumer gained.

But competition has two well-known failure modes, and both appear in this period.

It can be won. A firm that wins decisively ends up with no competitors, and then the pressure that made it efficient disappears. Standard Oil is the example in topic 9.7.

It can be avoided. Rival firms can agree among themselves to fix prices or divide up a market rather than compete, which gives them the profits of a monopoly without the effort of winning.

Both are reasons why governments came to regulate markets rather than leave them alone, and both are still the reasons competition law exists today.

CAREFUL

"Competition makes things better" is true and incomplete. It makes things better **while it lasts**, and a great deal of economic policy consists of trying to keep it going. Whenever you meet a claim that a market solves a problem by itself, ask what happens to that market once somebody has won it.

YOUR TURN

1. Give the three ways a firm can attract buyers from a rival.
2. Explain the two failure modes of competition.
3. Why does winning a market remove the pressure to be efficient?
4. Why do governments make competition law?

CHAPTER 10 · TOPIC 10.7

Inequality

The account

Industrial capitalism produced an enormous increase in the total amount of wealth in the societies that adopted it, and it distributed that wealth very unevenly. Both halves of that sentence are true and neither cancels the other.

Over the nineteenth century, in the industrialising countries, real wages eventually rose, diets improved and life expectancy increased. Those gains were real, and they came late and unevenly: for the first generations of factory workers, conditions were as chapter 7 described.

At the same time, the fortunes at the top reached a scale with no precedent outside royalty, and the gap between the owner of a steelworks and the person working in it was greater than the gap between a landowner and a farm labourer a century earlier.

The consequences were political. People who share a workplace, a city district and a grievance can organise, and workers did: first in local societies, then in trade unions, then in political parties. The employers responded, sometimes with concessions and sometimes with dismissals, blacklists and force.

That conflict is the reason chapter 11 exists. Every idea in it, liberalism, nationalism, socialism and Marxism, is in part an answer to the question this topic raises: **if industry makes a society richer and more unequal at the same time, what should be done about it?**

THE ENQUIRY: DID CAPITALISM BENEFIT EVERYONE?

Build the honest answer rather than the easy one.

1. List three benefits of industrial capitalism, with evidence from chapters 7, 8 and 9.
2. List three costs, with evidence from the same chapters.
3. For each benefit, say **when** it arrived. For each cost, say **who** paid it.
4. Write a conclusion that begins: "Industrial capitalism benefited most people eventually, but..." and finish the sentence with your strongest evidence.

YOUR TURN

1. Give the two halves of the sentence about wealth and distribution.
2. Give two gains for ordinary people over the nineteenth century.
3. Explain why the concentration of workers made organisation possible.
4. State, in your own words, the question this topic hands to chapter 11.

CHAPTER 10 · TOPIC 10.8

Portugal in the global story: the Regeneration and Fontismo

What was happening in Portugal?

After the political turmoil of the first half of the century, which chapter 11 sets out, Portugal entered a more stable period from **1851** known as the **Regeneration**. Its central idea was that the country's problem was material rather than constitutional: what Portugal needed was not another revolution but roads, railways, ports, bridges and telegraph lines.

The politician most closely associated with that programme was **Fontes Pereira de Melo**, an engineer by training who held office repeatedly across three decades, and the policy became known as **Fontismo**.

The record is substantial. The first railway opened in 1856, as chapter 8 described, and the network was extended steadily. Roads were built. Ports were improved. Telegraph lines were laid. The Maria Pia and Luis I bridges crossed the Douro at Porto. Regions that had been effectively separate began to trade with each other.

What part did geography play?

It made everything more expensive. Portuguese terrain, mountainous in the north and centre, meant cuttings, tunnels, embankments and viaducts, and there were few navigable rivers reaching inland to carry materials cheaply to the works. A kilometre of Portuguese railway cost more than a kilometre of railway across flat, well-watered country.

What was the connection?

This is topic 10.4 in practice. The works were paid for largely by **government borrowing**, much of it raised on foreign markets, and the interest had to be paid every year whether or not the new lines earned enough to cover it. Public debt rose substantially across the period, and by the end of the century the state's finances were under serious strain.

The judgement historians make is genuinely mixed, and you should hold both halves. Portugal at the end of the century was far better connected than at the beginning, and that connection was a necessary condition for anything else. Portugal was also more indebted, and it had not closed the gap with the industrial economies of northern Europe. Infrastructure was necessary. It was not, on its own, sufficient.

KEY QUESTION

Can infrastructure investment help a country overcome economic disadvantages?

Use Fontismo as your case, and answer in three steps.

1. What did the investment achieve that could not have been achieved without it?
2. What did it cost, and who carried that cost?
3. Look back at the seven conditions in chapter 6. Which of them does building a railway supply, and which does it leave untouched? Your conclusion should turn on that answer.

WHAT YOU CAN NOW DO

√ Define capital in the economic sense and explain why it must be paid for first.

√ Explain private property, and what follows from it for risk, profit and decisions.

√ Explain how banks, shares and bonds gather savings behind large projects.

√ Explain limited liability and the separation of ownership from management.

√ Judge, with evidence, whether industrial capitalism benefited everyone.

WORDS FOR THIS CHAPTER

capital, private property, entrepreneur, risk, profit, investment, bank, share, shareholder, bond, interest, corporation, limited liability, competition, monopoly, inequality, public debt, Regeneration, Fontismo

11

PART THREE · CHAPTER 11

Ideas that changed the nineteenth century

Four answers to one question: what should be done about all this?

THE QUESTION THIS CHAPTER ANSWERS

Why did industrial society produce new political ideas?

IN THIS CHAPTER, YOU WILL

- √ Explain what liberalism meant in the nineteenth century, and how that differs from today.
- √ Distinguish a state from a nation, and explain nationalism's two opposite effects.
- √ Describe the family of socialist ideas and the split inside it.
- √ Explain Marx's core terms, and say which of his claims events did not bear out.
- √ Explain how trade unions turned concentration into bargaining power.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, liberalism. The word has changed meaning more than once. Read what it meant in 1830 before you use it about 1830.

`www.britannica.com/topic/liberalism`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Write down one thing about the world you think is unfair, and one sentence saying what should be done about it. Keep the paper. Every idea in this chapter began as somebody doing exactly that, and the four answers in this chapter disagree with each other.

CHAPTER 11 · TOPIC 11.1

Liberalism

The account

Liberalism is the oldest of the four ideas in this chapter, and the word is a trap, because its meaning has shifted. In the nineteenth century it meant something quite specific, and that is the meaning you need.

Nineteenth-century liberals wanted:

Individual liberty. Freedom of speech, of the press, of religion and of association, protected against the government rather than granted by it.

Constitutional government. A written constitution setting out what the state may and may not do, so that power is limited by rules rather than by the character of whoever holds it.

Equality before the law. The same law for everybody, in place of separate legal privileges for nobility and clergy.

Representative government. A parliament, elected, that makes the laws and controls taxation.

Property rights and economic freedom. The right to own, buy, sell and trade with limited interference.

Two things are worth noticing, because they explain the next hundred years.

Liberalism was **revolutionary** in a Europe of absolute monarchies. Demanding a constitution in 1820 was not a mild request; it was a challenge to the basis on which kings ruled.

And early liberalism was **not democratic** in the modern sense. Most liberals wanted the vote restricted to men with property, on the argument that those with a stake in the country would vote responsibly. The extension of the vote to all adults was a later and separate fight, in which liberals were sometimes on the other side.

CAREFUL

When a source from 1830 says "liberal", it does not mean what a newspaper means today. Read it as: constitution, parliament, rights against the state, free trade, and a restricted franchise. Getting this wrong makes the whole of chapter 11 unintelligible.

YOUR TURN

1. List the five liberal demands in this topic.
2. Explain why demanding a constitution was revolutionary in 1820.
3. Explain the difference between liberal and democratic in this period.
4. Why is it dangerous to read a nineteenth-century source with a modern dictionary?

CHAPTER 11 · TOPIC 11.2

Nationalism

The account

Nationalism is the belief that the world is naturally divided into **nations**, that each nation should govern itself, and that the state and the nation should have the same borders.

Underneath it lies a distinction worth getting right. A **state** is a political unit with a government and borders. A **nation** is a group of people who believe they share something: a language, a history, a culture, sometimes a religion. The two do not automatically coincide, and in 1815 they frequently did not. Italy was a collection of separate states; Germany was a collection of separate states; while the Austrian and Ottoman empires each ruled many peoples who thought of themselves as distinct nations.

Nationalism therefore did two opposite things at once, and this is the heart of the topic.

It united. Between 1859 and 1871, Italy was assembled into a single kingdom out of separate states. Between 1864 and 1871, the German states were unified under Prussian leadership into a German Empire. In both cases nationalism was the argument that made unification desirable and unification the achievement that made nationalism powerful.

It divided. Within the multi-national empires, the same argument said that Czechs, Hungarians, Poles, Serbs and others should have states of their own, which meant breaking those empires up. Chapter 12 shows that argument becoming dangerous, and chapter 13 shows it becoming a war.

Nationalism also changed the state's relationship with its people. National armies, national schools, national languages and national holidays were all used deliberately to make people feel that they belonged to the nation, and they worked. That is why, in 1914, millions of men volunteered.

YOUR TURN

1. Give the difference between a state and a nation, with an example of each.
2. Explain how nationalism united Italy and Germany.
3. Explain why the same idea threatened Austria-Hungary and the Ottoman Empire.
4. Name three things a state can do to make people feel national.

CHAPTER 11 · TOPIC 11.3

Socialism

The account

Socialism is not one doctrine but a family of ideas that share a starting point: that the inequality produced by industrial capitalism is not acceptable and not inevitable.

Socialists differed enormously about what to do, and the differences are as important as the agreement.

Some wanted **cooperatives**, in which workers owned the enterprise between them and shared the profits.

Some wanted the **state** to own major industries so that they would be run for the public rather than for shareholders.

Some wanted **regulation and welfare** within a capitalist economy: limits on hours, safety laws, pensions, sickness insurance and free schooling.

Some wanted **revolution**, on the argument that those who hold economic power will never give it up to a vote.

The last two disagreed so sharply that the split between reformers and revolutionaries is the central division in the history of socialism, and it runs directly into chapter 21, where two blocs face each other, both calling themselves the friends of the worker.

What all of them shared was a diagnosis: that the wealth of an industrial society is produced collectively and distributed very unequally, and that the arrangement is a human choice rather than a law of nature.

YOUR TURN

1. What is the shared starting point of socialist ideas?
2. Give the four different socialist answers described here.
3. Why is the split between reformers and revolutionaries so important?
4. Name three welfare measures a reforming socialist might demand.

CHAPTER 11 · TOPIC 11.4

Marxism

The account

Karl Marx, working with **Friedrich Engels**, produced the most influential single analysis in this chapter. It is studied here as history: what the ideas were, what problem they addressed, and what they did in the world. Your task is to understand them, not to adopt or reject them.

The core of the analysis, at the level Year 9 needs:

Means of production. The factories, machines, land and raw materials with which things are made. Marx argued that who owns these decides the shape of a society.

Two classes. The **bourgeoisie** own the means of production; the **proletariat** own nothing but their capacity to work and must sell it.

Class struggle. Marx argued that history is driven by conflict between classes over the wealth that production creates, and that this conflict is built into the arrangement rather than caused by bad behaviour.

Surplus value. Workers produce more value than they receive in wages, and the difference is the source of profit. In Marx's account that is where inequality comes from.

Prediction. He expected capitalism to concentrate wealth, throw workers into crisis, and eventually be overthrown by them, producing a society without classes.

Two honest observations belong beside that summary.

The **analysis** was powerful and influenced economics, sociology and history permanently, including among people who rejected his conclusions entirely.

The **prediction** did not come true in the way he described. Revolution came first not in the most industrialised countries but in Russia, which was among the least, and in the industrial countries wages rose, unions were legalised and welfare states were built instead. Why the prediction failed is a genuine historical question, and part of the answer is topic 11.5.

GO FURTHER

States that called themselves Marxist in the twentieth century differed enormously from each other and from what Marx wrote, and you will meet several in part eight. Keep the distinction: what a thinker argued, and what later governments did in that thinker's name, are two separate historical subjects, and neither settles the other.

YOUR TURN

1. Define means of production, bourgeoisie, proletariat and class struggle.
2. Explain surplus value in one or two sentences.
3. What did Marx predict, and in what way did events differ?
4. Explain why the analysis can be influential even if the prediction failed.

CHAPTER 11 · TOPIC 11.5

Trade unions and workers' movements

The account

Ideas are one thing; organisation is another. The most immediate response of industrial workers to industrial conditions was to combine.

A single worker asking for shorter hours can be dismissed and replaced by tomorrow. All the workers in a mill asking together cannot be, because the mill would stop. **The union's power is collective and it comes from the concentration that the factory itself created**, which is one of the sharper ironies of the period.

Unions were resisted, and in many countries were illegal for a long time. Combining was treated as conspiracy; strikers were dismissed, evicted from company housing, blacklisted, and sometimes met with force. Legalisation came gradually and at different dates in different countries.

The methods were: **collective bargaining**, negotiating for all the workers at once; the **strike**, withdrawing labour; and eventually **political action**, forming or supporting parties to change the law rather than one employer's terms.

The results, across the later nineteenth and early twentieth centuries, were substantial: limits on the working day, restrictions on child labour, safety regulation, the beginnings of sickness and old-age insurance, and, in several countries, the extension of the vote.

Those gains are the best available answer to the question left over from topic 11.4. The industrial working class did not overthrow capitalism in western Europe. It organised inside it, and changed the terms.

READ THE SOURCE

A trade union rule book, later nineteenth century. Rule books of the period typically set out the weekly subscription, the benefits payable in sickness, accident, unemployment and death, the conditions under which the society would support a member in a dispute, and the fines for breaking the rules.

- **Origin:** written by the union itself, for its own members.
- **Purpose:** to govern the society and to reassure members that their money

was safe and their rules fair.

- **Value:** it shows that early unions were **insurance societies as well as

bargaining bodies**, which is easy to miss, and it shows what risks workers most feared.

- **Limitation:** it records rules, not behaviour. It cannot tell you whether

the rules were kept, or how a dispute actually went.

THE ENQUIRY: WHY DID INDUSTRIAL SOCIETY PRODUCE NEW POLITICAL IDEAS?

Set the four ideas of this chapter side by side and answer the chapter question properly.

1. For each of liberalism, nationalism, socialism and Marxism, write one sentence on the problem it was trying to solve.
2. Which two are most directly responses to industrialisation, and which is least? Defend your choice.
3. Choose the idea you think had the largest effect on the twentieth century. You will be able to test that judgement at the end of chapter 29.
4. Write a paragraph explaining why a society that changes quickly produces more new political ideas than one that changes slowly.

CHAPTER 11 · TOPIC 11.6

Portugal in the global story: liberalism, monarchy and Republic

What was happening in Portugal?

Portugal did not observe the age of ideas from outside. It lived through it, and its nineteenth century is a sequence of exactly the conflicts this chapter describes.

In **1820**, a liberal revolution began in **Porto**. Its demands were the liberal demands of topic 11.1: a constitution, a parliament, the return of the king from Brazil, and an end to government by royal decree.

In **1822** Portugal adopted its **first Constitution**, one of the earliest in Europe.

The victory was not settled. Supporters of absolute monarchy fought to reverse it, and the conflict became a civil war, the **Liberal Wars of 1828 to 1834**, between the liberal supporters of Pedro and the absolutist supporters of Miguel. The liberals won, and constitutional government was established, although Portuguese politics remained unstable for decades afterwards.

In the later part of the century, the other ideas of this chapter arrived too. **Republicanism** grew, arguing that a constitutional monarchy was not enough. **Socialist ideas and workers' organisations** appeared in Lisbon, Porto and the industrial districts of chapter 7.

Then, in **1890**, came the **British Ultimatum**. Portugal claimed a belt of territory linking Angola to Mozambique across southern Africa; Britain had its own plans for that corridor and demanded, under threat of war, that Portugal withdraw. Portugal complied, because it had no realistic choice. The humiliation was enormous and it was blamed on the monarchy.

On **5 October 1910** the monarchy was overthrown and Portugal became a **Republic**.

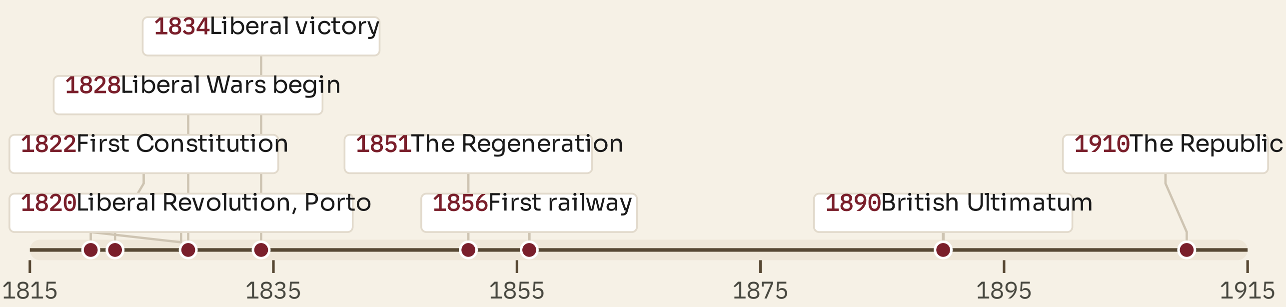


Figure 11.1 · Portugal's long nineteenth century, in nine dates.
Read it from the left. Notice how the cluster of the 1820s and 1830s is a single argument fought twice, once in a revolution and once in a war.

What was happening in the world?

The same decades saw constitutions demanded across Europe, revolutions in 1830 and 1848, the unification of Italy and Germany, and the growth of socialist parties and unions everywhere industry reached. Portugal's sequence is its own, and it is recognisably part of that European pattern rather than separate from it.

What part did geography play?

Two things. Portugal's Atlantic position and its empire meant that a colonial dispute in Africa became a domestic political crisis in Lisbon, which is the 1890 story exactly. And the concentration of industry and population on the coastal strip, from chapter 7, is where republicanism and the workers' movements were strongest.

What was the connection?

The ideas transforming Europe transformed Portugal too, and on the same timetable. Portugal was neither ahead nor behind: its first constitution came early, its civil war between liberals and absolutists is the same argument fought in Spain and elsewhere, and its republic of 1910 came at a moment when republicanism was rising across southern Europe.

KEY QUESTION

How did liberalism change Portugal's political system?

Answer in three parts. What did Portugal's system look like in 1815? What did it look like in 1834? And what had still not changed by 1910, given that the country had a constitution and a parliament for most of that time? The third part is the one that explains why a republic was demanded at all.

WHAT YOU CAN NOW DO

- √ Explain what liberalism meant in the nineteenth century, and how it differs from the modern use of the word.
- √ Distinguish a state from a nation, and explain nationalism's two opposite effects.
- √ Describe the family of socialist ideas and the split inside it.
- √ Explain Marx's core terms and say honestly which of his claims events did not bear out.
- √ Explain how trade unions turned concentration into bargaining power.

WORDS FOR THIS CHAPTER

liberalism, constitution, franchise, nationalism, state, nation, unification, self-determination, socialism, cooperative, welfare, Marxism, means of production, bourgeoisie, proletariat, class struggle, surplus value, trade union, collective bargaining, strike, republicanism, Liberal Wars, British Ultimatum

PART FOUR

Nationalism, empires and global rivalry

A war does not begin because somebody is shot. It begins because a great many arrangements have already been made that turn a shooting into a war. This part is about those arrangements: the nations that had been built, the empires that had been taken, the alliances that had been signed and the armies that had been enlarged, in the forty years before 1914.

CHAPTER 11 AT A GLANCE

Ideas that changed the nineteenth century

why did industrial society produce new political ideas?

THE TOPICS YOU HAVE COVERED

11.1	Liberalism
11.2	Nationalism
11.3	Socialism
11.4	Marxism
11.5	Trade unions and workers' movements
11.6	Portugal in the global story: liberalism, monarchy and Republic

WHAT YOU CAN NOW DO

- √ Explain what liberalism meant in the nineteenth century, and how it differs
- √ Distinguish a state from a nation, and explain nationalism's two opposite
- √ Describe the family of socialist ideas and the split inside it.
- √ Explain Marx's core terms and say honestly which of his claims events did not
- √ Explain how trade unions turned concentration into bargaining power.

THE WORDS THIS UNIT TAUGHT

liberalism · constitution · franchise · nationalism · state · nation · unification · self-determination · socialism · cooperative · welfare · Marxism · means of production · bourgeoisie · proletariat · class struggle · surplus value · trade union · collective bargaining · strike · republicanism · Liberal Wars · British Ultimatum

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART FOUR · CHAPTER 12

12

The world before 1914

What forty years of rivalry had built by the summer of 1914

THE QUESTION THIS CHAPTER ANSWERS

What had Europe built by 1914 that made a general war possible?

IN THIS CHAPTER, YOU WILL

- √ Explain what nationalism had become by 1900, and why that was dangerous.
- √ Explain how industrial capacity converts into military power.
- √ Give the motives, methods and consequences of imperial expansion.
- √ Explain how an alliance system turns a local quarrel into a general war.
- √ Explain why mobilisation timetables made a crisis harder to stop.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, imperialism. The motives it lists, and the resistance it records. Compare both with the map of 1914 in this chapter.

`www.britannica.com/topic/imperialism`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Two friends each promise to defend the other in any argument. A third and fourth pair do the same. Now one person from the first pair insults one person from the second. Work out, on paper, how many people are involved by the end. You have just modelled the alliance system of 1914.

CHAPTER 12 · TOPIC 12.1

Nationalism

The account

Chapter 11 introduced nationalism as an idea. This chapter is about what it had become by 1900, which was the most powerful political force in Europe and a distinctly more dangerous one.

Three changes had happened.

It had won. Italy and Germany existed as unified states. Nationalism was no longer the argument of frustrated professors; it was the founding principle of two great powers.

It had been adopted by governments. States that had once regarded national feeling as subversive now cultivated it deliberately, through compulsory schooling in a national language, conscription into a national army, national holidays, national histories and a press that reported the world as a contest between nations.

It had turned competitive. Where earlier nationalism said "our nation should govern itself", the later version increasingly said "our nation should be greater than theirs". National pride became national rivalry, and rivalry was measured in colonies, battleships and steel output.

The result was a Europe in which populations could be mobilised emotionally as well as militarily. In 1914 this mattered enormously: governments did not have to force reluctant subjects into a war, because large numbers of people believed the war was theirs.

CAREFUL

Nationalism is not the same thing as patriotism, although the words overlap. Loving your own country does not require believing it superior to others. Historians studying 1914 are interested in the second kind, and in what governments did to encourage it.

YOUR TURN

1. Give the three changes in nationalism between about 1850 and 1900.
2. Name three ways a state deliberately built national feeling.
3. Explain the difference between "our nation should govern itself" and "our nation should be greater".
4. Why did the change matter for what happened in 1914?

CHAPTER 12 · TOPIC 12.2

Industrial powers

The account

Industrialisation changed the balance of power between states, because a country that can make more can also arm more.

The connection is direct. Steel makes rails, ships and guns. Coal moves fleets. Chemicals make explosives and, later, gas. Railways move troops to a frontier faster than an enemy can react. Factories that make sewing machines in peacetime can make rifles in wartime. Industrial capacity **is** military capacity, once a war lasts more than a season.

The rankings shifted accordingly. Britain, the first industrial power, was overtaken in steel by Germany and by the United States. Germany, unified only in 1871, became the largest industrial economy in Europe within a generation. Russia was industrialising quickly from a low base and had by far the largest population. France was industrialised but growing more slowly.

Governments watched these figures the way they had once watched harvests, because everybody understood what they meant. A power that was falling behind had reason to be anxious; a power that was catching up had reason to be confident. Both feelings make a crisis harder to settle peacefully, and chapter 13 shows why.

GO FURTHER

Notice the argument shape here, because you will meet it again in chapter 24 with nuclear weapons. When a state's security depends on its position relative to another, an improvement by one is felt as a threat by the other, even if neither intends harm. That is why arms races are hard to stop once begun.

YOUR TURN

1. Give four ways industrial capacity converts into military capacity.
2. Which power had overtaken Britain in steel by 1900?
3. Why did Germany's rise so quickly change European calculations?
4. Explain why both a rising power and a falling one become harder to deal with.

CHAPTER 12 · TOPIC 12.3

Imperialism

The account

Between about 1870 and 1914 European states took control of most of Africa and large parts of Asia and the Pacific. This is imperialism, and it needs to be studied for its causes, its methods and its consequences, none of which is comfortable.

The **motives** were mixed and reinforced each other. Economic: raw materials such as rubber, cotton, palm oil, tin and later oil, and markets for manufactured goods. Strategic: coaling stations for the steamships of chapter 8, naval bases, and control of routes such as the Suez Canal. Political: prestige, and the belief that a great power must have colonies because other great powers did. And ideological: theories of racial and civilisational superiority which were used to justify the whole enterprise, and which were false.

The **methods** included treaties signed by rulers who often did not understand what they conceded, and, repeatedly, military conquest against people armed with far less. The industrial technology of chapter 8 is part of this story: repeating rifles, machine guns, steamships and telegraphs gave small European forces advantages that were decisive.

The **consequences** were profound and long-lasting: borders drawn for European convenience across communities and kingdoms, economies reorganised to export what Europe wanted, systems of forced or coerced labour, and the destruction of existing political structures. Resistance was continuous, took many forms, and is part of the record.

At the **Berlin Conference of 1884 to 1885**, European powers met to agree rules for their claims in Africa. No African ruler was present.

> **CAREFUL**
>
> Two failures of judgement to avoid. Do not present imperialism as a natural or inevitable stage of development; it was a set of choices, and people at the time argued about them. And do not present the colonised as passive; there was resistance everywhere, before, during and after conquest, and the histories of those societies did not begin when Europeans arrived.

YOUR TURN

1. Give the four kinds of motive for imperial expansion, with an example of each.
2. Explain how the technology of chapter 8 affected the balance of conquest.
3. Give three long-term consequences of colonial rule.
4. What does it tell you about the Berlin Conference that no African ruler was present?

CHAPTER 12 · TOPIC 12.4

Alliances

The account

By 1907 the great powers of Europe had sorted themselves into two blocs, each bound by treaty to support its members.

The **Triple Alliance** joined Germany, Austria-Hungary and Italy.

The **Triple Entente** joined France, Russia and Britain, though Britain's commitment was looser than the others.

Each alliance was defensive in its own eyes: it existed so that its members would not be attacked. But a defensive alliance has an effect its members do not always intend.

It **enlarges** a quarrel. A dispute between two states now potentially involves six.

It **removes room to manoeuvre**. A power that would otherwise stay out is committed in advance.

It **encourages recklessness**. A smaller state backed by a great power may take risks it would never take alone, knowing help is promised.

And it makes each side **fear the other's plans**, which is why the arms race of the next topic accelerated as the alliances hardened.

> **ON THE MAP**
>
> Draw a rough map of Europe in 1914 and colour the two blocs. Then mark Austria-Hungary and the Balkans. Look at where the Triple Alliance and the Triple Entente meet on the map, and write one sentence saying where you would expect a crisis to become dangerous.

YOUR TURN

1. Name the members of each alliance.
2. Give three effects an alliance has that its members may not intend.
3. Explain how an alliance can make a small state more reckless.
4. Why does an alliance system turn a local quarrel into a general war?

CHAPTER 12 · TOPIC 12.5

The arms race

The account

From the 1890s the powers expanded their armed forces steadily, and two features of that expansion mattered most.

The **naval race** was chiefly between Britain and Germany. Britain, an island that fed itself by importing, treated naval supremacy as a matter of survival. Germany, newly unified and newly industrial, built a large fleet as an expression of great-power status. In 1906 Britain launched a battleship so much better armed and faster than anything before it that every earlier battleship was made obsolete, which had the unintended effect of resetting the race and letting Germany start almost level.

Armies and plans grew alongside. Conscription meant that most continental powers could put millions of trained men into the field. Because railways moved those men, general staffs wrote detailed **mobilisation timetables** in advance, specifying which trains ran on which day.

That last detail matters more than it sounds, and chapter 13 turns on it. A timetable is fast and inflexible. Once mobilisation began, stopping it meant tearing up a plan that took years to write, and every general believed that the side which mobilised more slowly would lose. The machinery built to be ready for war made it far harder to stop a crisis short of war.

YOUR TURN

1. Why did Britain regard naval supremacy as a matter of survival?
2. What did the launch of a much superior battleship in 1906 unintentionally do?
3. What is a mobilisation timetable, and why did railways make one necessary?
4. Explain why the timetables reduced the chance of a peaceful settlement.

CHAPTER 12 · TOPIC 12.6

Europe in 1914

The account

Put the pieces together and the map of Europe in 1914 looks very different from the map you know.

Germany was larger than today, extending well to the east. **Austria-Hungary** was a great multi-national empire covering much of central Europe and the northern Balkans. The **Russian Empire** ran from Poland to the Pacific. The **Ottoman Empire** still held territory in the Middle East and a foothold in Europe. Poland did not exist as a state; Czechoslovakia and Yugoslavia did not exist; the Baltic states did not exist.

The **Balkans** were where the pressure concentrated. Ottoman power had been receding for a century, and new states had emerged from it: Serbia, Greece, Bulgaria, Romania, Montenegro, Albania. Each had national claims that overlapped with its neighbours'. Austria-Hungary feared that the nationalism of chapter 11 would pull its own multi-national empire apart, and watched Serbia in particular. Russia presented itself as the protector of the Slav peoples in the region. Two great powers therefore had a direct interest in every Balkan quarrel.

That is the situation into which chapter 13 drops a single assassination.

THE ENQUIRY: WAS A GENERAL WAR LIKELY BY 1914?

Do not answer yet. Build the evidence.

1. List the four conditions from this chapter: nationalism, industrial rivalry, imperial competition, alliances and the arms race.
2. For each, write one sentence on how it made war **more** likely, and one on how peace might still have been kept.
3. Note one thing that was still missing in 1913 which would be present in 1914.
4. Keep your notes. You will complete this enquiry at the end of chapter 13.

CHAPTER 12 · TOPIC 12.7

Portugal in the global story: Portugal, Africa and international rivalry

What was happening in Portugal?

Portugal was not a great power in 1890, and it held one of the older European empires in Africa, chiefly **Angola** on the Atlantic coast and **Mozambique** on the Indian Ocean coast.

As the scramble for Africa accelerated, Portugal advanced a claim to the territory **between** its two colonies, a belt running right across southern Africa. The claim was set out on a map printed in a distinctive colour, and the argument for it rested on long Portuguese presence and exploration.

Britain had a different plan for the same ground, and in **1890** delivered an **ultimatum**: withdraw Portuguese forces from the disputed territory, or face the consequences. Portugal, which could not fight Britain and was financially dependent on British goodwill, complied.

What was happening in the world?

Exactly what topic 12.3 describes. European states were partitioning a continent among themselves, and where two claims met, the stronger state prevailed. The Portuguese case is a clear example of the rule that operated throughout the scramble.

What part did geography play?

Portugal's two colonies sat on opposite coasts of the same continent, and the land between them was the natural connection. It was also the corridor Britain wanted for its own north-south ambitions. The same geography that made the claim obvious to Lisbon made it unacceptable to London.

What was the connection?

The consequence in Portugal was domestic, and it is the point of this section. The ultimatum was experienced as a **national humiliation**. The government had made claims it could not defend; the monarchy was seen to have failed the nation. Republican and nationalist feeling grew sharply, and *A Portuguesa*, the song written in the wave of protest that followed, later became the national anthem. Chapter 11 has already given you the end of the story: the monarchy fell in 1910.

An imperial dispute two continents away brought down a European monarchy within twenty years. That is exactly the kind of connection this book exists to make visible.

KEY QUESTION

How could competition for territory overseas create political instability at home?

Set out the chain in five steps, from the scramble for Africa in the 1880s to the fall of the monarchy in 1910. For each step, say whether it was caused by geography, by economics or by politics. Then answer this: could the Portuguese government of 1890 have acted differently, and what would it have cost?

WHAT YOU CAN NOW DO

- √ Explain what nationalism had become by 1900 and why that was dangerous.
- √ Explain how industrial capacity converts into military power.
- √ Give the motives, methods and consequences of imperial expansion.
- √ Explain how an alliance system enlarges a quarrel.
- √ Explain why mobilisation timetables made a crisis harder to stop.

WORDS FOR THIS CHAPTER

nationalism, patriotism, conscription, imperialism, colony, protectorate, scramble for Africa, Berlin Conference, Triple Alliance, Triple Entente, alliance, arms race, dreadnought, mobilisation, Balkans, British Ultimatum

PART FIVE

The First World War

Everything part four described came together in the summer of nineteen fourteen. Three chapters follow: why the war began, what fighting it was actually like, and what the peace that ended it did to the world. Portugal is in all three, as a young republic that chose to join.

13

PART FIVE · CHAPTER 13

Why did the First World War begin?

Five weeks from one murder to a war on three continents

> **THE QUESTION THIS CHAPTER ANSWERS**
>
> ## Was the First World War inevitable?

IN THIS CHAPTER, YOU WILL

- √ Explain why the Balkans were the most dangerous region in Europe.
- √ Describe the assassination at Sarajevo and why it was more than a crime.
- √ Distinguish a trigger from a cause, and defend the distinction.
- √ Set out the July crisis in order, with dates.
- √ Judge whether a general war was inevitable by 1914.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the First World War. Use its causes section to test the ranking you built from MAIN, and find one cause it gives that MAIN leaves out.

`www.britannica.com/event/World-War-I`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

A quarrel between two people becomes a quarrel between two families, then two villages, then two countries. At which step did it stop being about the original argument? Hold your answer. This chapter asks the same question about 1914.

CHAPTER 13 · TOPIC 13.1

The Balkans

The account

The Balkan peninsula in south-east Europe was where the pressures of chapter 12 were greatest, and it is worth understanding why before the crisis arrives.

For centuries the region had been ruled by the **Ottoman Empire**. Through the nineteenth century Ottoman power receded, and as it did, new states appeared: Greece, Serbia, Romania, Bulgaria, Montenegro and Albania. Each was built on the nationalism of chapter 11, and each had a view about which neighbouring territory contained people who belonged to it.

Two great powers had direct interests there.

Austria-Hungary was a multi-national empire ruling Germans, Hungarians, Czechs, Slovaks, Poles, Croats, Serbs, Slovenes, Romanians and others. National self-determination, applied consistently, would dissolve it. Serbia, an independent Slav state on its southern border, was therefore seen in Vienna not as a small neighbour but as a threat to the empire's existence. In 1908 Austria-Hungary annexed **Bosnia and Herzegovina**, a territory with a large Serb population, which Serbia bitterly resented.

Russia presented itself as the protector of Slav peoples and sought influence in the region and access to the Mediterranean.

Two Balkan wars in 1912 and 1913 redrew the map again and left every state dissatisfied. By 1914 the peninsula had national claims layered on top of one another, two great powers committed to opposite sides, and a recent history of war.

YOUR TURN

1. Why did the retreat of Ottoman power create instability rather than calm?
2. Explain why Austria-Hungary saw Serbia as a threat to its existence.
3. What was Russia's stated interest in the region?
4. Why did the Balkan wars of 1912 and 1913 make matters worse rather than better?

CHAPTER 13 · TOPIC 13.2

The assassination of Franz Ferdinand

The account

On **28 June 1914**, in **Sarajevo**, the capital of Austrian-ruled Bosnia, **Archduke Franz Ferdinand**, heir to the throne of Austria-Hungary, and his wife Sophie were shot dead by Gavrilo Princip, a young Bosnian Serb belonging to a group that wanted Bosnia joined to Serbia.

Three points make this more than a crime.

The target was symbolic. Killing the heir to the throne, in a province Austria-Hungary had annexed only six years earlier, on a day of national significance to Serbs, was an attack on the empire's claim to rule there.

The trail led toward Serbia. The assassins had been armed and helped by people connected to Serbian military intelligence, though the Serbian government itself did not order the killing. That distinction mattered a great deal to Belgrade and very little to Vienna.

Austria-Hungary wanted a reason. Influential figures in Vienna had already concluded that Serbia had to be dealt with, and the assassination provided the occasion.

So the murder did not cause the war in the way a spark causes a fire in an empty room. It caused a war because of everything that was already stacked around it, which is exactly the point of chapter 12.

CAREFUL

Beware the phrase "the assassination caused the First World War". It is the **trigger**, not the cause. A useful test: ask whether the same event in 1890, before the alliances hardened and the plans were written, would have produced the same result. Almost certainly not.

YOUR TURN

1. Give the date, the place and the victim.
2. Explain why the choice of target and place made this more than an ordinary crime.
3. What was the difference between Serbian officials being involved and the Serbian government ordering it?
4. Explain the difference between a trigger and a cause, using this event.

CHAPTER 13 · TOPIC 13.3

The July crisis

The account

Five weeks separate the assassination from a general European war. What happened in them is one of the best-documented sequences in modern history, and it shows how a crisis escalates.

Austria-Hungary sought German backing and received it, in terms so unconditional that historians call it a blank cheque. Vienna could now act knowing Germany would stand behind it.

Austria-Hungary sent Serbia an ultimatum with a very short deadline and terms deliberately severe. Serbia accepted most of it and qualified the rest. Austria-Hungary declared the reply unsatisfactory and declared war on 28 July.

Russia began to mobilise in support of Serbia. Mobilisation was not war, but under the assumptions of chapter 12 it was read as the start of one.

Germany's plan required speed. Facing the prospect of war with Russia in the east and France in the west, German planning assumed it must defeat France quickly before Russia's much larger army was ready. That plan required marching through **Belgium**, whose neutrality was guaranteed by treaty.

Germany declared war on Russia and then on France, and invaded Belgium. Britain, which had a treaty obligation to Belgium and a strong interest in not letting one power dominate the continent, declared war on Germany on 4 August.

Notice what did the damage. Each government acted in what it believed was a rational response to the others, on timetables that punished hesitation, inside alliances that removed the option of standing aside. A local dispute became a continental war in five weeks because the machinery was built to work that way.

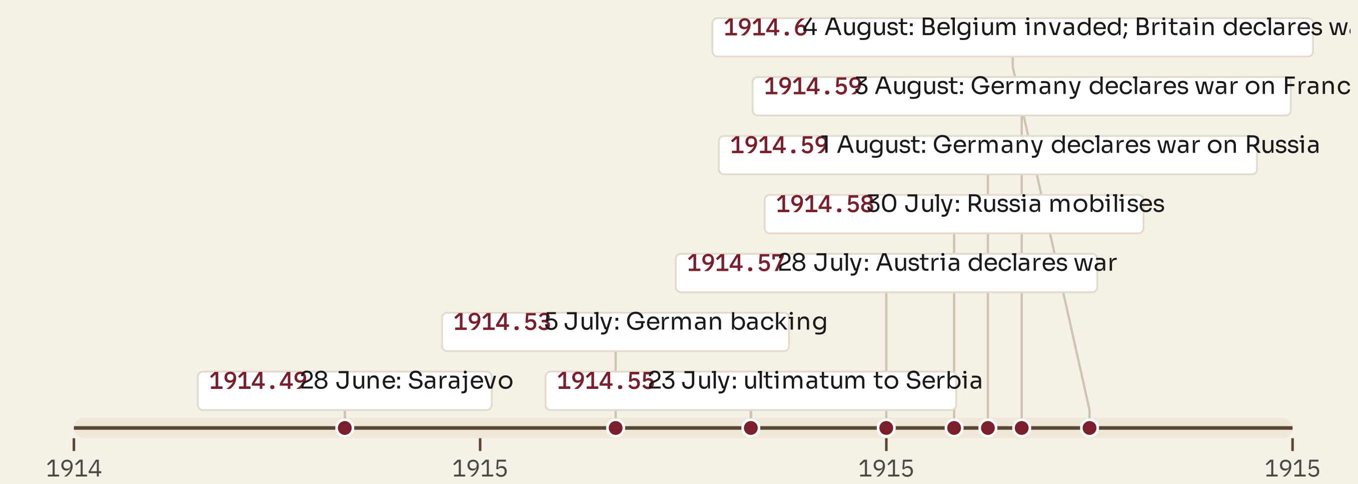


Figure 13.1 · Five weeks, step by step.
Read the dates in order and mark the point where you think it became impossible to stop. Historians disagree about which step that was, which is why the question is worth asking.

THE ENQUIRY: WAS THE FIRST WORLD WAR INEVITABLE?

Now complete the enquiry you began in chapter 12.

1. Take your list of long-term conditions from topic 12.6.
2. Add the short-term sequence from this topic.
3. Identify **three** moments where a different decision was genuinely available, and say who could have taken it.
4. Write a conclusion of one paragraph. It must contain the word "because" at least twice, and it must not begin with the word "yes" or the word "no".

WHAT YOU CAN NOW DO

- √ Explain why the Balkans were the most dangerous region in Europe by 1914.
- √ Describe the assassination and explain why it was more than a crime.
- √ Distinguish a trigger from a cause and defend the distinction.
- √ Set out the July crisis in order, with dates.
- √ Explain how alliances and timetables turned a local dispute into a general war.

WORDS FOR THIS CHAPTER

Balkans, Ottoman Empire, annexation, Bosnia, Sarajevo, assassination, trigger, cause, ultimatum, blank cheque, mobilisation, neutrality, escalation, July crisis

CHAPTER 12 AT A GLANCE

The world before 1914

what had Europe built by 1914 that made a general war possible?

THE TOPICS YOU HAVE COVERED

12.1	Nationalism
12.2	Industrial powers
12.3	Imperialism
12.4	Alliances
12.5	The arms race
12.6	Europe in 1914
12.7	Portugal in the global story: Portugal, Africa and international rivalry

WHAT YOU CAN NOW DO

- √ Explain what nationalism had become by 1900 and why that was dangerous.
- √ Explain how industrial capacity converts into military power.
- √ Give the motives, methods and consequences of imperial expansion.
- √ Explain how an alliance system enlarges a quarrel.
- √ Explain why mobilisation timetables made a crisis harder to stop.

THE WORDS THIS UNIT TAUGHT

nationalism · patriotism · conscription · imperialism · colony · protectorate · scramble for Africa · Berlin Conference · Triple Alliance · Triple Entente · alliance · arms race · dreadnought · mobilisation · Balkans · British Ultimatum

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART FIVE · CHAPTER 14

Fighting the First World War

Why the line did not move, and what that did to the men in it

THE QUESTION THIS CHAPTER ANSWERS

Why did soldiers go on fighting in the trenches?

IN THIS CHAPTER, YOU WILL

- √ Explain the stalemate on the Western Front in terms of technology.
- √ Describe trench conditions and read a soldier's letter critically.
- √ Explain how industrial production itself became a decisive weapon.
- √ Explain why the war was global, and who fought in it beyond Europe.
- √ Explain Portugal's entry, La Lys, and the consequences at home.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Imperial War Museums, the First World War. Photographs, letters and objects from the war itself. Choose one object and say what it is evidence of.

`www.iwm.org.uk/history/first-world-war`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Everyone in 1914 expected a short war. Write down two reasons an army might expect that, given what you learned about railways, industry and planning in chapters 8 and 12. Then read on and find out what they had not allowed for.

CHAPTER 14 · TOPIC 14.1

The Western Front

The account

The German plan required a fast victory in the west. It failed. German armies advanced through Belgium into northern France, were halted at the Marne in September 1914, and both sides then tried to outflank each other northward until they ran out of land at the sea.

What was left was a continuous line of **trenches** running roughly from the Belgian coast to the Swiss border, about seven hundred kilometres, and it barely moved for over three years.

Why it barely moved is the central military fact of the war, and it comes from chapter 8. Industrial technology in 1914 favoured **defence** overwhelmingly. Machine guns let a handful of defenders stop hundreds of attackers. Barbed wire slowed an advance to walking pace in front of those guns. Artillery could destroy an area but could not hold it, and the shelling churned the ground into mud that made advance harder still. Railways brought defenders' reinforcements to a threatened point faster than attackers could exploit a gap on foot.

So attacks gained a few kilometres at enormous cost, and were then pushed back. Both sides tried to break the deadlock with more shells, with gas, with tunnels and mines, and eventually with tanks, and none of it worked decisively until 1918.

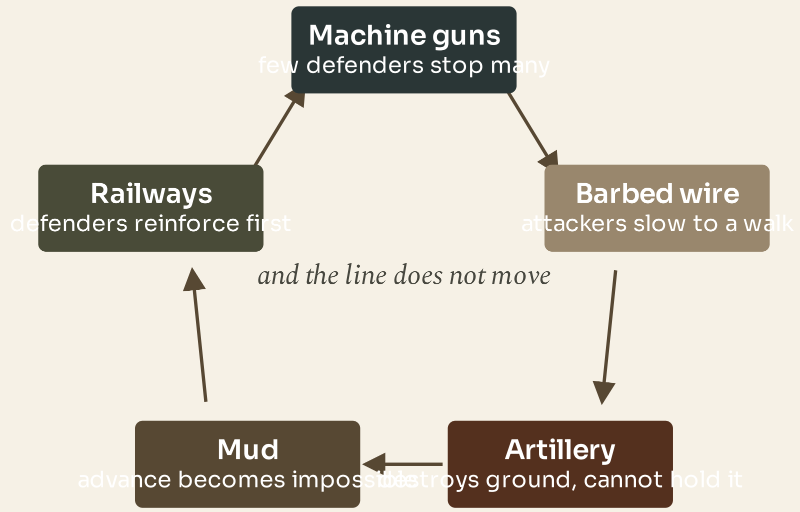


Figure 14.1 · Why the line did not move.
Read the boxes as a loop: each answer to the problem created the next problem. That loop is the shape of the whole Western Front.

YOUR TURN

1. How long was the Western Front, and roughly where did it run?
2. Give four reasons the defence was stronger than the attack.
3. Why could artillery destroy ground but not capture it?
4. Explain, using chapter 8, why railways helped defenders more than attackers.

CHAPTER 14 · TOPIC 14.2

Life in the trenches

The account

Trench systems were not a single ditch. They were layered: a front line, support trenches behind it, reserve trenches further back, and communication trenches running between them, dug in zigzags so that a shell or a raider could not sweep the whole length.

Soldiers rotated. A unit spent a period in the front line, then in support, then in reserve, then in rest, and the rotation was as much about endurance as about tactics.

Conditions were miserable in ways that had little to do with battle. Trenches filled with water and the ground turned to deep mud. Rats and lice were constant. Standing in cold water for days caused **trench foot**, a serious condition that could end in amputation, and armies eventually issued strict orders about dry socks and foot inspection because of it. Disease killed and disabled large numbers.

Most of the time there was no attack, and the danger was random: shelling, snipers, working parties sent out at night to repair wire. Soldiers described the combination of boredom and sudden terror as the defining experience, and the strain produced what was then called shell shock and is now understood as a psychological injury.

The letters, diaries and photographs left by soldiers are among the richest sources any historian of the period has, and they are worth reading in the original.

READ THE SOURCE

A soldier's letter home, Western Front. Letters from the front were written knowing that an officer would read and censor them, and knowing who would open them at the other end.

- **Origin:** a serving soldier, writing to family.
- **Purpose:** to reassure, to keep contact, and to ask for practical things.
- **Value:** it records daily life, food, weather, friendship and the small

details no official record keeps.

- **Limitation:** it is censored, and it is self-censored too, because most

soldiers deliberately spared their families the worst. Absence of horror in a letter is not evidence of absence of horror.

That last line is the most important lesson in source work: ask what a source had reason **not** to say.

YOUR TURN

1. Describe the layers of a trench system and say why the trenches zigzagged.
2. Give three causes of illness in the trenches.
3. Why did armies issue orders about socks?
4. Explain why a cheerful letter home is not evidence that conditions were good.

CHAPTER 14 · TOPIC 14.3

Industrial warfare

The account

The First World War is where the Industrial Revolution arrives on the battlefield. Every technology in chapter 8 has a military form here, and the scale is what makes it new.

Artillery did most of the killing. Industrial production meant shells could be fired in quantities no earlier war had seen, and factories at home worked to supply them, which is why a shortage of shells could become a political crisis.

Machine guns gave a small number of defenders the firepower of a company.

Poison gas was first used on a large scale in 1915. It was cruel, unreliable and dependent on the wind, and it produced respirators rather than breakthroughs.

Tanks appeared in 1916, designed to cross trenches and crush wire. Early ones broke down constantly; by 1918 they were a serious weapon.

Aircraft began as observation platforms, correcting artillery fire, and developed into fighting machines.

Submarines attacked merchant shipping, and nearly cut Britain's supply lines.

Two consequences follow.

The war became a contest of **production**, so factories, coal, steel, shipping and food supply were as decisive as armies. That is why governments took control of economies to a degree never seen before.

And civilians became **participants**: making the shells, and, through blockade and bombardment, suffering the consequences.

YOUR TURN

1. Name six technologies of industrial warfare and give one effect of each.
2. Why could a shortage of shells become a political crisis?
3. Explain the sentence "the war became a contest of production".
4. Give two ways civilians were drawn into the war.

CHAPTER 14 · TOPIC 14.4

A global war

The account

It is called a world war for good reasons, and a Europe-only account of it is wrong.

Empires brought their populations in. Troops and labourers from India, across Africa, Canada, Australia, New Zealand, the Caribbean and elsewhere served in Europe, the Middle East and Africa. Many were volunteers; many were conscripted or recruited under pressure; their contribution was large and was under-acknowledged for decades.

Fighting took place far outside Europe, in the Middle East against the Ottoman Empire, in East and West Africa, and at sea across the Atlantic, the Indian Ocean and the Pacific.

The sea war was global by nature. Blockade was a central strategy on both sides: Britain sought to cut Germany off from imports, and Germany used submarines against shipping to Britain.

Neutral countries were affected too, through trade, blockade, refugees and the loss of markets. Some, like Portugal, did not stay neutral.

ON THE MAP

On a world map, mark the main theatres: the Western Front, the Eastern Front, Italy, the Middle East, East Africa and the Atlantic. Then mark the sea routes into Britain. Write one sentence explaining why a war decided in France was fought in the Atlantic as well.

YOUR TURN

1. Name four parts of the world outside Europe involved in the fighting.
2. Explain what a blockade is and why both sides used one.
3. Why was the contribution of colonial troops under-acknowledged?
4. Give two ways a neutral country could still be badly affected.

CHAPTER 14 · TOPIC 14.5

The United States enters the war

The account

The United States stayed out of the war for nearly three years, and its entry in **1917** changed the outcome.

Two things brought it in.

Unrestricted submarine warfare. Germany resumed sinking merchant ships without warning, including neutral ones, calculating that it could starve Britain into surrender before America could act. American ships and lives were lost, and American opinion turned.

The Zimmermann telegram. A German diplomatic message proposing an alliance with Mexico against the United States was intercepted and published, which removed much of the remaining argument for staying out.

The effect was less immediate than it sounds, because an army had to be raised and shipped. But American **industrial and financial** weight was felt at once: food, steel, munitions, shipping and credit. By 1918 American troops were arriving in large numbers, and the balance of manpower shifted decisively.

Meanwhile Russia left. The **revolutions of 1917** ended Russian participation, and the new Soviet government made peace with Germany early in 1918. Germany could move divisions west, which is why 1918 opened with a huge German offensive and closed with German defeat: the last gamble failed, and the Allies, now reinforced from America, pushed forward until Germany sought an armistice on **11 November 1918**.

YOUR TURN

1. Give the two events that brought the United States into the war.
2. Why was the American contribution economic before it was military?
3. What happened to Russia's participation in 1917 and 1918?
4. Explain why 1918 saw both a great German offensive and a German defeat.

CHAPTER 14 · TOPIC 14.6

Portugal in the global story: Portugal goes to war

What was happening in Portugal?

Portugal entered the war as a **young and unstable republic**. The monarchy had fallen only in 1910, governments changed frequently, and the country was politically divided.

Portugal did not declare war in 1914. It was drawn in gradually. There was fighting between Portuguese and German forces on the borders of **Angola** and **Mozambique** from 1914 onward. Then in **February 1916**, at British request, Portugal seized German ships in Portuguese ports. Germany declared war on Portugal in **March 1916**.

Portugal sent the **Portuguese Expeditionary Corps**, the CEP, to the **Western Front** in Flanders. On **9 April 1918** the CEP, holding a section of the line and overdue for relief, was struck by a major German offensive at the **Battle of La Lys**. Portuguese losses were very heavy: thousands were killed, wounded or taken prisoner in a single day. It remains the most costly day in modern Portuguese military history.

What part did geography play?

Two things. Portugal's African colonies shared borders with German colonies, so the war reached Portuguese territory whether or not Lisbon wanted it. And Portugal's Atlantic ports and its long alliance with Britain made neutrality far harder to sustain than it was for Spain.

What was the connection?

The consequences at home were severe. The war brought **inflation**, shortages of food and fuel, and rationing. It deepened political division, because intervention had been contested from the start. The republic that had entered the war weak came out of it weaker.

The line runs directly into chapter 15: economic hardship and political instability after 1918 ended in the **military coup of 28 May 1926**, and the end of the parliamentary First Republic.

KEY QUESTION

Why did a war fought across Europe and Africa have such significant consequences for Portugal?

Answer in three parts: what Portugal committed, what it cost, and what changed politically. Then compare with Spain, which stayed neutral. Your last sentence should say what the comparison shows about the price of participation.

WHAT YOU CAN NOW DO

- √ Explain why the Western Front became a stalemate, in terms of technology.
- √ Describe trench conditions and use a soldier's letter critically.
- √ Explain how industrial production became a decisive weapon.
- √ Explain why the war was global, and who fought in it beyond Europe.
- √ Explain Portugal's entry, the CEP, La Lys, and the consequences at home.

WORDS FOR THIS CHAPTER

Western Front, stalemate, trench, no man's land, barbed wire, machine gun, artillery, poison gas, tank, submarine, blockade, attrition, armistice, Portuguese Expeditionary Corps, La Lys, rationing, inflation

CHAPTER 14 AT A GLANCE

Fighting the First World War

why did soldiers go on fighting in the trenches?

THE TOPICS YOU HAVE COVERED

- 14.1 The Western Front
- 14.2 Life in the trenches
- 14.3 Industrial warfare
- 14.4 A global war
- 14.5 The United States enters the war
- 14.6 Portugal in the global story: Portugal goes to war

WHAT YOU CAN NOW DO

- √ Explain why the Western Front became a stalemate, in terms of technology.
- √ Describe trench conditions and use a soldier's letter critically.
- √ Explain how industrial production became a decisive weapon.
- √ Explain why the war was global, and who fought in it beyond Europe.
- √ Explain Portugal's entry, the CEP, La Lys, and the consequences at home.

THE WORDS THIS UNIT TAUGHT

Western Front · stalemate · trench · no man's land · barbed wire · machine gun · artillery · poison gas · tank · submarine · blockade · attrition · armistice · Portuguese Expeditionary Corps · La Lys · rationing · inflation

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

15

PART FIVE · CHAPTER 15

The consequences of the First World War

The peace that ended one war and helped prepare the next

THE QUESTION THIS CHAPTER ANSWERS

Did the peace settlement make another European war more likely?

IN THIS CHAPTER, YOU WILL

- √ Describe the human and economic costs, and use estimates honestly.
- √ Explain how four empires ended and what replaced them.
- √ Compare the map of Europe before and after the war.
- √ Set out the terms of Versailles and argue both sides of the debate.
- √ Explain what the League of Nations achieved and why it failed.

THE TOPICS, IN THE ORDER YOU MEET THEM

15.1	The human and economic cost
15.2	The collapse of empires
15.3	New borders
15.4	The Treaty of Versailles
15.5	The League of Nations
15.6	Portugal in the global story: political instability after the war

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Treaty of Versailles. Read the terms, then decide which one a German voter in 1930 would have found hardest to accept, and why that matters.

www.britannica.com/event/Treaty-of-Versailles-1919

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

● **BEFORE YOU READ**

After a fight, who decides what happens next: the winner, or both sides together? Write one sentence on each option, saying what each is likely to produce. The peace of 1919 chose one of them, and chapter 16 is the result.

CHAPTER 15 · TOPIC 15.1

The human and economic cost

The account

The war ended on 11 November 1918. What it had cost is the first thing to establish, because everything that follows was decided by people living with it.

Millions of soldiers were killed, and millions more were wounded, many of them permanently: limbs lost, lungs damaged by gas, faces reconstructed by the new surgery the war forced into existence. Whole communities lost a large part of a generation of young men.

Civilians died too, from bombardment, from occupation, from hunger caused by blockade, and from the deportation and massacre of populations, most notably the Armenians in the Ottoman Empire. Then, in 1918 and 1919, an **influenza pandemic** swept a world already weakened, and killed on a scale comparable with the war itself.

The economic cost was equally severe. Governments had borrowed enormously, mostly from their own citizens and from the United States, which changed from a debtor into the world's creditor. Farmland, mines, factories, railways and whole towns in northern France and Belgium were destroyed. Trade patterns built over decades had been broken.

And there was a cost that does not appear in any table: millions of people who had been told the war would be short, then that it would be worth it, and who came home to find neither true.

◆ **CAREFUL**

Casualty figures for this war vary between reputable sources, because record keeping differed between armies, because "missing" is not the same as "killed", and because civilian deaths from hunger and disease are hard to attribute. When you use a figure, name your source and say it is an estimate. That is not weakness; it is how history is written.

■ **YOUR TURN**

1. Give three categories of human cost from this topic.
2. What happened in 1918 and 1919 that added enormously to the death toll?
3. How did the war change the financial position of the United States?
4. Why do casualty figures differ between sources, and what should you do about it?

CHAPTER 15 · TOPIC 15.2

The collapse of empires

The account

Four empires that had existed for centuries did not survive the war.

The **Russian Empire** fell first, in the revolutions of 1917, and was replaced after a civil war by the Soviet Union.

The **German Empire** ended in November 1918 with the abdication of the Kaiser and the proclamation of a republic.

The **Austro-Hungarian Empire** dissolved into its national components, which is the nationalism of chapter 11 arriving at its logical conclusion.

The **Ottoman Empire** was partitioned, and its Turkish core became a republic after a war of independence.

The consequence was a new map. States appeared or reappeared: Poland, Czechoslovakia, Yugoslavia, Hungary, Austria, Finland, Estonia, Latvia, Lithuania. In the Middle East, territories formerly Ottoman were placed under British and French administration under League of Nations mandates, with borders drawn largely by those powers.

Two problems were built into the new map from the start, and both mattered later.

No border could match the population exactly. Central and eastern Europe was too mixed for that. Every new state contained minorities, and every one had neighbours who claimed them.

Some peoples got a state and some did not. Self-determination was applied to the defeated empires far more than to the victors' colonies, which was noticed at the time and had long consequences.

YOUR TURN

1. Name the four empires that ended, and how each one ended.
2. Name four states that appeared or reappeared on the map.
3. Explain why no border in central Europe could match the population exactly.
4. Why did the uneven application of self-determination cause resentment?

CHAPTER 15 · TOPIC 15.3

New borders

The account

Compare a map of Europe in 1914 with one from 1920 and the difference is startling. It is worth doing slowly, because it is the clearest single demonstration in this book that **borders are historical**, which is the argument chapter 33 makes in full.

Germany lost territory in the east to the new Poland, including a corridor giving Poland access to the sea and separating East Prussia from the rest of Germany. Alsace and Lorraine returned to France. Austria-Hungary vanished, replaced by Austria, Hungary, Czechoslovakia and parts of Yugoslavia, Poland, Romania and Italy. Russia lost its western borderlands.

Each of those changes left somebody dissatisfied, and dissatisfaction with a border is one of the most durable political forces there is. German nationalists pointed to Germans now living in Poland and Czechoslovakia. Hungary had lost most of its former territory. Italy felt it had gained less than promised.

Hold this topic in mind when you reach chapters 16 and 17. The demand to redraw the 1919 map is the thread that runs through the whole interwar period.

ON THE MAP

Draw two rough outlines of central Europe, one for 1914 and one for 1920. Mark Germany, Austria-Hungary, Russia and the Ottoman Empire on the first, and the new states on the second. Then write down the three changes you think are most likely to cause future conflict, and why.

YOUR TURN

1. Give three specific territorial changes made after the war.
2. What was the Polish corridor, and what did it separate?
3. Why is dissatisfaction with a border so durable?
4. Explain the sentence "borders are historical" using this topic.

CHAPTER 15 · TOPIC 15.4

The Treaty of Versailles

The account

The treaty with Germany was signed at Versailles in June 1919. Its terms fell into four groups.

Territory. Germany lost land in Europe, as topic 15.3 described, and all its overseas colonies, which were distributed as mandates.

Military restrictions. The German army was capped at 100 000 men, conscription was forbidden, tanks, military aircraft and submarines were prohibited, and the Rhineland was demilitarised.

Reparations. Germany was required to pay for war damage, in a sum fixed later and very large.

War guilt. Article 231 required Germany to accept responsibility for causing the loss and damage of the war. Legally it was the basis for reparations. Politically it was the clause Germans resented most.

The debate about the treaty is one of the most useful in the whole course, because there is real evidence on both sides.

It was too harsh, said critics: it humiliated a country that had to be lived with, imposed payments that damaged the German economy and therefore Europe's, and handed German nationalists a permanent grievance.

It was not harsh enough, said others: Germany kept its industrial heartland and its unity, and the treaty was severe enough to enrage without being severe enough to disable.

It was constrained, say many historians: the negotiators faced impossible demands from their own publics, and any settlement acceptable at home in 1919 would have been resented in Germany.

Notice that Germany did not negotiate. It was presented with the terms and told to sign, which is why the treaty was called a dictated peace in Germany from the first day.

THE ENQUIRY: DID THE TREATY OF VERSAILLES MAKE ANOTHER WAR MORE LIKELY?

1. List the four groups of terms, with one example of each.
2. Write the strongest case that the treaty caused the next war.
3. Write the strongest case that it did not, and that later decisions mattered more.
4. Conclude. Your conclusion must name at least one thing that happened between 1919 and 1939 without which your answer would change. That is what makes it a historical judgement rather than an opinion.

YOUR TURN

1. Give three military restrictions imposed on Germany.
2. What was Article 231, and why did it matter more politically than legally?
3. Give one argument that the treaty was too harsh and one that it was too soft.
4. Why was it called a dictated peace?

CHAPTER 15 · TOPIC 15.5

The League of Nations

The account

The League of Nations was created in 1920, the first permanent international organisation intended to keep the peace. Its central idea was **collective security**: an attack on one member would be treated as a concern of all, and aggression would be met by the combined pressure of the rest.

Its methods were discussion, arbitration, and, if those failed, economic sanctions. It had no army of its own.

It did real work. It settled some smaller disputes, ran the mandate system, organised refugee relief and public health work, and gave states a permanent place to talk, which had never existed before.

It also failed at its main purpose, and the reasons are worth listing because chapter 20 builds the next attempt on them.

The United States never joined, although an American president had proposed it, because the Senate refused to ratify. The League therefore lacked the world's largest economy from the start.

Decisions required unanimity, so any member could block action.

It had no armed force, and sanctions only work if members apply them, which they frequently did not when their own trade was at stake.

Great powers left when it suited them. Japan and Germany both withdrew in the 1930s.

When aggression came, in Manchuria in 1931 and Abyssinia in 1935, the League condemned and did not stop it.

■ **YOUR TURN**

1. What is collective security, in one sentence?
2. Give two things the League genuinely achieved.
3. Give four structural weaknesses, and explain which you think was worst.
4. Why does an organisation without an army depend so completely on its members?

CHAPTER 15 · TOPIC 15.6

Portugal in the global story: political instability after the war

What was happening in Portugal?

Portugal came out of the war with the problems of chapter 14 unresolved and worse. **Inflation** was severe, the cost of living rose faster than wages, food and fuel were short, and the public debt had grown.

Politically, the First Republic could not produce stable government. Between 1910 and 1926 Portugal had dozens of governments, several presidents, attempted coups and periods of unrest. Parties fragmented; parliament could rarely sustain a majority; the army became increasingly involved in politics.

The war had made all of this worse. Intervention had been contested from the start, La Lys was experienced as a disaster, returning soldiers found little waiting for them, and the economic strain fell on people who had been promised better.

On **28 May 1926** a military coup ended the parliamentary First Republic. It was welcomed by many people who wanted order, and it opened a period of military government from which, as chapter 16 will show, the Estado Novo emerged.

What was happening in the world?

The same pattern appeared across Europe. Post-war economic distress, disillusion and weak parliamentary systems produced authoritarian government in Italy in 1922, and later in Poland, Spain, Germany and elsewhere. Portugal's 1926 coup belongs to that European sequence, and comes early in it.

What was the connection?

A parliamentary system that cannot deliver stability or bread loses its defenders. That is the general lesson, and it applies to Portugal, to Italy and, in chapter 16, to Germany. The people who welcomed the coup of 1926 were not mostly enthusiasts for dictatorship; many simply wanted a government that lasted more than a few months and prices that stopped rising.

■ **KEY QUESTION**

How can economic and social instability weaken democratic political systems?

Build the chain for Portugal from 1916 to 1926, step by step. Then test it: find one step that also appears in the Italian case, and one that is particular to Portugal. End with a sentence on what a democratic government would have needed to do, and when, to break the chain.

WHAT YOU CAN NOW DO

- √ Describe the human and economic costs of the war, and use estimates honestly.
- √ Explain how four empires ended and what replaced them.
- √ Compare the map of Europe before and after the war and explain the changes.
- √ Set out the terms of Versailles and argue both sides of the debate about it.
- √ Explain the League of Nations, what it achieved and why it failed.

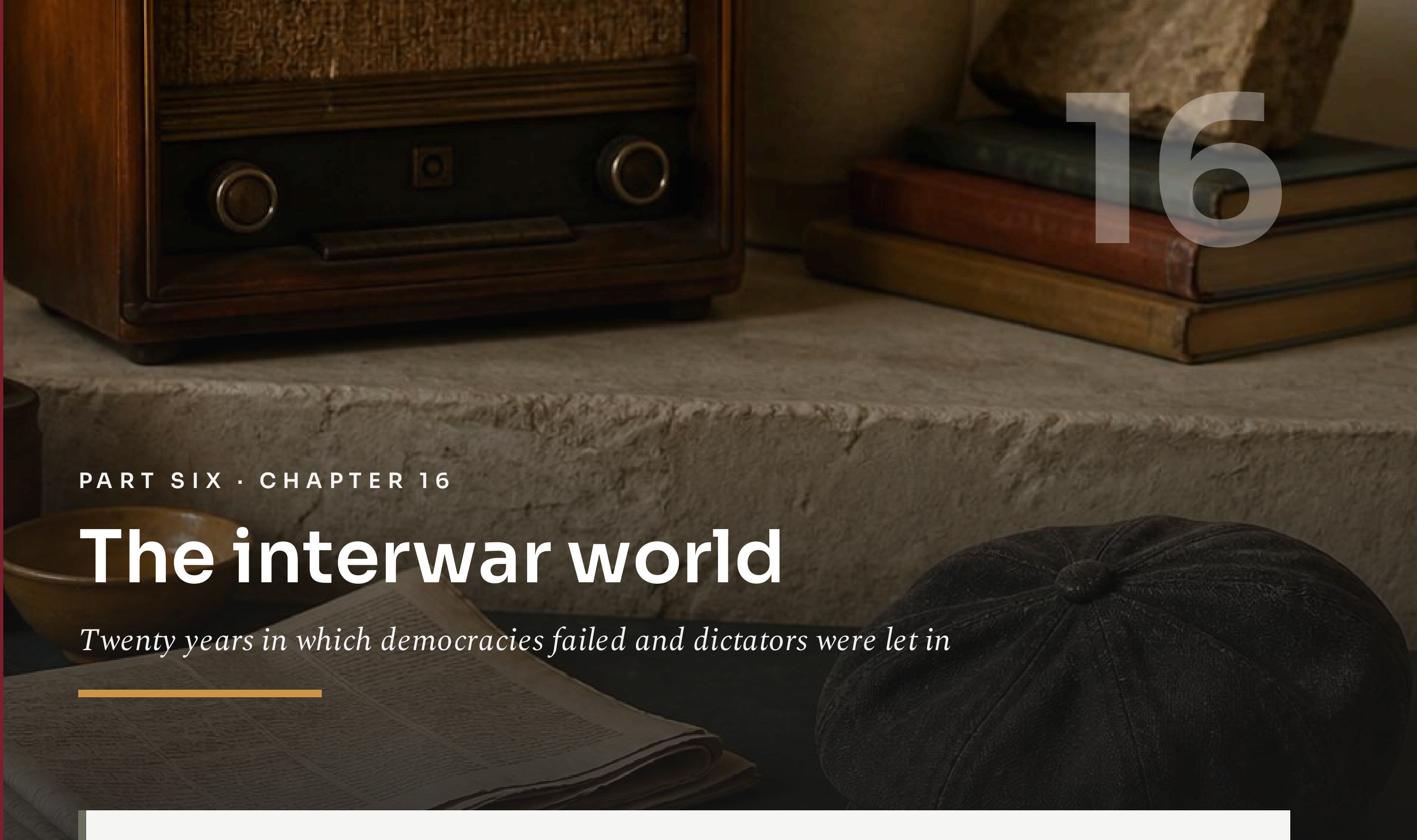
WORDS FOR THIS CHAPTER

armistice, reparations, war guilt, Treaty of Versailles, dictated peace, mandate, self-determination, minority, collective security, League of Nations, sanctions, unanimity, inflation, coup

PART SIX

Between two world wars

The twenty years between the wars are not a gap. They are the explanation. In them a peace settlement was resented, an economy collapsed, democracies failed and dictatorships were built, in Italy, in Germany, in the Soviet Union and in Portugal. Without this chapter the Second World War arrives from nowhere; with it, the war is the end of a sequence.



16

PART SIX · CHAPTER 16

The interwar world

Twenty years in which democracies failed and dictators were let in

THE QUESTION THIS CHAPTER ANSWERS

Why did dictatorship become more common in Europe between the wars?

IN THIS CHAPTER, YOU WILL

- √ Explain why the 1920s were unstable, then better, then catastrophic.
- √ Set out the mechanism of the Great Depression and how it spread.
- √ Explain the pattern by which crisis feeds extreme movements.
- √ Compare Fascist Italy, Nazi Germany and the Soviet Union on stated criteria.
- √ Argue both sides of the appeasement debate, and name the decisive step.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Great Depression. Follow how a financial collapse in one country became mass unemployment in most of them within three years.

www.britannica.com/event/Great-Depression

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

● **BEFORE YOU READ**

List three things a government could do if a quarter of the workforce lost their jobs in three years. Beside each, write who would have to pay for it. Keep the list. This chapter is about what governments actually did, and what happened where they did nothing.

CHAPTER 16 · TOPIC 16.1

The 1920s

The account

The decade after the war was not one thing. It began badly, improved in the middle, and ended in catastrophe.

The **early 1920s** were unstable. Europe had to demobilise millions of soldiers into economies that had been organised for war. Reparations poisoned relations between France and Germany, and when Germany fell behind on payments in 1923 French and Belgian troops occupied the Ruhr, Germany's industrial heartland. Germany responded with passive resistance, printed money to pay the striking workers, and produced a **hyperinflation** in which the currency became worthless. Savings held in cash were destroyed, which ruined a great many careful, middle-class families and left a lasting political scar.

The **middle 1920s** were genuinely better. Currencies were stabilised, an American plan restructured reparations and brought American loans into Germany, production recovered, and international agreements improved relations. Germany joined the League of Nations. There was real optimism, and in the cities of the period an outburst of new art, music, cinema and design.

But the recovery rested on a weak foundation: American loans. Germany paid reparations partly with money borrowed from America; the money returned to America as Allied war-debt payments. That circuit worked while American credit flowed, and only while it flowed.

■ **YOUR TURN**

1. Why did the early 1920s produce instability rather than relief?
2. What caused the German hyperinflation of 1923, and who lost most from it?
3. Give two reasons the middle 1920s were better.
4. Explain the weakness underneath the recovery.

CHAPTER 16 · TOPIC 16.2

The Great Depression

The account

In October **1929** share prices on the New York stock exchange collapsed. The crash did not by itself cause the depression that followed, but it triggered it and it is the date everybody remembers.

The mechanism ran roughly like this. Falling share prices destroyed wealth and confidence. Banks that had lent against shares failed, and when a bank fails its depositors lose their savings, so people stop spending. Businesses facing falling sales cut production and sacked workers. Sacked workers stop spending, so sales fall further. That is a **downward spiral**, and it does not stop by itself.

It spread internationally because the world economy was connected exactly as chapter 10 described. American loans to Europe stopped and were called in. Countries raised tariffs to protect their own producers, which cut world trade sharply and made everybody worse off. Commodity prices collapsed, ruining countries that exported raw materials.

The consequences were mass unemployment, at levels unimaginable today in industrial countries, and, because most countries had little or no unemployment insurance, mass destitution alongside it.

The political consequence is the one this chapter is about. Millions of people concluded that the existing system had failed them, and looked for something else. Where democracies responded with visible action, they largely survived. Where they appeared paralysed, the parties promising to sweep the system away grew fast.

GO FURTHER

Notice the connection to chapter 9. Mass production requires mass consumption: a factory built for very large runs is ruinous at half capacity. The most modern industrial economy in the world was therefore the most exposed when demand collapsed. Efficiency and vulnerability were the same feature seen from two sides.

YOUR TURN

1. Set out the downward spiral in four steps.
2. Give two ways the depression spread from America to the rest of the world.
3. Why did raising tariffs make things worse for everybody?
4. Explain the link between economic collapse and political extremism.

CHAPTER 16 · TOPIC 16.3

Political extremism

The account

Before looking at particular regimes, it is worth understanding the general pattern, because it recurs.

Crisis produces three things that extreme movements use. **Fear**, of losing work, savings, status or safety. **Anger**, at those held responsible. And a **demand for action**, which makes the slow compromises of parliamentary government look like paralysis.

Extreme movements of both left and right offered the same basic package: a simple explanation of who was to blame, a promise of decisive action, a disciplined organisation that looked capable of delivering it, and the claim that the nation or the people required unity rather than argument.

That last claim is the crucial one, because it is the point at which a movement stops being a party and becomes a threat to democracy. A party accepts that opponents are legitimate; a movement that says the nation must be united does not.

Two safeguards were weak in several countries: **new or fragile democracies**, with parliamentary systems only a few years old and little habit of compromise, and **elites who preferred the extremists to the alternative**, and who repeatedly helped them into power believing they could be controlled.

◆ **CAREFUL**

Do not treat the rise of dictatorship as automatic. Several democracies faced the same depression and survived it, and the difference lay in what governments did, how strong the democratic habit was, and what established elites chose. History here is about choices, not inevitability.

YOUR TURN

1. Give the three things a crisis produces that extreme movements use.
2. What is the difference between a party and a movement that demands unity?
3. Give the two weaknesses that made some democracies more vulnerable.
4. Why is it wrong to say the depression automatically produced dictatorship?

CHAPTER 16 · TOPIC 16.4

Fascism in Italy

The account

Italy came out of the First World War on the winning side and deeply dissatisfied: casualties had been heavy, the territorial gains were less than promised, the economy was in difficulty and politics were unstable, with strikes and factory occupations alarming property owners.

Benito Mussolini, a former socialist journalist, built a movement out of that. His squads attacked socialists and trade unionists in the streets, which won him the support of landowners and businessmen who feared revolution more than they feared him.

In 1922 the movement organised a march on Rome, and the king, rather than order the army to resist, appointed Mussolini prime minister. That is the pattern to notice: **he was let in**, not swept in.

Over the following years the regime dismantled the constitution: opposition parties banned, a free press ended, elections managed, opponents imprisoned or exiled, and the state, the party and the leader presented as one thing. Fascism in Italy also organised the economy into state-supervised corporations of employers and workers, and it used spectacle and propaganda intensively.

Its core claims were the primacy of the nation over the individual, the need for a single strong leader, hostility to both liberal democracy and communism, and a belief in national revival through discipline and, eventually, expansion.

YOUR TURN

1. Give three reasons Italy was unstable after 1918.
2. Why did some property owners support Mussolini's squads?
3. Explain the significance of the king's decision in 1922.
4. Name four steps by which a parliamentary state became a dictatorship.

CHAPTER 16 · TOPIC 16.5

Nazism in Germany

The account

Germany after 1918 was a republic with a democratic constitution, born in defeat and carrying three burdens: the **resentment of Versailles**, the **memory of 1923**, and, from 1929, the **depression**, which hit Germany harder than almost anywhere because its recovery had depended on American loans.

The Nazi party, led by **Adolf Hitler**, was tiny in the prosperous mid-1920s and grew very rapidly once unemployment rose. It offered work, national revival, the tearing up of Versailles, and enemies to blame. Its ideology was built on a racist theory of a superior German people, and on **antisemitism**, which was not a side element but central, and which chapter 19 follows to its conclusion.

In January 1933 Hitler was appointed chancellor by the president, in an arrangement made by conservative politicians who believed they could use and control him. Within eighteen months the state had been transformed: emergency powers after the Reichstag fire, a law allowing government by decree, other parties banned, trade unions dissolved, opponents imprisoned, the press and broadcasting controlled, and the party fused with the state.

Two instruments were central. **Propaganda** presented one version of events and removed the others, and used radio, film, rallies and schools. **Terror**, through the political police and the camps, made opposition dangerous. The two worked together: propaganda persuaded many, and terror silenced most of those it did not persuade.

READ THE SOURCE

A propaganda poster of the 1930s. Posters of this period are among the most studied sources in the course, and they must be read as arguments rather than as evidence of fact.

- **Origin:** produced by a state ministry, not by an individual.
- **Purpose:** to persuade, and often to make one answer look like the only

answer.

- **Value:** it tells you exactly what the regime wanted people to believe, and

what it thought would work on them.

- **Limitation:** it tells you nothing about whether people believed it. For

that you need diaries, police reports on public opinion, and what people did.

Ask of any propaganda source: what does this tell me about the **producer**, and what would I need in order to learn about the **audience**?

YOUR TURN

1. Give the three burdens carried by the German republic.
2. Why did Nazi support rise so sharply after 1929?
3. Explain how Hitler came to power in January 1933.
4. Explain how propaganda and terror worked together.

CHAPTER 16 · TOPIC 16.6

The Soviet Union under Stalin

The account

The Soviet Union was created after the revolutions of 1917 and a civil war, and was the first state to be built on Marxist principles, which you met in chapter

1. By the late 1920s **Joseph Stalin** had won the struggle for leadership.

His central policy was to industrialise very fast, from a largely agricultural base, and to do it without foreign investment. Two instruments carried it.

Five-year plans set output targets for coal, steel, oil, machinery and electricity, directed by the state rather than by markets. Heavy industry grew enormously, and the Soviet Union became a major industrial power within about a decade. That achievement is real, and it explains a great deal about chapter 18.

Collectivisation merged peasant farms into large state-directed units, to feed the new industrial cities and to fund industry by exporting grain. It was imposed against fierce resistance and was catastrophic in human terms. Millions died in the famine that accompanied it, above all in Ukraine, Kazakhstan and southern Russia, in a disaster whose causes and responsibility are extensively documented.

The regime was also a dictatorship of a severe kind. Opposition was treated as treason; the purges of the 1930s removed and killed party members, officials and army officers in very large numbers, and a system of forced-labour camps held millions.

> **CAREFUL**
>
> The Soviet Union and Nazi Germany were both dictatorships, and they were not the same thing: their ideologies, aims and victims differed, and they fought each other. Nor is either explained by saying it was extreme. Study each on its own terms, and compare afterwards, on stated criteria. That is how historians handle comparison.

YOUR TURN

1. What were the five-year plans, and what did they achieve?
2. What was collectivisation, and what were its consequences?
3. Why did the Soviet Union industrialise without foreign investment?
4. Why is "they were both extreme" an inadequate comparison?

CHAPTER 16 · TOPIC 16.7

Appeasement

The account

Through the 1930s Germany broke the Versailles terms step by step, and each time the other powers had to decide whether to act.

Conscription was reintroduced and rearmament announced in 1935. German troops reoccupied the demilitarised Rhineland in 1936. Austria was annexed in March

1. In September 1938, at **Munich**, Britain and France agreed that Germany

should take the Sudetenland, the German-speaking border region of Czechoslovakia, in return for a promise of no further demands. Czechoslovakia was not at the table. In March 1939 Germany occupied the rest of the Czech lands, which broke that promise completely.

Appeasement was the policy of granting demands in order to avoid war. It is easy to condemn and worth understanding, because the reasons for it were serious.

Memory of the First World War was overwhelming, and no government wanted to repeat it. Rearmament was incomplete, and Britain in particular wanted time. Public opinion was strongly against war. Some of the German demands looked defensible on the principle of self-determination, since the populations concerned were largely German-speaking. And several governments regarded the Soviet Union as the greater long-term danger.

The case against it is equally serious. Each concession increased German strength, weakened potential allies, and taught Hitler that threats worked. When Britain and France finally guaranteed Poland in 1939, the guarantee was not believed.

THE ENQUIRY: WAS APPEASEMENT A MISTAKE?

1. List the steps from 1935 to March 1939 in order, with dates.
2. Write the strongest case for appeasement, using only what was known at the time.
3. Write the strongest case against it, again using only what was known then.
4. Answer, and then answer a second question: at which step should the policy have changed? Naming the step is what turns an opinion into an argument.

CHAPTER 16 · TOPIC 16.8

Portugal in the global story: from military dictatorship to Estado Novo

What was happening in Portugal?

Chapter 15 left Portugal under the military government installed by the coup of **28 May 1926**. That government could restore order more easily than it could restore the public finances, and it turned to a professor of economics from Coimbra, **Antonio de Oliveira Salazar**, who became finance minister in 1928 on condition that he controlled spending across all departments.

He balanced the budget, and the success gave him great authority. He became prime minister in 1932, and in **1933** a new Constitution established the **Estado Novo**, the New State, which he led until 1968.

The regime's features are those of an authoritarian state: a single legal party, elections without genuine competition, **censorship** of press, books, theatre and film, a political police, and the banning of free trade unions in favour of state-supervised **corporations** of employers and workers. It presented itself as Catholic, rural, orderly and patriotic, summarised in slogans that set the family, the nation and traditional authority against the disorder of the republic.

What was happening in the world?

Exactly the same movement, in the same years. Italy 1922, Portugal 1926, Germany 1933, Spain after the civil war of 1936 to 1939. Portugal was **early** rather than late.

What was the connection, and what was different?

The Estado Novo belonged to the European authoritarian wave and it was **not the same** as Nazi Germany, and the difference matters. It was conservative rather than revolutionary: it wanted to preserve an order, not to build a new kind of society. It did not pursue territorial expansion in Europe. It was not built on a racial theory of the state. It was repressive, and it repressed in order to keep things as they were.

That difference explains something you will need in chapter 18 and chapter 21: a regime that wanted stability rather than conquest could stay neutral in a world war, and could later be accepted as an ally by democracies.

KEY QUESTION

Why did authoritarian governments become more common in Europe between the two world wars?

Use three cases: Italy, Germany and Portugal. For each, note the crisis, the weakness of the parliamentary system, and who helped the new regime into place. Then write a paragraph on what the three have in common, and one honest sentence on how the Portuguese case differs from the German one.

WHAT YOU CAN NOW DO

- √ Explain why the 1920s were unstable, then better, then catastrophic.
- √ Set out the mechanism of the Great Depression and how it spread.
- √ Explain the general pattern by which crisis feeds extreme movements.
- √ Compare Fascist Italy, Nazi Germany and the Soviet Union on stated criteria.
- √ Argue both sides of the appeasement debate, and name the decisive step.

WORDS FOR THIS CHAPTER

hyperinflation, reparations, Ruhr, Wall Street Crash, Great Depression, tariff, unemployment, propaganda, terror, fascism, Nazism, antisemitism, five-year plan, collectivisation, purge, appeasement, Munich, Estado Novo, censorship, corporatism

PART SEVEN

The Second World War

Four chapters: why it began, how it was fought across the whole planet, the Holocaust, and what the world built afterwards out of the wreckage. Portugal appears twice, once for its neutrality and the Azores, and once for a diplomat who disobeyed his government.

CHAPTER 16 AT A GLANCE

The interwar world

why did dictatorship become more common in Europe between the wars?

THE TOPICS YOU HAVE COVERED

16.1	The 1920s
16.2	The Great Depression
16.3	Political extremism
16.4	Fascism in Italy
16.5	Nazism in Germany
16.6	The Soviet Union under Stalin
16.7	Appeasement
16.8	Portugal in the global story: from military dictatorship to Estado Novo

WHAT YOU CAN NOW DO

- √ Explain why the 1920s were unstable, then better, then catastrophic.
- √ Set out the mechanism of the Great Depression and how it spread.
- √ Explain the general pattern by which crisis feeds extreme movements.
- √ Compare Fascist Italy, Nazi Germany and the Soviet Union on stated criteria.
- √ Argue both sides of the appeasement debate, and name the decisive step.

THE WORDS THIS UNIT TAUGHT

hyperinflation · reparations · Ruhr · Wall Street Crash · Great Depression · tariff · unemployment · propaganda · terror · fascism · Nazism · antisemitism · five-year plan · collectivisation · purge · appeasement · Munich · Estado Novo · censorship · corporatism

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART SEVEN · CHAPTER 17

Why did the Second World War begin?

How a grievance became a programme, and a programme became a war

THE QUESTION THIS CHAPTER ANSWERS

Could the Second World War have been prevented?

IN THIS CHAPTER, YOU WILL

- √ Explain how the resentment of Versailles became a political programme.
- √ Distinguish treaty revision from conquest, with an example of each.
- √ Set out the steps of German expansion from 1935 to 1939, in order.
- √ Explain why the guarantee to Poland failed to deter.
- √ Judge whether the war could have been prevented, naming a step and a cost.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Second World War. Compare its account of the causes with the one you built out of the interwar chapter.

`www.britannica.com/event/World-War-II`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Look back at your answer to the appeasement enquiry in chapter 16. Write down the step you named. This chapter tests it.

CHAPTER 17 · TOPIC 17.1

The unfinished business of Versailles

The account

Chapter 15 set out the Treaty of Versailles and chapter 16 the resentment it left. Here is how that resentment became a programme.

Three of the treaty's terms were the constant targets of German nationalist politics: the **territorial losses**, especially the corridor separating East Prussia from the rest of Germany; the **military restrictions**; and the **war guilt clause**. Undoing them was a demand shared far beyond the Nazi party, and that is precisely why it was so effective: it let a radical movement speak for a widely held grievance.

But there is a difference between revision and conquest, and it is the difference the 1930s turn on. Revising a border where the population is German-speaking is an argument that can be made under the principle of self-determination, and Britain and France partly accepted it. Occupying Prague in March 1939, where the population was Czech, was not that argument at all. It was conquest, and it destroyed the case that Germany was simply correcting an unjust peace.

So the honest formulation is this. **Versailles supplied the grievance; it did not supply the ambition.** The war came from both.

YOUR TURN

1. Name the three treaty terms German politics attacked most.
2. Why was it useful to a radical movement that the grievance was widely shared?
3. Explain the difference between revision and conquest, with an example of each.
4. Explain the sentence "Versailles supplied the grievance; it did not supply the ambition".

CHAPTER 17 · TOPIC 17.2

Nazi expansion

The account

The sequence of the 1930s, in order, because the order is the argument.

1935. Conscription reintroduced and rearmament announced openly, breaking the military terms of Versailles. No action was taken against it.

1936. German troops re-entered the demilitarised **Rhineland**. It was a gamble with a small force, and the response was protest only.

1936 to 1939. Germany and Italy intervened in the **Spanish Civil War** on the side of the Nationalists, testing aircraft and tactics, while Britain and France followed a policy of non-intervention.

March 1938. Austria was annexed. Many Austrians welcomed it, which made objection harder.

September 1938. At **Munich**, Britain, France, Germany and Italy agreed the transfer of the **Sudetenland** from Czechoslovakia to Germany. Czechoslovakia was not present. Its border defences were in the territory transferred.

March 1939. Germany occupied the remaining Czech lands. The promise made at Munich was broken, and appeasement collapsed as a policy.

August 1939. Germany and the Soviet Union signed a **pact of non-aggression**, with a secret protocol dividing eastern Europe between them. That removed Germany's fear of a war on two fronts, which chapter 13 showed had shaped German planning for a generation.

1 September 1939. Germany invaded **Poland**.

YOUR TURN

1. List the steps from 1935 to 1939 with their dates.
2. Why was the Rhineland move a gamble?
3. What did Czechoslovakia lose at Munich besides territory?
4. Why did the pact of August 1939 make a German attack far more likely?

CHAPTER 17 · TOPIC 17.3

From appeasement to the invasion of Poland

The account

After Prague, British and French policy changed. In March 1939 they guaranteed Poland: if Poland were attacked, they would go to war.

The guarantee failed to deter, for reasons worth understanding.

It came **after** a long record of not acting, so it was not believed.

It was **hard to make credible**: neither Britain nor France could physically defend Poland, which is in eastern Europe with Germany in between.

And the **Soviet pact** removed the one power that could have made a difference in the east.

On 1 September 1939 Germany invaded Poland. On 3 September Britain and France declared war on Germany. Later that month the Soviet Union invaded Poland from the east under the secret terms of the pact, and Poland was partitioned.

So the war began, in Europe, on 1 September 1939, over a country neither western power could reach.

THE ENQUIRY: COULD THE SECOND WORLD WAR HAVE BEEN PREVENTED?

1. Choose the step from topic 17.2 where you believe a firm response would have changed the outcome.
2. Say what that response would have had to be, and who would have had to make it.
3. Say what it would have cost, and what might have gone wrong.
4. Then answer the question. A good answer names a step, a cost and a risk. A weak answer only says that somebody should have been braver.

WHAT YOU CAN NOW DO

- √ Explain how the resentment of Versailles became a political programme.
- √ Distinguish treaty revision from conquest, with examples.
- √ Set out the steps of German expansion from 1935 to 1939, in order.
- √ Explain why the guarantee to Poland failed to deter.
- √ Judge whether the war could have been prevented, naming a step and a cost.

WORDS FOR THIS CHAPTER

revision, self-determination, rearmament, Rhineland, annexation, Sudetenland, Munich agreement, appeasement, guarantee, non-aggression pact, partition, deterrence

18

PART SEVEN · CHAPTER 18

The Second World War: a global conflict

A war fought on every ocean, and decided as much by supply as by battle

THE QUESTION THIS CHAPTER ANSWERS

How much of this war was decided by geography?

IN THIS CHAPTER, YOU WILL

- √ Set out the three phases of the war in Europe.
- √ Explain why North Africa and the Mediterranean mattered to a European war.
- √ Explain how the Battle of the Atlantic was fought and won.
- √ Explain why Japan attacked in 1941, and why it was a strategic disaster.
- √ Use six geographical factors to analyse any campaign.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Imperial War Museums, the Second World War. The war in objects and testimony. Find one item from the Battle of the Atlantic and say why the Azores mattered to it.

www.iwm.org.uk/history/second-world-war

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

● **BEFORE YOU READ**

Look at a world map. Find Britain, Germany, the Soviet Union, Egypt, Japan and the United States. Measure roughly how far apart they are. Now ask: how does a country fight another one that far away, and what does it need on the way? That question is this chapter.

CHAPTER 18 · TOPIC 18.1

Europe

The account

The war in Europe divides into three phases, and the shape is worth holding in mind before any detail.

Germany advancing, 1939 to 1941. Poland was defeated in weeks, partitioned with the Soviet Union. In 1940 Germany took Denmark and Norway, then attacked in the west, and France, whose army had been thought among the strongest in the world, was defeated in six weeks. British forces were evacuated from Dunkirk. Britain fought on alone in the west, and in the **Battle of Britain** in the summer of 1940 the Royal Air Force prevented the air superiority a German invasion would have required. Cities were then bombed heavily on both sides for years.

The war becomes enormous, 1941. In June Germany invaded the **Soviet Union**, opening a front of unprecedented size. The advance was rapid but did not achieve a decisive victory before winter, and the Soviet Union, having moved whole industries east, kept fighting. In December Japan attacked the United States, and Germany declared war on the United States days later. The war was now global, and the balance of production had turned decisively against Germany.

The Allies advancing, 1942 to 1945. Stalingrad, ending in early 1943, destroyed a German army and began a Soviet advance that continued for two years. The Allies landed in Italy in 1943, and in **Normandy on 6 June 1944**, opening the front in the west that the Soviet Union had long demanded. Germany was squeezed from both sides and surrendered in May 1945.

◆ **CAREFUL**

The scale of the fighting on the eastern front was far greater than in the west, and so were the losses, military and civilian. Any account of the European war that treats the east as a side theatre is misleading.

■ **YOUR TURN**

1. Give the three phases with their dates.
2. What did the Battle of Britain prevent?
3. Why did the events of 1941 change the balance of the war?
4. Name the two turning points that began the Allied advance, east and west.

CHAPTER 18 · TOPIC 18.2

North Africa and the Mediterranean

The account

Fighting in North Africa looks peripheral on a European map and was not, because of what lay at the eastern end of it.

The **Suez Canal** was the short route between Europe and Asia, carrying oil from the Middle East and troops and supplies from India and Australia to Britain. Losing it would have meant sending everything round Africa, adding weeks to every voyage at a time when shipping was the scarcest resource in the war.

So a campaign was fought back and forth across the desert of Egypt and Libya between 1940 and 1943. It was decided in the end by supply. Desert warfare is mobile, mechanised and utterly dependent on fuel, water, spare parts and ammunition brought hundreds of kilometres, and the side whose supply line worked could move while the other could not.

The **Mediterranean** as a whole was contested for the same reason: it is a sea route, and control of it decided whose supplies reached North Africa and whose did not. Malta, a small island in the middle of it, was besieged and bombed for years precisely because of where it sits.

Once North Africa was secured in 1943, the Allies could invade Sicily and then Italy, which is how the Mediterranean campaign fed back into the European one.

> **ON THE MAP**
>
> Draw the Mediterranean. Mark Gibraltar, Malta, Suez and the Libyan and Egyptian coast. Then draw the two possible routes from Britain to India, one through Suez and one round the Cape. Write one sentence on what the difference in distance means for a country short of ships.

YOUR TURN

1. Why was the Suez Canal so important to Britain?
2. What decided the desert campaign, in the end?
3. Why was Malta besieged?
4. Explain how the North African campaign led to the invasion of Italy.

CHAPTER 18 · TOPIC 18.3

The Battle of the Atlantic

The account

Britain is an island that could not feed itself and could not arm itself without imports. Everything, food, oil, steel, aircraft, ammunition and eventually American troops, arrived by sea. So the longest campaign of the war was the fight to keep the Atlantic sea lanes open.

German submarines attacked merchant shipping, hunting in groups, and in the worst periods sank ships faster than the Allies could build them. If that had continued, Britain would have been forced out of the war without a single enemy soldier landing.

It was won by an accumulation of measures rather than a battle.

Convoys, in which merchant ships sailed in escorted groups, which made them far harder to find and far more dangerous to attack.

Long-range air cover, which forced submarines to stay under water, where they were slow and nearly blind. For a long time there was a gap in the middle of the Atlantic that shore-based aircraft could not reach, and closing that gap was decisive.

Better detection, by radar and sonar, and the reading of encrypted German signals, which allowed convoys to be routed away from waiting submarines.

Shipbuilding, on a scale that eventually replaced losses faster than they occurred.

Topic 18.6 explains what the Azores had to do with the air-cover gap.

YOUR TURN

1. Why was the Atlantic the campaign Britain could least afford to lose?
2. Explain how a convoy protects merchant ships.
3. What was the mid-Atlantic air gap, and why did it matter?
4. Give two other measures that helped win the campaign.

CHAPTER 18 · TOPIC 18.4

Asia and the Pacific

The account

The war in Asia had begun before the war in Europe. Japan, an industrialised power short of raw materials, had invaded Manchuria in 1931 and gone to war with China from 1937.

When western embargoes cut Japan's access to oil and other materials, Japan attacked to seize the resources of south-east Asia, and struck the American fleet at **Pearl Harbor** in December 1941 to prevent interference. It was a tactical success and a strategic disaster, because it brought into the war the country with by far the largest industrial capacity on Earth.

The Pacific war was fought over enormous distances by naval and air forces, with aircraft carriers replacing battleships as the decisive weapon, and by island-to-island campaigns to gain airfields closer to Japan. Geography set the whole shape: distances so great that fuel, airfields and supply decided what was possible.

Fighting in China, Burma and south-east Asia continued throughout, and the war in Asia caused enormous civilian suffering, including famines in occupied territories.

YOUR TURN

1. When did the war in Asia begin, and how?
2. Why did Japan attack in 1941 after embargoes were imposed?
3. Why was Pearl Harbor a tactical success and a strategic disaster?
4. How did distance shape the way the Pacific war was fought?

CHAPTER 18 · TOPIC 18.5

The geography of war

The account

This topic is where the two halves of your Humanities course meet directly. Six geographical factors decided more of this war than any general did.

Distance. Everything has to reach the front. The further it travels, the more fuel, ships, trucks and time it consumes, and the more of the supply is consumed by the act of supplying.

Oil. Modern armies are mechanised. A country without oil must capture it, buy it or lose. Both German and Japanese strategy were shaped by that need.

Sea routes and choke points. Suez, Gibraltar, the Danish straits, the Panama Canal: narrow places through which a great deal must pass, and therefore places worth enormous effort to hold.

Climate and season. The Russian winter, the spring thaw that turns roads to mud, the monsoon in Asia, the weather window that decided the date of the Normandy landings.

Terrain. Mountains, rivers and marsh slow an advance; plains invite one. The north European plain is the corridor armies have used for centuries, and used again.

Industrial geography. Where the coal, steel and factories are, and whether they can be reached by an enemy. The Soviet decision to move factories east, beyond the range of attack, was one of the most consequential of the war.

Distance
Everything has to reach the front, and the journey itself consumes the supply.

Oil
A mechanised army that cannot fuel itself cannot move.

Choke points
Suez, Gibraltar, the Danish straits: narrow places through which a great deal must pass.

Climate and season
The Russian winter, the spring thaw, the monsoon, the weather window for a landing.

Terrain
Mountains and marsh slow an advance; a plain invites one.

Industrial geography
Where the factories are, and whether an enemy can reach them.

Figure 18.1 · Six geographical factors, and where each one decided something.
Use this as a checklist when you write about any campaign: at least three of the six will be doing work.

THE ENQUIRY: HOW MUCH OF THIS WAR WAS DECIDED BY GEOGRAPHY?

Choose one campaign from this chapter. For each of the six factors, say whether it mattered a great deal, a little, or not at all, and give your evidence. Then answer the question in one paragraph. Your last sentence should say what geography did **not** decide, because that matters too.

CHAPTER 18 · TOPIC 18.6

Portugal in the global story: neutral Portugal in the Second World War

What was happening in Portugal?

Portugal, under the Estado Novo of chapter 16, remained officially **neutral** throughout the war. Neutrality was the government's policy from the beginning and it held to the end, but the position was far more complicated than the word suggests.

Portugal had a very old alliance with **Britain**, dating back to the fourteenth century, and the government maintained it while avoiding belligerence. It also traded with both sides, most importantly in **wolfram**, the ore from which tungsten is obtained, which hardens steel for machine tools and armour-piercing shells. Both alliances wanted it, both paid heavily for it, and Portuguese wolfram exports became a subject of intense diplomatic pressure.

Lisbon became one of the most important cities in wartime Europe, for a geographical reason: it was a neutral capital with an Atlantic port and an airport, on the western edge of an occupied continent. It filled with diplomats, intelligence officers of every service, journalists and, above all, **refugees** trying to leave Europe. Topic 19.5 tells the part of that story that belongs to the Holocaust.

What part did geography play?

The **Azores**. The islands sit in the middle of the North Atlantic, and topic 18.3 explained why that mattered: they lie inside the air gap that shore-based aircraft could not otherwise reach. An airbase there closed the gap.

In **1943**, under the old alliance, Portugal granted Britain the use of facilities in the Azores; the United States obtained access later. Historians regard it as a significant contribution to the Battle of the Atlantic, and it was made by a neutral state.

What was the connection?

A small neutral country was strategically important because of where it was, not because of what it could do. Portugal had no capacity to affect the war militarily. It had wolfram, a neutral port and mid-ocean islands, and those three things made it matter to both sides.

The position also shaped what came next. Portugal ended the war with its regime intact, its alliance with Britain reaffirmed and a demonstrated strategic value to the western powers. Chapter 21 shows what that was worth in 1949.

KEY QUESTION

How could a small country become strategically important because of its location?

Answer using the geography you learned in part one. Give the latitude and longitude of the Azores, explain where that puts them in relation to the North Atlantic convoy routes, and then explain the air gap. Finish by naming one other small territory in this book whose importance came from position rather than power.

WHAT YOU CAN NOW DO

- √ Set out the three phases of the war in Europe.
- √ Explain why North Africa and the Mediterranean mattered to a European war.
- √ Explain how the Battle of the Atlantic was fought and won.
- √ Explain why Japan attacked in 1941 and why it was a strategic disaster.
- √ Use six geographical factors to analyse any campaign.

WORDS FOR THIS CHAPTER

blitzkrieg, Battle of Britain, eastern front, Stalingrad, Normandy, Suez Canal, choke point, convoy, air gap, radar, sonar, wolfram, tungsten, neutrality, Azores, aircraft carrier, embargo, supply line

PART SEVEN · CHAPTER 19

The Holocaust

What happened, how it was done, and how we know

THE QUESTION THIS CHAPTER ANSWERS

How can prejudice, propaganda and state power lead to genocide?

IN THIS CHAPTER, YOU WILL

- √ Explain what antisemitism is, and what the Nazi state did with it.
- √ Describe the steps from exclusion by law to open violence to mass murder.
- √ Explain what is meant by calling the killing bureaucratic and industrial.
- √ Name the other groups persecuted, and use survivor testimony critically.
- √ Explain the choice made by Aristides de Sousa Mendes, and what it cost him.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

United States Holocaust Memorial Museum, learning resources. A museum built to hold the record. Read one survivor testimony in full, slowly, and write down what it is evidence of.

www.ushmm.org/learn

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

This chapter is about the murder of millions of people, and it is studied carefully because it happened and because understanding how it happened matters. Read it slowly. Your teacher will tell you how the lesson will work, and there will be time to ask questions and to say what you think.

CHAPTER 19 · TOPIC 19.1

Nazi antisemitism and the Nuremberg Laws

The account

Antisemitism, hostility and discrimination directed at Jewish people, has a long history in Europe. It was not invented by the Nazi party. What the Nazi party did was make it central to the purpose of a state, and give it the machinery of a modern government.

Nazi ideology claimed that humanity was divided into races ranked by value, that Germans belonged to a superior one, and that Jewish people were not merely different but a danger to be removed. Every part of that claim is false. It has no basis in biology and none in history, and it was contested at the time.

The persecution proceeded by law, in public, step by step.

From 1933, Jewish people were removed from the civil service, from universities, from the professions, from the courts and from cultural life. Businesses were boycotted and later confiscated.

In **1935** the **Nuremberg Laws** stripped German Jews of citizenship and forbade marriage between Jews and other Germans. A person's legal status now depended on ancestry.

Each measure was announced, published and enforced by ordinary officials. Understanding that is important: this did not happen in secret, and it required the participation of a great many people who were not fanatics, simply doing their jobs.

CAREFUL

The racial theory used to justify all of this was pseudo-science: it borrowed the appearance of biology without any of its evidence, and it was rejected by scientists then as it is now. Explaining what a regime believed is not the same as granting the belief any standing, and this book grants it none.

YOUR TURN

1. Define antisemitism, and explain what was new about the Nazi version of it.
2. What did the Nuremberg Laws of 1935 do?
3. Why does it matter that the persecution proceeded openly and by law?
4. Explain why the racial theory is described here as pseudo-science.

CHAPTER 19 · TOPIC 19.2

Kristallnacht and the ghettos

The account

In November **1938**, across Germany and Austria, synagogues were burned, Jewish homes, shops and schools were attacked and looted, and tens of thousands of Jewish men were arrested and sent to camps. It is known as **Kristallnacht**, the night of broken glass, a name taken from the shattered shop windows. It was not spontaneous. It was organised, and the Jewish community was then fined for the damage done to it.

Kristallnacht marked a change from exclusion to open violence, and it made clear to many that there was no future in remaining. Emigration attempts increased sharply, and here a second fact belongs, uncomfortably: many countries limited how many refugees they would admit, and a great many people who tried to leave could not find anywhere that would take them. Topic 19.5 is about one exception.

After the invasion of Poland in 1939, and of the Soviet Union in 1941, the German state controlled territory containing very large Jewish populations. Jewish communities were forced into sealed districts of cities called **ghettos**, overcrowded and deliberately under-supplied with food, fuel and medicine. Enormous numbers died there of starvation and disease before any camp was reached.

There was resistance, in the ghettos and elsewhere, armed and unarmed, and it is part of the history: uprisings, escape networks, hidden schools, hidden archives and the deliberate recording of evidence by people who knew they would probably not survive to give it.

YOUR TURN

1. What happened in November 1938, and why is the name misleading about its organisation?
2. Why did Kristallnacht increase attempts to emigrate?
3. What was a ghetto, and why did so many people die in them?
4. Give two forms resistance took.

CHAPTER 19 · TOPIC 19.3

Deportation and the camps

The account

From 1941 the Nazi state moved from persecution and confinement to systematic mass murder.

Behind the advance into the Soviet Union, special units shot very large numbers of Jewish people, together with Roma and Soviet officials, in mass killings near the towns where they lived.

From 1942 the policy was extended and industrialised. Jewish people from across occupied Europe were **deported** by rail to camps in occupied Poland built for the purpose of killing, of which **Auschwitz-Birkenau** is the most widely known. Around six million Jewish people were murdered.

Two things about the method have to be understood, because they are what make this different from earlier massacres in history.

It was **bureaucratic**. Timetables, transport allocations, registers, budgets and correspondence. The killing was administered, and the paperwork survives, which is why it is among the most thoroughly documented crimes ever committed.

It was **industrial**. It used the organisation, transport and technology that chapters 8 and 9 described as the achievements of the modern age, and turned them to murder. That is the hardest and most important lesson of this chapter: modern capacity is neutral, and it does what the people directing it decide.

CAREFUL

The evidence for the Holocaust is overwhelming: German records, the testimony of survivors, the testimony of perpetrators at trial, Allied and Soviet documentation of the camps, and the physical sites themselves. Denial of it is not a historical position and is not treated as one here. Where sources disagree on a detail, historians say so; on whether it happened, they do not.

YOUR TURN

1. What changed in the treatment of Jewish people from 1941, and again from 1942?
2. Explain what is meant by calling the killing bureaucratic.
3. Explain what is meant by calling it industrial.
4. List four kinds of evidence historians use for these events.

CHAPTER 19 · TOPIC 19.4

The other groups the Nazi state persecuted

The account

Jewish people were the central target of Nazi racial policy and were murdered in by far the greatest numbers. Other groups were also persecuted, imprisoned and killed, and they belong in the record.

Roma and Sinti people were persecuted on the same racial grounds and murdered in very large numbers.

Disabled people were the first to be killed systematically, under a programme that murdered patients in hospitals and institutions and that developed methods later used more widely.

Soviet prisoners of war died in enormous numbers in captivity, from deliberate starvation and neglect.

Polish and other Slav civilians were subjected to mass killing, deportation and forced labour under a policy that treated them as inferior.

Political opponents, including communists, socialists and trade unionists, were among the first sent to camps in 1933.

Jehovah's Witnesses, **gay men**, and others who did not fit the regime's demands were imprisoned and killed.

READ THE SOURCE

Survivor testimony. A great many survivors recorded their accounts, in writing, in interviews and on film, and museums and archives hold them in large numbers.

- **Origin:** a person who was there, usually recorded years afterwards.
- **Purpose:** to record what happened, so that it is known and not repeated.
- **Value:** it carries what no document can: what it was like, and the names,

faces and choices of individual people rather than numbers.

- **Limitation:** memory is affected by time and by trauma, and one person sees

one part. Historians therefore read testimony alongside documents and other testimony, which is exactly how they treat every source.

Note carefully: a limitation on a source is not a reason to doubt the event. It is a reason to use more than one source, which is what historians always do.

THE ENQUIRY: HOW CAN PREJUDICE, PROPAGANDA AND STATE POWER LEAD TO GENOCIDE?

Build the chain, in order, using this chapter.

1. **Prejudice**: an existing hostility, available to be used.
2. **Ideology**: a claim that turns hostility into a theory of who counts.
3. **Law**: the theory written into rules that officials enforce.
4. **Separation**: the targeted group removed from ordinary life.
5. **Violence**: open, organised, and unpunished.
6. **Isolation**: nowhere to go, and few willing to help.
7. **Machinery**: transport, records and technology applied to killing.

For each step, write one sentence of evidence from this chapter. Then answer: which step do you think offered the last real opportunity for others, inside and outside Germany, to change what happened? Defend your choice.

CHAPTER 19 · TOPIC 19.5

Portugal in the global story: Aristides de Sousa Mendes

What was happening in Portugal?

Chapter 18 described Lisbon as a neutral Atlantic port on the edge of an occupied continent, and therefore a route out of Europe. To reach it, a refugee needed transit visas, and Portuguese policy on granting them had been tightened. A circular issued in 1939 instructed consuls to refer certain categories of applicant, including stateless people and, explicitly, Jewish people who could not return to their countries, to Lisbon for a decision, which in practice meant refusal or long delay.

In June **1940**, as German forces advanced into France, enormous numbers of refugees fled south toward the Spanish and Portuguese frontier. In **Bordeaux**, the Portuguese consul was **Aristides de Sousa Mendes**.

Faced with queues of families who had nowhere else to go, he decided to issue visas, against his instructions, and did so on a very large scale over several days, with his staff and his family helping, signing until he could no longer continue. He then continued at the frontier.

He was recalled, disciplined and removed from the diplomatic service. He lost his career and his income, and he and his family lived in serious hardship. He died in 1954 without having been reinstated. He was recognised much later, honoured in Israel as Righteous Among the Nations, and formally rehabilitated in Portugal after the restoration of democracy that you will meet in chapter 26.

What was the connection?

This section exists for a reason beyond the biography. Every other topic in this chapter is about systems: laws, bureaucracies, timetables, states. This one is about a **single official with a rubber stamp**, who had a choice, and whose choice is measurable in the number of people who got out.

It is also honest about the cost. He was not rewarded. He was punished, at once and for the rest of his life, and the recognition came long after he was dead. Studying him is not studying an easy decision.

KEY QUESTION

When should an individual refuse to follow official orders?

This is a real question and it does not have a tidy answer. Work in pairs.

1. State the case for obedience: why do governments need officials to follow instructions, and what happens when they do not?
2. State the case for refusal in this instance, as precisely as you can.
3. Identify what makes this case different from ordinary disagreement with a rule.
4. Write your own answer in no more than five sentences, and be prepared to hear somebody disagree with it well.

WHAT YOU CAN NOW DO

- √ Explain what antisemitism is and what the Nazi state did with it.
- √ Describe the steps from exclusion by law to open violence to mass murder.
- √ Explain what is meant by calling the killing bureaucratic and industrial.
- √ Name the other groups persecuted, and use survivor testimony critically.
- √ Explain the choice made by Aristides de Sousa Mendes and what it cost him.

WORDS FOR THIS CHAPTER

antisemitism, ideology, pseudo-science, Nuremberg Laws, citizenship, Kristallnacht, ghetto, deportation, Auschwitz, genocide, Holocaust, testimony, resistance, refugee, transit visa, Righteous Among the Nations

20

PART SEVEN · CHAPTER 20

The end and consequences of the Second World War

The end of the war, and the machinery built to prevent the next one

THE QUESTION THIS CHAPTER ANSWERS

What did the world build in 1945 that it had failed to build in 1919?

IN THIS CHAPTER, YOU WILL

- √ Describe how the war in Europe was won, and what the landings required.
- √ Set out the arguments for and against the use of the atomic bombs.
- √ Give the main consequences of the war for Europe and for the world.
- √ Explain the design of the United Nations as an answer to the League's failures.
- √ Explain why the veto was included deliberately, and what it costs.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

United Nations, the history of the UN. What the states of 1945 tried to build, in their own words, and what they thought the League of Nations had got wrong.

www.un.org/en/about-us/history-of-the-un

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Look back at your notes on the League of Nations in chapter 15. List its four weaknesses. This chapter shows what the world did about each of them in 1945, and whether it worked.

CHAPTER 20 · TOPIC 20.1

The defeat of Nazi Germany

The account

By the beginning of 1944 Germany was being pressed from the east by the Soviet Union and from the south in Italy, and its cities were being bombed heavily.

On **6 June 1944** the Allies landed in **Normandy**, in the largest seaborne invasion ever mounted, and opened a front in the west. It succeeded because of preparation on a scale the operation's geography demanded: artificial harbours, fuel pipelines, and a deception campaign that persuaded the German command the landing would come elsewhere. Weather and tide decided the date.

Through the second half of 1944 the Allies advanced across France and into the Low Countries, while Soviet forces advanced through eastern Europe. Germany counter-attacked in the Ardennes in December and failed.

In the spring of **1945** the two advances met inside Germany. Hitler killed himself in Berlin. Germany surrendered, and the war in Europe ended on **8 May 1945**.

As the armies advanced they reached the camps of chapter 19, and the evidence became public in a way it had not been before. Film and photographs taken then are among the most important documents of the century.

YOUR TURN

1. What was opened on 6 June 1944, and why did it matter?
2. Give three preparations the landings required, and say why each was needed.
3. Which two advances met inside Germany?
4. Give the date the war in Europe ended.

CHAPTER 20 · TOPIC 20.2

Hiroshima, Nagasaki and the Japanese surrender

The account

The war in Asia continued after Germany's surrender. American forces had advanced island by island toward Japan, and each island had been defended to the end, at very heavy cost to both sides. Planners expected an invasion of Japan itself to be enormously costly.

In August **1945** the United States dropped an atomic bomb on **Hiroshima** and three days later a second on **Nagasaki**. Each destroyed most of a city with a single weapon. Tens of thousands of people were killed immediately; many more died later from injuries and radiation. The Soviet Union declared war on Japan between the two bombings and invaded Manchuria. Japan surrendered, and the war ended in **September 1945**.

The decision is debated seriously by historians, and you should know the shape of the argument rather than a slogan.

The case made at the time: an invasion would have cost more lives, on both sides, than the bombs; Japan had not surrendered after months of conventional bombing; the war would otherwise have continued for many months.

The case against: Japan was already close to defeat and its position was hopeless; the Soviet declaration of war was itself a decisive blow; a warning or a demonstration might have been sufficient; and the weapon was used on cities containing very large civilian populations.

A third consideration raised by historians: the postwar position with respect to the Soviet Union was in the minds of some decision makers.

What is not disputed is the consequence. The world now contained a weapon that could destroy a city instantly, and chapter 24 is about living with that.

YOUR TURN

1. What did American planners expect an invasion of Japan to cost?
2. Give two arguments made in favour of using the bombs.
3. Give two arguments made against.
4. What consequence of the bombings is not disputed by anybody?

CHAPTER 20 · TOPIC 20.3

The consequences

The account

The scale of the Second World War is difficult to hold in mind. Estimates of the total dead are in the region of sixty million people, the majority of them civilians, and the largest national losses were suffered by the Soviet Union and by China. Historians give ranges rather than exact figures, for the reasons chapter 15 explained.

The consequences that shaped the rest of the century were these.

Europe was devastated and no longer dominant. Cities, industry, housing and transport were destroyed across the continent, and the powers that had run the world in 1900 could not do so in 1945.

Two superpowers emerged. The **United States**, whose territory was untouched and whose industry had expanded enormously, and the **Soviet Union**, which had suffered terribly and whose army now occupied eastern Europe.

Europe was divided. Where the armies stopped became a political line, and it stayed there for forty years.

Millions were displaced. Refugees, expelled populations, survivors with no home to return to, and prisoners of war moved across Europe for years.

Empires were weakened. The colonial powers had been defeated, occupied or exhausted, and the claim that they were permanent had been shown to be false. Independence movements accelerated everywhere, which is the story chapter 25 picks up.

■ **YOUR TURN**

1. Roughly how many people died, and why do historians give a range?
2. Why did the war leave the United States and the Soviet Union dominant?
3. Explain the sentence "where the armies stopped became a political line".
4. Why did the war weaken the European empires?

CHAPTER 20 · TOPIC 20.4

The United Nations

The account

The **United Nations** was founded in 1945, and it was designed by people who had studied the failure of the League of Nations very closely. Almost every feature is an answer to one of the weaknesses you listed at the start of this chapter.

The great powers were inside it. The United States joined this time, and so did the Soviet Union. An organisation without the strongest states cannot act.

The Security Council could take binding decisions, including authorising the use of force, which the League could not.

Five permanent members were given a veto: the United States, the Soviet Union, Britain, France and China. This was deliberate. The judgement was that an organisation which could act against a great power's will would simply be left by that power, as had happened before, and that keeping everybody inside was worth the cost of allowing each of them to block action.

Broader work was built in from the start: agencies for health, refugees, children, food and agriculture, education and science, and, in 1948, the **Universal Declaration of Human Rights**.

The honest assessment is mixed and worth stating. The veto has repeatedly prevented the United Nations from acting in exactly the crises it was created for, which is the price of the design. The agencies have done a great deal of work of a kind the League could never have attempted. And there has been no third world war, which is not proof that it worked, but is worth putting on the record beside the failures.

	League of Nations, 1920	United Nations, 1945
The great powers	The United States never joined. Japan and Germany left in the 1930s.	The United States and the Soviet Union were fo members.
Taking decisions	Unanimity required, so any member could block anything.	A Security Council that can take binding decis with a veto for five permanent members.
Force	None. It could condemn and recommend sanctions.	It can authorise the use of force, and has done.
Other work	Mandates, refugees and some health work.	Agencies for health, refugees, children, food, education and science, and a declaration of hu rights in 1948.
The cost of the design	Nothing could compel a great power, so aggression went unstopped.	The veto blocks action in exactly the crises it w built for.

Figure 20.1 · Two attempts at the same problem.
Read the rows across. Every difference in the second column is a lesson learned from a failure in the first.

KEY QUESTION FOR THE WHOLE PART

What did the world build in 1945 that it had failed to build in 1919?

Write a comparison of the two settlements, 1919 and 1945, under four headings: the treatment of the defeated, the international organisation, the role of the United States, and what happened to the territory of the defeated. Then answer: which single difference do you think mattered most, and why?

WHAT YOU CAN NOW DO

- √ Describe how the war in Europe was won, and what the landings required.
- √ Set out the arguments for and against the use of the atomic bombs.
- √ Give the main consequences of the war for Europe and the world.
- √ Explain the design of the United Nations as an answer to the League's

failures.

- √ Explain why the veto was included deliberately, and what it costs.

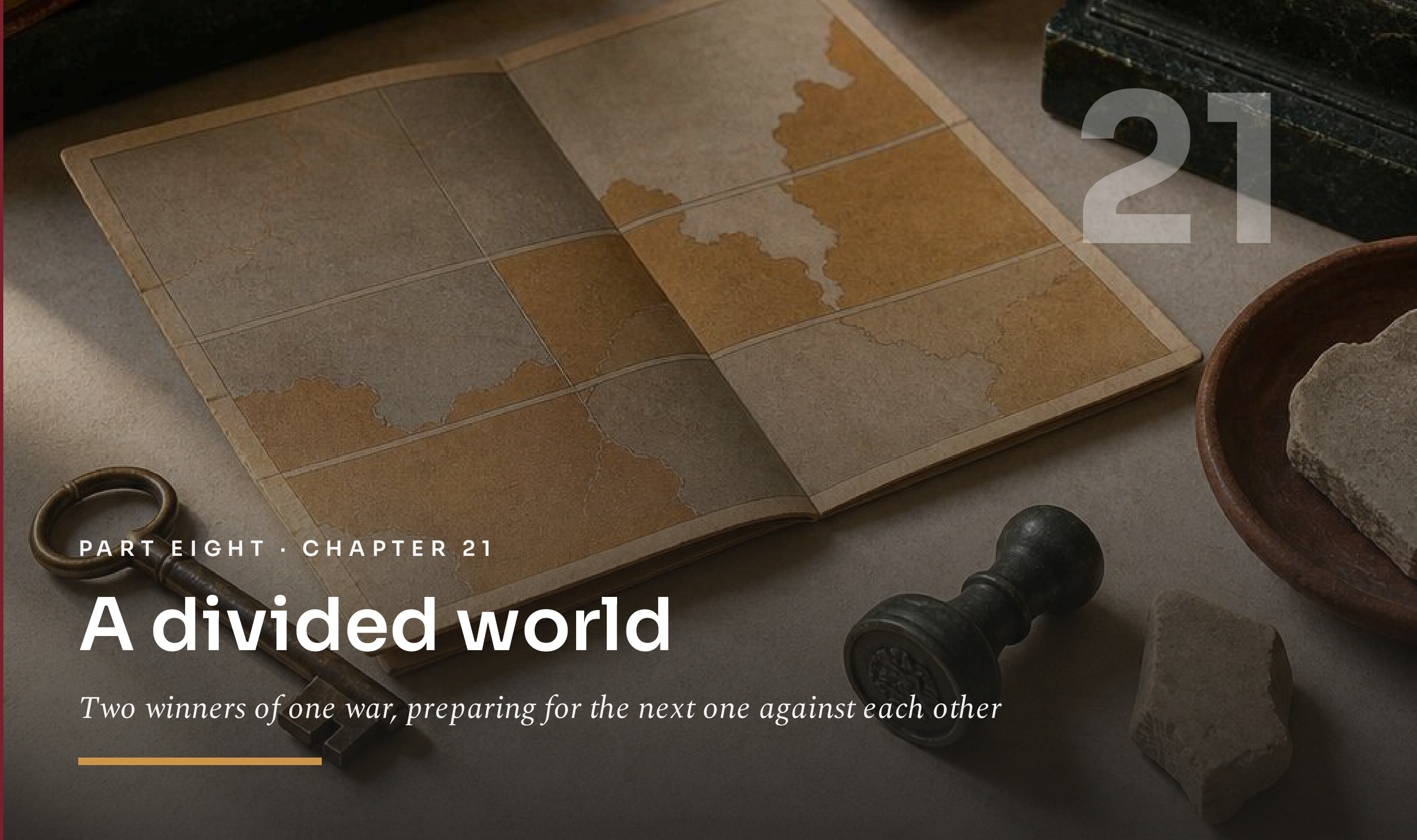
WORDS FOR THIS CHAPTER

Normandy, D-Day, deception, Ardennes, unconditional surrender, atomic bomb, Hiroshima, Nagasaki, superpower, displaced person, decolonisation, United Nations, Security Council, veto, Universal Declaration of Human Rights

PART EIGHT

The Cold War

Two states that had just won a war together spent the next forty years preparing to fight each other, and never did. Seven chapters follow: the division of the world, the city that came to stand for it, the crises, the weapons, the race to the Moon, a revolution in Portugal, and the decision by western Europe to pool its economies rather than fight over them.



21

PART EIGHT · CHAPTER 21

A divided world

Two winners of one war, preparing for the next one against each other

THE QUESTION THIS CHAPTER ANSWERS

Why did two states that had just won a war together become enemies?

IN THIS CHAPTER, YOU WILL

- √ Explain why the United States and the Soviet Union alone were superpowers.
- √ Describe both political and economic models fairly, without caricature.
- √ Explain how Europe was divided, and what the Marshall Plan was for.
- √ Explain collective defence and how NATO differed from the League.
- √ Explain why the two alliances were not symmetrical.

THE TOPICS, IN THE ORDER YOU MEET THEM

21.1	Two superpowers
21.2	Two political and economic models
21.3	Two blocs
21.4	NATO
21.5	The Warsaw Pact
21.6	Portugal in the global story: Portugal and the Western alliance

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Cold War. Note how many of its crises were fought by other countries on behalf of the two great powers.

www.britannica.com/event/Cold-War

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Two people share a house after a disagreement. Neither trusts the other, and each takes precautions. Describe how each precaution looks to the other person. That mechanism, and not any single decision, is most of what follows.

CHAPTER 21 · TOPIC 21.1

Two superpowers

The account

By 1945 the states that had run the world in 1900 could not do so any longer. Britain and France were exhausted and in debt; Germany, Italy and Japan were defeated and occupied. Two powers were left in a different class from everybody else, and the word **superpower** was coined for them.

The **United States** ended the war with its territory untouched, its industry larger than when it began, most of the world's gold reserves, and, until 1949, a monopoly of nuclear weapons.

The **Soviet Union** ended the war having suffered the greatest losses of any combatant, with enormous destruction across its western regions, and with the largest army in the world standing in the middle of Europe.

They had been allies against Germany, and the alliance had been one of necessity rather than agreement. Underneath it lay a genuine conflict: two states with opposite ideas about how a society should be organised, and each with reason to fear the other.

Two further things made suspicion worse. The Soviet Union had been invaded from the west twice within thirty years, and wanted governments in eastern Europe that could not be used against it again. The United States and Britain had seen eastern European countries taken over one by one, and read that as expansion rather than security.

Both readings were sincere. That is precisely what makes the Cold War hard, and interesting.

YOUR TURN

1. Why did the term superpower appear in 1945?
2. Give two strengths of the United States and two of the Soviet Union in 1945.
3. Explain the Soviet argument about eastern Europe.
4. Explain how the same events could be read as security by one side and expansion by the other.

CHAPTER 21 · TOPIC 21.2

Two political and economic models

The account

The two sides differed about how a country should be governed and how its economy should work, and it is worth setting the differences out plainly and without caricature.

The United States and its allies. Several parties competing in elections; courts independent of the government; a press not owned by the state; an economy based mainly on private ownership, with prices set by markets. In practice, the countries in this bloc varied a great deal, and included states that were not democracies at all, which is where chapter 21's Portuguese section comes in.

The Soviet Union and its allies. One party, holding a leading role written into the constitution; the state owning the factories, the land and the shops; production directed by plans rather than prices; and censorship of the press. The stated aim was a society without classes and with the wealth of industry held in common, which is the family of ideas you met in chapter 11.

Each side judged the other by its worst features and itself by its best, and each had real evidence to point to. Western states could point to the purges, the camps and the absence of elections. Soviet states could point to segregation and racial discrimination in the United States, to unemployment, and to the western support of dictatorships and colonial wars.

A historian's job here is not to award marks. It is to describe accurately what each system claimed, what each delivered, and how each looked to the other.

CAREFUL

Neither bloc was uniform. Western Europe built extensive welfare states and had large socialist and communist parties; Yugoslavia was communist and broke with Moscow; several members of the western alliance were dictatorships. Whenever you write "the West" or "the East" in this part, check whether the sentence would survive being applied to every member.

YOUR TURN

1. Give three features of each model, stated fairly.
2. Give one criticism each side made of the other that had real evidence behind it.
3. Why is it wrong to treat either bloc as uniform?
4. Explain what a historian's job is when describing two hostile systems.

CHAPTER 21 · TOPIC 21.3

Two blocs

The account

Where the armies had stopped in 1945 became the border of the two blocs, and it barely moved for forty years.

In eastern Europe, governments aligned with the Soviet Union were established between 1945 and 1948, in Poland, Hungary, Romania, Bulgaria, Czechoslovakia and the Soviet occupation zone of Germany. In western Europe, states aligned with the United States, encouraged by very large American economic aid under the **Marshall Plan**, which rebuilt industry and, deliberately, made communist parties less attractive by making life better.

Germany itself was divided into occupation zones and then into two states in 1949, and Berlin, deep inside the Soviet zone, was divided too. Chapter 22 is entirely about the consequences of that.

Winston Churchill described the dividing line in 1946 as an **iron curtain** descending across the continent, and the phrase stuck because it described something real: a frontier that was fortified, watched and, in most places, closed.

The division was not only European. Over the following decades the two blocs competed for influence across Asia, Africa, the Middle East and Latin America, frequently by supporting opposite sides in local conflicts, which is why the Cold War was cold only in Europe.

> **ON THE MAP**
>
> Draw Europe and mark the line between the two blocs in 1949. Note which of these were on which side: Poland, Austria, Yugoslavia, Greece, Finland, Portugal. Three of those six do not fit a simple two-colour map. Find out why, and write one sentence on each.

YOUR TURN

1. What decided where the line between the blocs fell?
2. What was the Marshall Plan, and give its two purposes.
3. Explain the phrase "iron curtain" and why it was accurate.
4. Explain the sentence "the Cold War was cold only in Europe".

CHAPTER 21 · TOPIC 21.4

NATO

The account

In **1949** twelve states signed the North Atlantic Treaty, creating **NATO**. The founder members were the United States, Canada, Britain, France, Italy, the Netherlands, Belgium, Luxembourg, Denmark, Norway, Iceland and **Portugal**.

Its central provision is the commitment that an armed attack on one member shall be considered an attack on them all. That is **collective defence**, and it is the League of Nations idea of chapter 15 with two differences that matter: it is a smaller group of states that actually agree with each other, and it has armed forces and joint plans behind it.

The purpose was to deter. If an attack on one member meant war with all of them, including the United States, then attacking any single one was unattractive.

The word **North Atlantic** is not decoration. The alliance depended on the United States being able to reinforce Europe across the ocean in a crisis, which made the sea lanes, and the islands and bases along them, part of the alliance's core geography. That is the whole reason for topic 21.6.

YOUR TURN

1. Give the year NATO was founded and four of its founder members.
2. Explain collective defence in one sentence.
3. How does NATO differ from the League of Nations in the two ways given?
4. Why does the name mention the Atlantic?

CHAPTER 21 · TOPIC 21.5

The Warsaw Pact

The account

The Soviet Union and its allies formed the **Warsaw Pact** in **1955**. The immediate occasion was the admission of West Germany to NATO ten years after the war, which was alarming in Moscow for obvious reasons.

Formally the Pact mirrored NATO: a treaty of mutual defence among sovereign states. In practice the two were not symmetrical, and the difference is worth being precise about.

NATO members were free to leave and to disagree, and did: France withdrew from the integrated military command in 1966 and remained a member of the alliance.

Warsaw Pact members were not free in the same way. When Hungary attempted to leave in **1956**, and when Czechoslovakia attempted major reforms in **1968**, Soviet forces intervened militarily. The Pact functioned partly as an alliance against the west and partly as an instrument of Soviet control within the bloc.

That is a fact with a long consequence. When Soviet leaders made clear in the late 1980s that they would no longer intervene, the governments of eastern Europe lost the thing that had kept several of them in place. Chapter 28 follows that directly.

	NATO, 1949	Warsaw Pact, 1955
What the treaty says	An attack on one member is an attack on all.	An attack on one member is an attack on all.
Who led it	The United States, with Canada and ten European states at the start.	The Soviet Union, with seven eastern Europear states.
Members who differed	France left the integrated command in 1966 and stayed in the alliance.	Hungary in 1956 and Czechoslovakia in 1968 w invaded.
Could a member leave?	Yes. Membership was, and remains, voluntary.	Not in practice, while Soviet forces were prepa to intervene.
What that meant in 1989	Nothing changed for the alliance.	Once Moscow would no longer intervene, the governments fell.

Figure 21.1 · Two alliances, and where the symmetry breaks.
Read the last two rows carefully: they are the difference that decides what happens in 1989.

YOUR TURN

1. Give the year the Warsaw Pact was formed and its immediate occasion.
2. In what formal respect did the two alliances resemble each other?
3. Give two examples showing the Pact was also an instrument of control.
4. Why did the Soviet decision to stop intervening matter so much later?

CHAPTER 21 · TOPIC 21.6

Portugal in the global story: Portugal and the Western alliance

What was happening in Portugal?

In **1949** Portugal became a **founder member of NATO**, alongside the United States, Britain, France and eight others.

At that date Portugal was governed by the **Estado Novo**, the authoritarian regime of chapter 16. There were no free elections, the press was censored, opposition was policed and there were no free trade unions. Portugal was, by any ordinary definition, not a democracy.

So a country that was not a democracy became a founding member of an alliance that described itself as defending democracy. That is not a slip in the record. It is one of the most useful facts in this part of the book, because it shows how the Cold War actually worked.

What part did geography play?

Everything. Return to topic 18.3 and topic 18.6. The alliance depended on reinforcement across the Atlantic, and the Atlantic is where Portugal is. The **Azores** sit in the middle of the northern Atlantic, mainland Portugal commands the approaches to the Mediterranean beside Gibraltar, and Portuguese airfields had already proved their value between 1943 and 1945.

For an alliance whose core problem was moving men, aircraft and supplies from North America to Europe, Portugal was not a marginal member. It was a stepping stone.

What was the connection?

Strategic value bought political tolerance. The western powers accepted a dictatorship inside their alliance because of where it was, and the regime gained international respectability and a powerful set of friends from the arrangement. Portugal joined the United Nations in 1955.

There was a price, and it fell due in chapter 25. The same friends were uncomfortable when Portugal fought to hold its African territories in the 1960s, and the regime's international position became steadily harder.

KEY QUESTION

Why would Western democracies form an alliance with authoritarian Portugal during the Cold War?

Answer in three steps: what the alliance needed, what Portugal had, and what each side gave up to get it. Then answer a harder question: was that a reasonable decision in 1949? Give one argument on each side, and say which you find stronger and why.

WHAT YOU CAN NOW DO

- √ Explain why the United States and the Soviet Union alone were superpowers.
- √ Describe both political and economic models fairly, without caricature.
- √ Explain how and where Europe was divided, and what the Marshall Plan did.
- √ Explain collective defence, and how NATO differed from the League.
- √ Explain why the two alliances were not symmetrical, and why that mattered in

1989.

WORDS FOR THIS CHAPTER

superpower, ideology, bloc, iron curtain, Marshall Plan, occupation zone, NATO, collective defence, deterrence, Warsaw Pact, satellite state, sphere of influence, Estado Novo, Azores

CHAPTER 21 AT A GLANCE

A divided world

why did two states that had just won a war together become enemies?

THE TOPICS YOU HAVE COVERED

21.1	Two superpowers
21.2	Two political and economic models
21.3	Two blocs
21.4	NATO
21.5	The Warsaw Pact
21.6	Portugal in the global story: Portugal and the Western alliance

WHAT YOU CAN NOW DO

- √ Explain why the United States and the Soviet Union alone were superpowers.
- √ Describe both political and economic models fairly, without caricature.
- √ Explain how and where Europe was divided, and what the Marshall Plan did.
- √ Explain collective defence, and how NATO differed from the League.
- √ Explain why the two alliances were not symmetrical, and why that mattered in

THE WORDS THIS UNIT TAUGHT

superpower · ideology · bloc · iron curtain · Marshall Plan · occupation zone · NATO · collective defence · deterrence · Warsaw Pact · satellite state · sphere of influence · Estado Novo · Azores

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

22

PART EIGHT · CHAPTER 22

Berlin: symbol of the Cold War

One city, four sectors, and the border that ran down the street

THE QUESTION THIS CHAPTER ANSWERS

Why did one city come to stand for the whole Cold War?

IN THIS CHAPTER, YOU WILL

- √ Explain how Germany and Berlin were divided, and why it became permanent.
- √ Explain the blockade of 1948 and why it was so hard to answer.
- √ Describe the airlift and give three of its consequences.
- √ Explain why the Berlin Wall was built and what it did to ordinary lives.
- √ Compare two photographs of the same object critically.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Berlin Wall. Read what the wall was built to stop. That is what its opening in November 1989 announced.

www.britannica.com/topic/Berlin-Wall

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Imagine a city where a border runs down the middle of a street, and your school is on one side and your grandmother's flat on the other. List five ordinary things that become complicated. Berlin was that city for twenty-eight years.

CHAPTER 22 · TOPIC 22.1

The division of Germany

The account

At the end of the war Germany was divided into four **occupation zones**, administered by the United States, Britain, France and the Soviet Union. Berlin, the capital, lay deep inside the Soviet zone and was itself divided into four sectors by the same four powers.

That arrangement was meant to be temporary, while a peace settlement was agreed. No settlement was agreed, because the occupying powers wanted different Germanies: the western allies increasingly wanted a rebuilt, self-supporting Germany integrated into western Europe, and the Soviet Union wanted a Germany that could never threaten it again and, ideally, one friendly to Moscow.

In **1949** the division became permanent in fact. The three western zones became the **Federal Republic of Germany**, and the Soviet zone became the **German Democratic Republic**. Two German states existed where one had been.

Berlin, however, stayed as it was: a four-sector city, with the western sectors more than a hundred and fifty kilometres inside the eastern state, reachable only by agreed road, rail and air corridors.

That is a geographical fact with enormous consequences, and the next two topics are both about them. **West Berlin was an island.**

YOUR TURN

1. How was Germany divided in 1945, and how was Berlin divided?
2. Why was no permanent settlement agreed?
3. What happened in 1949?
4. Explain why West Berlin is described as an island.

CHAPTER 22 · TOPIC 22.2

The Berlin blockade

The account

In **1948** the western allies introduced a new currency in their zones, part of rebuilding a functioning economy. The Soviet authorities objected, and in June 1948 closed the road, rail and canal routes into West Berlin.

The purpose was straightforward. West Berlin held over two million people who depended on supplies brought in from the west: food, coal, medicine, everything. Cut the routes, and the western allies would have to leave the city or watch its population starve. Either outcome would remove a western presence from the middle of the Soviet zone.

It was a very effective form of pressure precisely because it involved no shooting. Forcing the road open would have meant a western army driving through Soviet-controlled territory, which risked war. Doing nothing meant losing the city.

The blockade lasted almost a year, and the next topic is the answer that was found.

YOUR TURN

1. What triggered the blockade in June 1948?
2. What did the Soviet authorities close, and what was the aim?
3. Why did the blockade put the western allies in a very difficult position?
4. Why was cutting supplies more effective than an attack would have been?

CHAPTER 22 · TOPIC 22.3

The Berlin airlift

The account

The western allies supplied the city **by air**, because the air corridors were covered by a separate written agreement and closing them would have meant firing on unarmed transport aircraft.

The scale is the point. For nearly eleven months, aircraft landed in West Berlin day and night, at intervals of a few minutes, carrying food, medicine, machinery and, above all, **coal**, since the city heated and powered itself with it. Crews flew fixed routes at fixed heights and speeds; an aircraft that missed its approach did not circle, because circling would break the pattern, but returned to base fully loaded.

In May **1949** the blockade was lifted. Supply had held.

The consequences went well beyond Berlin.

It showed that the two sides would take enormous risks and enormous trouble to avoid actually shooting at each other, which is the pattern of the whole Cold War.

It transformed how West Berliners and West Germans saw the western powers, from occupiers to protectors, within a single year.

And it hardened the division. NATO was signed during the blockade, and the two German states were founded at the end of it.

YOUR TURN

1. Why could the western allies use the air corridors when the roads were shut?
2. What was the single largest cargo, and why?
3. Why did an aircraft that missed its approach return to base fully loaded?
4. Give three consequences of the airlift beyond the supply of the city.

CHAPTER 22 · TOPIC 22.4

The Berlin Wall

The account

Through the 1950s the division of Germany created a problem for the eastern state. Living standards rose faster in the west, and people could still move between the sectors of Berlin relatively easily. So East Germans who wanted to leave travelled to East Berlin, crossed into a western sector, and flew out.

Millions left over a decade, and they were disproportionately the young, the qualified and the professionally trained: doctors, engineers, teachers. A state losing exactly those people has a serious and worsening problem.

On the night of **13 August 1961**, East German authorities closed the border across the city, first with wire and then with concrete. The **Berlin Wall** eventually became a fortified system: two walls with a patrolled strip between them, watchtowers, lights and guards with orders to prevent crossings. It ran for many kilometres around the western sectors.

The immediate effect was to end the outflow. The human effect was that families, neighbours and workplaces were divided overnight, and people who attempted to cross were arrested, and some were shot.

The propaganda effect ran the other way from what its builders intended. It was presented in the east as protection against the west. But a wall built to keep people in, in a city where everybody could see both sides, was an argument that made itself. That is why the pictures from 1989, in chapter 29, meant so much.

READ THE SOURCE

Two photographs of the same wall, 1961 and 1989. Set a photograph of the wall being built beside one of people standing on it as it was opened.

- **What can you see?** List what is physically present in each.
- **Who took each one, and for whom?** A press photographer, an official

photographer and a private citizen all frame the same event differently.

- **What does the pairing tell you** that neither photograph tells you alone?
- **What can neither photograph tell you?** Why the wall was built, why it was

opened, and what either crowd was thinking.

Pairing sources is one of the most useful things a historian does, and it costs nothing.

YOUR TURN

1. Why were people leaving East Germany through Berlin?
2. Why was the loss of qualified people especially damaging?
3. Give the date the border was closed and describe what the wall became.
4. Explain why the wall was a propaganda problem for the state that built it.

WHAT YOU CAN NOW DO

√ Explain how Germany and Berlin were divided, and why the division became

permanent.

√ Explain the blockade of 1948 and why it was so difficult to answer.

√ Describe the airlift and give three of its consequences.

√ Explain why the Berlin Wall was built and what it did to ordinary lives.

√ Compare two photographs of the same object critically.

WORDS FOR THIS CHAPTER

occupation zone, sector, corridor, blockade, airlift, Federal Republic, German Democratic Republic, defection, Berlin Wall, checkpoint, propaganda

CHAPTER 22 AT A GLANCE

Berlin: symbol of the Cold War

why did one city come to stand for the whole Cold War?

THE TOPICS YOU HAVE COVERED

22.1	The division of Germany
22.2	The Berlin blockade
22.3	The Berlin airlift
22.4	The Berlin Wall

WHAT YOU CAN NOW DO

- √ Explain how Germany and Berlin were divided, and why the division became
- √ Explain the blockade of 1948 and why it was so difficult to answer.
- √ Describe the airlift and give three of its consequences.
- √ Explain why the Berlin Wall was built and what it did to ordinary lives.
- √ Compare two photographs of the same object critically.

THE WORDS THIS UNIT TAUGHT

occupation zone · sector · corridor · blockade · airlift · Federal Republic · German Democratic Republic · defection · Berlin Wall · checkpoint · propaganda

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

23

PART EIGHT · CHAPTER 23

Cold War crises

Three wars fought by other people, on behalf of two

THE QUESTION THIS CHAPTER ANSWERS

How did a rivalry between two states become a danger to everybody else?

IN THIS CHAPTER, YOU WILL

- √ Explain how superpower rivalry was fought through other countries.
- √ Describe the Korean War and the three lessons drawn from it.
- √ Set out the Cuban missile crisis and its settlement.
- √ Explain why arms control agreements followed that crisis.
- √ Explain why the Vietnam War was so difficult for the intervening power.

THE TOPICS, IN THE ORDER YOU MEET THEM

23.1	The Korean War
23.2	The Cuban Missile Crisis
23.3	The Vietnam War

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Cuban missile crisis. Follow the thirteen days hour by hour, and mark the moment you think the crisis stopped getting worse.

www.britannica.com/event/Cuban-missile-crisis

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Why might two countries fight a war in a third country rather than each other? Write two reasons before you read on. This chapter is about three wars fought that way.

CHAPTER 23 · TOPIC 23.1

The Korean War

The account

Korea had been occupied by Japan until 1945, and at the end of the war it was divided along the **38th parallel**, Soviet forces accepting the Japanese surrender in the north and American forces in the south. As in Germany, a temporary line became two states.

In **1950** North Korean forces invaded the south. The **United Nations**, in a Security Council meeting the Soviet Union was boycotting and therefore could not veto, authorised a force to resist, and a coalition led by the United States intervened. When that force advanced far north toward the Chinese border, **China entered the war** in enormous numbers and pushed it back.

The fighting stabilised near where it had begun, and an armistice in **1953** left Korea divided close to the original line. No peace treaty was ever signed.

Three lessons, and all three shaped the rest of the Cold War.

Superpower conflict would be fought by other people, in other countries.

Limits would be observed. Nuclear weapons were not used, and the war was not allowed to spread to China itself or to Soviet territory, deliberately.

Divisions made in 1945 hardened into permanent states. Korea is still divided along that line today, which is why it is the clearest surviving relic of this period.

YOUR TURN

1. How and when was Korea divided, and by whom?
2. Why was the United Nations able to authorise a force in 1950?
3. What happened when the coalition advanced toward the Chinese border?
4. Give the three lessons and explain one of them in your own words.

CHAPTER 23 · TOPIC 23.2

The Cuban Missile Crisis

The account

Cuba, a large island about 150 kilometres from the coast of Florida, had a revolution in 1959 and its new government moved into alliance with the Soviet Union.

In **October 1962** American reconnaissance aircraft photographed sites in Cuba being prepared for Soviet nuclear missiles. Missiles there could reach much of the United States with very little warning.

The American government considered its options, which included an air strike and an invasion, and chose a **naval quarantine**: a blockade to stop further military shipments, combined with a public demand that the missiles be removed. For several days Soviet ships approached the line while the world waited.

It was resolved by negotiation, some of it public and some of it not. The Soviet Union withdrew the missiles. The United States publicly promised not to invade Cuba, and privately agreed to withdraw its own missiles from Turkey, which were close to Soviet territory in the same way.

Two things make this the most studied crisis of the Cold War.

It came **closer to nuclear war than any other moment**, and both leaderships understood it at the time.

And it changed behaviour afterwards. A **direct communication link** was established between Washington and Moscow so that leaders could talk in a crisis without delay, and within a year a treaty banned nuclear testing in the atmosphere. Fright produced arms control.

YOUR TURN

1. Why did Soviet missiles in Cuba alarm the United States so much?
2. What was the quarantine, and why was it chosen over an air strike?
3. Give the three elements of the settlement, including the private one.
4. Give two things that changed afterwards because of the crisis.

CHAPTER 23 · TOPIC 23.3

The Vietnam War

The account

Vietnam had been a French colony. After the Second World War an independence movement led by communists fought the French and defeated them in 1954, and the country was divided, again temporarily and again permanently, into a communist north and a non-communist south.

The United States became progressively more involved in support of the south, from advisers to, by the later 1960s, very large numbers of combat troops. The reasoning was the **domino theory**: the belief that if one country in a region became communist its neighbours would follow.

The war was extremely difficult for the intervening power, and the reasons are largely geographical and political rather than military.

The **terrain** was jungle, mountain and paddy field, which suits an opponent who knows the ground and does not suit heavy mechanised forces.

The opponent used **guerrilla methods**: avoiding pitched battle, moving among the population, striking and dispersing. Superior firepower cannot easily find a target.

Supplies reached the north's forces along routes through neighbouring countries that were extremely difficult to cut.

And it was the **first television war**. Film reached homes within a day or two, and public support in the United States fell as the war continued, producing large protest movements.

American forces withdrew in 1973, and the north completed the unification of the country in 1975. The human cost, above all to the Vietnamese population, was enormous, and the war also spread severely into Cambodia and Laos.

GO FURTHER

Compare the three wars in this chapter. In each, a superpower intervened in a divided country, and the results differed: a stalemate that froze the division in Korea, a negotiated retreat in Cuba without fighting, and a defeat in Vietnam. Ask what was different in each case about geography, about the local population's support, and about what the superpower was willing to spend. That comparison is worth more than any one of the three on its own.

YOUR TURN

1. How did Vietnam come to be divided?
2. Explain the domino theory in one sentence.
3. Give three reasons the war was so difficult for the intervening power.
4. What difference did television make?

WHAT YOU CAN NOW DO

- √ Explain how a superpower rivalry was fought through other countries.
- √ Describe the Korean War and the three lessons drawn from it.
- √ Set out the Cuban Missile Crisis and its settlement, including what was private.
- √ Explain why arms control agreements followed the crisis.
- √ Explain why the Vietnam War was so difficult for the intervening power.

WORDS FOR THIS CHAPTER

proxy war, 38th parallel, armistice, containment, quarantine, blockade, brinkmanship, hotline, test ban, domino theory, guerrilla warfare, escalation, withdrawal

PART EIGHT · CHAPTER 24

The nuclear arms race

A weapon nobody could use, and forty years of not using it

THE QUESTION THIS CHAPTER ANSWERS

Can weapons prevent a war by making it too dangerous to fight?

IN THIS CHAPTER, YOU WILL

- √ Explain the difference between atomic and thermonuclear weapons.
- √ Explain how each delivery system changed the strategic problem.
- √ Explain why a missile submarine makes a first strike pointless.
- √ Give the three conditions deterrence requires.
- √ Argue both sides of whether deterrence made the world safer.

THE TOPICS, IN THE ORDER YOU MEET THEM

24.1	From atomic to thermonuclear
24.2	Missiles and submarines
24.3	Deterrence and mutually assured destruction

WATCH, READ AND LISTEN

Encyclopaedia Britannica, arms races. The pattern, with examples from before 1914 as well as after 1945. Decide whether the two races worked the same way.

`www.britannica.com/topic/arms-race`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Can a weapon make a war less likely? Write yes or no and one sentence of reasoning. Keep the paper. You will be asked again at the end of the chapter, and you are allowed to change your mind.

CHAPTER 24 · TOPIC 24.1

From atomic to thermonuclear

The account

Chapter 20 ended with two atomic bombs and the world changed. This topic is what happened to the weapon over the following fifteen years, because the change in scale is what makes the rest of the chapter necessary.

The United States tested the first atomic weapon in 1945 and used two. The Soviet Union tested its own in **1949**, four years later, which ended the American monopoly and is the moment the arms race properly begins. Britain, France and China followed over the next fifteen years.

Then the weapons themselves changed. The bombs of 1945 released energy by splitting heavy atoms, a process called fission. From the early 1950s both superpowers developed **thermonuclear** weapons, which use a fission explosion to trigger fusion, and which are not slightly more powerful but hundreds of times more powerful.

A single one of the later weapons could destroy a large city completely. Both sides eventually held them in thousands.

That combination, unlimited destructive power on both sides, is the fact everything else in this chapter is a response to.

YOUR TURN

1. Give the year of the first Soviet atomic test, and say why it mattered.
2. What is the difference between a fission and a thermonuclear weapon?
3. Name three other countries that developed nuclear weapons in this period.
4. State in one sentence the fact that the rest of this chapter responds to.

CHAPTER 24 · TOPIC 24.2

Missiles and submarines

The account

A weapon is only as useful as its ability to reach the target, so the arms race was as much about **delivery** as about warheads, and each advance changed the strategic problem.

Bombers carried the first weapons. Aircraft can be tracked by radar, intercepted by fighters, and recalled after being launched, which means a decision can be reversed.

Intercontinental ballistic missiles, from the late 1950s, could cross between continents in about thirty minutes, could not be intercepted with the technology of the time, and could not be recalled. The consequence was that leaders would have minutes rather than hours to decide, on incomplete information.

Submarine-launched missiles were the most important development of all, and the reason is not obvious. A submarine at sea cannot be found reliably, so it cannot be destroyed in a first strike. That means a country that is attacked can always strike back, whatever happens to its land.

That last sentence is the whole of the next topic. The submarine did not make attack easier; it made attack pointless.

YOUR TURN

1. Give one advantage and one disadvantage of the bomber as a delivery system.
2. What two things changed when missiles replaced bombers?
3. Why is a missile submarine so hard to destroy?
4. Explain why the submarine made a first strike pointless.

CHAPTER 24 · TOPIC 24.3

Deterrence and mutually assured destruction

The account

Deterrence means preventing an action by making its consequences unacceptable. Nuclear deterrence rests on one idea: if attacking you means being destroyed myself, I will not attack you.

For that to work, three conditions must hold.

Capability. The weapons must exist and must work.

Survivability. Enough of them must survive an attack to strike back. That is what submarines, hardened silos and airborne patrols were for.

Credibility. The other side must believe you would actually use them, which is why a great deal of Cold War behaviour was about appearing determined.

When both sides had all three, the situation was called **mutually assured destruction**: neither could attack without being destroyed in return. It is a grim arrangement, and it has an argument in its favour and an argument against.

In favour: no war between the superpowers took place. Both sides took extraordinary care to avoid direct fighting, as chapters 22 and 23 showed, and several historians argue that the weapons are why.

Against: the arrangement depended on nobody making a mistake, ever. There were false alarms caused by faulty equipment, misread radar and misinterpreted exercises. It also cost enormous sums, spread weapons to more states, and left the world with stockpiles it still has to manage.

From the 1960s the two sides negotiated **arms control**: treaties limiting testing, limiting numbers, and limiting defensive systems, on the argument that a defence which worked would tempt the side that had it to strike first.

THE ENQUIRY: CAN WEAPONS PREVENT A WAR BY MAKING IT TOO DANGEROUS TO FIGHT?

1. Set out the three conditions for deterrence, with an example of each.
2. Give the strongest evidence that deterrence worked between 1945 and 1991.
3. Give the strongest evidence that it was dangerous, using this chapter and chapter 23.
4. Look at the answer you wrote before the chapter. Write your answer now, and say plainly whether you have changed your mind and what changed it.

WHAT YOU CAN NOW DO

- √ Explain the difference between atomic and thermonuclear weapons.
- √ Explain how each delivery system changed the strategic problem.
- √ Explain why a missile submarine makes a first strike pointless.
- √ Give the three conditions for deterrence.
- √ Argue both sides of the question of whether deterrence made the world safer.

WORDS FOR THIS CHAPTER

fission, thermonuclear, warhead, delivery system, intercontinental ballistic missile, first strike, second strike, survivability, deterrence, mutually assured destruction, arms control, test ban, stockpile

25

PART EIGHT · CHAPTER 25

The Space Race

Why getting to the Moon was a political question

THE QUESTION THIS CHAPTER ANSWERS

Why was getting to the Moon a political question?

IN THIS CHAPTER, YOU WILL

- √ Explain why Sputnik produced a reaction out of proportion to the object.
- √ Describe Gagarin's flight and the American response.
- √ Describe Apollo 11 and the effort behind it.
- √ Give four reasons the Space Race mattered, and separate them clearly.
- √ Explain the Colonial War and why it turned the army against the regime.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

NASA, the Apollo programme. The mission record from the agency that flew it, including the failures. Find out what Apollo 8 was for.

www.nasa.gov/the-apollo-program/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Why would a government spend an enormous sum on reaching a place where nobody can live? Write two possible reasons. Only one of them is about science.

CHAPTER 25 · TOPIC 25.1

Sputnik

The account

On **4 October 1957** the Soviet Union launched **Sputnik**, the first artificial satellite. It was a metal sphere with four aerials, and it did almost nothing: it transmitted a repeating radio signal that anybody with a receiver could hear as it passed overhead.

The reaction in the United States was out of all proportion to the object, and that reaction is the historically interesting part.

It was a **shock to assumptions**. The United States had believed itself technologically ahead. A country widely portrayed as backward had got there first.

It was a **military signal**. A rocket that can put a satellite into orbit can deliver a warhead between continents, which is exactly the point topic 24.2 made. Sputnik was therefore read as a demonstration of missile capability.

And it produced a **response**. The United States created a national space agency in 1958, and poured money into science and mathematics teaching in schools and universities on the argument that the country was losing an educational race.

Notice what this shows about the Cold War: a competition in prestige had practical consequences for schoolchildren on the other side of the world.

YOUR TURN

1. Give the date of Sputnik and describe what it actually did.
2. Give the three elements of the American reaction.
3. Explain why a satellite launch was read as a military signal.
4. What does the response tell you about how the two sides competed?

CHAPTER 25 · TOPIC 25.2

Yuri Gagarin

The account

On **12 April 1961**, **Yuri Gagarin** became the first human being to travel into space, completing one orbit of the Earth and returning safely. He was twenty-seven.

The Soviet Union had achieved a second and much larger first, and again the significance was political as much as technical. Gagarin toured the world and was received enormously warmly, and the point being made was that the Soviet system could do what the richest country on Earth had not yet done.

The American response came a few weeks later, when President Kennedy committed the United States, publicly and with a deadline, to landing a person on the Moon and returning them safely before the end of the decade.

That commitment is worth pausing on. It named a goal that could be measured, it set a date, and it chose a target the United States believed it could reach first because it required capabilities nobody yet had. It is a political decision taken in the language of engineering.

YOUR TURN

1. Give the date and describe what Gagarin did.
2. Why was the achievement politically significant as well as technically?
3. What did the United States commit to in response, and with what deadline?
4. Why choose a target that nobody had the capability for yet?

CHAPTER 25 · TOPIC 25.3

Apollo 11

The account

On **20 July 1969**, **Apollo 11** landed on the Moon. **Neil Armstrong** and **Buzz Aldrin** walked on the surface while **Michael Collins** remained in orbit above them, and all three returned safely.

The scale of the effort is easy to lose behind the pictures. The programme employed hundreds of thousands of people, consumed an enormous share of public spending, and required advances in rocketry, materials, navigation, computing, communications and life support, most of which had to be invented for the purpose. Earlier missions had tested each stage in sequence, and a launchpad fire in 1967 had killed three astronauts and forced a redesign.

The moment itself was watched live by an estimated six hundred million people, which was then the largest audience in history. That number matters: this was an achievement designed to be witnessed.

Five more landings followed to 1972, and then the programme ended. The Soviet Union never landed a person on the Moon, and turned instead toward space stations, at which it achieved a great deal.

YOUR TURN

1. Give the date, and name the three astronauts and their roles.
2. Name four fields in which advances were required.
3. What happened in 1967, and what did it change?
4. Why does the size of the television audience matter historically?

CHAPTER 25 · TOPIC 25.4

Why the Space Race mattered

The account

Four reasons, and they are worth separating because they are usually mixed up.

Prestige. Each side wanted to show that its system produced better results. Space was ideal for that: highly visible, unambiguous, and impossible to fake.

Military capability. Rockets, satellites and re-entry vehicles are the same technologies as missiles and warheads. Reconnaissance satellites, which both sides developed, also made arms control possible, because a treaty can be verified from orbit rather than by inspection on the ground.

Science and technology. Real advances came out of it, in computing, materials, weather forecasting, navigation and communications. Communication satellites in particular changed the world you live in, and chapter 32 picks that up.

Education. Both sides invested heavily in scientific education because of it.

Now put the race where it belongs in this book. Chapter 8 traced a chain of technologies. It continues here:

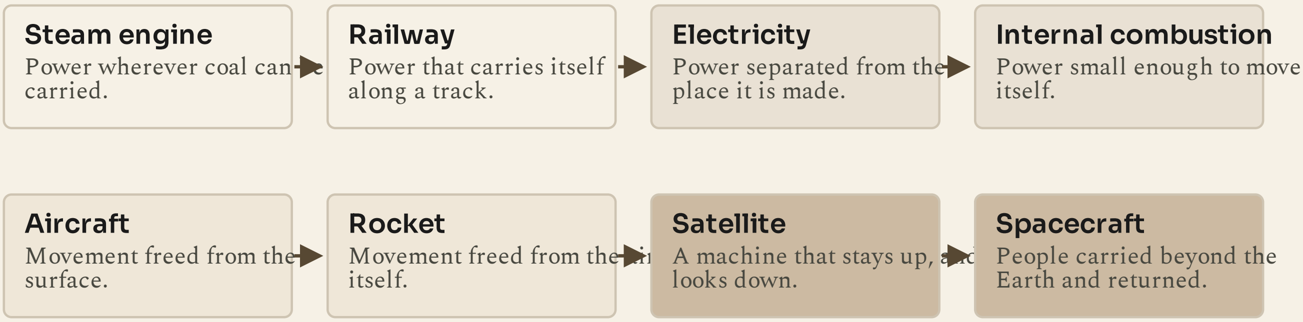


Figure 25.1 · Two hundred years of moving faster.
Every step needed the one before it. The last step in this figure was watched live by the generation before yours.

The people who watched the Moon landing in 1969 were the grandchildren of people who had watched the first aeroplanes. That is how fast the chain in chapter 8 was still running.

YOUR TURN

1. Give the four reasons the Space Race mattered.
2. Explain how reconnaissance satellites made arms control easier.
3. Name three everyday technologies that came from this effort.
4. Using the figure, explain why each step depended on the one before it.

CHAPTER 25 · TOPIC 25.5

Portugal in the global story: the Colonial War and the end of the Estado Novo

What was happening in the world?

After 1945 the European empires came apart. India and Pakistan became independent in 1947, Indonesia in 1949, and from the late 1950s a rapid wave of independence swept Africa: Ghana in 1957, and in 1960 alone some seventeen African states. Britain, France and Belgium withdrew, sometimes peacefully and sometimes after war. The United Nations, with a growing membership of newly independent states, pressed for decolonisation.

What was happening in Portugal?

Portugal did not withdraw. The Estado Novo held that its African territories were not colonies at all but **overseas provinces**, parts of a single, indivisible Portugal, and that Portuguese rule there was not comparable to other empires. On that argument, independence was not something to negotiate.

Independence movements formed nonetheless, and from **1961** fighting began, first in **Angola**, then in **Guinea-Bissau** and **Mozambique**. The **Portuguese Colonial War** lasted thirteen years, until 1974.

The cost to Portugal was very heavy.

Human. Hundreds of thousands of young men were conscripted and sent to Africa, many for repeated tours. Thousands died, many more were wounded, and the consequences reached every family.

Economic. Defence consumed a very large share of public spending, in a country that was among the poorest in western Europe.

Demographic. Emigration rose sharply in these years, to France, Germany, Luxembourg and elsewhere, driven both by poverty and by conscription.

Political. Portugal became increasingly isolated internationally, criticised at the United Nations and awkward for its NATO allies, whose own empires were being dismantled.

What part did geography played, and what was the connection?

The territories were in Africa, thousands of kilometres away, and had to be supplied and reinforced by sea and air. Portugal was fighting three separate wars on three separate fronts, in very different terrain, from a small European base. That is an enormous logistical burden for a small country, and it is a geographical explanation for an economic and political outcome.

And the political consequence is the crucial one, because it leads directly into chapter 26. The pressure fell hardest on the **army**, and above all on career officers who could see no end to a war they were being asked to fight indefinitely. Discontent inside the officer corps, which began over conditions and promotion, became political.

The regime was brought down by the institution it depended on most.

KEY QUESTION

How did global decolonisation and the Cold War affect Portugal?

Answer in three parts. First, describe the international pattern after 1945. Second, explain why Portugal's response differed from Britain's and France's. Third, list the four costs above and say which you think mattered most in producing the events of chapter 26. Defend your choice with evidence.

WHAT YOU CAN NOW DO

- √ Explain why Sputnik produced a reaction out of proportion to the object.
- √ Describe Gagarin's flight and the American response to it.
- √ Describe Apollo 11 and the effort behind it.
- √ Give four reasons the Space Race mattered, and separate them clearly.
- √ Explain the Colonial War, its four costs, and why it turned the army against

the regime.

WORDS FOR THIS CHAPTER

satellite, orbit, Sputnik, space race, prestige, reconnaissance satellite, Apollo, decolonisation, overseas province, conscription, Colonial War, emigration, isolation, officer corps

PART EIGHT · CHAPTER 26

A democratic revolution

How a dictatorship of forty-eight years ended in a single morning

THE QUESTION THIS CHAPTER ANSWERS

Why was Portugal able to move from dictatorship to democracy in the 1970s?

IN THIS CHAPTER, YOU WILL

- √ Explain what the MFA was and why officers turned against the regime.
- √ Describe 25 April 1974 and explain why so little blood was shed.
- √ Explain the decolonisation of 1974 and 1975 and its consequences.
- √ Explain who the retornados were and why their arrival was so demanding.
- √ Set out the sequence from revolution to constitution.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, Portugal: the Carnation Revolution. 25 April 1974 described in English, inside the European decade it belongs to rather than on its own.

www.britannica.com/place/Portugal/The-Carnation-Revolution

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Most revolutions in this book involve fighting. This one involved a radio signal, a column of soldiers and a flower seller. As you read, keep asking: why did almost nobody defend a government that had lasted forty-eight years?

CHAPTER 26 · TOPIC 26.1

The Carnation Revolution

The account

By 1974 the Estado Novo had been in place for four decades. Salazar had been replaced in 1968, after illness, by Marcelo Caetano, who promised a measure of liberalisation and did not end the war in Africa. Chapter 25 explained the pressure that war had built inside the army.

Career officers formed the **Movimento das Forcas Armadas**, the Armed Forces Movement, or MFA. It began over professional grievances and became political, reaching one conclusion: the war could not be won, the government would not end it, so the government had to go.

On the night of **24 to 25 April 1974** the movement acted. Radio stations broadcast agreed songs as signals, and units moved on Lisbon, taking the ministries, the airport, the radio and television stations and the barracks. The regime's forces largely did not resist. Caetano surrendered to the MFA at the Carmo barracks and left the country.

Then the part that gave the day its name. Despite appeals on the radio for people to stay at home, crowds came into the streets, and **red carnations** were handed to soldiers and placed in the barrels of their rifles. It became, almost immediately, the image of the day.

Almost nobody died. For a coup that ended a forty-eight-year regime, that is remarkable, and it is the first thing to explain rather than to admire.

Censorship ended. Political prisoners were released. Exiles returned. Parties that had been illegal organised openly, and within a year Portugal held the freest elections in its history.

READ THE SOURCE

Photographs of 25 April 1974. Some of the most reproduced photographs in Portuguese history were taken that day: carnations in rifle barrels, crowds on armoured vehicles, soldiers smiling.

- **What can you see?** Soldiers, civilians, flowers, vehicles, a city street.
- **Who took them, and why?** Press photographers, working for newspapers that

had been censored the previous day.

- **What do they tell a historian?** That the army and the crowd were not

opposed to each other, which is the single most unusual feature of the day.

- **What can they not tell you?** Why the regime collapsed, what the MFA

intended, or what would happen next. For that you need the war, the economy and the two years that followed.

An image can prove a mood. It cannot prove a cause.

YOUR TURN

1. What was the MFA, and how did its aims change?
2. Give the date and describe how the movement took control.
3. Explain how the day got its name.
4. Give three things that changed immediately.

CHAPTER 26 · TOPIC 26.2

Decolonisation

The account

The revolution ended the war, which is what it had been made for. Negotiations with the independence movements began quickly, and in **1974 and 1975** the Portuguese territories in Africa became independent states: **Guinea-Bissau**, **Mozambique**, **Cape Verde**, **Sao Tome and Principe** and **Angola**. **East Timor** declared independence in 1975 and was then occupied by Indonesia, and did not become independent in fact until 2002.

Decolonisation was rapid and its consequences were severe.

In **Angola** and **Mozambique**, rival independence movements that had fought the Portuguese turned on each other, with support from opposite sides of the Cold War, and long civil wars followed. That is the Cold War of chapter 23 operating again, in new states.

In **Portugal**, something happened that has no parallel in the other European decolonisations at this scale. Around half a million people, the **retornados**, arrived from Africa within about two years, into a country of some nine million with an economy in crisis. They were housed, absorbed and employed, with great difficulty and, on the whole, successfully. It remains one of the largest movements of people relative to population that any European country has managed in peacetime.

YOUR TURN

1. Name four territories that became independent in 1974 and 1975.
2. Why did civil wars follow in Angola and Mozambique?
3. Who were the retornados, and roughly how many arrived?
4. Why was absorbing them so difficult in those particular years?

CHAPTER 26 · TOPIC 26.3

The Constitution of 1976

The account

Between April 1974 and the end of 1975 Portugal was not yet a settled democracy. It was a country in which the direction of the revolution itself was being contested. There were successive provisional governments, large-scale nationalisations of banks and industry, land occupations in the Alentejo, and serious disagreement between those who wanted a parliamentary democracy on the western European model and those who wanted a revolutionary socialist state. There were attempted coups from both directions.

The decisive step came through elections. On **25 April 1975**, exactly a year after the revolution, Portugal elected a constituent assembly, on a turnout of over ninety per cent. Its task was to write a constitution.

The **Constitution of 1976** established a parliamentary democracy: regular free elections, a president elected by universal suffrage, an assembly, an independent judiciary, and an extensive list of civil, political and social rights. It also reflected the revolution in its economic provisions, several of which were revised in later amendments as Portugal's position changed, particularly with the European accession of chapter 27.

So the sequence is: **dictatorship**, **revolution**, **contested transition**, **constitution**, **democracy**. The contested transition in the middle is the part usually left out, and it is the part that explains why the constitution looks the way it does.

THE ENQUIRY: WHY WAS PORTUGAL ABLE TO MOVE FROM DICTATORSHIP TO DEMOCRACY IN THE 1970S?

1. List the pressures on the regime by 1974, using chapter 25 as well as this one.
2. Explain why the army, rather than a political party or a popular rising, was the instrument.
3. Explain why the transition took two years rather than two days.
4. Compare with one other country in this book that moved from dictatorship to democracy, and name one thing Portugal had that it did not, or the reverse.

WHAT YOU CAN NOW DO

- √ Explain what the MFA was and why officers turned against the regime.
- √ Describe 25 April 1974 and explain why so little blood was shed.
- √ Explain the decolonisation of 1974 and 1975 and its consequences on both sides.
- √ Explain who the retornados were and why their arrival was so demanding.
- √ Set out the sequence from revolution to constitution, including the contested middle.

WORDS FOR THIS CHAPTER

Estado Novo, Movimento das Forcas Armadas, coup, Carnation Revolution, censorship, political prisoner, decolonisation, independence, civil war, retornados, provisional government, nationalisation, constituent assembly, Constitution of 1976, universal suffrage

27

PART EIGHT · CHAPTER 27

European integration

States that had fought twice decide to pool what wars are made of

THE QUESTION THIS CHAPTER ANSWERS

Why did states that had fought two wars decide to pool their economies?

IN THIS CHAPTER, YOU WILL

- √ Explain why coal and steel were the first industries pooled.
- √ Give the motives for integration beyond preventing war.
- √ Describe the common market and the four European institutions.
- √ Explain how accession became an argument for democracy in southern Europe.
- √ Explain what membership changed for Portugal after 1986.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

European Union, the history of the EU. The official timeline, decade by decade. Find 1986 and read what joining committed Portugal to.

european-union.europa.eu/principles-countries-history/history-eu_en

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

France and Germany fought three wars in seventy years, in 1870, 1914 and 1939. Ten years after the last one they were building a common institution together. Suggest one reason why. Then read on and see whether you were right.

CHAPTER 27 · TOPIC 27.1

Why cooperate after two world wars?

The account

In 1945 western Europe faced a question it had never answered successfully: how to stop France and Germany going to war again. Every previous answer had been tried and had failed. Defeating Germany had been tried in 1918 and produced 1939. Occupying and dividing it was being tried and could not be permanent.

A different answer was proposed: make it **materially impossible** for them to arm against each other without the other knowing, by placing the industries of war under a common authority.

That was the reasoning behind the first institution, in 1951, which pooled the **coal and steel** production of France, West Germany, Italy, the Netherlands, Belgium and Luxembourg under a shared authority. Read that list of materials against chapter 6 and chapter 12 and the logic is unmistakable: coal and steel are what armies are made of. Six countries agreed that none of them would control its own war-making materials alone.

Three further motives ran alongside.

Economic recovery. A larger market meant cheaper production and faster rebuilding, which is the argument of topic 9.2 applied to a continent.

The Cold War. A prosperous, integrated western Europe was more resistant to communist parties, and the United States encouraged integration for that reason.

Loss of empire. European states were losing their colonies, and a bloc of states together carried weight that none of them carried alone.

YOUR TURN

1. What question did western Europe face in 1945, and which two earlier answers had failed?
2. Why were coal and steel chosen as the first industries to pool?
3. Give three motives beyond preventing war.
4. Explain why the United States encouraged European integration.

CHAPTER 27 · TOPIC 27.2

From the European Communities to the EEC

The account

The coal and steel authority worked, and the six states extended the method. In **1957** they signed the Treaty of Rome, creating the **European Economic Community**, the EEC.

Its central instrument was the **common market**: goods, and progressively services, capital and people, moving between member states without tariffs or restrictions. That is a very large idea, and its consequences are the argument of chapter 31. A common external tariff was set, and common policies were developed, most importantly for agriculture.

The institutions created then are the ancestors of those that exist now: a commission to propose and administer, a council of member governments to decide, an assembly which became the European Parliament, and a court to interpret the treaties, whose judgments bind member states. That last one matters: the members accepted a court above their own.

The Community enlarged. Britain, Ireland and Denmark joined in 1973. Greece joined in 1981. **Portugal and Spain joined in 1986**, which is topic 27.3.

Notice the pattern in those dates. Greece, Portugal and Spain had all been dictatorships in the previous decade and all joined shortly after becoming democracies. Membership was not open to a dictatorship, and that made accession an argument for democracy inside each of those countries. **The Community became a machine for consolidating democracy**, which was not among the reasons it was founded.

YOUR TURN

1. What was created in 1957, and what was its central instrument?
2. Name the four kinds of institution created, and what each does.
3. Give the accession dates of Britain, Greece, and Portugal and Spain.
4. Explain how membership became an argument for democracy.

CHAPTER 27 · TOPIC 27.3

Portugal in the global story: Portugal joins Europe

What was happening in Portugal?

Portugal applied to join the European Community in **1977**, one year after the Constitution of 1976 and three years after the revolution. That timing is not a coincidence: the application was made as soon as the country was a democracy, and it was a deliberate statement about which Europe Portugal now belonged to.

Negotiations took eight years. On **1 January 1986**, Portugal and Spain became members.

Portugal in 1986 was the poorest country in the Community. Its income per person, its road and rail network, its telephone system, its schools and its water and sanitation were all well behind the western European average. The Colonial War, the loss of the empire, the return of half a million people and two oil crises had all fallen within twelve years.

What changed?

Markets. Portuguese producers gained access to a very large market, and Portuguese consumers gained access to competing goods, which was harder for some firms than for others.

Investment. Community structural funds paid for a great deal of infrastructure: motorways, water and sanitation, ports, schools and universities. Anyone travelling in Portugal today is using a good deal of it.

Movement. Portuguese people could live and work across the Community, and migration that had been informal became a right.

Politics. Membership tied Portugal into a framework of law and obligation that was itself a guarantee of the democratic settlement of 1976.

Portugal went on to take part in the **Maastricht Treaty of 1992**, which chapter 29 covers, and later joined the single currency.

What was the connection?

Go back to chapter 6 and to topic 10.8. Portugal's difficulty for two centuries had been a shortage of capital, a small internal market and expensive internal transport. **Accession addressed all three at once**, from outside. It did not make Portugal rich, and it did not remove the differences between the coast and the interior that chapter 4 described. It changed the terms.

KEY QUESTION

Why was European integration important for Portugal after decades of dictatorship?

Answer under the four headings above: markets, investment, movement, politics. Then write a final paragraph on the fourth one specifically. Chapter 26 showed a democracy created in 1976 after a contested transition. Explain how membership of a community of democracies made that settlement harder to reverse.

WHAT YOU CAN NOW DO

- √ Explain why coal and steel were the first industries to be pooled.
- √ Give the motives for European integration beyond preventing war.
- √ Describe the EEC, the common market and the four institutions.
- √ Explain how accession became an argument for democracy in southern Europe.
- √ Explain what membership changed for Portugal after 1986.

WORDS FOR THIS CHAPTER

integration, supranational, coal and steel authority, Treaty of Rome, European Economic Community, common market, tariff, common external tariff, commission, council, parliament, court, accession, structural funds, single currency

PART NINE

The end of the Cold War

In nineteen eighty the division of Europe looked permanent. Eleven years later one of the two superpowers had ceased to exist, and it happened without a war between them. Two chapters ask why, and what replaced the world that ended.

CHAPTER 10 AT A GLANCE

The expansion of industrial capitalism

industry changed the machines, but what did it change about money, work and ownership?

THE TOPICS YOU HAVE COVERED

10.1	What is capital?
10.2	Private property
10.3	Entrepreneurship
10.4	Investment
10.5	Corporations
10.6	Competition
10.7	Inequality
10.8	Portugal in the global story: the Regeneration and Fontismo

WHAT YOU CAN NOW DO

- √ Define capital in the economic sense and explain why it must be paid for
- √ Explain private property, and what follows from it for risk, profit and
- √ Explain how banks, shares and bonds gather savings behind large projects.
- √ Explain limited liability and the separation of ownership from management.
- √ Judge, with evidence, whether industrial capitalism benefited everyone.

THE WORDS THIS UNIT TAUGHT

capital · private property · entrepreneur · risk · profit · investment · bank · share · shareholder · bond · interest · corporation · limited liability · competition · monopoly · inequality · public debt · Regeneration · Fontismo

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART NINE · CHAPTER 28

Why did the Cold War end?

How a system that looked permanent ran out of supports

THE QUESTION THIS CHAPTER ANSWERS

Why did a system that looked permanent in 1980 not exist by 1991?

IN THIS CHAPTER, YOU WILL

- √ Give four pressures on the Soviet system and explain each.
- √ Define glasnost and perestroika, and the reasoning behind them.
- √ Explain why reforms meant to save the system helped end it.
- √ Describe the sequence of change across eastern Europe in 1989.
- √ Explain why 1989 was mostly peaceful and why it spread so fast.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, Mikhail Gorbachev. A leader who set out to save a system and helped end it. Find the point where reform stopped being controllable.

www.britannica.com/biography/Mikhail-Gorbachev

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Name something that looked permanent to you a few years ago and is now gone. Then ask what had to be true for it to end. Historians ask exactly that about 1989.

CHAPTER 28 · TOPIC 28.1

Pressures inside the Soviet bloc

The account

The Soviet Union in 1980 was a superpower with a formidable army, a space programme and enormous natural resources. It also had problems that had been accumulating for years.

The economy was slowing. Central planning had industrialised the country rapidly, as chapter 16 described, and it was much less good at the next stage. Planners could order more steel; they could not easily order better shoes, faster computers or shops that had what people wanted. Consumer goods were scarce, queues were normal, and quality lagged.

Technology was falling behind. The western economies were entering the computing age. A system that controlled information tightly, including photocopiers and later computers, found that difficult, because the same technologies that raise productivity also spread information.

Military spending was very heavy, in a smaller economy than its rival's, and from 1979 the Soviet Union was fighting a long and unsuccessful war in Afghanistan, which cost lives and money and was unpopular at home.

Eastern Europe was restless. Hungary in 1956 and Czechoslovakia in 1968 had been suppressed by force. In **Poland** from 1980 an independent trade union, **Solidarity**, gained enormous membership before being suppressed under martial law, and it did not disappear.

None of these alone would have ended the system. Together they meant that whoever led the Soviet Union next would have to attempt something.

YOUR TURN

1. Give four pressures on the Soviet system by the early 1980s.
2. Explain why central planning was better at steel than at shoes.
3. Why was controlling information a problem in the computing age?
4. What was Solidarity, and why did it matter that it did not disappear?

CHAPTER 28 · TOPIC 28.2

Gorbachev, glasnost and perestroika

The account

Mikhail Gorbachev became leader of the Soviet Union in **1985**. He was younger than his predecessors and believed the system could be saved by reform. That belief is the key to everything that followed: **he was trying to preserve it, not to end it.**

His two central policies became known worldwide by their Russian names.

Glasnost, usually translated as openness. Censorship was relaxed, the press was allowed to report failures, past crimes of the Soviet state were discussed publicly, and criticism became possible. The reasoning was that problems cannot be fixed if nobody may mention them.

Perestroika, restructuring. Limited market mechanisms, some private enterprise, and more independence for factory managers, to address the economic problems of topic 28.1.

He also changed foreign policy. He negotiated arms reductions with the United States, withdrew from Afghanistan, and, most consequentially, made clear that the Soviet Union would **no longer use force to keep governments in power in eastern Europe**.

The results were not what he intended, and the reason is worth understanding.

Glasnost let people say what had been unsayable, and much of what they said was that they wanted a different system, or, in the Soviet republics, a country of their own.

Perestroika disrupted the planned economy without replacing it, so shortages got worse rather than better in the short run.

And the foreign policy change removed the one thing holding several eastern European governments in place.

A reform intended to strengthen the system removed the supports it rested on.

YOUR TURN

1. What did Gorbachev believe he was doing?
2. Define glasnost and perestroika, and give the reasoning behind each.
3. What did he change about the use of force in eastern Europe?
4. Explain why each of his three policies produced an unintended result.

CHAPTER 28 · TOPIC 28.3

Change across eastern Europe

The account

Once it was clear that Soviet forces would not intervene, the governments of eastern Europe faced their own populations alone, and in **1989** they gave way one after another, in months.

Poland. Solidarity was legalised, talks were held, and partly free elections in June were won overwhelmingly by the opposition. A non-communist prime minister took office.

Hungary. The government opened its border with Austria in the summer, cutting the iron curtain. East Germans on holiday in Hungary began leaving through it, which is how the pressure reached Berlin.

East Germany. Mass demonstrations grew week by week, notably in Leipzig, where crowds marched and were not fired on. The government changed leader and then, in November, the wall was opened. Chapter 29 tells that part.

Czechoslovakia. Demonstrations in November brought down the government within weeks, in what became known as the Velvet Revolution.

Bulgaria and Romania. Bulgaria's leader was removed by his own party. In Romania alone the change was violent, and the ruling couple were tried and executed in December.

Two features of 1989 are worth noticing, and both matter beyond eastern Europe.

It was mostly peaceful, in a way almost nobody predicted, and the moment in Leipzig where the crowds were not fired on is where several historians locate the turning point.

It spread by example. Each success made the next more likely, because it demonstrated that these governments would not be rescued.

YOUR TURN

1. What single change made 1989 possible?
2. Explain how the Hungarian border decision affected East Germany.
3. Name the countries where change was peaceful and the one where it was not.
4. Explain what is meant by saying the change spread by example.

WHAT YOU CAN NOW DO

- √ Give four pressures on the Soviet system, and explain each.
- √ Define glasnost and perestroika and explain the reasoning behind them.
- √ Explain why reforms intended to save the system helped end it.
- √ Describe the sequence of change across eastern Europe in 1989.
- √ Explain why 1989 was mostly peaceful and why it spread so fast.

WORDS FOR THIS CHAPTER

central planning, consumer goods, stagnation, Afghanistan, Solidarity, martial law, glasnost, perestroika, reform, non-intervention, Velvet Revolution, demonstration

PART NINE · CHAPTER 29

The fall of the Berlin Wall

Eleven months from an open barrier to one Germany

THE QUESTION THIS CHAPTER ANSWERS

Why did the fall of one wall become the picture of an era ending?

IN THIS CHAPTER, YOU WILL

- √ Explain how the wall came to be opened on 9 November 1989.
- √ Use a pair of photographs from different dates as evidence.
- √ Explain why reunification was fast, and what it cost.
- √ Explain the failed coup of 1991 and why its failure accelerated collapse.
- √ Compare Portugal's transition to democracy with eastern Europe's.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, German reunification. Eleven months from the opening of the wall to one German state. Note who had to agree before it could happen.

www.britannica.com/event/German-reunification

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Find a photograph of the Berlin Wall in 1989. Before reading, write down three things you can see in it. You will use them in the source work later in this chapter.

CHAPTER 29 · TOPIC 29.1

The ninth of November 1989

The account

By the autumn of 1989 the East German government was under enormous pressure. Citizens were leaving through Hungary and Czechoslovakia, and demonstrations in Leipzig and other cities were growing weekly.

On **9 November 1989** the government announced new travel rules intended to relieve the pressure by allowing regulated travel. At a televised press conference the official announcing them was asked when they took effect, and, without clear instructions, answered that as far as he knew, immediately.

That answer was broadcast. East Berliners went to the crossing points and asked to pass. The guards had received no orders, could not reach anybody with authority, and were faced with growing crowds. At **Bornholmer Strasse** the officer in charge decided to open the barrier rather than use force. The others followed.

By that night people were crossing freely in both directions, standing on the wall and beginning to break pieces out of it.

The detail matters, and pupils often miss it. **The wall was not opened by a decision to open it.** It was opened by an ambiguous announcement, an absence of orders and a series of individual decisions by people on the spot not to shoot. Chapter 19 asked you to think about officials who obey and officials who refuse. This is the same question, with a happier answer.

READ THE SOURCE

Photographs and film of 9 to 11 November 1989. Take one image of people on the wall and set it beside the image of it being built in 1961 from chapter 22.

- **What can you see?** Compare the two crowds, the two lighting conditions and

what people are holding.

- **Who made each image, and for what audience?**
- **What does the pair show** that neither shows alone?
- **What can neither show?** Why the government made the announcement, what the

guards were told, or what the Soviet Union had decided months earlier.

Then answer: which of the causes in chapter 28 does the photograph make visible, and which does it hide completely?

YOUR TURN

1. Give the date and explain what the press conference announced.
2. Why did the guards at the crossing points open the barriers?
3. Explain the sentence "the wall was not opened by a decision to open it".
4. Why is this a useful example of the difference between a cause and an event?

CHAPTER 29 · TOPIC 29.2

German reunification

The account

Once the wall was open, the two German states moved together far faster than almost anyone expected. Free elections in East Germany in March 1990 produced a government committed to unification. Currency was unified in July. On **3 October 1990** the German Democratic Republic ceased to exist and Germany was one state again, eleven months after the wall opened.

Speed was possible for a reason that is easy to overlook: this was not two countries merging on new terms but the eastern states joining the existing Federal Republic under its existing constitution.

It also required international agreement, because Germany was still formally subject to the rights of the four occupying powers of 1945. A treaty negotiated between the two German states and the four powers settled the outstanding questions, including Germany's borders and its membership of NATO.

The costs were considerable and lasted for decades. Eastern industry was largely uncompetitive and much of it closed; unemployment in the east rose sharply; and very large transfers were made from west to east for infrastructure, pensions and rebuilding. Reunification was popular, necessary and expensive, all at once, and a good answer says all three.

YOUR TURN

1. Give the date of reunification and the interval since the wall opened.
2. Why was unification legally quick?
3. Why was an international treaty required?
4. Give two costs of reunification.

CHAPTER 29 · TOPIC 29.3

The collapse of the Soviet Union

The account

The changes reached the Soviet Union itself. Glasnost had allowed national movements to speak in the republics, and several, beginning with the Baltic states, demanded independence.

In **August 1991** hardliners in the party, the army and the security services attempted a coup against Gorbachev to stop the process. It failed within days, partly because of public resistance in Moscow and partly because the plotters could not rely on the forces they commanded. Its failure destroyed the authority of the institutions that had attempted it, and accelerated exactly what it had been intended to prevent.

Over the following months the republics declared independence, and in **December 1991** the Soviet Union was formally dissolved. Fifteen independent states replaced it, the largest being the Russian Federation.

The Cold War was over. The structure of chapter 21, two blocs and two superpowers, no longer existed, and one superpower remained.

The 1990s in the former Soviet states were extremely difficult: output fell sharply, savings were destroyed by inflation, and life expectancy in some places declined. In parts of the former Soviet Union and in Yugoslavia, which broke up in the same years, the end of the old order was accompanied by war. **The end of the Cold War was not the end of conflict**, which is the argument chapter 30 begins with.

■ **YOUR TURN**

1. What did glasnost make possible in the Soviet republics?
2. What happened in August 1991, and why did its failure matter so much?
3. Give the date of dissolution and the number of states that replaced it.
4. Give two reasons the 1990s were difficult in the former Soviet states.

CHAPTER 29 · TOPIC 29.4

Portugal in the global story: Portugal in a new Europe

What was happening in Portugal?

Set 1989 beside 1949, the year Portugal joined NATO in chapter 21, and the distance travelled is the point of this section.

In **1949** Portugal was an authoritarian state with censorship, no free elections, a political police, an empire in Africa and one of the lowest incomes in western Europe.

In **1989** Portugal was a parliamentary democracy under the Constitution of 1976, a member of NATO, a member of the European Community since 1986, with a free press, competitive elections and no empire.

In **1992** Portugal took part in the signing of the **Maastricht Treaty**, which created the European Union out of the earlier Communities, established European citizenship and set out the path to a single currency. Portugal later adopted the euro.

What was happening in the world?

Eastern Europe in 1989 was beginning a transition from a one-party state to democracy and from a planned economy to a market one. Portugal had made an equivalent transition fifteen years earlier, from a different starting point.

What was the connection?

The comparison is the reason this section exists, and it should be made carefully rather than loosely.

What is similar. Both were transitions from an authoritarian state to a democracy, both had to build political parties and free institutions quickly, and both looked to western European integration as the way to make the change permanent. Several eastern European states joined the European Union in the 2000s, following the same reasoning that had taken Portugal in.

What is different. Portugal's dictatorship was conservative rather than communist, and Portugal already had private property and a market economy, so it did not have to build one. Portugal's transition was triggered by a colonial war; eastern Europe's was triggered by the withdrawal of Soviet power. Portugal's change came from inside the country's own institutions; theirs followed a decision made in Moscow.

The sequence for Portugal, which chapter 34 will ask you to place on a timeline beside the world:

dictatorship → revolution → democratisation → European integration

KEY QUESTION

How had Portugal's position in Europe changed between 1945 and 1992?

Write two short profiles of Portugal, one for 1945 and one for 1992, each covering: form of government, empire, international alliances, economy and freedom of the press. Then write a paragraph naming the three events that did most to produce the difference, and say which of the three you would put first.

WHAT YOU CAN NOW DO

- √ Explain how the wall came to be opened on 9 November 1989.
- √ Use a pair of photographs from different dates as historical evidence.
- √ Explain why German reunification was fast, and what it cost.
- √ Explain the failed coup of 1991 and why its failure accelerated the collapse.
- √ Compare Portugal's transition to democracy with eastern Europe's, on stated

criteria.

WORDS FOR THIS CHAPTER

travel regulations, crossing point, reunification, Federal Republic, occupying powers, treaty, coup, dissolution, Russian Federation, transition, Maastricht Treaty, European Union, single currency

CHAPTER 29 AT A GLANCE

The fall of the Berlin Wall

why did the fall of one wall become the picture of an era ending?

THE TOPICS YOU HAVE COVERED

29.1	The ninth of November 1989
29.2	German reunification
29.3	The collapse of the Soviet Union
29.4	Portugal in the global story: Portugal in a new Europe

WHAT YOU CAN NOW DO

- √ Explain how the wall came to be opened on 9 November 1989.
- √ Use a pair of photographs from different dates as historical evidence.
- √ Explain why German reunification was fast, and what it cost.
- √ Explain the failed coup of 1991 and why its failure accelerated the collapse.
- √ Compare Portugal's transition to democracy with eastern Europe's, on stated

THE WORDS THIS UNIT TAUGHT

travel regulations · crossing point · reunification · Federal Republic · occupying powers · treaty · coup · dissolution · Russian Federation · transition · Maastricht Treaty · European Union · single currency

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART TEN

The modern connected world

The last five chapters bring the book to the world you live in, and then back to the beginning. What replaced a world organised around two superpowers, how goods and information now move, why the borders on your map are where they are, and, finally, how much of all this history was decided by the geography you studied in part one.

PART ONE AT A GLANCE

Understanding our global position

Four chapters, and the question each one answers. Come back to this page when you want to know where you are.

Chapter 1

The global grid

how do a few imaginary lines let us say exactly where anything on Earth is?

Chapter 2

Latitude, longitude and time zones

if the Earth turns once in twenty-four hours, what does that do to where a place is and to what time it is there?

Chapter 3

Climate zones and biomes

why is the living world so different from one latitude to another?

Chapter 4

Hemispheres and global patterns

how alike are the two halves of the planet, whichever way you cut it?

30

PART TEN · CHAPTER 30

The world after the Cold War

What replaced a world organised around two superpowers

THE QUESTION THIS CHAPTER ANSWERS

What replaced a world organised around two superpowers?

IN THIS CHAPTER, YOU WILL

- √ Explain what replaced the bipolar structure of the Cold War.
- √ Explain why some conflicts broke open after 1991.
- √ Describe the border changes of the 1990s and classify them.
- √ Explain when and why borders change.
- √ Describe the main international organisations and the argument about them.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

United Nations, about us. What the organisation actually does now, and which of its bodies a small country such as Portugal sits on.

www.un.org/en/about-us

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Find a map or an atlas printed before 1990, if your school has one. Compare its map of Europe with a current one and list every difference you can find. That list is this chapter.

CHAPTER 30 · TOPIC 30.1

A different international environment

The account

For forty-five years international politics had a simple organising structure: two superpowers, two alliances, and most questions understood as belonging to one side or the other. After 1991 that structure was gone, and three things changed.

One superpower remained. The United States was, for a period, without a rival of comparable military and economic weight. That is unusual in modern history, and it did not last indefinitely: other powers, notably China, grew enormously over the following decades.

Conflicts stopped being read through one lens. During the Cold War almost every local conflict was interpreted, and often supplied, as part of the global contest. Afterwards, disputes over territory, resources, religion and national identity were seen for what they were, and some of them, previously frozen by the larger contest, broke open. **Yugoslavia**, which had held together several national groups within one federal state, broke apart in the 1990s in a series of wars in Europe itself, forty-five years after 1945.

International institutions were asked to do more. The United Nations, freed from the routine veto of two hostile permanent members, authorised more operations than before. It was not always successful, and chapter 20 explained why the design has that limit built in.

The lesson worth carrying out of this topic is the one chapter 29 ended with. **The end of a confrontation is not the same as the arrival of peace.**

YOUR TURN

1. What was the organising structure of international politics before 1991?
2. Give three ways the situation changed afterwards.
3. What happened in Yugoslavia, and why is it significant that it was in Europe?
4. Explain why the United Nations could do more after 1991, and what still limited it.

CHAPTER 30 · TOPIC 30.2

Changing borders

The account

The 1990s produced the largest change to the map of Europe since 1945, and it is worth listing because the map in your classroom is the result.

The **Soviet Union** became fifteen states.

Yugoslavia became, over about fifteen years, seven.

Czechoslovakia divided into the Czech Republic and Slovakia in 1993, peacefully and by agreement, which is why it is called the velvet divorce.

Germany went the other way, from two states to one.

Compare those four. Two were separations after violence, one was a separation by agreement, and one was a unification. So a border change is not one kind of event, and asking which kind you are looking at is the first useful question.

Two general points follow, and both prepare chapter 33.

Borders change when **the state that held them changes**: when it loses a war, loses its authority, or agrees to alter itself. They rarely change on their own.

And a border drawn on the map is not the same as a border in people's heads. The national feeling of chapter 11 does not disappear when a line is redrawn, which is why minority questions outlived every one of these settlements.

YOUR TURN

1. Give the four European border changes listed, with what each produced.
2. Explain the difference between the Yugoslav and Czechoslovak separations.
3. State the general rule about when borders change.
4. Explain the difference between a border on a map and a border in people's heads.

CHAPTER 30 · TOPIC 30.3

International organisations

The account

The world after 1991 is organised by a dense network of institutions, and it is worth knowing what the main kinds do, because they appear in every news report you will ever read.

Global political. The **United Nations**, described in chapter 20: a general assembly of nearly all states, a security council that can take binding decisions, and specialised agencies for health, refugees, children, food, education and labour.

Regional political and economic. The **European Union** above all, described in chapter 27, which is unusual because its members have transferred real powers to common institutions and to a court. Other regions have bodies that cooperate without going as far.

Military. NATO, which outlived the alliance it was formed against, and enlarged eastward as former Warsaw Pact members joined.

Economic. Bodies that set the rules of trade, lend to states in difficulty and coordinate financial policy.

Two honest observations, because a list of institutions is not an argument.

They exist because states found that some problems, trade rules, disease, pollution, refugees, cannot be solved by one country acting alone.

And they are argued about, because every one of them involves a state accepting limits on what it may do by itself. That argument, between what a country decides alone and what it decides with others, is one of the central political questions of the present, and it is the direct descendant of the nationalism you studied in chapter 11.

YOUR TURN

1. Give the four kinds of international organisation and an example of each.
2. What makes the European Union unusual among regional organisations?
3. Give two reasons such organisations exist.
4. Explain the argument that surrounds all of them, and link it to chapter 11.

WHAT YOU CAN NOW DO

- √ Explain what replaced the bipolar structure of the Cold War.
- √ Explain why some conflicts broke open after 1991.
- √ Describe the main border changes in Europe in the 1990s and classify them.
- √ Explain when and why borders change.
- √ Describe the main kinds of international organisation and the argument about them.

WORDS FOR THIS CHAPTER

bipolar, unipolar, sovereignty, federation, dissolution, velvet divorce, reunification, minority, peacekeeping, veto, regional organisation, integration

PART TEN · CHAPTER 31

Globalisation

Everything you own, and the journey it made to reach you

THE QUESTION THIS CHAPTER ANSWERS

Is globalisation new, or only faster?

IN THIS CHAPTER, YOU WILL

- √ Define globalisation and give the three mechanisms that carry it.
- √ Explain why supply chains are efficient and fragile for the same reason.
- √ Distinguish a migrant from a refugee, and give effects on both countries.
- √ Explain the consequences of digital communication, including for truth.
- √ Show, with examples from this book, that globalisation has a history.

THE TOPICS, IN THE ORDER YOU MEET THEM

31.1	Trade, companies and supply chains
31.2	Migration
31.3	Digital communication
31.4	Globalisation has a history

WATCH, READ AND LISTEN

Encyclopaedia Britannica, globalisation. Read its definition before you argue about the word, then decide which of its strands your own week depends on.

www.britannica.com/topic/globalization

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Check the labels of five things near you and write down where each was made. Then try to work out how each one reached this room. Keep the list; you will use it at the end of the chapter.

CHAPTER 31 · TOPIC 31.1

Trade, companies and supply chains

The account

Globalisation is the increasing connection of economies and societies across the world: goods, money, people and information moving between countries in greater volumes, faster, and at lower cost.

Three mechanisms carry it.

International trade grew enormously after 1945, encouraged by agreements to lower tariffs. Countries specialise in what they produce relatively well and buy the rest, which raises total output and makes each country dependent on the others.

Multinational companies operate in many countries at once: designing in one, manufacturing in another, servicing customers from a third. That is possible only because transport and communication became cheap, which is chapters 8 and 32.

Supply chains are the consequence, and they are the part worth understanding properly. A single manufactured object today is typically assembled from components made in several countries, from materials mined in others. Nobody plans the whole chain; it emerges from thousands of separate decisions, each of which is looking for a lower cost.

Two consequences follow, and both are visible in the news constantly.

Efficiency. Each stage is done wherever it is done best or most cheaply, so goods are cheaper than they could otherwise be.

Fragility. A chain with a hundred links can be broken at any one of them. A blocked canal, a closed port, a shortage of one component or a pandemic can stop production on the other side of the world. Chapter 18 called Suez a choke point in war; it is a choke point in peace too.

Materials
Metals and minerals mined in several countries.

→ **Components**
Parts made where each is made most cheaply.

→ **Assembly**
Put together in one place, often far from both.

Shipping
Containers by sea through a small number of ports and canals.

→ **Distribution**
Warehouses, lorries and rail to the shops.

→ **The weak point**
Any single link can stop the whole chain, and the narrow ones stop it first.

Figure 31.1 · One object, several countries.
Follow the arrows and count the border crossings before the object reaches a shop. Then find the single point where a delay would stop everything.

■ YOUR TURN

1. Define globalisation in one sentence.
2. Give the three mechanisms and explain each briefly.
3. Explain why supply chains are efficient and fragile for the same reason.
4. Using the figure, name the point where a delay would do the most damage.

CHAPTER 31 · TOPIC 31.2

Migration

The account

People move, and they always have. What changes over time is how many, how far, how quickly and under what rules.

The **reasons** are conventionally divided into push and pull, and the division is useful as long as you remember that most decisions involve both. People are pushed by war, persecution, poverty, unemployment, disaster and lack of opportunity. They are pulled by work, higher wages, safety, education, family already settled and, sometimes, simply the existence of a route.

Two distinctions matter and are frequently confused.

A **migrant** is anybody who moves to live in another country, for any reason.

A **refugee** is somebody who has fled their country because of a well-founded fear of persecution or of war, and who has a specific status in international law established after the Second World War, for reasons chapter 19 makes very clear.

The **effects** run in both directions and belong to both places. Receiving countries gain workers, skills, taxes and population, particularly where the existing population is ageing, and face real pressures on housing, schools and services in the places where arrivals concentrate. Sending countries lose working-age people, sometimes including those they most need, and receive **remittances**, money sent home, which in some economies is a very large share of national income.

WHERE PORTUGAL FITS

Portugal is one of the clearest examples in Europe of a country that has been both. For most of the period covered by this book it was a country of **emigration**: to Brazil and the United States in the nineteenth century, as chapter 9 described; to France, Germany, Luxembourg and Switzerland in the 1960s and early 1970s, as chapter 25 described. Since the later twentieth century it has also been a country of **immigration**, receiving people from its former territories in Africa, from Brazil, from eastern Europe and from Asia. The same country, within one lifetime, on both sides of the same process.

■ YOUR TURN

1. Give three push factors and three pull factors.
2. Explain the difference between a migrant and a refugee.
3. Give two effects on a receiving country and two on a sending country.
4. What are remittances, and why do they matter to some economies?

CHAPTER 31 · TOPIC 31.3

Digital communication

The account

Chapter 8 explained that the telegraph separated information from transport for the first time. Digital communication completed that separation and then made it almost free.

The technical chain is short. Computers became small and cheap enough to be everywhere. Networks were built to connect them, and connected to each other. **Fibre-optic cables**, most of them under the sea, carry enormous volumes of data between continents at very high speed. Mobile networks connect people who are not near a cable at all.

Three consequences change how societies work.

Cost. Sending a message across the world costs effectively nothing, where in 1850 it cost a substantial sum per word.

Volume and speed. Text, pictures, sound and video move constantly and in quantities no earlier system could carry.

Who can publish. Until recently, reaching a large audience required a newspaper, a printing works or a broadcast licence. Now anybody can publish to the world. That has enlarged what people can find out and made it far harder to tell what is true, because the checks that came with a publisher do not automatically come with a post.

Notice a geographical detail that surprises people: the internet has a very physical geography. It runs on cables that follow particular routes across ocean floors, landing at particular places, and on data centres built where power and cooling are cheap. A map of the world's undersea cables looks remarkably like a map of the world's shipping routes, for the same reasons.

YOUR TURN

1. Explain how digital communication completes what the telegraph began.
2. Give the three consequences described, with an example of each.
3. Why is it harder to tell what is true when anybody can publish?
4. Explain what is meant by saying the internet has a physical geography.

CHAPTER 31 · TOPIC 31.4

Globalisation has a history

The account

It is tempting to treat globalisation as something recent. It is not, and the whole of this book is the evidence.

Look back at what you have studied. The **steamship and the telegraph** of chapter 8 created a world market in ordinary goods and prices that moved together across oceans. The **migrations** of chapter 9 moved tens of millions of people between continents. The **empires** of chapter 12 tied producers in one hemisphere to factories in another. Two **world wars** were fought across every ocean. The **Cold War** divided the entire planet into two economic systems. Every one of those is globalisation, in an earlier form.

What is genuinely new is not connection but its **speed, volume and cheapness**, and the fact that information now moves independently of everything else.

There is also a lesson in the interruptions. Connection is not a one-way process. The period between 1914 and 1945 saw trade collapse, tariffs rise and borders close, and it took decades to recover the level of integration that had existed before the First World War. **Globalisation can go backwards**, and it has.

THE ENQUIRY: IS GLOBALISATION NEW, OR ONLY FASTER?

1. Find one example of each of the following from earlier in this book: a global market in goods; mass migration between continents; information moving faster than people; production in one country for sale in another.
2. Say what is genuinely new about the present, in one sentence.
3. Explain, using 1914 to 1945, why the process is not inevitable.
4. Look at the list of five objects you made before this chapter. Choose one and write a paragraph tracing what had to exist, historically, for it to be in this room.

WHAT YOU CAN NOW DO

- √ Define globalisation and give the three mechanisms that carry it.
- √ Explain why supply chains are efficient and fragile for the same reason.
- √ Distinguish a migrant from a refugee, and give effects on both countries.
- √ Explain the consequences of digital communication, including for truth.
- √ Show, with examples from this book, that globalisation has a history.

WORDS FOR THIS CHAPTER

globalisation, tariff, specialisation, multinational company, supply chain, choke point, push factor, pull factor, migrant, refugee, remittance, fibre-optic cable, data centre, integration, protectionism

CHAPTER 31 AT A GLANCE

Globalisation

is globalisation new, or only faster?

THE TOPICS YOU HAVE COVERED

31.1	Trade, companies and supply chains
31.2	Migration
31.3	Digital communication
31.4	Globalisation has a history

WHAT YOU CAN NOW DO

- √ Define globalisation and give the three mechanisms that carry it.
- √ Explain why supply chains are efficient and fragile for the same reason.
- √ Distinguish a migrant from a refugee, and give effects on both countries.
- √ Explain the consequences of digital communication, including for truth.
- √ Show, with examples from this book, that globalisation has a history.

THE WORDS THIS UNIT TAUGHT

globalisation · tariff · specialisation · multinational company · supply chain · choke point · push factor · pull factor · migrant · refugee · remittance · fibre-optic cable · data centre · integration · protectionism

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

32

PART TEN · CHAPTER 32

Transport and communication

The map has not changed; the distances have

THE QUESTION THIS CHAPTER ANSWERS

What happens to geography when the time between two places falls?

IN THIS CHAPTER, YOU WILL

- √ Explain time-space convergence with examples across five centuries.
- √ Explain why falling transport costs change what is worth moving.
- √ Explain why the container mattered more than its appearance suggests.
- √ Explain how transport costs shape where people and industry are.
- √ Explain how a place can become more central without moving.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Our World in Data, transport. How people and goods actually move, measured. Find the mode that carries most of world trade by weight.

ourworldindata.org/transport

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

How long would it take you to reach Lisbon from your school today? Now estimate the same journey in 1750, in 1900 and in 1970. The distance has not changed at all. That gap is what this chapter is about.

CHAPTER 32 · TOPIC 32.1

From the Roman road to the steamship

The account

Geographers use a phrase for the central idea of this chapter: **time-space convergence**. Places do not move, but the time between them shrinks, and when it shrinks enough the practical geography of the world is different even though the map is identical.

Run the sequence.

The Roman road. Well-built, drained, direct, and travelled on foot or by horse. It made an empire administrable, because messages and soldiers moved predictably. Its speed was the speed of a horse, and that was the ceiling for another fifteen hundred years.

The medieval trade route. Overland caravans and coastal shipping, moving goods valuable enough to justify months of travel: silk, spices, silver. Bulk goods did not travel far, because they could not pay for their own carriage.

The sailing ship. From the fifteenth century, ocean voyages connected continents, and Portuguese and Spanish navigation is the beginning of that. It was transformative and it was unreliable: the wind decided the schedule, as chapter 3 showed with the monsoon.

The steamship. From the nineteenth century, chapter 8, ships kept schedules. Combined with iron hulls and falling freight rates, that made it possible to carry heavy, cheap goods across oceans: grain, ore, coal, timber. **That is the moment ordinary things start to travel**, and it is more important than the increase in speed.

Notice the pattern. Each step lowers the cost per tonne-kilometre, and each time the cost falls, a new class of goods becomes worth moving.

YOUR TURN

1. Define time-space convergence in your own words.
2. Why did bulk goods not travel far before the nineteenth century?
3. Why was reliability more important than speed for the steamship?
4. State the pattern that links all four steps.

CHAPTER 32 · TOPIC 32.2

From the railway to the digital network

The account

The sequence continues, and it accelerates.

The railway, chapter 8, did overland what the steamship did at sea, and created national markets and standard time.

The motor vehicle, chapter 8 again, added flexibility. A lorry goes from door to door without a track, which lets settlement and industry spread away from railway lines and reshapes cities.

The aeroplane collapsed long-distance passenger travel from weeks to hours, and made high-value, low-weight goods, and people, movable across the world in a day.

The container, from the 1950s, is the least glamorous item in this chapter and arguably the most consequential. A standard steel box that fits a ship, a train and a lorry, and that is never unpacked in between, cut the cost and the time of loading a ship enormously. Loading had been slow, expensive and labour-intensive; containers made it fast and mechanical. Global supply chains of chapter 31 are not possible without it.

The digital network carries information at the speed of light through cables and switches, and separates information entirely from physical movement.

Two conclusions for the whole book.

Human geography is written by transport costs. Where people live, where industry is, which places are central and which are peripheral, all follow from what it costs to move things, and all change when that cost changes.

The map does not change; the world does. Lisbon has not moved since chapter

1 Its distance from London, measured in hours or in euros, has changed

enormously, and that is why the same place can be peripheral in one century and central in another.

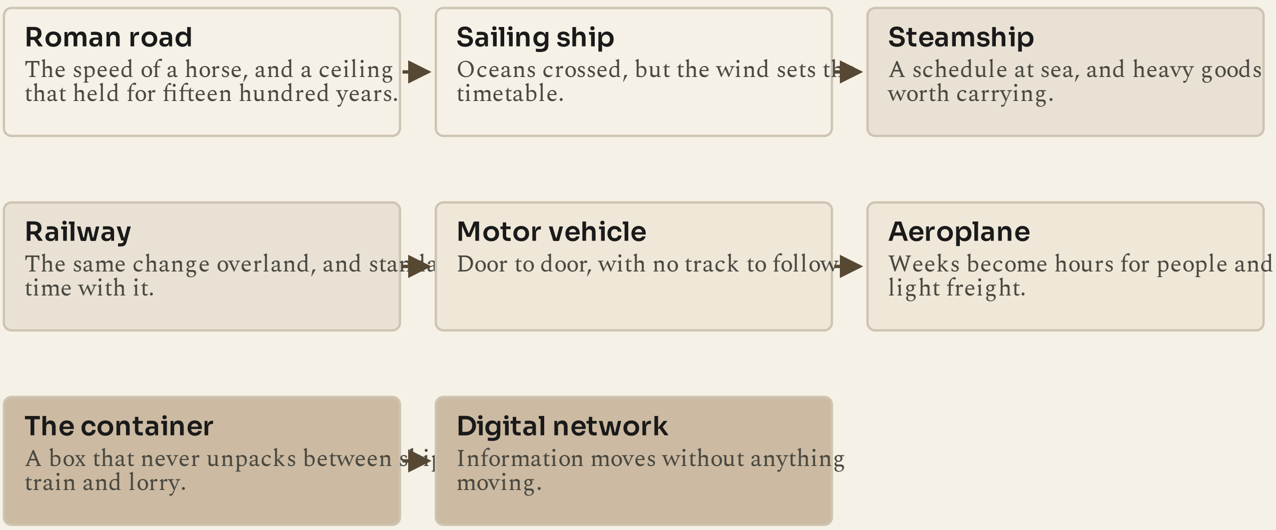


Figure 32.1 · Nine steps, and the same journey each time.
Read along the chain and notice that the biggest change is not always the fastest one. The container did more for the world economy than the aeroplane.

THE ENQUIRY: WHAT HAPPENS TO GEOGRAPHY WHEN THE TIME BETWEEN TWO PLACES FALLS?

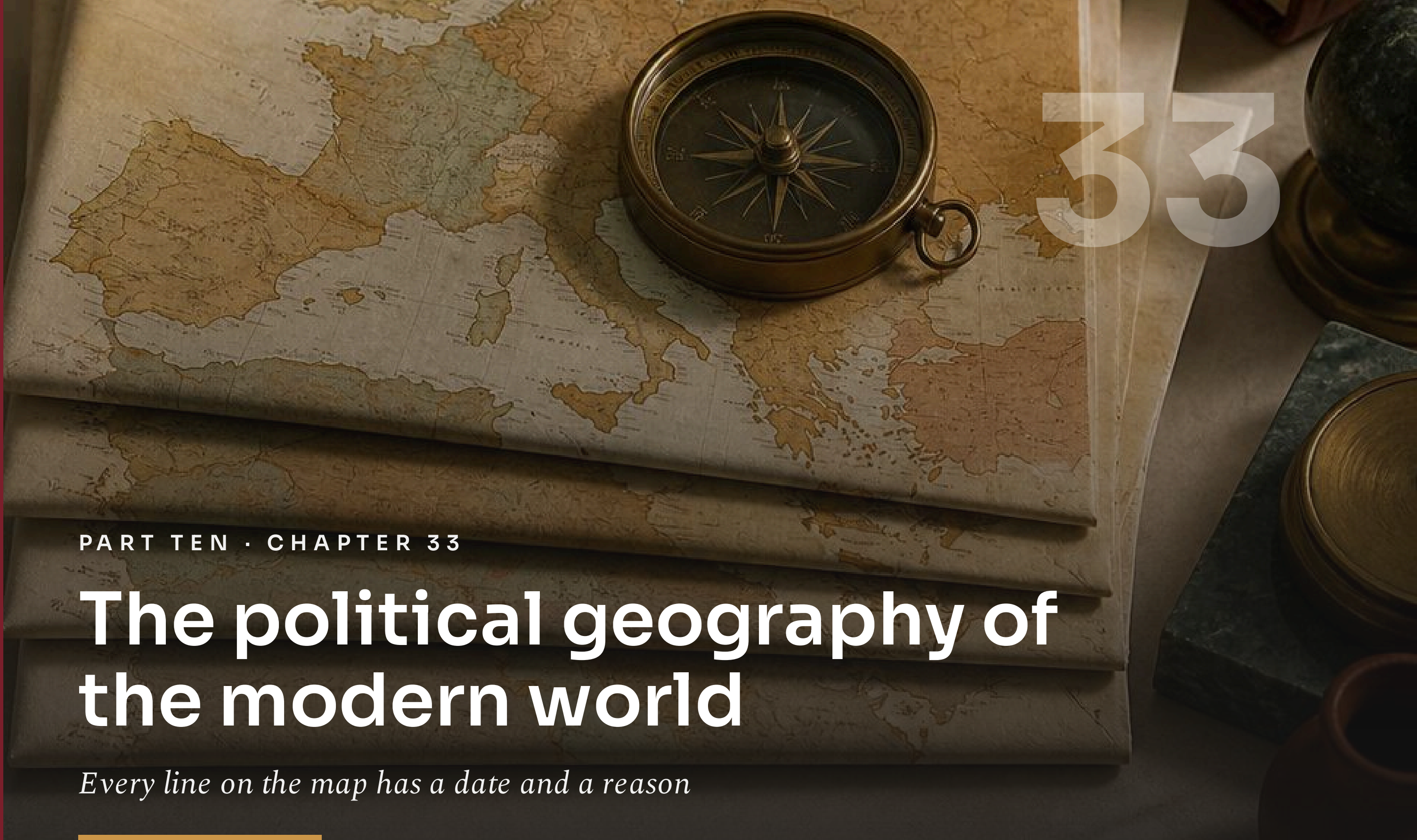
1. Choose two places, one Portuguese and one on another continent.
2. Estimate the journey time between them in 1500, 1850, 1950 and today.
3. For each period, say what could reasonably be traded between them, and why.
4. Write a conclusion explaining how a place can become more central without moving. Use the Azores or Lisbon as your example, and refer back to chapter 18.

WHAT YOU CAN NOW DO

- √ Explain time-space convergence and give examples across five centuries.
- √ Explain why falling transport costs change what is worth moving.
- √ Explain why the container mattered more than its appearance suggests.
- √ Explain how transport costs shape where people and industry are.
- √ Explain how a place can become more central without moving.

WORDS FOR THIS CHAPTER

time-space convergence, freight cost, bulk goods, caravan, sailing ship, steamship, railway, standard time, motor vehicle, aeroplane, container, intermodal, digital network, periphery, centrality



33

PART TEN · CHAPTER 33

The political geography of the modern world

Every line on the map has a date and a reason

THE QUESTION THIS CHAPTER ANSWERS

Why are borders where they are?

IN THIS CHAPTER, YOU WILL

- √ Define state, nation, nation-state and border precisely.
- √ Explain why true nation-states are uncommon.
- √ Describe five maps of Europe and name the event behind each.
- √ Give the five causes of border change, with an example of each.
- √ Explain a current border historically rather than as natural.

THE TOPICS, IN THE ORDER YOU MEET THEM

33.1	States, nations and borders
33.2	Five maps of Europe
33.3	Borders are historical

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the nation-state. The difference between a state and a nation, which is where most arguments about borders begin.

www.britannica.com/topic/nation-state

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Look at a political map of Europe. Choose one border and ask: why is it exactly there? If your answer is "it just is", this chapter will give you a better one.

CHAPTER 33 · TOPIC 33.1

States, nations and borders

The account

Three words are used loosely in conversation and precisely in geography, and the difference matters for everything in this chapter.

A **state** is a political unit: a defined territory, a permanent population, a government, and the capacity to enter relations with other states. It is a legal and political fact.

A **nation** is a group of people who believe they share something: a language, a history, a culture, a sense of belonging together. It is a fact about how people feel, which does not make it less real.

A **nation-state** is a state whose borders match a nation. It is the ideal that nineteenth-century nationalism aimed at, and in practice it is uncommon. Most states contain more than one national group, and several nations are spread across more than one state.

A **border** is where one state's authority ends and another's begins. Some follow physical features, rivers, mountain ranges, coastlines. Some are drawn straight, especially where they were imposed by outsiders on land whose inhabitants were not consulted, which chapter 12 described in Africa. Some follow where an army stopped, which chapter 20 described in Europe and Korea.

Portugal's border with Spain is one of the oldest in Europe, largely settled by the thirteenth century, which is genuinely unusual on this continent and is part of why Portugal has such a strong sense of itself as a nation.

YOUR TURN

1. Give precise definitions of state, nation and nation-state.
2. Why is a true nation-state uncommon?
3. Give three different kinds of thing a border can follow.
4. Why is the age of the Portuguese border significant?

CHAPTER 33 · TOPIC 33.2

Five maps of Europe

The account

This is the most important activity in the chapter, and it should be done with five maps side by side.

Europe in 1914. Four empires: German, Austro-Hungarian, Russian, Ottoman. No Poland, no Czechoslovakia, no Yugoslavia, no Baltic states, no Ireland.

Europe in 1919. The empires gone. New states across central and eastern Europe, built on self-determination and containing large minorities. Germany smaller.

Europe in 1945. Borders moved west in the east: Poland shifted bodily, losing territory to the Soviet Union and gaining it from Germany. Germany occupied and about to divide. Enormous populations moved or expelled.

Cold War Europe. Two blocs, an iron curtain, two German states, and a border that stayed put for forty years.

Europe after 1991. The Soviet Union gone, replaced by fifteen states. Yugoslavia breaking into seven. Czechoslovakia into two. Germany one again. Then, through the 2000s, most of the former eastern bloc inside the European Union.

Five maps, one continent, less than eighty years.

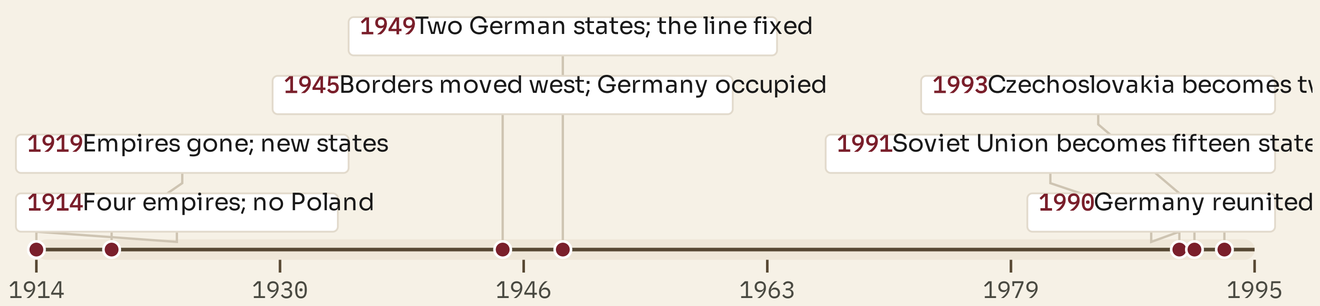


Figure 33.1 · One continent, five maps.
For each date, note what caused the change: a war, a treaty, an occupation, a collapse or an agreement. Every border on the last map is the result of at least one of them.

YOUR TURN

1. Name the four empires on the 1914 map.
2. What happened to Poland's position between 1919 and 1945?
3. What stayed still between 1945 and 1989, and what does that tell you?
4. For each of the five maps, name the single event that produced it.

CHAPTER 33 · TOPIC 33.3

Borders are historical

The account

This is the conclusion the whole chapter exists for, and it is one of the most useful ideas a Humanities pupil can take away.

Borders are not natural features of the world. They are the outcomes of past events, and every one has a date and a reason.

The causes are the ones you have studied.

War, which redraws by force: 1919, 1945.

Treaty, which records a settlement, whether negotiated between equals or imposed on the defeated.

Independence movements, which create states from empires: eastern Europe in 1919, Africa and Asia after 1945, the Soviet republics in 1991.

Collapse, when a state ceases to exist and its internal boundaries become international ones, as in the Soviet Union and Yugoslavia.

Agreement, which is rarest and most encouraging: Czechoslovakia in 1993, and the unification of Germany in 1990.

Two consequences follow for how you read the news.

Every border dispute has a history, and understanding it usually requires knowing which of the five causes produced the line and who was not consulted at the time.

Borders will change again. They have changed throughout the period of this book, and there is no reason to think the current map is the final one. That is not a prediction about any particular place; it is an observation about the evidence in front of you.

THE ENQUIRY: WHY ARE BORDERS WHERE THEY ARE?

Choose one European border on a current map.

1. Find out when it took its present form.
2. Identify which of the five causes produced it.
3. Find out who lived there before and whether they were consulted.
4. Write a paragraph explaining the border to somebody who has never seen a map of Europe. Your last sentence should say what would have to happen for it to change again.

WHAT YOU CAN NOW DO

- √ Define state, nation, nation-state and border precisely.
- √ Explain why true nation-states are uncommon.
- √ Describe the five maps of Europe and name the event behind each.
- √ Give the five causes of border change, with an example of each.
- √ Explain a current border historically rather than treating it as natural.

WORDS FOR THIS CHAPTER

state, nation, nation-state, sovereignty, border, frontier, natural boundary, imposed boundary, self-determination, minority, partition, collapse, enlargement

34

PART TEN · CHAPTER 34

Bringing geography and history together

Back to the beginning: how much of this was decided by where things are?

THE QUESTION THIS CHAPTER ANSWERS

How much of the history in this book was decided by the geography in it?

IN THIS CHAPTER, YOU WILL

- √ Explain how geography shaped industrialisation, and where that stops.
- √ Apply six geographical factors to both world wars.
- √ Explain why Portugal's position was worth different amounts in different centuries.
- √ Read the world timeline and the Portuguese timeline together.
- √ Explain the whole chain from place to the connected world.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Royal Geographical Society. The learned society for geography. Find a case study where the physical setting shaped what people were able to do.

www.rgs.org/

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Turn back to chapter 1 and read the first paragraph of topic 1.1 again. You have now travelled from a line drawn round a planet to the world you live in. This chapter asks how much of the journey was decided by where things are.

CHAPTER 34 · TOPIC 34.1

Geography and industrialisation

The account

Chapter 6 asked why the Industrial Revolution began in Britain and gave seven conditions. Ask the same question again, now, as a geographer.

Coal had to be in the ground, near the surface, and near water. Geology put it there; nobody chose it.

Iron ore had to be close to the coal, because moving several tonnes of both to make one tonne of iron is only affordable over short distances.

Rivers and coastline determined what could be moved and at what cost. An island with navigable rivers and good harbours has cheap transport built in.

Rainfall and fast-flowing streams powered the first mills, which is why the earliest factories are where they are rather than where the customers were.

Position gave access to Atlantic trade routes and to markets and materials overseas.

Now the honest half. Geography did not decide it. Chapter 6 listed capital, labour, markets and institutions alongside the physical conditions, and other places had comparable geology without industrialising first. China had coal. Portugal, as topic 6.8 showed, had an Atlantic position that had been an enormous advantage for four centuries and was worth much less for this particular transformation.

The accurate formulation is the one chapter 3 gave you: **geography sets the price; people decide whether to pay it.**

YOUR TURN

1. Give five geographical conditions that favoured British industrialisation.
2. Give one place with comparable geology that did not industrialise first.
3. Explain what Portugal's geography was excellent for, and what it was not.
4. Explain the formula about price, in your own words.

CHAPTER 34 · TOPIC 34.2

Geography and the United States

The account

Chapter 9 explained the rise of American industry. Read it geographically and four features do most of the work.

Size. A territory of continental scale, with the resources of a continent inside one set of borders and one customs area.

Internal waterways. The Mississippi system drains the interior to the sea, and the Great Lakes connect the iron ore of the north-west to the coal of the east. Ore and coal joined by water, again, exactly as in chapter 6, but across two thousand kilometres.

Agricultural land. The prairies fed the industrial cities and produced exports, so industry did not have to compete with food for the same land.

Two ocean coasts, giving access to Europe and to Asia.

Set that against Portugal, as topic 9.10 did, and the contrast is not about effort. It is about the size of the endowment.

But note the same qualification as before. Argentina and Brazil also had continental scale, resources and coastlines, and neither followed the same path. Geography made American industrialisation **possible on that scale**; capital, immigration, institutions and a very large internal market made it happen.

YOUR TURN

1. Give the four geographical features and explain each.
2. Explain the Great Lakes point by reference to chapter 6.
3. Name two other large countries with comparable endowments.
4. Why does the comparison show that geography is necessary but not sufficient?

CHAPTER 34 · TOPIC 34.3

Geography and the world wars

The account

Chapter 18 gave you six factors. Apply them across both wars.

Distance and supply. The German plan of 1914 required speed because of the distance between two fronts. The German advance into the Soviet Union in 1941 outran its own supply. The Allies in 1944 had to build artificial harbours because they had no port.

Choke points. Suez, Gibraltar, the Danish straits, the Turkish straits: narrow places worth enormous effort to hold, in both wars.

Sea lanes. Britain could be starved by submarines in either war, and both times the Atlantic became the campaign that had to be won first.

Terrain and climate. Trenches followed the ground in Flanders. The Russian winter and the spring thaw set the timetable in the east. The Normandy landing date was chosen from tide and moon.

Resources. Oil shaped strategy in the second war: it is why Germany drove toward the Caucasus and why Japan moved into south-east Asia.

Industrial geography. The side with more coal, steel and factories, and whose factories could not be reached, won both wars. The Soviet decision to move industry east is among the most consequential geographical decisions ever taken.

And a small country, twice: **the Azores**, mid-Atlantic, closing the air gap in one war and anchoring an alliance in the next.

■ **YOUR TURN**

1. Give one example of each of the six factors, from either war.
2. Why did the Allies need artificial harbours in 1944?
3. Explain the role of oil in shaping German and Japanese strategy.
4. Explain the significance of the Azores in one sentence, twice: for 1943 and for 1949.

CHAPTER 34 · TOPIC 34.4

Geography and Portugal

The account

Portugal has appeared in seventeen sections of this book, and one geographical fact runs through all of them.

Portugal is on the western edge of Europe, facing the Atlantic.

For the age of ocean exploration and empire, that position was among the best in the world, and Portugal built a global presence on it.

For the age of coal and iron, chapters 5 to 8, it was worth far less. The Industrial Revolution rewarded coalfields, navigable interiors and large internal markets, and Portugal had none of the three. The same position, a different century, a completely different value.

For the age of world wars and the Cold War, chapters 18 and 21, it became extremely valuable again, because the Atlantic itself became the decisive space: the convoy routes, the air gap, the reinforcement of Europe from North America. The Azores are the clearest single case in this book of a place whose importance comes entirely from where it is.

For the age of European integration, chapter 27, the calculation changed once more. Portugal's future was decided as part of a continental market rather than an Atlantic empire.

The same geography was an asset, then a limitation, then an asset again. That is the single best answer this book can give to the question of how geography and history interact: position does not have a fixed value, and what a place is worth depends on what the world is doing.

	The age	What the Atlantic edge was worth
1400 to 1700, ocean exploration	The same coastline, the same harbours, the same winds.	An enormous advantage. The route to Africa, A Brazil began here.
1750 to 1900, coal and iron	Unchanged. Still Atlantic, still on the edge of Europe.	Worth much less. Industry rewarded coalfields navigable interiors and large home markets.
1939 to 1974, world war and Cold War	Unchanged, and now with the Azores in the middle of the ocean.	Extremely valuable. The Atlantic was the decis space, and the islands closed the air gap.
1986 onward, European integration	Unchanged, but the frame of reference moved to the continent.	Valuable differently. The future is a continenta market rather than an Atlantic empire.

Figure 34.1 · One position, four centuries, four different values.
Read the rows and notice that the geography does not change at all. Only the column on the right changes.

YOUR TURN

1. State the one geographical fact about Portugal in a single sentence.
2. Explain why it was worth a great deal in 1500 and much less in 1850.
3. Explain why it became valuable again after 1940.
4. Explain, in your own words, why position does not have a fixed value.

CHAPTER 34 · TOPIC 34.5

Geography and the Cold War

The account

The Cold War was named after a temperature and organised by a map.

The line was where the armies stopped. Not a negotiated border, not an ethnic boundary, not a river. Where soldiers halted in 1945 became the frontier of two systems for forty-four years, and chapter 33 showed it was the most stable line on the continent while it lasted.

The blocs had geographical names, west and east, and those names were only approximately geographical: Greece is east of Prague and was in the western bloc; Australia is about as far east as it is possible to be and was counted in the West. Chapter 4 warned you about exactly this confusion.

Bases and position decided alliances. Iceland, Turkey, the Azores, Cuba, Berlin: places whose importance was entirely a matter of location. Turkey mattered because of the straits and its border. Cuba mattered because of its distance from Florida, which is what made 1962 a crisis.

Distance defined the weapons. Chapter 24's whole argument, intercontinental missiles and submarines, is about crossing distance and about hiding in the ocean.

And geography ended it, partly. The Hungarian decision to open a border with Austria in 1989 let East Germans leave. A line on a map, opened, and the wall in Berlin became pointless before it was opened.

■ YOUR TURN

1. What decided where the line between the blocs fell?
2. Give two examples showing that "the West" was not a geographical term.
3. Explain why Cuba and Turkey mattered for the same kind of reason.
4. Explain how a border decision in Hungary affected Berlin.

CHAPTER 34 · TOPIC 34.6

Portugal in the global story: the whole timeline, side by side

The two lines

This is the section the whole book has been building toward. The world above, Portugal below, on one spread, so that the two questions can be answered at once: **what was happening in the world, and what was happening in Portugal?**

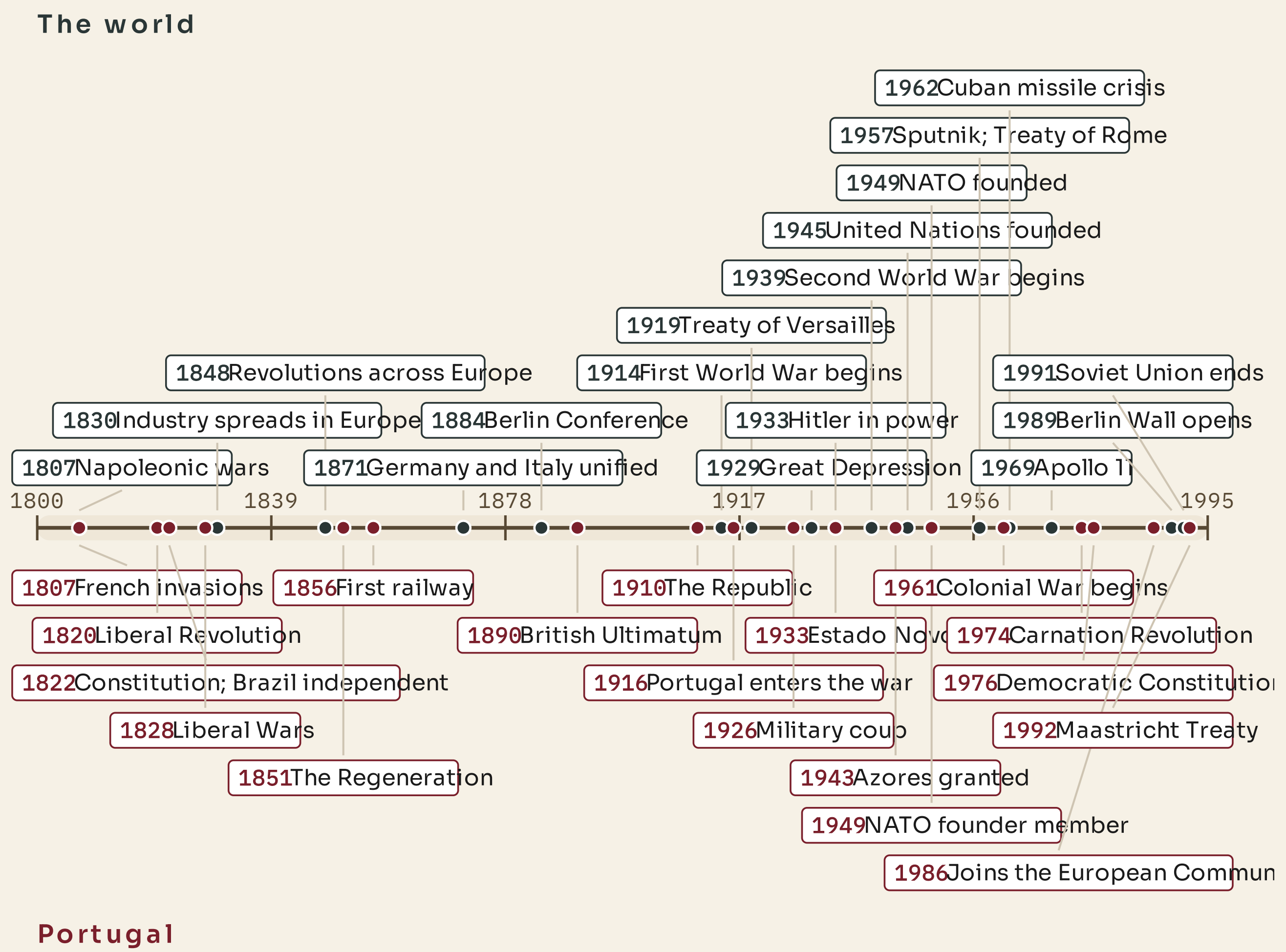


Figure 34.2 · Two centuries, two lines, one chronology.
Read the upper line first, then the lower. Then read them together, vertically: pick any year and ask what was happening in both places at once.

Four vertical pairs are worth reading carefully, because in each case the Portuguese event cannot be explained without the world event above it.

1807 to 1822. Napoleon closes Europe to British trade; a French army invades Portugal; the court sails for Brazil; Brazil becomes independent.

1914 to 1918. A European war becomes global; Portugal, a four-year-old republic, enters in 1916 and loses heavily at La Lys in 1918; the strain helps end the First Republic in 1926.

1939 to 1945. A war is decided partly in the Atlantic; Portugal stays neutral and grants the Azores in 1943; Lisbon becomes a refugee route; the regime survives.

1961 to 1986. Empires are dismantled worldwide; Portugal fights to hold its territories, the army turns against the regime, 25 April 1974 follows, and democracy leads to the European Community in 1986.

What the pairing shows

Portugal was never outside the story, and it was never simply following it. Sometimes it was ahead: its liberal constitution of 1822 was early, its republic of 1910 was early, its dictatorship of 1926 was early. Sometimes it was behind: it industrialised late and democratised after the rest of western Europe. Sometimes it did something no one else did: it kept an empire for a decade after everybody else had let go, and then let go of all of it at once.

KEY QUESTION

What was happening in Portugal while the world was changing?

Choose any three years from the timeline. For each, write two sentences: one on the world and one on Portugal. Then answer the four questions this book has asked of every Portugal section:

1. What was happening in Portugal?
2. What was happening in the world?
3. What part did geography play?
4. Was Portugal participating, resisting, following later, developing differently, or influencing events elsewhere?

CHAPTER 34 · TOPIC 34.7

The modern world has a history, and that history has a geography

The account

You began this book with five imaginary lines and a question about where things are. You are ending it with the world you live in and a question about how it got that way.

Here is the whole argument, in one chain. Every link was a chapter.

Place decided what resources were available. **Resources** and **capital** made industry possible where it was possible. **Industry** built cities, produced new classes and made countries powerful. **Technology** shrank distance and made markets global. **Ideas** grew out of industrial society and demanded that power be shared or seized. **Nationalism** and **imperial competition** turned rivalry into war. **Two world wars** destroyed the old order and produced two superpowers. **Ideological rivalry** divided the planet. **Technological competition** carried people to the Moon. **Political and economic pressure** ended that division. **Transport and technology** produced the connected world you live in.

And underneath every link, position: which country had coal, which had ocean, which was on the route, which was in the way.

So the two things you should be able to do now are these.

Given any event in this book, you should be able to ask **where** it happened and **why there**.

And given any place, you should be able to ask what happened there, and what it left behind.

That is what it means to study Humanities rather than two separate subjects. Your school, your town, the borders of your country, the language you are reading this in, the shops you buy from and the alliances your country belongs to are all outcomes of the processes in this book. **None of them was inevitable, and all of them have a date.**

WHAT YOU CAN NOW DO

√ Explain how geography shaped industrialisation, and where the explanation stops.

√ Explain the American case geographically and say why geography is not enough.

√ Apply six geographical factors to both world wars.

√ Explain why Portugal's position was worth different amounts in different centuries.

√ Read the world timeline and the Portuguese timeline together, vertically.

√ Explain the whole chain from place to the connected world.

WORDS FOR THIS CHAPTER

position, endowment, necessary condition, sufficient condition, choke point, sea lane, air gap, periphery, centrality, chronology, parallel timeline, continuity, change, causation

Answers

Answers to every closed question in this book. Open questions, the enquiries and the Go further tasks have acceptance criteria instead: what a good answer must contain, rather than the words it must use. Attempt the question before you look.

Chapter 1: The global grid

1.1 1. The line half way between the two poles, at zero degrees latitude, fixed by the shape of the planet. 2. Because the Earth is a ball, rays arrive close to vertical near the Equator so a beam covers a small patch, and at a shallow angle near the poles so the same beam is spread over a much larger one.

1 South America, Africa and Asia. 4. The Earth's radius is about 6 400 km against a distance of about 150 million km, so being on the Equator brings you closer by a fraction far too small to matter; the cause is angle, not distance.

1.2 1. About 23.5 degrees, and it keeps pointing the same way all year.

1 The furthest north the Sun can ever be directly overhead at noon. 3. Because both are set by the same axial tilt, measured in opposite directions. 4. Nairobi and Singapore, because both lie between the Tropics; Lisbon and Sydney lie outside them.

1.3 1. About 66.5 degrees north and south, which is ninety minus the tilt of 23.5. 2. Inside the Arctic Circle there is at least one day a year when the Sun does not set, and the further north you go the more such days there are. 3. The Arctic is an ocean surrounded by continents; Antarctica is a continent surrounded by ocean. Also: people have lived around the Arctic for thousands of years; nobody lives permanently in Antarctica. 4. Antarctica is a continent covered by an ice sheet kilometres thick with no land able to support settlement or hunting, and it is separated from every other continent by open ocean.

1.4 1. A meridian runs pole to pole and measures longitude; a parallel runs east to west and measures latitude. Meridians are all the same length and meet at the poles; parallels never meet and shrink toward the poles. 2. The Equator is fixed by the shape of the Earth, so latitude has a natural zero; no meridian is different from any other, so a zero has to be agreed. 3. Charts measured from different prime meridians could not be combined, so a position reported by one ship did not mean the same to another. 4. That Paris had a long tradition of astronomical measurement and French charts already used it; it lost because far more of the world's shipping already used Greenwich.

1.5 Discussed in the text; see 2.2 and 2.6 for the time consequences.

Chapter 2: Latitude, longitude and time zones

2.1 1. How far north or south of the Equator a place is, from 0 to 90 degrees. 2. Parallels are circles drawn round a ball and get smaller toward the poles; meridians are all half-circles of the same size. 3. Lower latitude means more direct solar energy and generally warmer. Exception: altitude, so Quito sits on the Equator and is cool because it is high. 4. One is in the northern hemisphere and one in the southern, so their seasons are opposite.

2.2 1. How far east or west of the Prime Meridian, up to 180 degrees. 2. It is the single meridian on the opposite side of the Earth from Greenwich, reached by going 180 degrees either way. 3. Latitude can be read from the height of the Sun or the Pole Star; longitude requires comparing local time with the time at a known meridian, so it needs an accurate clock carried on the voyage. 4. Longitude is measured as a difference in time, so solving one solves the other.

2.3 1. Absolute location is a coordinate pair naming one point; relative location describes a place by its relation to another. 2. Ponta Delgada at 25.7 W and Tokyo at 139.7 E. 3. 16.6 degrees. 4. Because a description depends on the other person already knowing the reference point, and a coordinate does not.

2.4 1. West to east, so the eastern side of you turns to face the Sun first.

1 Exactly half. 3. Because the Earth turns, so the Sun is highest at different

moments in different places. 4. A timetable states one time for a departure, and that is meaningless if every station keeps its own clock.

2.5 1. 360 divided by 24 equals 15 degrees per hour. 2. Lisbon keeps UTC in January and Tokyo UTC+9, so 09:00 plus 9 hours gives 18:00. 3. New York is UTC−5 and London UTC, a difference of +5 from New York, so 18:00 plus 5 gives 23:00.

1 Boundaries follow national borders so a country shares one working day, and

they are bent for convenience; some offsets are half or three-quarters of an hour.

2.6 1. A standard is a reference scale kept by atomic clocks; a zone is what clocks in a place are set to. 2. Atomic clocks keep it accurate, and leap seconds keep it close to the Sun's position over Greenwich. 3. Because mainland Portugal lies close to the Prime Meridian. 4. Because everybody involved uses one unambiguous scale, and because local times change with summer time while UTC does not.

2.7 1. Going east clocks run to 12 hours ahead and going west to 12 hours behind, which is 24 hours apart at the same meridian. 2. You skip forward a day.

1. To keep island groups and countries on a single date. 4. Open answer: it must

contain a coordinate pair, and may add a date and a time with its zone.

Chapter 3: Climate zones and biomes

3.1 1. Weather is what the atmosphere is doing now; climate is the long-run pattern; a biome is a large region defined by climate, vegetation and animal life. 2. Latitude, solar energy, temperature, climate, vegetation, biome, human activity. 3. Steep rays concentrate energy on a small area and pass through less atmosphere; shallow rays spread the same energy over more ground. 4. Altitude, distance from the sea, ocean currents, prevailing winds, mountain ranges.

3.2 1. Temperature is constant and high; rainfall varies enormously.

1. Rainforest has heavy rain in every month; savanna has a marked wet and dry

season. The difference is rainfall pattern, not temperature. 3. Wettest March with 362 mm, driest August with 96 mm, annual total 2 694 mm. 4. Nutrients are held in the living plants and heavy rain washes minerals from the soil, so cleared land gives one or two harvests and is then exhausted.

3.3 1. Real seasons: a distinct summer and winter differing in temperature and day length. 2. Temperate forest, western and central Europe; temperate grassland, the Eurasian steppe or North American prairie; Mediterranean, southern Portugal.

1. Driest July with 2 mm, annual total 558 mm. 4. To lose as little water as

possible during a hot dry summer, which is when they are growing.

3.4 1. A region receiving under about 250 mm of rain a year. Rainfall is the word doing the work. 2. Air that rose at the Equator and lost its moisture sinks at about 20 to 30 degrees, and sinking air warms and dries. 3. Distance from any ocean, and rain shadow behind mountains. 4. 13 mm, against Lisbon's 558 mm, so about one fortieth.

3.5 1. Ground that has stayed below freezing for at least two years. Water cannot drain through it, so the thawed surface layer is waterlogged. 2. They cannot draw water through frozen ground, and the growing season is too short and the wind too strong. 3. Yakutsk, because it is far from any ocean, so it has both the colder winter and the warmer summer. 4. The growing season is too short for crops, so food had to come from animals and fish.

3.6 1. Rice, tropical, needs warmth and standing water; wheat, temperate, needs a cool dry ripening season; olives, Mediterranean, need dry summers.

1. Steep roofs shed snow, so snow falls there; flat roofs mean it does not.
2. The monsoon winds reverse twice a year, so ships sailed one way in one season

and returned in the other. 4. Acceptance criteria: the rewritten sentence must say that climate sets costs or constraints, and that technology and money can change what those costs allow.

Chapter 4: Hemispheres and global patterns

4.1 1. The northern hemisphere holds most of the land; the southern holds mostly ocean. 2. Because people live on land, and most land is northern.

1. Because ocean heats and cools far more slowly than land, so a hemisphere that

is mostly ocean has milder extremes. 4. Wholly southern: Australia or Antarctica. Wholly northern: Europe or North America.

4.2 1. The Prime Meridian and the 180th meridian. 2. Europe, Africa, Asia, and Australia. 3. Western, because mainland Portugal lies west of the Prime Meridian. 4. Because the Equator is fixed by the planet's shape and the choice of a prime meridian was an agreement made in 1884.

4.3 1. The western hemisphere is a geographical term set by two meridians; the West is a political and cultural term with no fixed definition. 2. Australia is usually counted in the West and lies far east; Portugal is in the western hemisphere and was a dictatorship inside the western bloc. 3. Because one term describes position and the other describes politics, and no map records politics.

1 Acceptance criteria: the rewrite must separate the geographical claim from the

political one, and should date the political claim.

4.4 1. Too dry, such as the Sahara; too cold, such as northern Siberia; too high, such as the Himalaya; too wet or forested, such as the Amazon basin.

1 Water, growing season, flat workable land, work, connection, history. Ranking

is the pupil's, with reasons. 3. Cities founded for a reason that no longer applies still stand there, such as ports founded for sailing ships. 4. Open; credit any answer that names the direction of the error and offers a reason.

Chapter 5: The world before industrialisation

5.1 1. About four in five. 2. Seasonal, local, precarious and slow to change, each with an example from the text. 3. Because food was expensive to move, so a surplus could not easily reach a region in shortage. 4. Because output rose by adding land or workers rather than by raising what each worker produced.

5.2 1. Merchant buys raw material, carries it to cottages; family spins and weaves at home; merchant collects and pays by the piece; merchant sells on.

1 Control of hours, ownership of tools, and income that cushions a bad farming

year. 3. Output could only grow by finding more cottages. 4. No control over speed or quality, and losses of material along the way.

5.3 1. Better rotations, which removed the bare year; selective breeding, which produced heavier animals; enclosure, which allowed new methods. 2. Because about a third of the land grew nothing that year. 3. More food, more people, fewer workers needed on the land, so people move to the towns. 4. For: output rose. Against: many families lost access to common land they had relied on.

Chapter 6: Why did the Industrial Revolution begin in Britain?

6.1 1. Muscle, weak; wind, unreliable; water, only beside fast water; wood, runs out. 2. It was shallow, near iron ore, and near navigable water. 3. Because coal is heavy and cheap, so land transport could cost more than the coal was worth. 4. Mines flooded as they went deeper, and the first steam engines were built to pump the water out.

6.2 1. Smelting used charcoal, which is made from wood. 2. Iron production was tied to coal instead of to timber. 3. Because several tonnes of coal and ore are needed for one tonne of iron, and all of it is heavy. 4. Acceptance criteria: the answer must say that heavy industry locates near its raw materials and light industry near its market.

6.3 1. Roads were poor and a cart carried very little, so cost per tonne was very high. 2. They brought in raw cotton and sent out finished cloth. 3. It fell sharply. 4. Because other islands did not industrialise first and non-islands had excellent water transport, so it is one condition among several.

6.4 1. Land, building, machinery and wages must all be paid for before anything is sold. 2. Profitable agriculture, long-distance and colonial trade including the slave trade, and a banking system. 3. Wealth is a stock; a habit of investment is a willingness to risk it in ventures that may not pay for years.

1 Because historians disagree about the size of the share, while agreeing it was part of the picture.

6.5 1. Population growth, fewer workers needed per hectare, and enclosure removing access to common land. 2. Because a machine is useless without people who can build, install and repair it. 3. Because people used to seasonal, task- based work had to be brought to accept fixed hours under another person's control. 4. Open; credit either answer if it explains why skill takes longer to build than numbers.

6.6 1. Home, which is steady and cheap to supply; overseas, which is large.

1 Machines lower cost, lower cost lowers price, lower price widens the market, a wider market justifies more machines. 3. Because each turn of that loop makes the next easier. 4. Because Indian textile production was undercut and restricted, which was part of how the British market was secured.

6.7 1 to 4 are open; see the enquiry criteria in the teachers' notes.

6.8 Enquiry; acceptance criteria in the teachers' notes. The Portuguese strength pupils should find is position and ports.

Chapter 7: Factories, cities and social change

7.1 1. Tools stop belonging to the worker; the machine sets the pace; hours become fixed; work is divided into small repeated tasks. 2. Because a powered machine costs far more than a household can raise. 3. Because an expensive engine is run for as long as possible, and everyone tending it must be present.

1 Advantage: output per worker rises. Disadvantage: the work is monotonous and the worker learns a task rather than a trade.

7.2 1. Because there was no cheap transport, so workers walked to the mill.

1 Manchester, cotton; Birmingham, metalworking; Glasgow, shipbuilding. 3. Water supply, sewers, and schools or hospitals. 4. Because everything a city needs takes years to build, and the population arrived in months.

7.3 1. Terraces sharing side and rear walls so a house had windows on one side only, built cheaply and quickly. 2. Standpipes and cesspits were close together, so waste contaminated drinking water. 3. He mapped cholera deaths street by street and showed they clustered round one pump; it is geography because he found the cause from the pattern of where cases were. 4. Value: systematic evidence across many towns. Limitation: it argues a case, and argues from cost.

7.4 1. Twelve hours or more, six days a week, timed by the employer's clock with fines for lateness. 2. Unguarded machinery, and cotton dust in warm damp air damaging lungs. 3. Because wages were too low for one to support a household.

1 The concentration: thousands under one roof, on one clock, at the pace of machinery.

7.5 1. Cheap; families needed the wages; children had always worked; and they were small enough for some tasks. 2. Because it explains why factory work did not at first seem like a new kind of demand. 3. They limited hours, set minimum ages and required schooling; inspectors mattered because an unenforced law changes little. 4. Evidence, law, enforcement, change.

7.6 1. Middle class: income from profit on capital owned. Working class: income from wages for hours worked. 2. Because their economic weight was not matched by political power. 3. A skilled engineer and a child piecer; a great mill owner and a small shopkeeper. 4. Because landowning aristocracies kept great influence throughout the century.

Chapter 8: Inventions that changed the world

8.1 1. Spinning and weaving; weaving became faster first, so spinning could not keep up. 2. It let one worker turn many spindles at once. 3. Because it was driven by a water wheel, so it had to stand beside a river in a mill. 4. Solving a bottleneck at one stage creates one at the next.

8.2 1. It exists only where there is fast-flowing water. 2. It condensed steam in a separate vessel so the working cylinder stayed hot, instead of being cooled and reheated every stroke. 3. Because fuel was underfoot at the pit head.

1 Watt did not invent it; he made it efficient enough to be used away from a

coalfield.

8.3 1. Very low friction between smooth iron surfaces. 2. Cheap overland freight, mass passenger travel, and standard time. 3. Because an inland town could now be supplied as cheaply as a port. 4. Because a timetable is impossible if every town keeps its own clock.

8.4 1. Reliability: it keeps a schedule instead of waiting for wind. 2. Because they allowed much larger ships, and a larger ship carries cargo more cheaply per tonne. 3. Heavy, low-value goods such as grain, ore, coal and timber, because falling freight costs made them worth moving. 4. Because a coal-burning ship must refuel, so islands on a route acquired new value.

8.5 1. A message travelled at the speed of the person or vehicle carrying it.

1 Markets, government and war, with the examples from the text. 3. Because a

price difference could be seen and acted on immediately. 4. Open; credit an answer that gives a reason and a counter-argument.

8.6 1. It needed trained operators and charged by the word. 2. You can ask a question and get an answer while still connected. 3. Because it needed no operator and no code, so an untrained person could use it at home. 4. Because several inventors were working on it at once and priority was disputed in law.

8.7 1. Cast iron is hard and brittle; wrought iron is tough and soft; steel is both hard and tough. 2. The process: methods that made it in tonnes rather than kilograms, cheaply. 3. Rails, bridges, ship hulls, boilers, machine tools, tall building frames, any four. 4. Because a stronger boiler holds higher pressure, and higher pressure gives more power.

8.8 1. One engine drove a long shaft, and belts connected each machine to it.

1 Machines can be arranged for the work; small workshops can have power; one

machine can stop without stopping the rest. 3. It made evening work and study ordinary, and streets safer. 4. Electric trams and underground railways, and electrolysis for refining metals.

8.9 1. External burns fuel outside the cylinder to boil water; internal burns it inside, so the gases push the piston directly. 2. Because there is no boiler and no water to carry. 3. The motor car, the lorry and the aeroplane. 4. Because a coal country could usually find its own coal, and an oil economy must have oil or obtain it.

8.10 Open; acceptance criteria in the teachers' notes.

8.11 Enquiry; the required point is that time changed while distance did not.

Chapter 9: The United States becomes an industrial power

9.1 1. Coal, iron ore, timber, oil and agricultural land. 2. They connected ore in the north-west with coal in the east, and drained the interior to the sea.

1 Because moving both to one place is the expensive part, and water made it

cheap. 4. Because several countries with comparable resources did not become industrial powers, so resources are necessary and not sufficient.

9.2 1. Goods move without tariffs or customs, so one large factory can serve everybody. 2. A larger production run lowers the cost of each item. 3. Because people with income above subsistence buy manufactured goods. 4. The German lands had customs posts between states until unification; the United States had none.

9.3 1. Labour, skills and consumers. 2. Push: hunger, poverty, land shortage, persecution. Pull: wages, land, and family already settled. 3. Because they had least choice and faced hostility from those who had arrived earlier. 4. That its economy could not employ its own population.

9.4 1. Resources reached factories; produce reached cities; settlement followed the track; the railway was itself a huge industry. 2. Because they had been granted land along the route and wanted traffic. 3. It consumed steel, coal, timber and engineering on an enormous scale and raised capital on a new scale.

1 Because the same lines built a national market and carried the process that

displaced the Native American nations.

9.5 1. Chicago, railways meeting the Great Lakes; Pittsburgh, coal and rivers; Detroit, lakes and engineering; New York, port and finance. 2. Crowded housing and poor sanitation. 3. Cheap steel frames and the safety lift. 4. Because it could not spread far across water and land was expensive at the centre.

9.6 1. Owning every stage of production; he owned ore fields, coke works, transport and mills. 2. Every stage's profit became his, and no supplier could hold him up. 3. Very long shifts in dangerous conditions, and broken unions; the workers paid them. 4. Open; credit both sides.

9.7 1. He controlled refining, the stage everything passed through, rather than every stage. 2. A market with effectively one seller; it worries governments because the seller can set prices. 3. Buying rivals, and rail rates competitors could not match. 4. It was ordered to break into separate companies in 1911.

9.8 1. He organised invention as a business, in a laboratory with teams and funding. 2. Because it applies many people systematically to a chosen target.

1 Generation, cables, switches, measurement, billing: any three. 4. Because others

were working on it too and his contribution was a practical lamp plus the system to run it.

9.9 1. It moves the work past the worker, so each does one small operation repeatedly. 2. Because any part must fit any car without adjustment. 3. Lower cost, lower price, more buyers, more investment in the line. 4. Benefit: higher wages. Cost: repetitive, tightly paced work with little control.

9.10 Enquiry; the condition where Portugal was not at a disadvantage is position and access to the Atlantic.

Chapter 10: The expansion of industrial capitalism

10.1 1. The things used to produce other things: a loom, a factory, a lorry.

1. Because it must be built and equipped before the first item is sold.
2. Because a worker with a machine produces many times what the same worker

produces without one. 4. Because the worker needs the capital and the owner can choose among workers.

10.2 1. Factories, machines, land and materials belong to individuals or companies. 2. Risk, profit and decisions, each with an example. 3. Because property that might be seized will not be built. 4. The owner takes the risk, receives the profit and makes the decisions; the worker supplies labour for a wage.

10.3 1. Capital, a product, labour, materials and responsibility. 2. Owning is holding; combining is putting things together in a new arrangement. 3. Because history remembers the successes and most ventures failed. 4. Available capital, a legal system, education, a market, and a society that lets people rise: any three.

10.4 1. It takes deposits from savers and lends them to builders, taking the difference in interest. 2. A part-ownership of a company, which lets many people each fund a small part of a large project. 3. Because a railway costs more than any individual or family could provide. 4. Because many savers are exposed to the same distant enterprises, so a loss of confidence spreads at once.

10.5 1. A shareholder loses only what was invested; it makes it safe to buy a small part of a large firm. 2. Owners are scattered shareholders; managers are employees who run it. 3. Advantage: cheaper production and research. Danger: domination of a market. 4. Because it does not die or retire and can raise money from strangers.

10.6 1. Sell more cheaply, sell something better, or sell something nobody else offers. 2. It can be won, leaving a monopoly; or avoided, by agreement between rivals. 3. Because there is no longer anybody to be undercut by.

1. Because both failure modes remove the benefit competition was supposed to

produce.

10.7 1. Total wealth rose enormously, and it was distributed very unevenly.

1. Real wages eventually rose, and diets and life expectancy improved. 3. Because

people who share a workplace, a district and a grievance can organise. 4. If industry makes a society richer and more unequal at once, what should be done about it?

10.8 Enquiry; the required point is that infrastructure supplies transport and market access but not coal, capital or skills.

Chapter 11: Ideas that changed the nineteenth century

11.1 1. Individual liberty, constitutional government, equality before the law, representative government, property and economic freedom. 2. Because it challenged the basis on which absolute monarchs ruled. 3. Most liberals wanted the vote restricted to men with property; democracy came later and separately.

1. Because the meanings of political words change, and reading 1830 with a modern

dictionary gets the argument backwards.

11.2 1. A state is a political unit with borders and a government; a nation is a group who believe they share a language, history or culture. 2. It supplied the argument that separate states sharing a culture should be one country. 3. Because applied inside them it required those empires to break up. 4. Compulsory schooling in a national language, conscription, national holidays, national histories: any three.

11.3 1. That the inequality produced by industrial capitalism is neither acceptable nor inevitable. 2. Cooperatives, state ownership, regulation and welfare, and revolution. 3. Because it divided the movement permanently and produced two very different kinds of state in the twentieth century. 4. Limits on hours, safety laws, pensions, sickness insurance, free schooling: any three.

11.4 1. Means of production: the factories, machines and land used to make things. Bourgeoisie: those who own them. Proletariat: those who must sell their labour. Class struggle: conflict between them over the wealth production creates.

1 Workers produce more value than they receive in wages, and the difference is

the source of profit. 3. He predicted revolution in the most industrialised countries; it came first in Russia, which was among the least, and western countries reformed instead. 4. Because the analysis supplied concepts historians and economists still use, whatever they think of the prediction.

11.5 1. A single worker can be dismissed and replaced; all the workers cannot.

1 Combining was treated as conspiracy, and strikers were dismissed, evicted and

blacklisted. 3. Collective bargaining, the strike, and political action. 4. Limits on hours, restrictions on child labour, safety regulation, insurance, and the extension of the vote: any three.

11.6 Enquiry; the required point for the third part is that Portugal had a constitution and a parliament but a narrow franchise and unstable government.

Chapter 12: The world before 1914

12.1 1. It had won, in Italy and Germany; it had been adopted by governments; and it had turned competitive. 2. Schooling in a national language, conscription, national holidays, a national press: any three. 3. The first asks for independence; the second asks for superiority over others. 4. Because populations could be mobilised emotionally, so governments did not have to force reluctant subjects to fight.

12.2 1. Steel for guns and ships, coal for fleets, chemicals for explosives, railways for moving troops, factories convertible to war production: any four.

1 Germany. 3. Because a state unified only in 1871 had become the largest

industrial economy in Europe within a generation. 4. Because a falling power feels anxious and a rising one feels confident, and both make a crisis harder to settle.

12.3 1. Economic, raw materials and markets; strategic, coaling stations and routes; political, prestige; ideological, false theories of superiority.

1 Repeating rifles, machine guns, steamships and telegraphs gave small European

forces decisive advantages. 3. Borders drawn for European convenience, economies reorganised for export, coerced labour, destruction of existing political structures: any three. 4. That the participants regarded Africa as something to be divided among themselves rather than negotiated with.

12.4 1. Triple Alliance: Germany, Austria-Hungary, Italy. Triple Entente: France, Russia, Britain. 2. It enlarges a quarrel, removes room to manoeuvre, and encourages recklessness. 3. Because it can take risks knowing a great power has promised support. 4. Because a dispute between two members automatically involves all the others.

12.5 1. Because it was an island that fed itself by importing. 2. It made every earlier battleship obsolete, so the race restarted almost level. 3. A detailed plan of which trains carry which troops on which day; railways made speed decisive and therefore made planning necessary. 4. Because stopping meant tearing up a plan years in the making, and every staff believed the slower side would lose.

12.6 Enquiry; acceptance criteria in the teachers' notes.

12.7 Enquiry; the required chain runs from the scramble, through the claim and the ultimatum, to national humiliation, republican growth and 1910.

Chapter 13: Why did the First World War begin?

13.1 1. Because new states emerged with overlapping national claims, and two great powers had opposing interests there. 2. Because national self-determination applied inside a multi-national empire would dissolve it. 3. Protection of Slav peoples, and influence toward the Mediterranean. 4. Because they redrew the map and left every state dissatisfied.

13.2 1. 28 June 1914, Sarajevo, Archduke Franz Ferdinand. 2. The heir to the throne, in a province annexed six years earlier, on a day of national significance to Serbs. 3. Officials being involved allowed Vienna to blame Serbia, while the government not ordering it meant Serbia denied responsibility. 4. A trigger is the occasion; a cause is what made the occasion dangerous. In 1890 the same event would not have produced a general war.

13.3 Enquiry; the sequence and dates are in the figure, and the acceptance criteria are in the teachers' notes.

Chapter 14: Fighting the First World War

14.1 1. About seven hundred kilometres, from the Belgian coast to the Swiss border. 2. Machine guns, barbed wire, artillery that destroyed but could not hold ground, and railways that reinforced defenders faster. 3. Because shells cannot occupy a position, and the shelling made the ground harder to cross. 4. Because defenders moved reinforcements by rail while attackers advanced on foot over broken ground.

14.2 1. Front, support, reserve and communication trenches; they zigzagged so that a shell or a raider could not sweep the whole length. 2. Standing water and trench foot, lice, rats, contaminated conditions: any three. 3. Because standing in cold water caused trench foot, which could end in amputation. 4. Because letters were censored and most soldiers deliberately spared their families the worst.

14.3 1. Artillery, machine guns, gas, tanks, aircraft, submarines, with an effect for each. 2. Because armies could not fire without shells, so factory output became a political question. 3. That factories, coal, steel, shipping and food mattered as much as armies. 4. Making munitions, and suffering blockade and bombardment.

14.4 1. The Middle East, East Africa, West Africa, the Atlantic, the Indian Ocean, the Pacific: any four. 2. Cutting an enemy's imports by sea; both sides used one because both depended on trade. 3. Open; credit answers noting that colonial troops had least political voice at home. 4. Loss of markets, blockade, refugees, and pressure to abandon neutrality.

14.5 1. Unrestricted submarine warfare, and the Zimmermann telegram. 2. Because an army had to be raised and shipped, while food, steel, munitions and credit arrived at once. 3. Revolution in 1917 ended Russian participation, and the new government made peace in 1918. 4. Because Germany could move divisions west for a final offensive, and when it failed the reinforced Allies advanced.

14.6 Enquiry; the comparison with Spain should conclude that neutrality avoided the human and economic cost that helped end the First Republic.

Chapter 15: The consequences of the First World War

15.1 1. Deaths, permanent wounds, and civilian deaths from bombardment, occupation, hunger and deportation. 2. The influenza pandemic of 1918 and 1919.

1 It changed from a debtor into the world's creditor. 4. Because record keeping

differed, missing is not the same as killed, and civilian deaths are hard to attribute; name the source and say it is an estimate.

15.2 1. Russian, by revolution; German, by abdication; Austro-Hungarian, by dissolution; Ottoman, by partition. 2. Poland, Czechoslovakia, Yugoslavia, Finland, the Baltic states, Austria, Hungary: any four. 3. Because the populations were too mixed for any line to separate them. 4. Because it was applied to the defeated empires but not to the victors' colonies.

15.3 1. German losses in the east and the Polish corridor; Alsace and Lorraine to France; the dissolution of Austria-Hungary. 2. A strip giving Poland access to the sea, separating East Prussia from the rest of Germany. 3. Because national feeling does not disappear when a line moves. 4. Acceptance criteria: the answer must say that each border was produced by a datable event.

15.4 1. Army capped at 100 000, conscription forbidden, tanks, military aircraft and submarines prohibited, Rhineland demilitarised: any three. 2. The war guilt clause; legally it justified reparations, politically it was the term Germans resented most. 3. Too harsh: it humiliated a country that had to be lived with. Too soft: Germany kept its industry and its unity. 4. Because Germany did not negotiate and was presented with the terms to sign.

15.5 1. An attack on one member is treated as a concern of all. 2. Settling smaller disputes, running mandates, refugee relief, public health: any two. 3. The United States never joined; unanimity was required; it had no force; great powers left. The choice is the pupil's, with a reason. 4. Because it can only act through what its members are willing to do.

15.6 Enquiry; the chain runs from war costs through inflation and instability to the coup of 1926.

Chapter 16: The interwar world

16.1 1. Because millions of soldiers had to be absorbed into economies organised for war, and reparations poisoned relations. 2. Germany printed money to pay workers during the Ruhr occupation; people holding savings in cash lost them.

1 Currencies stabilised, and American loans and restructured reparations

restored production. 4. Recovery depended on American credit, which could stop.

16.2 1. Confidence falls, sales fall, production is cut, workers are sacked, so spending falls further. 2. American loans stopped and were called in, and tariffs cut world trade. 3. Because if every country protects its own producers, every country's exports fall. 4. Because people who conclude the system has failed look for parties promising to replace it.

16.3 1. Fear, anger and a demand for action. 2. A party accepts that opponents are legitimate; a movement demanding unity does not. 3. New or fragile democracies, and elites who preferred extremists to the alternative. 4. Because several democracies faced the same depression and survived it.

16.4 1. Heavy casualties, gains less than promised, economic difficulty and unstable politics: any three. 2. Because they feared revolution more than they feared him. 3. He was appointed rather than seizing power, which is the pattern to notice. 4. Opposition parties banned, free press ended, elections managed, opponents imprisoned.

16.5 1. The resentment of Versailles, the memory of 1923, and the depression.

1 Because unemployment rose sharply and the party offered work, revival and

enemies to blame. 3. He was appointed chancellor by the president in an arrangement made by conservatives who believed they could control him.

1 Propaganda persuaded many; terror silenced most of those it did not persuade.

16.6 1. State-directed output targets; heavy industry grew enormously and the Soviet Union became a major industrial power within about a decade. 2. Merging peasant farms into state-directed units; it was imposed against fierce resistance and millions died in the famine that accompanied it. 3. Because it had little access to foreign capital and chose to fund industry internally. 4. Because their ideologies, aims and victims differed, and they fought each other.

16.7 Enquiry; acceptance criteria in the teachers' notes.

16.8 Enquiry; the required difference is that the Estado Novo was conservative rather than revolutionary and did not pursue expansion in Europe.

Chapter 17: Why did the Second World War begin?

17.1 1. Territorial losses, military restrictions and the war guilt clause.

1 Because a radical movement could speak for a grievance far beyond its own

supporters. 3. Revision: the Sudetenland, where the population was largely German speaking. Conquest: Prague in March 1939, where it was not. 4. That the treaty supplied a widely shared grievance, but the ambition to conquer came from the regime.

17.2 1. 1935 rearmament; 1936 Rhineland; 1936 to 1939 Spain; March 1938 Austria; September 1938 Munich; March 1939 Prague; August 1939 the pact; 1 September 1939 Poland. 2. Because the force sent was small and could have been pushed back. 3. Its border defences, which lay in the territory transferred.

1 Because it removed the risk of a war on two fronts.

17.3 Enquiry; acceptance criteria in the teachers' notes.

Chapter 18: The Second World War: a global conflict

18.1 1. Germany advancing 1939 to 1941; the war becomes global in 1941; the Allies advancing 1942 to 1945. 2. The air superiority a German invasion would have required. 3. Because the Soviet Union and the United States both entered, and the balance of production turned decisively. 4. Stalingrad in the east, and the Normandy landings in the west.

18.2 1. Because it was the short route carrying oil and troops from Asia, Africa and Australia. 2. Supply: fuel, water, spare parts and ammunition brought hundreds of kilometres. 3. Because it sits in the middle of the Mediterranean and controlled which supplies reached North Africa. 4. Securing North Africa allowed the invasion of Sicily and then Italy.

18.3 1. Because Britain could not feed or arm itself without imports.

1 Merchant ships sail in escorted groups, which makes them harder to find and

more dangerous to attack. 3. The area in mid-ocean beyond the range of shore-based aircraft; it mattered because submarines could operate there on the surface. 4. Radar and sonar, signals intelligence, and shipbuilding: any two.

18.4 1. In 1931 with the invasion of Manchuria, and from 1937 in China.

1 Because embargoes cut its access to oil and materials, so it attacked to seize

them. 3. It disabled part of a fleet, and it brought into the war the largest industrial economy on Earth. 4. Distances made fuel, airfields and supply decisive, and carriers replaced battleships.

18.5 Enquiry; acceptance criteria in the teachers' notes.

18.6 Enquiry; the answer must give the Azores' mid-Atlantic position and link it to the air gap in topic 18.3.

Chapter 19: The Holocaust

19.1 1. Hostility and discrimination directed at Jewish people; the Nazi state made it central to its purpose and gave it the machinery of a modern government.

1 They stripped German Jews of citizenship and forbade marriage between Jews and

other Germans. 3. Because it shows that it required the participation of many ordinary officials, and that it was not hidden. 4. Because it borrowed the appearance of biology without any of its evidence, and was rejected by scientists.

19.2 1. Organised attacks on synagogues, homes, shops and schools, and mass arrests; the name suggests spontaneous damage, and it was organised. 2. Because it made clear there was no future in staying. 3. A sealed, overcrowded district deliberately under-supplied with food, fuel and medicine; people died of starvation and disease. 4. Armed uprisings, escape networks, hidden schools, hidden archives: any two.

19.3 1. From 1941 mass shootings behind the advance; from 1942 systematic deportation to camps built for killing. 2. That it was administered with timetables, registers, budgets and correspondence. 3. That it used the transport, organisation and technology of the modern age. 4. German records, survivor testimony, perpetrator testimony at trial, Allied and Soviet documentation, and the sites themselves.

19.4 Open; see the source panel and the enquiry criteria.

19.5 Open; this is a discussion question with no single answer, and the acceptance criteria are in the teachers' notes.

Chapter 20: The end and consequences of the Second World War

20.1 1. A front in the west, which the Soviet Union had long demanded.

1 Artificial harbours, because no port was captured; fuel pipelines; a deception

campaign about where the landing would come. 3. The Allied advance from the west and the Soviet advance from the east. 4. 8 May 1945.

20.2 1. Enormously costly in lives on both sides. 2. That an invasion would have cost more lives, and that Japan had not surrendered after months of conventional bombing. 3. That Japan was already close to defeat, and that the cities held very large civilian populations. 4. That the world now contained a weapon able to destroy a city instantly.

20.3 1. About sixty million; ranges are given because records differ and civilian deaths are hard to attribute. 2. Because American territory was untouched and its industry had grown, while the Soviet army occupied eastern Europe.

1 That the line between the occupying armies became the political frontier of

Europe. 4. Because the colonial powers had been defeated, occupied or exhausted, and their claim to permanence had been disproved.

20.4 1. That an attack on one member is a concern of all. 2. The great powers joined; the Security Council can take binding decisions; five permanent members have a veto; specialised agencies were built in: any two. 3. The four listed in the text; the choice is the pupil's with a reason. 4. Because sanctions and force can only be applied by member states.

Chapter 21: A divided world

21.1 1. Because the powers that had run the world in 1900 were exhausted, defeated or occupied, leaving two states in a different class. 2. United States: untouched territory, larger industry, gold reserves, a nuclear monopoly until

1 Soviet Union: the largest army in the world, standing in the middle of

Europe. 3. That it had been invaded from the west twice in thirty years and wanted governments that could not be used against it again. 4. Because the same action, installing friendly governments, is security to the state doing it and expansion to the state watching.

21.2 1. Western: competing parties, independent courts, a press not owned by the state, private ownership, prices set by markets. Eastern: one party with a leading role, state ownership, planned production, censorship. 2. Western criticism: the purges, the camps, no elections. Eastern criticism: segregation and racial discrimination, unemployment, support for dictatorships and colonial wars. 3. Because western Europe had large socialist parties and several dictatorships, and Yugoslavia was communist and broke with Moscow. 4. To describe accurately what each claimed, what each delivered and how each looked to the other, rather than to award marks.

21.3 1. Where the armies stopped in 1945. 2. American aid to rebuild western European industry, and to make communist parties less attractive by improving living standards. 3. A fortified, watched and largely closed frontier across the continent. 4. Because the two blocs competed by supporting opposite sides in conflicts elsewhere, which were not cold at all.

21.4 1. 1949; any four of the United States, Canada, Britain, France, Italy, the Netherlands, Belgium, Luxembourg, Denmark, Norway, Iceland, Portugal. 2. An armed attack on one member is treated as an attack on all. 3. It is a smaller group of states that actually agree, and it has forces and joint plans behind it.

1 Because the alliance depended on reinforcing Europe across the Atlantic.

21.5 1. 1955; the admission of West Germany to NATO. 2. Both were treaties of mutual defence among sovereign states. 3. Hungary in 1956 and Czechoslovakia in

1 4. Because it removed what had kept several eastern European governments in

place.

21.6 Enquiry; the answer must give the Azores and the Atlantic position as what Portugal had, and note that each side gave up something to get it.

Chapter 22: Berlin, symbol of the Cold War

22.1 1. Germany into four occupation zones, and Berlin into four sectors inside the Soviet zone. 2. Because the occupying powers wanted different Germanies. 3. The western zones became the Federal Republic and the Soviet zone the German Democratic Republic. 4. Because the western sectors lay more than a hundred and fifty kilometres inside the eastern state, reachable only by agreed corridors.

22.2 1. The introduction of a new currency in the western zones. 2. Road, rail and canal routes; the aim was to force the western allies out of the city.

1 Because forcing the routes open risked war, and doing nothing meant losing the

city. 4. Because it put the choice of starting a war onto the other side.

22.3 1. Because the air corridors were covered by a separate written agreement.

1 Coal, because the city heated and powered itself with it. 3. Because circling would break the fixed pattern of approaches. 4. It showed both sides would take great trouble to avoid shooting; it changed how West Germans saw the western powers; and it hardened the division, with NATO signed and two German states founded.

22.4 1. Because living standards rose faster in the west and the sector border in Berlin was still crossable. 2. Because those leaving were disproportionately young, qualified and professionally trained. 3. 13 August 1961; it became two walls with a patrolled strip, watchtowers, lights and guards. 4. Because a wall built to keep people in, in a city where both sides were visible, made the argument against the state that built it.

Chapter 23: Cold War crises

23.1 1. In 1945, along the 38th parallel, by Soviet and American forces accepting the Japanese surrender. 2. Because the Soviet Union was boycotting the Security Council and so did not veto. 3. China entered the war in large numbers and pushed the coalition back. 4. Superpower conflict fought by others; limits observed, including no nuclear use; and 1945 divisions hardening into permanent states.

23.2 1. Because missiles there could reach much of the United States with very little warning. 2. A naval blockade to stop further military shipments; it was chosen because an air strike or invasion risked immediate war. 3. Soviet withdrawal of the missiles; an American promise not to invade Cuba; and a private agreement to withdraw American missiles from Turkey. 4. A direct communication link between the capitals, and a treaty banning atmospheric nuclear testing.

23.3 1. After the defeat of France in 1954 it was divided into a communist north and a non-communist south. 2. That if one country in a region became communist its neighbours would follow. 3. Terrain unsuited to mechanised forces; guerrilla methods that presented no target; supply routes through neighbouring countries that were hard to cut. 4. Film reached homes within a day or two, and public support fell as the war continued.

Chapter 24: The nuclear arms race

24.1 1. 1949; it ended the American monopoly and began the arms race proper.

1 Fission splits heavy atoms; a thermonuclear weapon uses a fission explosion to trigger fusion and is hundreds of times more powerful. 3. Britain, France and China. 4. That both sides held, in thousands, weapons able to destroy a city each.

24.2 1. Advantage: it can be recalled after launch. Disadvantage: it can be tracked and intercepted. 2. Decision time fell from hours to minutes, and launch could not be reversed. 3. Because a submarine at sea cannot be found reliably.

1 Because the attacked side can always strike back, so attacking gains nothing.

24.3 1. Capability, survivability and credibility, with an example of each.

1 That no war between the superpowers took place, and that both took extraordinary care to avoid direct fighting. 3. False alarms from faulty equipment and misread radar, and the Cuban crisis of 1962. 4. Open; the answer must state whether it has changed and what changed it.

Chapter 25: The Space Race

25.1 1. 4 October 1957; it transmitted a repeating radio signal and nothing else. 2. A shock to assumptions, a military signal, and a large response in funding and education. 3. Because a rocket that can orbit a satellite can deliver a warhead between continents. 4. That prestige competition had practical effects on ordinary life, including school curricula.

25.2 1. 12 April 1961; he completed one orbit of the Earth and returned safely. 2. Because it showed that the Soviet system could do what the richest country on Earth had not. 3. To land a person on the Moon and return them safely before the end of the decade. 4. Because it reset the contest at a point where the United States believed it could arrive first.

25.3 1. 20 July 1969; Armstrong and Aldrin walked on the surface, Collins remained in orbit. 2. Rocketry, materials, navigation, computing, communications, life support: any four. 3. A launchpad fire killed three astronauts and forced a redesign. 4. Because the achievement was designed to be witnessed, and about six hundred million people watched.

25.4 1. Prestige, military capability, science and technology, and education.

1. Because a treaty can be verified from orbit rather than by inspection on the

ground. 3. Weather forecasting, satellite navigation, satellite communications: any three. 4. Each step supplied a capability the next one needed, from steam through electricity and aircraft to rockets.

25.5 Enquiry; the four costs are human, economic, demographic and political, and the choice among them is the pupil's with evidence.

Chapter 26: A democratic revolution

26.1 1. The Armed Forces Movement, formed by career officers; it began over professional grievances and became political. 2. The night of 24 to 25 April 1974; radio signals, and units taking the ministries, airport, broadcasters and barracks. 3. Crowds came into the streets and put red carnations in the barrels of soldiers' rifles. 4. Censorship ended, political prisoners were released, and exiles returned and parties organised openly.

26.2 1. Guinea-Bissau, Mozambique, Cape Verde, Sao Tome and Principe, Angola: any four. 2. Because rival independence movements turned on each other with support from opposite sides of the Cold War. 3. About half a million people who came to Portugal from Africa within about two years. 4. Because the economy was in crisis and the country had about nine million people.

26.3 Enquiry; the required point is that the transition was contested between parliamentary and revolutionary directions and was settled through elections.

Chapter 27: European integration

27.1 1. How to stop France and Germany going to war again; defeating Germany in 1918 and occupying and dividing it had both been tried. 2. Because they are what armies are made of, so pooling them made rearming secretly impossible.

1. Economic recovery, the Cold War, and the loss of empires. 4. Because a

prosperous, integrated western Europe was more resistant to communist parties.

27.2 1. The European Economic Community; the common market. 2. A commission to propose and administer, a council of governments to decide, an assembly, and a court to interpret the treaties. 3. Britain, Ireland and Denmark in 1973; Greece in 1981; Portugal and Spain in 1986. 4. Because membership was not open to a dictatorship, so joining required becoming and staying a democracy.

27.3 Enquiry; the fourth heading requires the point that membership tied the 1976 settlement into external obligations that made reversal harder.

Chapter 28: Why did the Cold War end?

28.1 1. A slowing economy, technology falling behind, heavy military spending including Afghanistan, and restlessness in eastern Europe. 2. Because planners could order quantities of standard goods but not variety, quality or responsiveness. 3. Because the technologies that raise productivity also spread information, which the system tried to control. 4. An independent trade union in Poland with enormous membership; it mattered because suppression did not destroy it.

28.2 1. Saving the system by reforming it, not ending it. 2. Glasnost, openness, so that problems could be named and fixed; perestroika, restructuring, to address the economic problems. 3. That the Soviet Union would no longer use force to keep governments in power in eastern Europe. 4. Openness produced demands for a different system; restructuring disrupted planning without replacing it; and non-intervention removed the support several governments rested on.

28.3 1. That Soviet forces would not intervene. 2. Hungary opened its border with Austria, so East Germans could leave through it. 3. Peaceful: Poland, Hungary, East Germany, Czechoslovakia, Bulgaria. Violent: Romania. 4. That each success made the next more likely by showing these governments would not be rescued.

Chapter 29: The fall of the Berlin Wall

29.1 1. 9 November 1989; new travel rules, which an official said, without clear instructions, took effect immediately. 2. Because they had no orders, could not reach anybody with authority, and faced growing crowds. 3. Because it followed an ambiguous announcement and individual decisions on the spot, not a decision to open it. 4. Because the causes lie in chapter 28 and the event was produced by an accident of communication.

29.2 1. 3 October 1990, eleven months after the wall opened. 2. Because the eastern states joined the existing Federal Republic under its existing constitution. 3. Because Germany was still formally subject to the rights of the four occupying powers of 1945. 4. Eastern industry largely closed and unemployment rose; very large transfers were made from west to east.

29.3 1. National movements in the republics could speak and organise.

1 Hardliners attempted a coup against Gorbachev; its failure destroyed the

authority of the institutions behind it and accelerated the collapse. 3. December 1991; fifteen states. 4. Output fell sharply and savings were destroyed by inflation; in places the end of the old order was accompanied by war.

29.4 Enquiry; the two profiles must differ on government, empire, alliances, economy and press freedom.

Chapter 30: The world after the Cold War

30.1 1. Two superpowers, two alliances, and most questions read as belonging to one side or the other. 2. One superpower remained; conflicts stopped being read through one lens; international institutions were asked to do more.

1 Yugoslavia broke apart in a series of wars; it is significant because it

happened in Europe, forty-five years after 1945. 4. Because the routine veto of two hostile permanent members ended; the veto itself still limits it.

30.2 1. The Soviet Union became fifteen states; Yugoslavia became seven; Czechoslovakia became two; Germany became one. 2. Yugoslavia separated after violence; Czechoslovakia separated by agreement. 3. Borders change when the state holding them loses a war, loses its authority, or agrees to change.

1 A line can be redrawn while national feeling stays where it was.

30.3 1. Global political, the United Nations; regional, the European Union; military, NATO; economic, the bodies that set trade rules. 2. Its members have transferred real powers to common institutions and to a court. 3. Because some problems cannot be solved by one country acting alone, and because states gain weight by acting together. 4. That every one involves accepting limits on what a state may do alone, which is the descendant of the nationalism of chapter 11.

Chapter 31: Globalisation

31.1 1. The increasing connection of economies and societies across the world.

1 International trade, multinational companies and supply chains. 3. Because each

stage happens wherever it is cheapest, which lowers cost and adds links that can each break. 4. The narrow point through which everything passes, such as a canal or a single component supplier.

31.2 1. Push: war, persecution, poverty, unemployment, disaster. Pull: work, higher wages, safety, education, family already settled. 2. A migrant moves to live in another country for any reason; a refugee has fled a well-founded fear of persecution or war and has a specific status in international law. 3. Receiving: workers and taxes; pressure on housing and services. Sending: loss of working-age people; remittances. 4. Money sent home by people working abroad; in some economies it is a very large share of national income.

31.3 1. The telegraph separated information from transport; digital communication completed the separation and made it nearly free. 2. Cost, volume and speed, and who can publish. 3. Because the checks that came with a publisher do not automatically come with a post. 4. It runs on cables following particular routes across ocean floors and on data centres built where power is cheap.

31.4 Enquiry; acceptance criteria in the teachers' notes.

Chapter 32: Transport and communication

32.1 1. Places do not move, but the time between them shrinks, so the practical geography changes while the map does not. 2. Because carriage cost more than the goods were worth. 3. Because a merchant can plan around a ship that keeps a schedule, but not around one that is fast and unpredictable. 4. Each step lowered the cost per tonne-kilometre, and each time a new class of goods became worth moving.

32.2 Enquiry; the conclusion must explain that centrality changes with transport cost while position does not.

Chapter 33: The political geography of the modern world

33.1 1. A state is a territory with a population, a government and the capacity for external relations; a nation is a group who believe they share something; a nation-state is a state whose borders match a nation. 2. Because most states contain more than one national group and several nations are spread across more than one state. 3. Physical features; straight lines imposed by outsiders; where an army stopped. 4. Because it has been largely settled since the thirteenth century, which is unusual in Europe.

33.2 1. German, Austro-Hungarian, Russian and Ottoman. 2. It was created in 1919 and then shifted bodily westward in 1945, losing land to the Soviet Union and gaining land from Germany. 3. The line between the blocs; it shows that a frontier held by two alliances can be more stable than one held by treaty. 4. 1914 empires; 1919 Versailles and the successor states; 1945 war and occupation; Cold War division; 1991 collapse and agreement.

33.3 Enquiry; the answer must identify one of the five causes and say who was not consulted.

Chapter 34: Bringing geography and history together

34.1 1. Coal near the surface, iron ore near the coal, navigable rivers and harbours, rainfall and fast streams, Atlantic position. 2. China, which had coal.

1. Excellent for ocean exploration and empire; poor for coal-based industry.
2. Geography makes some things cheaper and others dearer; people decide whether to

pay.

34.2 1. Continental size, internal waterways, agricultural land, two ocean coasts. 2. Because the Lakes joined northern ore to eastern coal by water, which is what made cheap iron possible in chapter 6. 3. Argentina and Brazil.

1. Because those countries had comparable endowments and followed different paths.

34.3 1. Any six examples from the text. 2. Because no port had been captured and supply had to come across open beaches. 3. Germany drove toward the Caucasus and Japan into south-east Asia to secure oil. 4. In 1943 the Azores closed the mid-Atlantic air gap; in 1949 they made Portugal valuable enough to be a founder member of NATO.

34.4 1. Portugal is on the western edge of Europe, facing the Atlantic. 2. In 1500 the Atlantic was the route to everywhere; in 1850 industry rewarded coal, navigable interiors and large home markets, none of which Portugal had. 3. Because the Atlantic became the decisive space in the Battle of the Atlantic and then in the reinforcement of Europe. 4. Because what a position is worth depends on what the world is doing at the time.

34.5 1. Where the armies stopped in 1945. 2. Greece is east of Prague and was in the western bloc; Australia is far east and was counted in the West.

1. Both mattered because of distance to a superpower's territory. 4. Hungary

opened its border with Austria, so East Germans could leave, which made the wall in Berlin pointless before it was opened.

34.6 and 34.7 Enquiries; acceptance criteria in the teachers' notes.

Glossary

Every word this book teaches, defined as it is used here, with the unit that introduces it. 243 entries.

absolute location	A position given as a coordinate pair, naming one point on Earth and no other. Unit 2
accession	The process by which a state joins an organisation such as the European Community. Unit 27
Agricultural Revolution	The changes in farming, including rotations, breeding and enclosure, that raised output before industrialisation. Unit 5
air gap	The area of mid-ocean beyond the range of shore-based aircraft, closed in 1943 partly from the Azores. Unit 18
airlift	The supply of West Berlin by air during the blockade of 1948 and 1949. Unit 22
annexation	Taking territory into a state, usually without the agreement of those who live there. Unit 17
Antarctic Circle	The parallel at about 66.5 degrees south, the southern equivalent of the Arctic Circle. Unit 1
antisemitism	Hostility and discrimination directed at Jewish people. Unit 19
appeasement	The policy of granting an aggressor's demands in order to avoid war. Unit 16
Arctic Circle	The parallel at about 66.5 degrees north, inside which the Sun does not set on at least one day a year. Unit 1
armistice	An agreement to stop fighting, which is not the same as a peace treaty. Unit 14
arms control	Agreements limiting the testing, numbers or types of weapons. Unit 24
arms race	A competition in which each state increases its forces because the other is doing so. Unit 12
assembly line	A system moving the work past the worker, each doing one small operation, which cut production time enormously. Unit 9
attrition	A strategy of wearing an enemy down by inflicting continuous losses. Unit 14
axial tilt	The lean of the Earth's axis, about 23.5 degrees, which does not change through the year and causes the seasons. Unit 1
axis	The imaginary line through the Earth about which it turns, tilted at about 23.5 degrees. Unit 1
Berlin Conference	The meeting of 1884 to 1885 at which European powers agreed rules for their claims in Africa, with no African ruler present. Unit 12

Berlin Wall	The fortified barrier built from 1961 around the western sectors of Berlin, to stop people leaving. Unit 22
biodiversity	The variety of living things in a place, counted as species or as the range of forms. Unit 3
biome	A large region of the Earth defined by its climate, its vegetation and the animals that live in it. Unit 3
bloc	A group of states aligned with one another against another such group. Unit 21
blockade	Cutting off an enemy's supplies, usually by sea. Unit 14
bond	A loan to a company or a government, repaid with interest over an agreed period. Unit 10
border	The line where one state's authority ends and another's begins. Unit 33
bottleneck	The stage in a process that limits the whole of it, and which the next invention is usually built to clear. Unit 8
bourgeoisie	In Marxist analysis, those who own the means of production. Unit 11
brinkmanship	Pushing a crisis close to war in order to force the other side to give way. Unit 23
British Ultimatum	The British demand of 1890 that Portugal withdraw from territory between Angola and Mozambique. Unit 11
capital	In economics, the things used to produce other things, and the money used to buy them. Unit 6
Carnation Revolution	The Portuguese revolution of 25 April 1974, named for the flowers put in soldiers' rifles. Unit 26
causation	The relationship between an event and what produced it, which historians argue about by weight rather than by fact. Unit 34
censorship	Official control of what may be printed, broadcast or performed. Unit 16
central planning	An economic system in which the state, rather than prices, decides what is produced. Unit 28
centrality	How well connected a place is to the rest of a network. Unit 32
choke point	A narrow place through which a great deal of traffic must pass, and which is therefore worth holding. Unit 18
chronometer	An accurate clock built to keep reference time at sea, which is what made longitude measurable on a voyage. Unit 2
class struggle	Conflict between classes over the wealth that production creates. Unit 11

climate	The pattern weather makes over a long period, usually thirty years or more. It is described, not forecast. Unit 3
coaling station	A port or island where a coal-burning ship refuelled, which gave some small places great strategic value. Unit 8
coke	Fuel made by baking coal, which replaced charcoal in smelting iron and freed it from the supply of timber. Unit 6
collective bargaining	Negotiating terms for all the workers at once rather than one at a time. Unit 11
collective defence	The commitment of allies to treat an attack on one as an attack on all. Unit 21
collective security	The principle that an attack on one member of an organisation is a concern of all. Unit 15
collectivisation	The merging of peasant farms into large state-directed units, imposed against resistance and accompanied by famine. Unit 16
Colonial War	The wars fought by Portugal in Angola, Guinea-Bissau and Mozambique from 1961 to 1974. Unit 25
colony	A territory controlled and settled or administered by another state. Unit 12
common land	Land that villagers had the right to use, for grazing or fuel, without owning it. Unit 5
common market	An area in which goods, services, capital and people move without tariffs or restrictions. Unit 27
competition	Rivalry between sellers for buyers, which pushes firms to lower prices or improve products. Unit 10
conscription	Compulsory service in the armed forces. Unit 12
constituent assembly	A body elected to write a constitution. Unit 26
constitution	A document setting out what a state may and may not do, and how it is governed. Unit 11
container	A standard steel box that moves between ship, train and lorry without being unpacked. Unit 32
containment	The policy of preventing the spread of an opposing system rather than attacking it. Unit 23
continental climate	A climate far from the sea, with hot summers and cold winters. Unit 4
convoy	A group of merchant ships sailing together under escort. Unit 18
Coordinated Universal Time	The world time standard, kept by atomic clocks, from which every time zone is described as an offset. Unit 2
corporation	A company treated in law as a body separate from the people who own it. Unit 10

corporatism	The organisation of employers and workers into state-supervised bodies in place of free trade unions. Unit 16
crop rotation	Growing different crops in a field in sequence so that the soil is not exhausted and no year is wasted. Unit 5
decolonisation	The process by which colonies became independent states, mostly after 1945. Unit 25
density	How many of something there are in a given area, such as people per square kilometre. Unit 4
deportation	Forced removal of people from their homes to another place, in this period by rail to camps. Unit 19
desert	A region receiving under about 250 millimetres of rain a year. The definition is about rainfall, not heat. Unit 3
deterrence	Preventing an action by making its consequences unacceptable. Unit 17
displaced person	Somebody forced from their home and unable to return, of whom there were millions after 1945. Unit 20
dissolution	The formal ending of a state, as of the Soviet Union in December 1991. Unit 29
distribution	How something, usually population, is spread across an area. Unit 4
division of labour	Splitting work into small repeated tasks done by different people. Unit 7
domestic system	Manufacturing done in people's homes for a merchant who supplies the materials and collects the product. Unit 5
domino theory	The belief that if one country in a region became communist, its neighbours would follow. Unit 23
emigration	People leaving a country to live elsewhere. Unit 9
enclosure	The division and fencing of open fields and common land into private farms. Unit 5
entrepreneur	Somebody who brings together capital, labour, materials and a product idea, and takes the risk. Unit 10
Equator	The imaginary line half way between the poles, at zero degrees latitude, fixed by the shape of the Earth. Unit 1
escalation	The process by which a dispute becomes larger and harder to stop. Unit 13
Estado Novo	The Portuguese authoritarian regime established by the Constitution of 1933 and led by Salazar. Unit 16
Factory Acts	Laws limiting hours, setting minimum ages and eventually appointing inspectors to enforce them. Unit 7

factory system	Production concentrated in one building, with machines owned by the employer and hours set by the employer. `Unit 7`
fascism	A movement placing the nation above the individual, demanding a single leader and rejecting both liberal democracy and communism. `Unit 16`
first strike	An attack intended to destroy an enemy's weapons before they can be used. `Unit 24`
fission	Splitting heavy atoms to release energy, as in the first atomic weapons. `Unit 24`
five-year plan	A Soviet plan setting output targets for industry, directed by the state rather than by markets. `Unit 16`
Fontismo	The policy of material modernisation associated with Fontes Pereira de Melo. `Unit 10`
franchise	The right to vote, and the rules deciding who has it. `Unit 11`
generator	A machine that produces electricity, allowing power to be made in one place and used in another. `Unit 8`
genocide	The deliberate destruction of a people. `Unit 19`
ghetto	A sealed and overcrowded district into which Jewish communities were forced, deliberately under-supplied. `Unit 19`
glasnost	Openness: the relaxation of censorship in the Soviet Union from 1985. `Unit 28`
globalisation	The increasing connection of economies and societies across the world. `Unit 31`
grassland	A biome of continental interiors where rainfall is too low for forest, and the soils are often very productive. `Unit 3`
Great Depression	The worldwide economic collapse that began in 1929, producing mass unemployment. `Unit 16`
Greenwich Mean Time	The time zone kept by clocks in Britain in winter, and the ancestor of the modern world standard. `Unit 2`
growing season	The part of the year in which conditions allow plants to grow. `Unit 3`
guerrilla warfare	Fighting by avoiding pitched battle, moving among the population, striking and dispersing. `Unit 23`
hemisphere	Half of the Earth, divided either by the Equator or by a pair of meridians. `Unit 1`
Holocaust	The systematic murder of about six million Jewish people by the Nazi state and its collaborators. `Unit 19`
hyperinflation	Extremely rapid loss of a currency's value, which destroys savings held in cash. `Unit 16`
immigration	People arriving in a country to live. `Unit 9`

imperialism	The extension of one state's control over the territory, people and resources of another. Unit 12
industrial middle class	Factory owners, merchants, bankers and professionals, whose income came from profit on what they owned. Unit 7
industrial working class	People who sold their labour for a wage and owned no productive property. Unit 7
inequality	The uneven distribution of income and wealth within a society. Unit 10
integration	The joining of separate economies or political systems into a shared framework. Unit 27
interchangeable parts	Parts made accurately enough that any one will fit any product, without which an assembly line cannot work. Unit 9
intermodal	Moving the same load by more than one kind of transport without repacking it. Unit 32
internal combustion engine	An engine that burns fuel inside the cylinder, light enough to move itself and a load. Unit 8
internal market	The buyers inside one country, and inside one customs area, which lets a firm build one very large factory. Unit 9
International Date Line	The line, roughly along the 180th meridian, where the calendar date changes. Unit 2
investment	Putting money into an enterprise in the expectation of a return later. Unit 10
iron curtain	The fortified and largely closed frontier between the two blocs in Europe. Unit 21
Kristallnacht	The organised attacks of November 1938 on synagogues, homes, shops and schools, followed by mass arrests. Unit 19
latitude	How far north or south of the Equator a place is, measured as an angle from 0 to 90 degrees. Unit 1
League of Nations	The international organisation founded in 1920 to keep the peace, which lacked the United States and had no force. Unit 15
liberalism	In the nineteenth century, the demand for constitutions, parliaments, equality before the law and rights against the state. Unit 11
limited liability	The rule that a shareholder can lose only what was invested, which is what makes raising money from strangers possible. Unit 10
locomotive	A steam engine on wheels, carrying its own power along a track. Unit 8
longitude	How far east or west of the Prime Meridian a place is, measured as an angle up to 180 degrees. Unit 1
Maastricht Treaty	The treaty of 1992 that created the European Union and set out the path to a single currency. Unit 29

mandate	A territory administered by one state on behalf of the League of Nations after the First World War. Unit 15
maritime climate	A climate near the sea, whose extremes are moderated by water. Unit 4
Marshall Plan	Large American economic aid to western Europe, to rebuild industry and to reduce the appeal of communist parties. Unit 21
Marxism	The analysis of Marx and Engels, built on the means of production, class and class struggle. Unit 11
mass production	Making goods in very large quantities at low cost per item. Unit 8
means of production	The factories, machines, land and materials with which things are made. Unit 11
Mediterranean climate	A climate with mild wet winters and hot dry summers, in which plants are adapted to drought in the growing season. Unit 3
meridian	A line running from the North Pole to the South Pole. All meridians are the same length. Unit 1
midnight Sun	A period inside a polar circle when the Sun does not set at all. Unit 1
migrant	Somebody who moves to live in another country, for any reason. Unit 31
minority	A national or linguistic group living inside a state governed by another group. Unit 15
mobilisation	Bringing armed forces to readiness for war, usually on a fixed railway timetable. Unit 12
monopoly	A market with effectively one seller, who can therefore set the price. Unit 9
Movimento das Forcas Armadas	The Portuguese Armed Forces Movement, which overthrew the Estado Novo on 25 April 1974. Unit 26
multinational company	A company operating in several countries at once. Unit 31
mutually assured destruction	The situation in which neither side can attack without being destroyed in return. Unit 24
nation	A group of people who believe they share a language, a history or a culture. Unit 11
nation-state	A state whose borders match a nation, which is less common than the phrase suggests. Unit 33
nationalisation	The taking of private industry into state ownership. Unit 26
nationalism	The belief that the world is divided into nations, and that each should govern itself. Unit 11
NATO	The North Atlantic Treaty Organisation, founded in 1949, in which an attack on one member is an attack on all. Unit 21

navigable	A river or channel deep and clear enough for boats to use, which makes heavy transport cheap. `Unit 6`
Nazism	The German movement that added a racial theory of the state, and antisemitism, to the features of fascism. `Unit 16`
necessary condition	Something without which an outcome could not happen, but which is not enough on its own. `Unit 34`
neutrality	The position of a state that takes no side in a war. `Unit 13`
no man's land	The ground between two opposing trench systems. `Unit 14`
non-aggression pact	A treaty in which two states agree not to attack each other. `Unit 17`
Nuremberg Laws	The German laws of 1935 that stripped Jews of citizenship and forbade marriage between Jews and other Germans. `Unit 19`
occupation zone	One of the four areas into which Germany was divided in 1945. `Unit 22`
offset	The number of hours a time zone differs from Coordinated Universal Time. `Unit 2`
orbit	The path of an object around another under gravity. `Unit 25`
overseas province	The Estado Novo's term for its African territories, used to argue they were not colonies. `Unit 25`
parallel	A line of equal latitude. Parallels never meet and get smaller toward the poles. `Unit 1`
partition	The division of a territory into two or more parts, usually imposed. `Unit 33`
patent	A legal right giving an inventor control of an invention for a period, so that it can be profited from. `Unit 6`
peacekeeping	The deployment of international forces to hold a ceasefire or protect a settlement. `Unit 30`
perestroika	Restructuring: the attempted economic reform of the Soviet system from 1985. `Unit 28`
periphery	A place on the edge of a network, far from its centres of activity. `Unit 32`
permafrost	Ground that has stayed below freezing for at least two years, through which water cannot drain. `Unit 3`
polar night	A period inside a polar circle when the Sun does not rise at all. `Unit 1`
Portuguese Expeditionary Corps	The Portuguese force sent to the Western Front, which fought at La Lys in April 1918. `Unit 14`
power loom	A machine that mechanised weaving, so cloth could keep pace with the new supply of thread. `Unit 8`

Prime Meridian	The meridian at zero degrees longitude, running through Greenwich, agreed internationally in 1884. `Unit 1`
private property	Ownership of productive things by individuals or companies rather than by the state or the community. `Unit 10`
proletariat	In Marxist analysis, those who own no productive property and must sell their labour. `Unit 11`
propaganda	Material produced to persuade rather than to inform, often by a state. `Unit 16`
property rights	The legal protection of ownership, without which nobody builds anything they might lose. `Unit 6`
protectionism	The use of tariffs and restrictions to shield domestic producers from foreign competition. `Unit 31`
provisional government	A temporary government holding office until a permanent settlement is agreed. `Unit 26`
proxy war	A conflict in which two powers support opposite sides rather than fighting each other directly. `Unit 23`
public debt	Money owed by a state, often borrowed to pay for infrastructure, and carrying interest every year. `Unit 10`
purge	The removal, imprisonment or killing of members of a party, government or army by their own state. `Unit 16`
rain shadow	The dry area behind a mountain range, where air has already lost its moisture. `Unit 3`
rainforest	A tropical biome with heavy rain in every month, a closed canopy and the highest biodiversity on land. `Unit 3`
refugee	Somebody who has fled their country from a well-founded fear of persecution or war, with a status in international law. `Unit 31`
Regeneration	The period of Portuguese politics from 1851 that prioritised roads, railways, ports and communications. `Unit 10`
relative location	A position described by its relationship to another place rather than by coordinates. `Unit 2`
remittance	Money sent home by somebody working abroad. `Unit 9`
reparations	Payments required from a defeated state for war damage. `Unit 15`
retornados	The roughly half a million people who came to Portugal from the former African territories in 1974 and 1975. `Unit 26`
reunification	The joining of East and West Germany into one state on 3 October 1990. `Unit 29`

rotation	The turning of the Earth about its axis, once in about twenty-four hours, from west to east. `Unit 2`
sanitation	The removal of waste and the supply of clean water, on which urban health depends. `Unit 7`
satellite	An object placed in orbit around the Earth. `Unit 25`
satellite state	A nominally independent state whose government is controlled from outside. `Unit 21`
savanna	A tropical biome of grassland with scattered trees, produced by a marked wet season and dry season. `Unit 3`
second strike	The capacity to strike back after being attacked, which is what makes deterrence work. `Unit 24`
Security Council	The body of the United Nations that can take binding decisions, on which five states hold a veto. `Unit 20`
self-determination	The principle that a people should decide its own government. `Unit 11`
separate condenser	Watt's improvement, condensing steam in a second vessel so the working cylinder stayed hot and less fuel was used. `Unit 8`
share	A part-ownership of a company, which lets many people each fund a small part of a large enterprise. `Unit 10`
smelting	Heating ore to extract the metal from it. `Unit 6`
socialism	A family of ideas holding that the inequality produced by industrial capitalism is neither acceptable nor inevitable. `Unit 11`
solar energy	The energy arriving from the Sun, which is concentrated where its rays strike steeply and spread where they strike at an angle. `Unit 3`
Solidarity	The independent Polish trade union of 1980, suppressed under martial law and not destroyed. `Unit 28`
sovereignty	A state's authority to govern itself without outside interference. `Unit 30`
Space Race	The competition between the superpowers to achieve firsts in space, from 1957 to 1969. `Unit 25`
sphere of influence	A region in which one great power expects its interests to prevail. `Unit 21`
spinning jenny	A machine that let one worker turn many spindles at once, easing the shortage of thread. `Unit 8`
stalemate	A position in which neither side can make progress. `Unit 14`

state	A political unit with a territory, a population, a government and the capacity for relations with other states. Unit 11
steam engine	An engine that boils water and uses the pressure of the steam to drive a piston. Unit 8
steamship	A ship driven by steam, which could keep a schedule instead of waiting for the wind. Unit 8
steel	Iron with a controlled small amount of carbon, both hard and tough, made cheaply from the nineteenth century. Unit 8
strike	The withdrawal of labour by workers acting together. Unit 11
structural funds	European money spent on infrastructure and development in poorer member regions. Unit 27
subsistence	Producing mainly enough to live on, with little surplus to sell. Unit 5
sufficient condition	Something which, by itself, is enough to produce an outcome. Unit 34
superpower	A state whose military and economic weight puts it in a different class from all others. Unit 20
supply chain	The sequence of places and firms through which materials, components and products pass before sale. Unit 31
supply line	The route by which an army is fed, fuelled and armed, and the limit on how far it can advance. Unit 18
supranational	Above the level of a single state, describing an institution whose decisions bind its members. Unit 27
taiga	The great coniferous forest of the north, the largest forest on Earth. Unit 3
tariff	A tax on imported goods. Unit 9
telegraph	A system sending messages as electrical pulses along a wire, which separated information from transport. Unit 8
telephone	A system carrying the voice itself, which made instant communication possible without trained operators. Unit 8
temperate forest	A biome of the middle latitudes with rain through the year and trees that often drop their leaves in autumn. Unit 3
testimony	A first-hand account given by somebody who was there, used alongside documents by historians. Unit 19
thermonuclear	A weapon using a fission explosion to trigger fusion, hundreds of times more powerful. Unit 24

time zone	A band of longitude within which clocks are set to the same time. Unit 2
time-space convergence	The shrinking of the time between places while the distance between them stays the same. Unit 32
trade union	An organisation of workers acting together to bargain over wages, hours and conditions. Unit 11
transit visa	Permission to pass through a country on the way to another, which is what Sousa Mendes issued in 1940. Unit 19
Treaty of Rome	The treaty of 1957 that created the European Economic Community. Unit 27
trench	A dug and revetted ditch from which soldiers fought and lived. Unit 14
trigger	The event that begins a crisis, as distinct from the causes that made the crisis dangerous. Unit 13
Triple Alliance	The pre-war alliance of Germany, Austria-Hungary and Italy. Unit 12
Triple Entente	The pre-war alignment of France, Russia and Britain. Unit 12
Tropic of Cancer	The parallel at about 23.5 degrees north, the furthest north the Sun can be directly overhead. Unit 1
Tropic of Capricorn	The parallel at about 23.5 degrees south, the furthest south the Sun can be directly overhead. Unit 1
tundra	A polar biome of low mosses, lichens and dwarf shrubs above permanently frozen ground. Unit 3
ultimatum	A final demand with a deadline, and a stated consequence if it is refused. Unit 13
unconditional surrender	Surrender without terms, which was the Allied requirement in the Second World War. Unit 20
United Nations	The international organisation founded in 1945, designed to correct the weaknesses of the League. Unit 20
urbanisation	The growth of the share of a population living in towns and cities. Unit 7
vertical integration	Owning every stage of production, from raw material to finished product. Unit 9
veto	The power of a permanent member of the Security Council to block a decision. Unit 20
war guilt	The clause of the Treaty of Versailles requiring Germany to accept responsibility for the loss and damage of the war. Unit 15
Warsaw Pact	The alliance of the Soviet Union and its allies, formed in 1955. Unit 21

water frame	A spinning machine driven by a water wheel, which forced production out of the cottage and into the mill. `Unit 8`
weather	What the atmosphere is doing now or this week. It is forecast, and it changes hour by hour. `Unit 3`
Western Front	The continuous line of trenches from the Belgian coast to the Swiss border, held for most of the First World War. `Unit 14`
wolfram	The ore from which tungsten is obtained, used to harden steel and exported by neutral Portugal to both sides. `Unit 18`

Sources and references

Every figure, date and statistic in this book was checked against a source named here before it was printed. The QR destinations are listed too, so the book works without a device.

Written and data sources

1. **European Centre for Medium-Range Weather Forecasts**, ERA5 reanalysis, daily 2 m temperature and total precipitation, 1991 to 2020. Retrieved through the Open-Meteo historical archive API, 2026-08-08, and aggregated to monthly means here. *The five climate graphs in chapter 3: Manaus, Lisbon, In Salah, Yakutsk and Utqiagvik*
2. **Open-Meteo**, Historical weather archive API. archive-api.open-meteo.com, accessed 2026-08-08. *The delivery route for the ERA5 daily series above*
3. **Royal Museums Greenwich**, The Royal Observatory and the Prime Meridian. rmg.co.uk. *Chapters 1 and 2: the choice of a prime meridian and the 1884 conference*
4. **Encyclopaedia Britannica**, Equator, biome, imperialism, capitalism, liberalism, nation-state, globalisation and the period entries. britannica.com. *Definitions and period summaries checked against the text of chapters 1 to 34*
5. **Imperial War Museums**, First World War and Second World War collections. iwm.org.uk. *Chapters 14 and 18: trench conditions, the Battle of the Atlantic, and the use of objects as evidence*
6. **United States Holocaust Memorial Museum**, Learning resources, survivor testimony and documentation. ushmm.org. *Chapter 19 throughout, and the treatment of testimony as a source*
7. **United Nations**, History of the United Nations, and the Universal Declaration of Human Rights. un.org. *Chapter 20: the design of the United Nations as an answer to the League*
8. **European Union**, The history of the EU, decade by decade. european-union.europa.eu. *Chapter 27: the Treaty of Rome, enlargement, and the accession of Portugal and Spain in 1986*
9. **UK Parliament**, Early factory legislation, living heritage. parliament.uk. *Chapter 7: the sequence of Factory Acts and the appointment of inspectors*
10. **NASA**, The Apollo programme mission record. nasa.gov. *Chapter 25: Apollo 11 and the missions before it*
11. **Our World in Data**, Economic growth, population growth and transport. ourworldindata.org. *Chapters 5, 4 and 32: long-run output, population distribution and how goods actually move*
12. **Royal Geographical Society**, Geographical case studies and resources. rgs.org. *Chapter 34: the relationship between physical setting and human activity*
13. **Portuguese national record**, Dates of the first railway, the Regeneration, the British Ultimatum, the Republic, the coup of 1926, the Estado Novo constitution and the accession of 1986. Checked against the general reference works above, not against the issuing body. *The Portugal in the global story sections in chapters 6 to 29*

Scanned destinations

1. Encyclopaedia Britannica, the Equator. www.britannica.com/science/Equator
2. Royal Observatory Greenwich, Royal Museums Greenwich. www.rmg.co.uk/royal-observatory
3. Encyclopaedia Britannica, biomes. www.britannica.com/science/biome
4. Our World in Data, world population growth. ourworldindata.org/world-population-growth
5. Our World in Data, economic growth. ourworldindata.org/economic-growth
6. Encyclopaedia Britannica, the Industrial Revolution. www.britannica.com/event/Industrial-Revolution
7. UK Parliament, early factory legislation. www.parliament.uk/about/living-heritage/transformingsociety/livinglearning/19thcentury/overview/earlyfactorylegislation/
8. Encyclopaedia Britannica, the steam engine. www.britannica.com/technology/steam-engine
9. Encyclopaedia Britannica, the rise of big business in the United States. www.britannica.com/place/United-States/The-rise-of-big-business
10. Encyclopaedia Britannica, capitalism. www.britannica.com/topic/capitalism
11. Encyclopaedia Britannica, liberalism. www.britannica.com/topic/liberalism
12. Encyclopaedia Britannica, imperialism. www.britannica.com/topic/imperialism

13. Encyclopaedia Britannica, the First World War. www.britannica.com/event/World-War-I
14. Imperial War Museums, the First World War. www.iwm.org.uk/history/first-world-war
15. Encyclopaedia Britannica, the Treaty of Versailles. www.britannica.com/event/Treaty-of-Versailles-1919
16. Encyclopaedia Britannica, the Great Depression. www.britannica.com/event/Great-Depression
17. Encyclopaedia Britannica, the Second World War. www.britannica.com/event/World-War-II
18. Imperial War Museums, the Second World War. www.iwm.org.uk/history/second-world-war
19. United States Holocaust Memorial Museum, learning resources. www.ushmm.org/learn
20. United Nations, the history of the UN. www.un.org/en/about-us/history-of-the-un
21. Encyclopaedia Britannica, the Cold War. www.britannica.com/event/Cold-War
22. Encyclopaedia Britannica, the Berlin Wall. www.britannica.com/topic/Berlin-Wall
23. Encyclopaedia Britannica, the Cuban missile crisis. www.britannica.com/event/Cuban-missile-crisis
24. Encyclopaedia Britannica, arms races. www.britannica.com/topic/arms-race
25. NASA, the Apollo programme. www.nasa.gov/the-apollo-program/
26. Encyclopaedia Britannica, Portugal: the Carnation Revolution. www.britannica.com/place/Portugal/The-Carnation-Revolution
27. European Union, the history of the EU. european-union.europa.eu/principles-countries-history/history-eu_en
28. Encyclopaedia Britannica, Mikhail Gorbachev. www.britannica.com/biography/Mikhail-Gorbachev
29. Encyclopaedia Britannica, German reunification. www.britannica.com/event/German-reunification
30. United Nations, about us. www.un.org/en/about-us
31. Encyclopaedia Britannica, globalisation. www.britannica.com/topic/globalization
32. Our World in Data, transport. ourworldindata.org/transport
33. Encyclopaedia Britannica, the nation-state. www.britannica.com/topic/nation-state
34. Royal Geographical Society. www.rgs.org/

For teachers

What this book assumes, how it is built, and what it deliberately does not do. This section is for the adult teaching from it.

What this book assumes

It assumes Year 7 and Year 8 of this course, or equivalent teaching. Year 7 asked how civilisations begin; Year 8 asked how states, empires and networks develop. Year 9 asks how the modern world was created, and the introduction says so to the pupil on the first page.

It assumes no prior knowledge of any particular period. Every chapter defines its own terms, and the glossary carries every word the book teaches.

How the year is structured

Ten parts, thirty-four chapters, 176 numbered topics.

Part one, chapters 1 to 4, is geography: global position, coordinates and time, climate and biomes, and the hemispheres. It exists so that the rest of the book can ask **where** and **why there** of every event.

Parts two to nine, chapters 5 to 29, are one unbroken causal line from the world before industrialisation to the collapse of the Soviet Union. The line is printed in the introduction, and no link in it is skipped. Chapters 10, 11 and 12 in particular exist so that the First World War is explained rather than merely dated.

Part ten, chapters 30 to 34, brings the book to the present and then back to its own first chapter.

Topics run on within a chapter; a new page begins a chapter, not a topic.

Portugal in the global story

Seventeen chapters end with a section under this name, plus chapter 26, which is a Portuguese case study of democratisation and decolonisation. Each section answers the same four questions: what was happening in Portugal, what was happening in the world, what part geography played, and what the connection was.

Portugal is never taught as a parallel national history. It is the same chronology seen from here, and the closing figure of the book, in topic 34.6, places the two lines on one axis so that pupils can read any year vertically.

Assessment

Closed questions have answers at the back. Open questions and the chapter enquiries have acceptance criteria instead, and the criteria are what should be marked.

A good enquiry answer, at this age, does four things: it takes a position, it gives evidence from the book, it states the strongest objection to itself, and it says what would have to be true for it to be wrong. Several enquiries in this book explicitly ask for the third and fourth of those, and pupils should be credited for changing their minds with a reason.

Three enquiries are deliberately unresolved and should be discussed rather than marked: the atomic bomb decision in 20.2, the appeasement debate in 16.7, and the question of obedience and refusal in 19.5.

How the Holocaust is taught here

Chapter 19 is a full chapter, not a section inside a military narrative. It sets out the ideology, the steps of persecution, the ghettos, the deportations and the camps, and it names the other groups the Nazi state persecuted. It is explicit that the racial theory was pseudo-science, and that the evidence for the events is overwhelming and denial is not a historical position.

The illustration programme in that chapter is deliberately restrained: place, absence and record rather than atrocity. The chapter opens with a note to the pupil saying what the chapter is about and that there will be time to ask questions.

What this book does not do

It does not teach the Renaissance, the Age of Discovery, absolutism or the French Revolution. Those belong to Year 8 in this course, and the first edition of this title covered them; the second edition was rebuilt to the current scheme of work.

It does not pretend that history is settled. Where historians disagree, the book says so and gives both cases: Versailles, appeasement, the atomic bomb, and whether deterrence made the world safer.

It does not use invented data. Every plotted number came from a named source, and where a figure is an estimate the book says so and explains why estimates differ.

A note on the sources

The five climate graphs are 1991 to 2020 monthly means computed from the ERA5 reanalysis through the Open-Meteo historical archive, aggregated from daily data rather than copied from a summary table. Every QR destination was opened in a real browser and returned HTTP 200 before the build, and each is used exactly once in the book. The full list is printed in the sources section.

Index

Where each term is taught, read off the printed pages of this edition. 241 entries.

A · 17

B · 12

C · 32

K · 1

L · 6

M · 16

N · 12

O · 4

P · 20

R · 10

S · 29